

Petrucci's

GENERAL CHEMISTRY

Principles and Modern Applications

TWELFTH EDITION

Petrucci | Herring | Madura | Bissonnette



1
1A
18
8A

Period

*Lanthanide series	57	La	58	Ce	59	Pr	60	Nd	61	Pm	62	Sm	63	Eu	64	Gd	65	Tb	66	Dy	67	Ho	68	Er	69	Tm	70	Yb	71	Lu	
	138.91		140.12		140.91		144.24		(145)		150.36		151.96		157.25		158.93		162.50		164.93		167.26		168.93		173.01		174.97		
	†Actinide series	89	Ac	90	Th	91	Pa	92	U	93	Np	94	Pu	95	Am	96	Cm	97	Bk	98	Cf	99	Es	100	Fm	101	Md	102	No	103	Lr
		(227)		232.04		231.04		238.03		(237)		(244)		(243)		(247)		(247)		(251)		(252)		(257)		(258)		(259)		(262)	

The scheme for numbering of groups is explained on page 80. The metals are and the nonmetals are . Metalloids are indicated by . The noble gases are .

Notes:

1. Atomic masses are from Pure Appl. Chem. Vol. 85, No. 5, pp. 1047–1078, 2013. They are given here with five significant figures where possible.
2. For H, Li, B, C, N, O, Mg, Si, S, Cl, Br, and Ti, the *conventional atomic mass*, a representative value from the *atomic mass interval*, is provided. (See page 77.)

PETRUCCI'S

GENERAL CHEMISTRY

PRINCIPLES AND MODERN APPLICATIONS

TWELFTH EDITION

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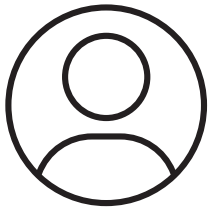
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DEDICATION

Pearson would like to dedicate this edition to Ralph H. Petrucci, F. Geoffrey Herring, Jeffry D. Madura, and Carey Bissonnette, who have passed away since the publication of the eleventh edition. Thank you for your wonderful contributions to the world of chemistry.

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Ralph Petrucci received his B.S. in Chemistry from Union College, Schenectady, NY, and his Ph.D. from the University of Wisconsin–Madison. Following ten years of teaching, research, consulting, and directing the NSF Institutes for Secondary School Science Teachers at Case Western Reserve University, Cleveland, OH, Dr. Petrucci joined the planning staff of the new California State University campus at San Bernardino in 1964. There, in addition to his faculty appointment, he served as Chairman of the Natural Sciences Division and Dean of Academic Planning. Professor Petrucci also coauthored *General Chemistry* with John W. Hill, Terry W. McCreary, and Scott S. Perry.

F. Geoffrey Herring

Geoff Herring received both his B.Sc. and his Ph.D. in Physical Chemistry, from the University of London. He served as a Professor Emeritus in the Department of Chemistry of the University of British Columbia, Vancouver. Dr. Herring had research interests in biophysical chemistry and published more than 100 papers in physical chemistry and chemical physics. Dr. Herring also pursued studies in the use of information technology and interactive engagement methods in teaching general chemistry with a view to improving student comprehension and learning. Dr. Herring taught chemistry from undergraduate to graduate levels for 30 years and was the recipient of the Killam Prize for Excellence in Teaching twice.

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Jeffry D. Madura was Professor and the Lambert F. Minucci Endowed Chair in Computational Sciences and Engineering in the Department of Chemistry and Biochemistry at Duquesne University located in Pittsburgh, PA. He earned a B.A. from Thiel College in 1980 and a Ph.D. in Physical Chemistry from Purdue University in 1985 under the direction of Professor William L. Jorgensen. The Ph.D. was followed by a postdoctoral fellowship in computational biophysics with Professor J. Andrew McCammon at the University of Houston. Dr. Madura's research interests were in computational chemistry and biophysics. He published more than 100 peer-reviewed papers in physical chemistry and chemical physics. Dr. Madura taught chemistry to undergraduate and graduate students for 24 years and was the recipient of a Dreyfus Teacher-Scholar Award. Dr. Madura was the recipient of the 2014 American Chemical Society Pittsburgh Section Award and received the Bayer School of Natural and Environmental Sciences and the Duquesne University Presidential Award for Excellence in Scholarship in 2007. Dr. Madura was an ACS Fellow and a Fellow of the Royal Society of Chemistry. He also worked with high school students and teachers as part of the ACS Science Coaches program.

Carey Bissonnette

Carey Bissonnette was Continuing Lecturer in the Department of Chemistry at the University of Waterloo, Ontario. He received his B.Sc. from the University of Waterloo in 1989 and his Ph.D. in 1993 from the University of Cambridge in England. His research interests were in the development of methods for

modeling dynamical processes of polyatomic molecules in the gas phase. He won awards for excellence in teaching, including the University of Waterloo's Distinguished Teacher Award in 2005. Dr. Bissonnette made extensive use of technology in both the classroom and the laboratory to create an interactive environment for his students to learn and explore. He was also engaged in undergraduate curriculum development, high-school liaison activities, and the coordination of the university's high-school chemistry contests, written each year by students around the world.

Preface

“Know your audience.” We have tried to follow this important advice by attending even more to the needs of students who are taking a serious journey through this material. We also know that most general chemistry students have career interests not in chemistry but in other areas such as biology, medicine, engineering, environmental science, and agricultural sciences. And we understand that general chemistry will be the only university or college chemistry course for some students, and thus their only opportunity to learn some practical applications of chemistry. We have designed this book for all these students.

Students of this text should have already studied some chemistry. But those with no prior background and those who could use a refresher will find that the early chapters develop fundamental concepts from the most elementary ideas. Students who do plan to become professional chemists will also find opportunities in the text to pursue their own special interests.

The typical student may need help identifying and applying principles and visualizing their physical significance. The pedagogical features of this text are designed to provide this help. At the same time, we hope the text serves to sharpen students’ skills in problem solving and critical thinking. Thus, we have tried to strike the proper balances between principles and applications, qualitative and quantitative discussions, and rigor and simplification.

Throughout the text and in Mastering (www.mlm.pearson.com/global/) we provide real-world examples to enhance the discussion. Examples relevant to the biological sciences, engineering, and the environmental sciences are found in numerous places. This should help to bring chemistry alive for these students and help them understand its relevance to their career interests. It also, in most cases, should help them master core concepts.

ORGANIZATION

In this edition we retain the core organization of the previous edition with one exception. Based on extensive feedback from subject experts, we have moved the chapter on chemical kinetics forward in the text to Chapter 15.

Like the previous edition, this edition begins with a brief overview of core concepts in Chapter 1. Then, we introduce atomic theory, including the periodic table, in Chapter 2. The periodic table is an extraordinarily useful tool, and presenting it early allows us to use the periodic table in different ways throughout the early chapters of the text. In Chapter 3, we introduce chemical compounds and their stoichiometry. Organic compounds are included in this presentation. The early introduction of organic compounds allows us to use organic examples throughout the book. Chapters 4 and 5 introduce chemical reactions. We discuss gases in Chapter 6, partly because they are familiar to students (which helps them build confidence), but also because some instructors prefer to cover this material early to better integrate their lecture and lab programs. (Chapter 6 can easily be deferred for coverage with the other states of matter, in Chapter 12.)

In Chapter 7, we introduce thermochemistry and discuss the energy changes that accompany physical and chemical transformations. Chapter 8 introduces quantum mechanical concepts that are needed to understand the energy changes we encounter at the atomic level. This chapter includes a discussion of wave mechanics, although this topic may be omitted at the instructor’s discretion.

Collectively, Chapters 8 through 11 provide the conceptual basis for describing the electronic structure of atoms and molecules, and the physical and chemical properties of these entities. The properties of atoms and molecules are then used in Chapter 12 to rationalize the properties of liquids and solids. Chapter 13 introduces the concept of entropy, the criteria for predicting the direction of spontaneous change, and the thermodynamic equilibrium condition. In Chapters 14–20, we apply and extend concepts introduced in Chapter 13. However, Chapters 14–20 can be taught without explicitly covering, or referring back to, Chapter 13.

As with previous editions, we have emphasized real-world chemistry in the final chapters that cover descriptive chemistry (Chapters 21–24), and we have tried to make this material easy to bring forward into earlier parts of the text. Moreover, many topics in these chapters can be covered selectively, without requiring the study of entire chapters. The text ends with comprehensive chapters on nuclear chemistry (Chapter 25) and organic chemistry (Chapters 26 and 27). Please note that an additional chapter on biochemistry (Chapter 28) is available online.

NEW TO THIS EDITION

- Based on suggestions from subject experts, the table of contents has been reorganized to improve flow.
- Twenty new profiles of eminent scientists (listed on page 15) cover their notable contributions and contextualize concepts discussed in the surrounding text. These profiles have been added to better represent diversity in the scientific community.
- The text has been updated to reflect the latest revisions to the periodic table.
- The artwork has been updated with many new or modified illustrations and photographs.
- The overall design of the text has been updated and enhanced for a more modern, accessible layout.
- The end-of-chapter assessment materials have been refreshed, with over 400 of them revised or updated.
- The Item Library in Mastering™ Chemistry has been expanded to offer greater coverage of the end-of-chapter assessment materials.

Accessibility

Pearson works continuously to ensure our products are as accessible as possible to all students. Currently, we are working toward achieving WCAG 2.0 AA for our existing products (2.1 AA for future products) and Section 508 standards, as expressed in the Pearson Guidelines for Accessible Educational Web Media (<https://www.pearson.com/uk/accessibility.html>).

OVERALL APPROACH

The pedagogical apparatus and overall approach in this edition continue to reflect contemporary thoughts on how best to teach general chemistry. We have retained the following key features of the text:

- *Logical approach to solving problems.* All worked examples are presented consistently throughout the text by using a tripartite structure of Analyze–Solve–Assess. This presentation not only encourages students to use a logical approach in solving problems but also provides them with a way to start when they are trying to solve a problem that may seem, at first, impossibly difficult. The approach is used implicitly by those who have had plenty of practice solving problems, but for those who are just starting out, the Analyze–Solve–Assess structure will serve to remind students to (1) analyze the information and plan a strategy, (2) implement the strategy, and (3) check or assess their answer to ensure that it is a reasonable one.
- *Integrative Practice Examples and End-of-Chapter Exercises.* Users of previous editions have given us very positive feedback about the quality of the integrative examples at the end of each chapter and the variety of the end-of-chapter exercises. Two practice examples (Practice Example A and Practice Example B) are available with every Integrative Example in the text. In several chapters, at least 10 new exercises have been added; and in many chapters, 20 or more exercises have been added.
- *Use of IUPAC recommendations.* We are pleased that our book serves the needs of instructors and students around the globe. Because communication among scientists in general, and chemists in particular, is made easier when we agree to use the same terms and notations, we have decided to follow—with relatively few exceptions—recommendations made by the International Union of Pure and Applied Chemistry (IUPAC). In particular, the version of the periodic table that now appears throughout the text is based on the one currently endorsed by IUPAC. An important feature is the use of IUPAC-recommended symbols and units for thermodynamic quantities. For example, standard enthalpies of reaction are represented by the symbol $\Delta_r H^\circ$ (not ΔH_r°) and are expressed in kJ mol^{-1} (not kJ).

FEATURES OF THIS EDITION

We have made a careful effort with this edition to incorporate features that will facilitate the teaching and learning of chemistry.

Chapter Opener

Each chapter opens with listing of the main headings to provide a convenient overview of the chapter's **Contents**. The opener also contains a list of numbered **Learning Objectives** that correspond with the main sections of the chapter.

Key Terms

Key terms are boldfaced where they are defined in the text. A **Glossary** of key terms with their definitions is presented in Appendix F.

Matter: Its Properties and Measurement

1

From the clinic that tests chemical dependency to a theoretical physicist's search for the last remaining "no chemical address," chemistry and chemicals were an integral part of life, even if it was only in the mind. A chemical is a substance that has the ability to change the chemical nature of a local system no matter. All kinds of chemical changes are possible, even if it is only in the mind. A chemical is a substance that has the ability to change the chemical nature of a local system no matter. All kinds of chemical changes are possible, even if it is only in the mind. A chemical is a substance that has the ability to change the chemical nature of a local system no matter. All kinds of chemical changes are possible, even if it is only in the mind.



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 1.1 The Scientific Method
 1.2 Matter and Change
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 1.99 Describing Matter
 2.00 Describing Change

Highlighted Boxes

Significant equations, concepts, and rules are highlighted against a color background for easy reference.

The result of multiplication or division may contain only as many significant figures as the least precisely known quantity in the calculation.

Concept Assessment

Concept Assessment questions (many of which are qualitative) are distributed throughout the body of the chapters. They enable students to test their understanding of basic concepts before proceeding further. Full solutions are provided in Appendix H.

2.4 Concept Assessment

What is the single exception to the statement that all atoms comprise protons, neutrons, and electrons?

EXAMPLE 14.5 Using Henry's Law

At 0 °C and an O_2 pressure of 1.00 atm, the aqueous solubility of $O_2(g)$ is 4.69 mL O_2 per liter. What is the molarity of O_2 in a saturated water solution when the O_2 is under its normal partial pressure in air, 0.209 atm?

Analyze Think of this as a two-part problem. (1) Determine the molarity of the saturated O_2 solution at 0 °C and 1 atm. (2) Use Henry's law in the manner just outlined.

Solve Determine the molarity of O_2 at 0 °C when $P_{O_2} = 1$ atm. We are given the information that, at an O_2 pressure of 1.00 atm, a saturated solution of O_2 in water contains 4.69 mL (0.0469 L) of O_2 . We also know that, at 0 °C and 1.00 atm, 1 mol O_2 occupies a volume of 22.4 L. Therefore,

$$\text{molarity} = \frac{0.0469 \text{ L}}{1 \text{ L water}} \times \frac{1 \text{ mol}}{22.4 \text{ L}} = 2.1 \times 10^{-3} \frac{\text{mol}}{\text{L}} = 2.1 \times 10^{-3} \text{ M}$$

Evaluate the Henry's law constant:

$$k = \frac{C}{P_{O_2}} = \frac{2.1 \times 10^{-3} \text{ M}}{1.00 \text{ atm}}$$

Apply Henry's law:

$$C = k \times P_{O_2} = 2.1 \times 10^{-3} \frac{\text{M}}{\text{atm}} \times 0.209 \text{ atm} = 4.37 \times 10^{-4} \text{ M}$$

Assess When working problems involving gaseous solutes in a solution in which the solute is at very low concentration, use Henry's law.

PRACTICE EXAMPLE A: Use data from Example 14.5 to determine the partial pressure of O_2 above an aqueous solution at 0 °C known to contain 5.50 mg O_2 per 100.0 mL of solution.

PRACTICE EXAMPLE B: A handbook lists the solubility of carbon monoxide in water at 0 °C and 1 atm pressure as 0.035 mL CO per milliliter of H_2O . What pressure of $CO(g)$ must be maintained above the solution to obtain 0.030 M CO ?

► Other atomic symbols not based on English names include Cu, Ag, Sn, Sb, Au, and Hg.

Examples with Practice Examples A and B

Worked-out Examples throughout the text illustrate how to apply the concepts. In many instances, a drawing or photograph is included to help students visualize what is going on in the problem. More importantly, all worked-out Examples now follow a tripartite structure of **Analyze–Solve–Assess** to encourage students to adopt a logical approach to problem solving.

Two **Practice Examples** are provided for each worked-out Example. The first, **Practice Example A**, provides immediate practice in a problem very similar to the given Example. The second, **Practice Example B**, often takes the student one step further than the given Example and is similar to the end-of-chapter problems in terms of level of difficulty. Answers to all the Practice Examples are given in Appendix G.

Marginal Notes

Marginal notes help clarify important points.

Keep In Mind Notes

Keep In Mind margin notes remind students about ideas introduced earlier in the text that are important to an understanding of the topic under discussion. In some instances they also warn students about common pitfalls.

Keep in Mind

that all we know is that the second oxide is twice as rich in oxygen as the first. If the first is CO , the possibilities for the second are CO_2 , C_2O_4 , C_3O_6 , and so on. (See also Exercise 18.)

Are You Wondering?

Are You Wondering? boxes pose and answer good questions that students often ask. Some are designed to help students avoid common misconceptions; others provide analogies or alternate explanations of a concept; and still others address apparent inconsistencies in the material that the students are learning. These topics can be assigned or omitted at the instructor's discretion.

1-1 ARE YOU WONDERING?

Why is it so important to attach units to a number?

In 1993, NASA started the Mars Surveyor program to conduct an ongoing series of missions to explore Mars. In 1995, two missions were scheduled that would be launched in late 1998 and early 1999. The missions were the Mars Climate Orbiter (MCO) and the Mars Polar Lander (MPL). The MCO was launched December 11, 1998, and the MPL, January 3, 1999.

What is the most abundant element? This seemingly simple question does not have a simple answer. To learn more about the abundances of elements in the universe and in the Earth's crust, go to the Focus On feature for Chapter 2, entitled Occurrence and Abundances of the Elements, in Mastering.

Summary

2-1 Early Chemical Discoveries and the Atomic Theory—Modern chemistry began with eighteenth-century experiments that showed that the mass and proportions of chemical combination, the law of conservation of mass, and the law of definite proportions were constant and proportional. These discoveries led to Dalton's atomic theory, which states that matter is made of small particles called atoms. The atoms of an element are identical to one another but different from atoms of all other elements. The chemical compounds are combination of atoms of different elements based on their mass. Dalton proposed that the relative mass and chemical combination of multiple proportions.

2-2 Electrons and Other Discoveries in Atomic Structure—The cathode ray experiment was carried out through the discovery and characterization of cathode rays by J. J. Thomson. Cathode rays are made of negatively charged particles called electrons. The mass-to-charge ratio (p. 2-7) and then the charge on electron (p. 2-8) were found. Important accidental discoveries were the discovery of cathode rays, radioactivity, and the principal types of radiation emitted from radioactive substances, α particles, β particles, and γ rays (p. 2-10).

2-3 The Nuclear Atom—Studies on the scattering of particles by the metal foil (pp. 2-11) led to the concept of the nuclear atom—a tiny, but massive, positively charged nucleus surrounded by lightweight negatively charged electrons (p. 2-12). A complete description of the nuclear atom is given in terms of its size, mass, and nature and structure. An individual atom is characterized in terms of its atomic number (proton number) Z and mass number A (pp. 2-13 and 2-14). The mass number is the sum of the masses of individual atoms and their component parts.

Integrative Example

An **Integrative Example** is provided near the end of each chapter. These challenging examples show students how to link various concepts from the chapter and earlier chapters to solve complex problems. Each Integrative Example is now accompanied by a **Practice Example A** and **Practice Example B**. Answers to these Practice Examples are given in Appendix G.

End-of-Chapter Questions and Exercises

Each chapter ends with four categories of questions:

Exercises are organized by topic subheads and are presented in pairs. Answers to selected questions (i.e., those numbered in red) are given in Appendix G.

Focus On Discussions

References are given near the end of each chapter to a **Focus On** essay that is found in Mastering (www.mlm.pearson.com/global/). These essays describe interesting and significant applications of the chemistry discussed in the chapter. They help show the importance of chemistry in all aspects of daily life.

Summary

A prose **Summary** is provided for each chapter. The Summary is organized by the main headings in the chapter and incorporates the key terms in boldfaced type.

[illegible][illegible]

Feature Problems

113. *Cinnamylaldehyde* is the chief constituent of cinnamon oil, which is obtained from the twigs and leaves of cinnamon tree grows in tropical regions. Cinnamylal is used in the manufacture of food flavors, perfumes, and cosmetics. The normal boiling point of a cinnamylal, $\text{C}_9\text{H}_8\text{O}$, is 248.3°C at 1 atm. At this temperature it begins to decompose. As a result, this book lists the boiling point of cinnamylal as a hypothetical normal boiling point. A method that can be used to avoid this difficulty is a variation of the Clausius-Clapeyron equation. A sample of cinnamylal is heated until the sum of the vapor pressure and the liquid pressure is equal to 1 atm. At this point, the temperature remains constant as the liquid vaporizes. The initial and final pressures to produce two immiscible liquids are equal is essentially the same and the other curve is a straight line. The following vapor pressures of cinnamylal are available: 10°C : 0.0001 mm Hg; 50°C : 0.001 mm Hg; 100°C : 0.005 mm Hg; 150°C : 0.01 mm Hg; and 220°C : 0.05 mm Hg of vapor pressure of water are

Temperature ($^\circ\text{C}$)	Pressure (mm Hg)
10	0.0001
50	0.001
100	0.005
150	0.01
220	0.05

Integrative and Advanced Exercises are more advanced than the preceding Exercises. They are not grouped by topic or type. They integrate material from sections of the chapter and sometimes from multiple chapters. In some instances, they introduce new ideas or pursue specific ideas further than is done in the chapter. Answers to selected questions (i.e., those numbered in red) are given in Appendix G.

Feature Problems require the highest level of skill to solve. Some deal with classic experiments; some require students to interpret data or graphs; some suggest alternative techniques for problem solving; some are comprehensive in their scope; and some introduce new material. These problems are a resource that can be used in several ways: for discussion in class, for individually assigned homework, or for collaborative group work. Answers to selected questions (i.e., those numbered in red) are given in Appendix G.

Exercises

Homogeneous and Heterogeneous Mixtures

- Which of the following do you expect to be the most water soluble, and why? C_4H_{10} , $Si(CH_3)_4$, C_4H_9Cl , $CaCl_2$, CH_3OH .
- Which of the following is moderately soluble both in water and in benzene, C_4H_{10} , and why? (a) C_4H_{10} is moderately soluble in benzene, but not in water. (b) C_4H_{10} is moderately soluble in water, but not in benzene. (c) C_4H_{10} is moderately soluble in both benzene and water. (d) C_4H_{10} is not soluble in either.
- Substance A is not soluble in water generally, but does dissolve in both solvents, however. One of the following is such a substance. Which do you think it is and why?



1,2-dichlorobenzene
(*o*-DCB)



1,2-cyclohexanediol
(*o*-CCHD)



Vitamin C

Self-Assessment Exercises are designed to help students review and prepare for some of the types of questions that often appear on quizzes and exams. Students can use these questions to decide whether they are ready to move on to the next chapter or first spend more time working with the concepts in the current chapter. Answers with explanations to selected questions (i.e., those numbered in red) are given in Appendix G.

Self-Assessment Exercises

77. For the following pairs of molecular orbitals, indicate for one you expect to have lower energy:
(a) σ_{2s} or σ_{2s}^* (b) σ_{2s} or σ_{2p}
(c) σ_{2p} or σ_{2p}^* (d) σ_{2p} or σ_{2p}^*

78. Match the central atom in each of the following species with one of the hybridization schemes: sp , sp^2 , sp^3 , and sp^3d .
(a) PF_5 (b) NO_2 (c) CS_2 (d) $SiCl_4$
(e) NO_3^- (f) SO_2 (g) SO_3 (h) SO_3^{2-}

79. Which of the following does $NOCl$ involve any delocalized π bonds?
(a) acetate anion (b) carbonyl anion
(c) nitric acid (d) nitrous acid
(e) none

80. A molecule in which sp^2 hybrid orbitals are used by the central atom in forming covalent bonds is (a) PCl_5 , (b) N_2 , (c) NO_2 , (d) H_2O , (e) CH_4 , (f) BF_3 , (g) SO_2 , (h) SO_3 , (i) NO_3^- , (j) CO_2 , (k) $CH_2=CH_2$, (l) $H_2C=O$, (m) $H_2C=CH_2$, (n) $H_2C=CH_2$, (o) $H_2C=CH_2$, (p) $H_2C=CH_2$, (q) $H_2C=CH_2$, (r) $H_2C=CH_2$, (s) $H_2C=CH_2$, (t) $H_2C=CH_2$, (u) $H_2C=CH_2$, (v) $H_2C=CH_2$, (w) $H_2C=CH_2$, (x) $H_2C=CH_2$, (y) $H_2C=CH_2$, (z) $H_2C=CH_2$.

81. The bond angle in NO_2 is best described as (a) between 120° and 125° , (b) less than 120° , (c) less than 180° , (d) less than 90° , (e) 180° , (f) 120° , (g) 125° , (h) 120° , (i) 125° , (j) 120° , (k) 125° , (l) 120° , (m) 125° , (n) 120° , (o) 125° , (p) 120° , (q) 125° , (r) 120° , (s) 125° , (t) 120° , (u) 125° , (v) 120° , (w) 125° , (x) 120° , (y) 125° , (z) 120° .

82. The hybridization scheme for the central atom in NO_2 is (a) sp , (b) sp^2 , (c) sp^3 , (d) sp^3d , (e) sp^3d^2 , (f) sp^3d^2 , (g) sp^3d^2 , (h) sp^3d^2 , (i) sp^3d^2 , (j) sp^3d^2 , (k) sp^3d^2 , (l) sp^3d^2 , (m) sp^3d^2 , (n) sp^3d^2 , (o) sp^3d^2 , (p) sp^3d^2 , (q) sp^3d^2 , (r) sp^3d^2 , (s) sp^3d^2 , (t) sp^3d^2 , (u) sp^3d^2 , (v) sp^3d^2 , (w) sp^3d^2 , (x) sp^3d^2 , (y) sp^3d^2 , (z) sp^3d^2 .

83. Of the following, the species with a bond order of 1 is (a) H_2 , (b) He_2 , (c) Li_2 , (d) Be_2 , (e) B_2 , (f) C_2 , (g) N_2 , (h) O_2 , (i) F_2 , (j) Ne_2 , (k) Na_2 , (l) Mg_2 , (m) Al_2 , (n) Si_2 , (o) P_2 , (p) S_2 , (q) Cl_2 , (r) Br_2 , (s) I_2 , (t) H_2 , (u) He_2 , (v) Li_2 , (w) Be_2 , (x) B_2 , (y) C_2 , (z) N_2 .

84. The hybridization scheme for Na_2 is (a) sp , (b) sp^2 , (c) sp^3 , (d) sp^3d , (e) sp^3d^2 , (f) sp^3d^2 , (g) sp^3d^2 , (h) sp^3d^2 , (i) sp^3d^2 , (j) sp^3d^2 , (k) sp^3d^2 , (l) sp^3d^2 , (m) sp^3d^2 , (n) sp^3d^2 , (o) sp^3d^2 , (p) sp^3d^2 , (q) sp^3d^2 , (r) sp^3d^2 , (s) sp^3d^2 , (t) sp^3d^2 , (u) sp^3d^2 , (v) sp^3d^2 , (w) sp^3d^2 , (x) sp^3d^2 , (y) sp^3d^2 , (z) sp^3d^2 .

85. Delocalized molecular orbitals are found in (a) H_2 , (b) He_2 , (c) Li_2 , (d) Be_2 , (e) B_2 , (f) C_2 , (g) N_2 , (h) O_2 , (i) F_2 , (j) Ne_2 , (k) Na_2 , (l) Mg_2 , (m) Al_2 , (n) Si_2 , (o) P_2 , (p) S_2 , (q) Cl_2 , (r) Br_2 , (s) I_2 , (t) H_2 , (u) He_2 , (v) Li_2 , (w) Be_2 , (x) B_2 , (y) C_2 , (z) N_2 .

86. Explain why the molecular structure of NO_2 cannot be adequately described through resonance involving sp and sp^2 orbitals.

87. Why does the hybridization sp^3d not account for bonding in the molecule NO_2 ? When hybridization scheme does work? Explain.

88. What is the total number of (a) σ bonds and (b) π bonds in the molecule CH_3CN ?

89. Which of the following species are paramagnetic? (a) H_2 , (b) He_2 , (c) Li_2 , (d) Be_2 , (e) B_2 , (f) C_2 , (g) N_2 , (h) O_2 , (i) F_2 , (j) Ne_2 , (k) Na_2 , (l) Mg_2 , (m) Al_2 , (n) Si_2 , (o) P_2 , (p) S_2 , (q) Cl_2 , (r) Br_2 , (s) I_2 , (t) H_2 , (u) He_2 , (v) Li_2 , (w) Be_2 , (x) B_2 , (y) C_2 , (z) N_2 .

90. Use the valence molecular orbital configuration to determine which of the following species is expected to have the lowest ionization energy: (a) C_2 , (b) C_2^+ , (c) C_2^- , (d) C_2^{2+} , (e) C_2^{2-} .

91. Use the valence molecular orbital configuration to determine which of the following species is expected to have the greatest electron affinity: (a) C_2 , (b) C_2^+ , (c) C_2^- , (d) C_2^{2+} , (e) C_2^{2-} .

92. Which of these diatomic molecules do you think has the greatest bond energy, I_2 , or C_2 ? Explain.

93. For each of the following sets of molecules, decide whether the structure is best described by a single Lewis structure or by resonance structures: (a) CO_2 , (b) SO_2 , (c) NO_2 , (d) NO_3^- , (e) SO_3 , (f) SO_3^{2-} , (g) CO_3^{2-} , (h) NO_2^+ , (i) NO_2^- , (j) NO_3^- , (k) NO_3^- , (l) NO_3^- , (m) NO_3^- , (n) NO_3^- , (o) NO_3^- , (p) NO_3^- , (q) NO_3^- , (r) NO_3^- , (s) NO_3^- , (t) NO_3^- , (u) NO_3^- , (v) NO_3^- , (w) NO_3^- , (x) NO_3^- , (y) NO_3^- , (z) NO_3^- .

94. Draw Lewis structures for the NO_2^+ and NO_2^- ions, and determine the hybrid geometry for each by using VSEPR theory. How does the hybridization of N differ as these ions appear?

95. In which of the following is the central atom sp hybridized? (a) $BeCl_2$, (b) BCl_3 , (c) CCl_4 , (d) NO_2 , (e) NO_3^- , (f) SO_2 , (g) SO_3 , (h) SO_3^{2-} , (i) CO_2 , (j) CH_4 , (k) $CH_2=CH_2$, (l) $H_2C=O$, (m) $H_2C=CH_2$, (n) $H_2C=CH_2$, (o) $H_2C=CH_2$, (p) $H_2C=CH_2$, (q) $H_2C=CH_2$, (r) $H_2C=CH_2$, (s) $H_2C=CH_2$, (t) $H_2C=CH_2$, (u) $H_2C=CH_2$, (v) $H_2C=CH_2$, (w) $H_2C=CH_2$, (x) $H_2C=CH_2$, (y) $H_2C=CH_2$, (z) $H_2C=CH_2$.

96. Which of the following can be used to explain why all bond distances and angles in methane, CH_4 , are the same? (a) resonance, (b) delocalization of electrons, (c) bond polarity, (d) electronegativity, (e) orbital hybridization.

97. According to molecular orbital theory, the O_2^{2+} ion has which of the following? (a) two unpaired electrons, (b) a bond order of two, (c) the highest-energy electron in a $2p$ orbital, (d) no π electrons, (e) all of these.

98. What is the angle between the hybrid orbitals obtained by combining the $2s$ and two $2p$ orbitals of an atom? (a) 90° , (b) 120° , (c) 180° , (d) 120° , (e) 180° , (f) 120° , (g) 180° , (h) 120° , (i) 180° , (j) 120° , (k) 180° , (l) 120° , (m) 180° , (n) 120° , (o) 180° , (p) 120° , (q) 180° , (r) 120° , (s) 180° , (t) 120° , (u) 180° , (v) 120° , (w) 180° , (x) 120° , (y) 180° , (z) 120° .

99. Compare the molecule with the Lewis structure given below.

(a) How many σ and π bonds are there?
(b) What is the appropriate hybridization scheme for each of C , N , and O ?

Appendices

The Appendices at the back of the book provide important information:

Appendix A succinctly reviews of some basic **Mathematical Operations**.

Appendix B concisely describes **Some Basic Physical Concepts**.

Appendix C summarizes the conventions of **SI Units**.

Appendix D provides five useful **Data Tables**.

Appendix E provides guidelines, along with an example, for constructing **Concept Maps**.

Appendix F consists of a **Glossary** of all the key terms in the book.

Appendix G provides **Answers to Practice Examples and Selected Exercises**.

Appendix H provides **Answers to Concept Assessment Questions**.

For easy reference, the **Periodic Table of Elements** is presented on the inside of the front cover.

For convenience, a Tabular Listing of Elements along with listings of **Selected Physical Constants**, **Some Common Conversion Factors**, **Some Useful Geometric Formulas**, and **Location of Important Data and Other Useful Information** are presented at the back of the textbook.

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- The **Selected Solutions Manual**, available in the Study Area of Mastering, provides full solutions to all the end-of-chapter exercises and problems that are numbered in red.

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WARNING: Many of the compounds and chemical reactions described or pictured in this book are hazardous. Do not attempt any experiment pictured or implied in the text except with permission in an authorized laboratory setting and under adequate supervision.

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Matter: Its Properties and Measurement

1

From the clinic that treats chemical dependency to a theatrical performance with good chemistry to the food label stating “no chemicals added,” chemistry and chemicals seem an integral part of life, even if everyday references to them are often misleading. A label implying the absence of chemicals in a food makes no sense. All foods consist entirely of chemicals, even if organically grown. In fact, all material objects—whether living or inanimate—are made up only of chemicals, and we should begin our study with that thought clearly in mind.

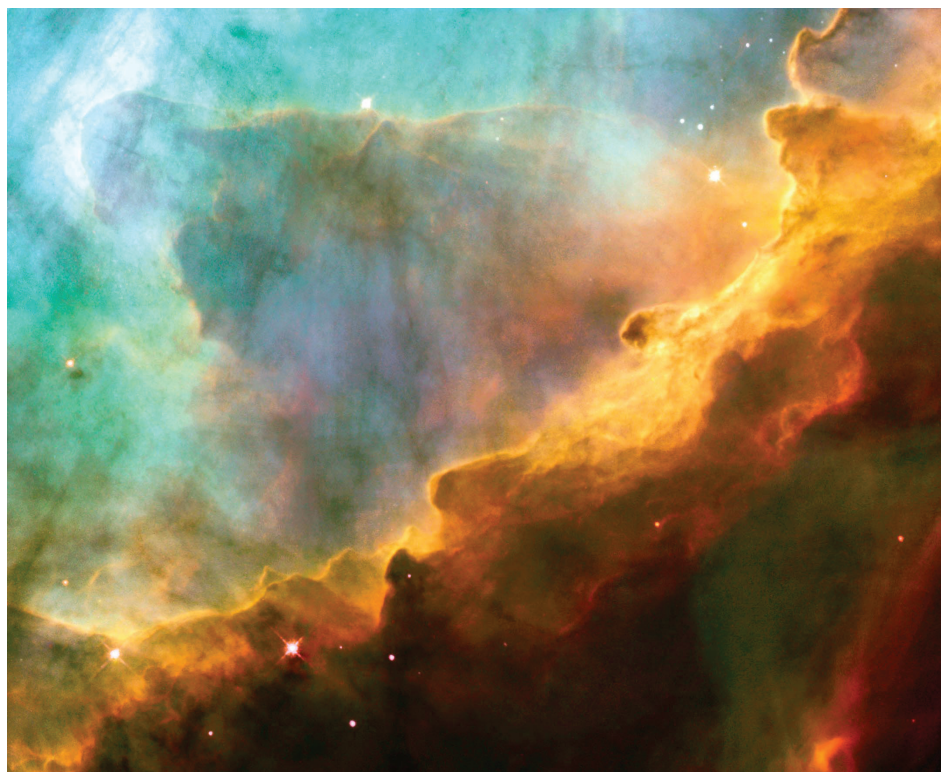
By manipulating materials in their environment, people have always practiced chemistry. Among the earliest applications were glazing pottery, smelting ores to produce metals, tanning hides, dyeing fabrics, and making cheese, wine, beer, and soap. With modern knowledge, though, chemists

Contents

- 1-1 The Scientific Method
- 1-2 Properties of Matter
- 1-3 Classification of Matter
- 1-4 Measurement of Matter: SI (Metric) Units
- 1-5 Density and Percent Composition: Their Use in Problem Solving
- 1-6 Uncertainties in Scientific Measurements
- 1-7 Significant Figures

Learning Objectives

- 1.1 Describe the purpose and process of the scientific method.
- 1.2 Discuss the meaning of *matter* and the changes it can undergo physically and chemically.
- 1.3 Classify matter based on its basic building blocks (atoms), and identify the three states of matter.
- 1.4 Identify the SI unit for length, mass, time, temperature, amount of substance, electric current, and luminous intensity.
- 1.5 Use percent composition and the relationship among density, volume, and mass as conversion factors in problem solving.
- 1.6 Differentiate between precision and accuracy.
- 1.7 Use the standard rules for significant figures to determine the number of significant figures needed at the end of a calculation.



ESA, J. Hester (ASU)/NASA

▲ A Hubble Space Telescope image of a cloud of hydrogen gas and dust (lower right half of the image) that is part of the Swan Nebula (M17). The colors correspond to light emitted by hydrogen (green), sulfur (red), and oxygen (blue). The chemical elements discussed in this text are those found on Earth and, presumably, throughout the universe.

can decompose matter into its smallest components (atoms) and reassemble those components into materials that do not exist naturally and that often exhibit unusual properties. Thus, motor fuels and thousands of chemicals used in the manufacture of plastics, synthetic fabrics, pharmaceuticals, and pesticides can all be made from petroleum. Modern chemical knowledge is also needed to understand the processes that sustain life and to understand and control processes that are detrimental to the environment, such as the formation of smog and the destruction of stratospheric ozone. Because it relates to so many areas of human endeavor, chemistry is sometimes called the *central science*.

Early chemical knowledge consisted of the “how to” of chemistry, discovered through trial and error. Modern chemical knowledge answers the “why” as well as the “how to” of chemical change. It is grounded in principles and theory, and mastering the principles of chemistry requires a systematic approach to the subject. Scientific progress depends on the way scientists do their work—asking the right questions, designing the right experiments to supply the answers, and formulating plausible explanations of their findings. We begin with a closer look into the scientific method.

1-1 The Scientific Method

Science differs from other fields of study in the *method* that scientists use to acquire knowledge and the special significance of this knowledge. Scientific knowledge can be used to explain natural phenomena and, at times, to *predict* future events.

The ancient Greeks developed some powerful methods of acquiring knowledge, particularly in mathematics. The Greek approach was to start with certain basic assumptions or premises. Then, by the method known as *deduction*, certain conclusions must logically follow. For example, if $a = b$ and $b = c$, then $a = c$. Deduction alone is not enough for obtaining scientific knowledge, however. The Greek philosopher Aristotle *assumed* four fundamental substances: air, earth, water, and fire. All other materials, he believed, were formed by combinations of these four elements. Chemists of several centuries ago (more commonly referred to as alchemists) tried, in vain, to apply the four-element idea to turn lead into gold. They failed for many reasons, one being that the four-element assumption is false.

The modern scientific method originated in the seventeenth century with such people as Galileo, Francis Bacon, Robert Boyle, and Isaac Newton. The key to the method is to make no initial assumptions, but rather to make careful observations of natural phenomena. When enough observations have been made so that a pattern begins to emerge, a generalization or natural law can be formulated describing the phenomenon. **Natural laws** are concise statements, often in mathematical form, about natural phenomena. The form of reasoning in which a general statement or natural law is inferred from a set of observations is called *induction*. For example, early in the sixteenth century, Polish astronomer Nicolaus Copernicus (1473–1543), through careful study of astronomical observations, concluded that Earth revolves around the sun in a circular orbit, although the general teaching of the time, not based on scientific study, was that the sun and other heavenly bodies revolved around Earth. We can think of Copernicus’s statement as a natural law. Another example of a natural law is the radioactive decay law, which dictates how long it takes for a radioactive substance to lose its radioactivity.

The success of a natural law depends on its ability to explain, or account for, observations and to predict new phenomena. Copernicus’s work was a great success because he was able to predict future positions of the planets more accurately than his contemporaries. We should not think of a natural law as an *absolute* truth, however. Future experiments may require us to modify the law. For example, Copernicus’s ideas were refined a half-century later by Johannes Kepler, who showed that planets travel in elliptical, not circular, orbits. To verify

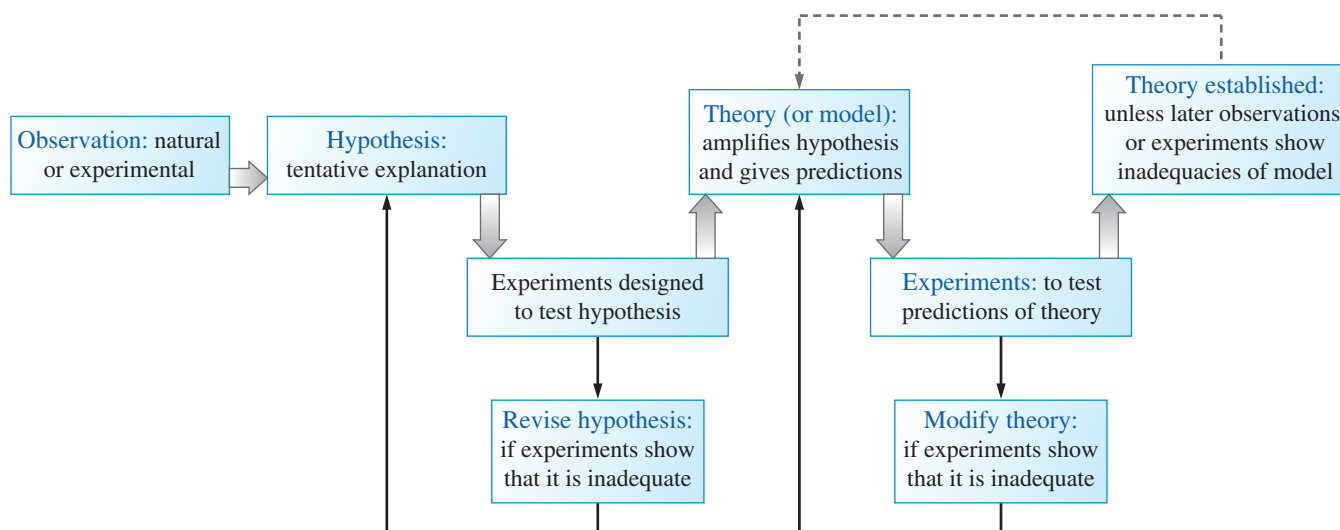


Figure 1-1 The scientific method illustrated

a natural law, a scientist designs *experiments* that show whether the conclusions deduced from the natural law are supported by experimental results.

A **hypothesis** is a tentative explanation of a natural law. If a hypothesis survives testing by experiments, it is often referred to as a theory. In a broader sense, a **theory** is a model or way of looking at nature that can be used to explain natural laws and make further predictions about natural phenomena. When differing or conflicting theories are proposed, the one that is most successful in its predictions is generally chosen. Also, the theory that involves the smallest number of assumptions—the simplest theory—is preferred. Over time, as new evidence accumulates, most scientific theories undergo modification, and some are discarded.

The **scientific method** is the combination of observation, experimentation, and the formulation of laws, hypotheses, and theories. The method is illustrated by the flow diagram in Figure 1-1. Scientists may develop a pattern of thinking about their field, known as a *paradigm*. Some paradigms may be successful at first but then become less so. When that happens, a new paradigm may be needed or, as is sometimes said, a *paradigm shift* occurs. In a way, the method of inquiry that we call the scientific method is itself a paradigm, and some people feel that it, too, is in need of change. That is, the varied activities of modern scientists are more complex than the simplified description of the scientific method presented here.* In any case, merely following a set of procedures, rather like using a cookbook, will not guarantee scientific success.

Another factor in scientific discovery is chance, or serendipity. Many discoveries have been made by accident. For example, in 1839, American inventor Charles Goodyear was searching for a treatment for natural rubber that would make it less brittle when cold and less tacky when warm. During this work, he accidentally spilled a rubber–sulfur mixture on a hot stove and found that the resulting product had exactly the properties he was seeking. Other chance discoveries include X-rays, radioactivity, and penicillin. So scientists and inventors always need to be alert to unexpected observations. Perhaps no one was more aware of this than Louis Pasteur, who wrote, “Chance favors the prepared mind.”



PirStamps/Alamy Stock Photo

▲ Louis Pasteur (1822–1895). This great practitioner of the scientific method was the developer of the germ theory of disease, the sterilization of milk by pasteurization, and vaccination against rabies. He has been called the greatest physician of all time by some. He was not a physician but a chemist—by training and by profession.



1-1 Concept Assessment

Is the common saying “The exception proves the rule” a good statement of the scientific method? Explain.

◀ Answers to Concept Assessment questions are given in Appendix H.

*W. Harwood, *JCST*, 33, 29 (2004). *JCST* is an abbreviation for *Journal of College Science Teaching*.

1-2 Properties of Matter

Dictionary definitions of chemistry usually include the terms *matter*, *composition*, and *properties*, as in the statement that “chemistry is the science that deals with the composition and properties of matter.” In this and the next section, we will consider some basic ideas relating to these three terms in hopes of gaining a better understanding of what chemistry is all about.

Matter is anything that occupies space and displays the properties of *mass* and *inertia*. Every human being is a collection of matter. We all occupy space, and we describe our mass in terms of weight, a related property. (Mass and weight are described in more detail in Section 1-4. Inertia is described in Appendix B.) All the objects that we see around us consist of matter. The gases of the atmosphere, even though they are invisible, are matter—they occupy space and have mass. Sunlight is *not* matter; rather, it is a form of energy. Energy is discussed in later chapters.

Composition refers to the parts or components of a sample of matter and their relative proportions. Ordinary water is made up of two simpler substances—hydrogen and oxygen—present in certain fixed proportions. A chemist would say that the composition of water is 11.19% hydrogen and 88.81% oxygen by mass. Hydrogen peroxide, a substance used in bleaches and antiseptics, is also made up of hydrogen and oxygen, but it has a different composition. Hydrogen peroxide is 5.93% hydrogen and 94.07% oxygen by mass.

Properties are those qualities or attributes that we can use to distinguish one sample of matter from others; and, as we consider next, the properties of matter are generally grouped into two broad categories: physical and chemical.

Physical Properties and Physical Changes

A **physical property** is one that a sample of matter displays without changing its composition. Thus, we can distinguish between the reddish brown solid, copper, and the yellow solid, sulfur, by the physical property of *color* (Fig. 1-2).

Another physical property of copper is that it can be hammered into a thin sheet of foil (see Figure 1-2). Solids having this ability are said to be *malleable*. Sulfur is not malleable. If we strike a chunk of sulfur with a hammer, it crumbles into a powder. Sulfur is *brittle*. Another physical property of copper that sulfur does not share is the ability to be drawn into a fine wire (ductility). Also, sulfur is a far poorer conductor of heat and electricity than is copper.

Sometimes a sample of matter undergoes a change in its physical appearance. In such a **physical change**, some of the physical properties of the sample may change, but its composition remains unchanged. When liquid water freezes into solid water (ice), it certainly looks different and, in many ways, it is different. Yet the water remains 11.19% hydrogen and 88.81% oxygen by mass.

Chemical Properties and Chemical Changes

In a **chemical change**, or **chemical reaction**, one or more kinds of matter are converted to new kinds of matter with different compositions. The key to

Figure 1-2 Physical properties of sulfur and copper

A lump of sulfur (left) crumbles into a yellow powder when hammered. Copper (right) can be obtained as large lumps of native copper, formed into pellets, hammered into a thin foil, or drawn into a wire.



Paul Silverman/Fundamental Photographs

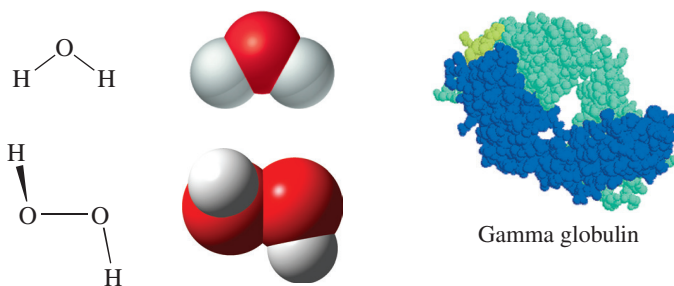
identifying chemical change, then, comes in observing a *change in composition*. The burning of paper involves a chemical change. Paper is a complex material, but its principal constituents are carbon, hydrogen, and oxygen. The chief products of the combustion are two gases, one consisting of carbon and oxygen (carbon dioxide) and the other consisting of hydrogen and oxygen (water, as steam). The ability of paper to burn is an example of a chemical property. A **chemical property** is the ability (or inability) of a sample of matter to undergo a change in composition under stated conditions.

Zinc reacts with hydrochloric acid solution to produce hydrogen gas and a solution of zinc chloride in water (Fig. 1-3). This reaction is one of zinc's distinctive chemical properties, just as the inability of gold to react with hydrochloric acid is one of gold's chemical properties. Sodium reacts not only with hydrochloric acid but also with water. In some of their physical properties, zinc, gold, and sodium are similar. For example, each is malleable and a good conductor of heat and electricity. In most of their chemical properties, though, zinc, gold, and sodium are quite different. Knowing these differences helps us to understand why zinc, which does not react with water, is used in roofing nails, roof flashings, and rain gutters, and sodium is not. Also, we can appreciate why gold, because of its chemical inertness, is prized for jewelry and coins: It does not tarnish or rust. In our study of chemistry, we will see why substances differ in properties and how these differences determine the ways in which we use them.

1-3 Classification of Matter

Matter is made up of very tiny units called **atoms**. Each different type of atom is the building block of a different chemical **element**. Presently, the International Union of Pure and Applied Chemistry (IUPAC) recognizes 118 elements. The known elements range from common substances, such as carbon, iron, and silver, to uncommon ones, such as lutetium and thulium. About 90 of the elements can be obtained from natural sources. The remainder do not occur naturally and have been created only in laboratories. On the inside front cover you will find a complete listing of the elements and also a special tabular arrangement of the elements known as the *periodic table*. The periodic table is the chemist's directory of the elements. We will describe it in Chapter 2 and use it throughout most of the text.

Chemical **compounds** are substances comprising atoms of two or more elements joined together. Scientists have identified millions of different chemical compounds. In some cases, we can isolate a molecule of a compound. A **molecule** is the smallest entity having the same proportions of the constituent atoms as does the compound as a whole. A molecule of water consists of three atoms: two hydrogen atoms joined to a single oxygen atom. A molecule of hydrogen peroxide has two hydrogen atoms and two oxygen atoms; the two oxygen atoms are joined together and one hydrogen atom is attached to each oxygen atom. By contrast, a molecule of the blood protein gamma globulin is made up of 19,996 atoms, but they are of just four types: carbon, hydrogen, oxygen, and nitrogen.



▲ Structures of water, hydrogen peroxide, and gamma globulin. Gamma globulin consists of three subunits (shown in different colors). Each subunit consists of carbon, hydrogen, oxygen, and nitrogen.



Diane Hirsch/Fundamental Photographs

Figure 1-3 A chemical property of zinc and gold: reaction with hydrochloric acid

The zinc-plated (galvanized) nail reacts with hydrochloric acid, producing the bubbles of hydrogen gas seen on its surface. The gold bracelet is unaffected by hydrochloric acid. In this photograph, the zinc plating has been consumed, exposing the underlying iron nail. The reaction of iron with hydrochloric acid imparts some color to the acid solution.

◀ The International Union of Pure and Applied Chemistry (IUPAC) is recognized as the world authority on chemical nomenclature, terminology, standardized methods for measurement, atomic mass, and more. Along with many other activities, IUPAC publishes journals, technical reports, and chemical databases, most of which are available at www.iupac.org.

◀ The identity of an atom is established by a feature called its atomic number (see Section 2-3). Characterizing “superheavy” elements is a daunting challenge; they are produced only a few atoms at a time and the atoms disintegrate almost instantaneously.



▲ Is it homogeneous or heterogeneous? When viewed through a microscope, homogenized milk is seen to consist of globules of fat dispersed in a watery medium. Homogenized milk is a *heterogeneous* mixture.

► It is composition, particularly its variability, that helps us distinguish the several classifications of matter.

► Solutions can be gaseous and liquids as described here, but they can also be solids. Some alloys are examples of solid solutions.

The composition and properties of an element or a compound are uniform throughout a given sample and from one sample to another. Elements and compounds are called **substances**. (In the chemical sense, the term *substance* should be used only for elements and compounds.) A *mixture* of substances can vary in composition and properties from one sample to another. One that is uniform in composition and properties throughout is said to be a **homogeneous mixture** or a *solution*. A given solution of sucrose (cane sugar) in water is uniformly sweet throughout the solution, but the sweetness of another sucrose solution may be rather different if the sugar and water are present in different proportions. Ordinary air is a homogeneous mixture of several gases, principally the *elements* nitrogen and oxygen. Seawater is a solution of the *compounds* water, sodium chloride (salt), and a host of others. Gasoline is a homogeneous mixture or solution of dozens of compounds.

In **heterogeneous mixtures**—sand and water, for example—the components separate into distinct regions. Thus, the composition and physical properties vary from one part of the mixture to another. Salad dressing, a slab of concrete, and the leaf of a plant are all heterogeneous. It is usually easy to distinguish heterogeneous from homogeneous mixtures. A scheme for classifying matter into elements and compounds and homogeneous and heterogeneous mixtures is summarized in Figure 1-4.

Separating Mixtures

A mixture can be separated into its components by appropriate physical means. Consider again the heterogeneous mixture of sand in water. When we pour this mixture into a funnel lined with porous filter paper, the water passes through and sand is retained on the paper. This process of separating a solid from the liquid in which it is suspended is called *filtration* (Fig. 1-5a). You will probably use this procedure in the laboratory. Conversely, we cannot separate a homogeneous mixture (solution) of copper(II) sulfate in water by filtration because all components pass through the paper. We can, however, boil the solution of copper(II) sulfate and water. In the process of *distillation*, a pure liquid is condensed from the vapor given off by a boiling solution. When all

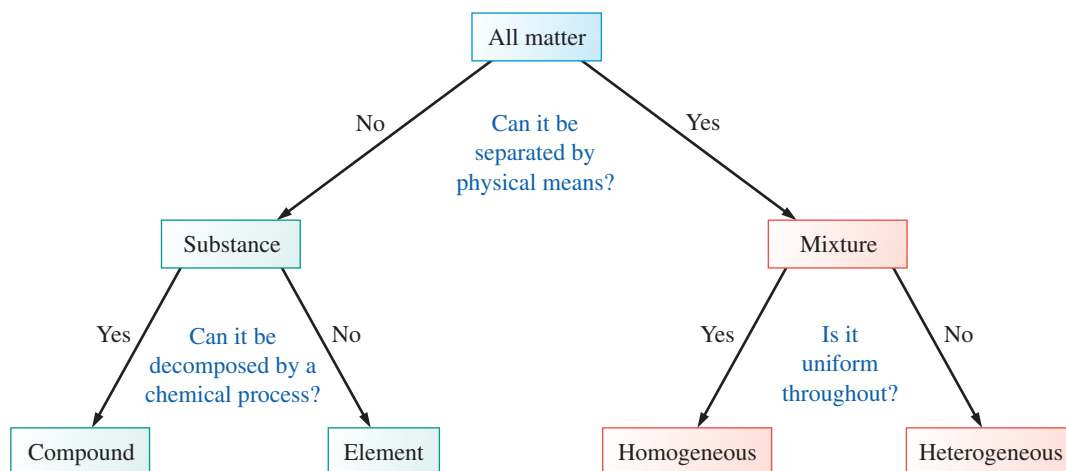


Figure 1-4 A classification scheme for matter

Every sample of matter is either a single substance (an element or compound) or a mixture of substances. At the molecular level, an element consists of atoms of a single type and a compound consists of two or more different types of atoms, usually joined into molecules. In a homogeneous mixture, atoms or molecules are randomly mixed at the molecular level. In heterogeneous mixtures, the components are physically separated, as in a layer of octane molecules (a constituent of gasoline) floating on a layer of water molecules.

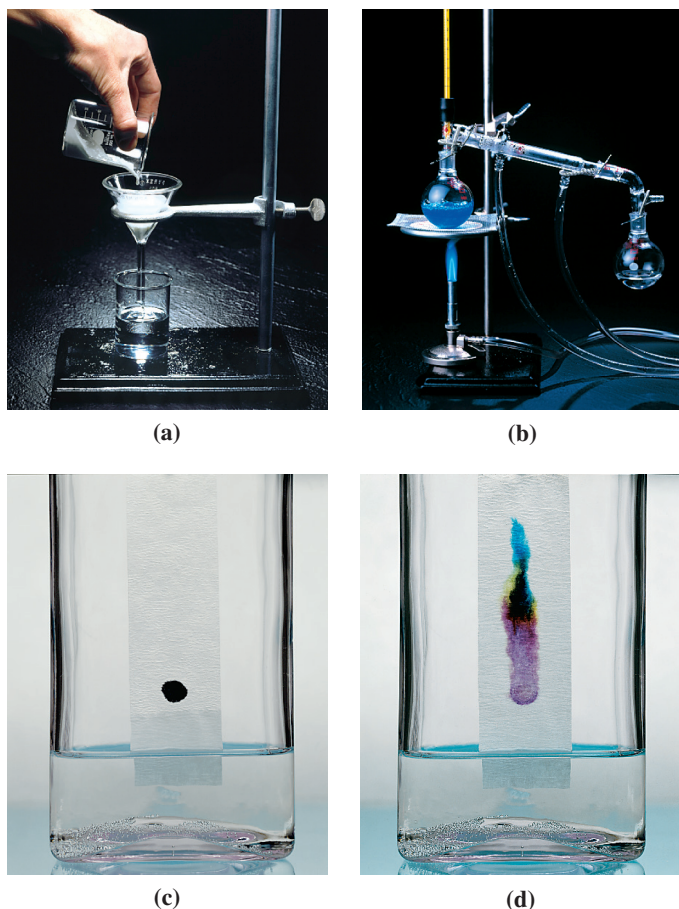


Figure 1-5 Separating mixtures:
a physical process

(a) Separation of a heterogeneous mixture by filtration: Solid copper(II) sulfate is retained on the filter paper, while liquid hexane passes through. (b) Separation of a homogeneous mixture by distillation: Copper(II) sulfate remains in the flask on the left as water passes to the flask on the right, by first evaporating and then condensing back to a liquid. (c) Separation of the components of ink by using chromatography: A dark spot of black ink can be seen just above the water line as water moves up the paper. (d) Water has dissolved the colored components of the ink, and these components are retained in different regions on the paper according to their differing tendencies to adhere to the paper.

(a) Carey B. Van Loon; (b) Carey B. Van Loon; (c) Richard Megna/Fundamental Photographs; (d) Richard Megna/Fundamental Photographs

the water has been removed by boiling a solution of copper(II) sulfate in water, solid copper(II) sulfate remains behind (Fig. 1-5b).

Another method of separation available to modern chemists depends on the differing abilities of compounds to adhere to the surfaces of various solid substances, such as paper and starch. The technique of *chromatography* relies on this principle. The dramatic results that can be obtained with chromatography are illustrated by the separation of ink on a filter paper (Fig. 1-5c, d).

Decomposing Compounds

A chemical compound retains its identity during physical changes, but it can be decomposed into its constituent elements by *chemical changes*. The decomposition of compounds into their constituent elements is a more difficult matter than the mere physical separation of mixtures. The extraction of iron from iron oxide ores requires a blast furnace. The industrial production of pure magnesium from magnesium chloride requires electricity. It is generally easier to convert a compound into other compounds by a chemical reaction than it is to separate a compound into its constituent elements. For example, when heated, ammonium dichromate decomposes into the substances chromium(III) oxide, nitrogen, and water. This reaction, once used in movies to simulate a volcano, is illustrated in Figure 1-6.

States of Matter

Matter is generally found in one of three *states*: solid, liquid, or gas. In a **solid**, atoms or molecules are in close contact, sometimes in a highly organized arrangement called a *crystal*. A solid has a definite shape. In a **liquid**, the atoms or molecules are usually separated by somewhat greater distances than in a solid. Movement of these atoms or molecules gives a liquid its most distinctive property—the ability to flow, covering the bottom and assuming the shape of its container. In a **gas**, distances between atoms or molecules are much greater



Figure 1-6 A chemical
change: decomposition of
ammonium dichromate

Carey B. Van Loon

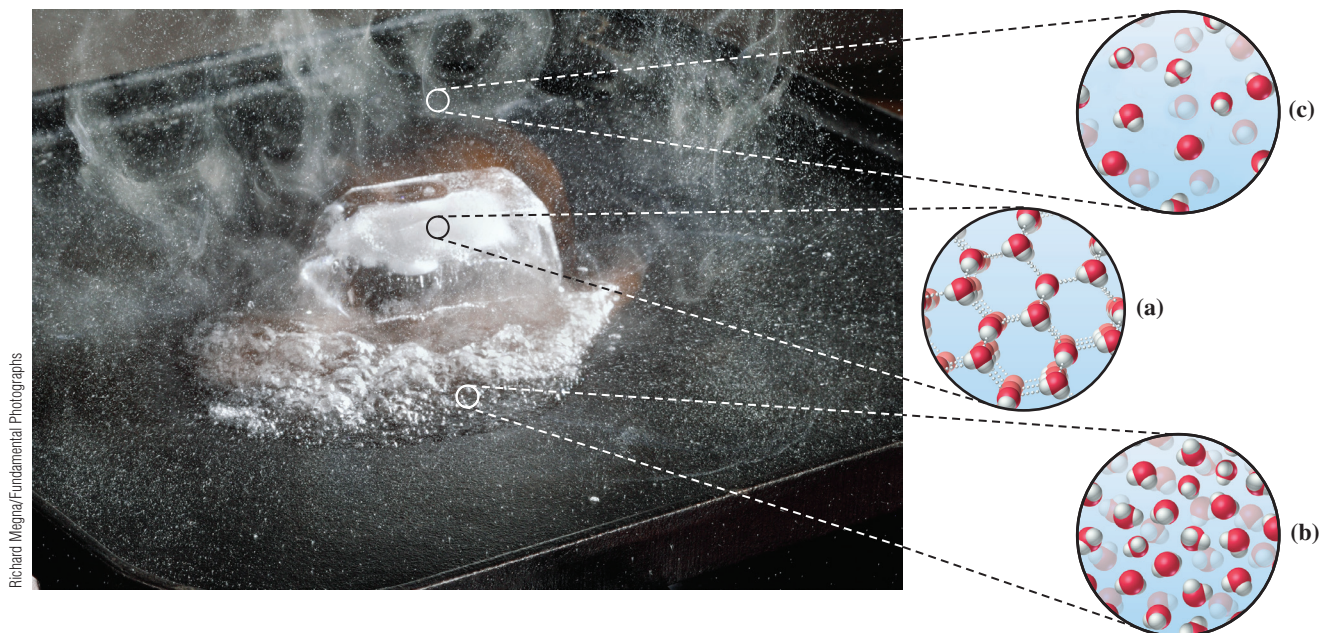


Figure 1-7 Macroscopic and microscopic views of matter

The picture shows a block of ice on a heated surface and the three states of water. The circular insets show how chemists conceive of these states microscopically, in terms of molecules with two hydrogen atoms joined to one of oxygen. In ice **(a)**, the molecules are arranged in a regular pattern in a rigid framework. In liquid water **(b)**, the molecules are rather closely packed but move freely. In gaseous water **(c)**, the molecules are widely separated.

than in a liquid. A gas always expands to fill its container. Depending on conditions, a substance may exist in only one state of matter, or it may be present in two or three states. Thus, as the ice in a small pond begins to melt in the spring, water is in two states: solid and liquid (actually, three states if we also consider water vapor in the air above the pond). The three states of water are illustrated at two levels in Figure 1-7.

The *macroscopic level* refers to how we perceive matter with our eyes, through the outward appearance of objects. The *microscopic level* describes matter as chemists conceive of it—in terms of atoms and molecules and their behavior. In this text, we will describe many macroscopic, observable properties of matter, but to explain these properties, we will often shift our view to the atomic or molecular level—the microscopic level.

1-4 Measurement of Matter: SI (Metric) Units

► Nonnumerical information is *qualitative*, such as the color blue.

► The *definition of the meter*, formerly based on the atomic spectrum of ^{86}Kr , was changed to the speed of light in 1983. Effectively, the speed of light is now defined as $2.99792458 \times 10^8 \text{ m/s}$.

Chemistry is a *quantitative* science, which means that in many cases we can measure a property of a substance and compare it with a standard having a known value of the property. We express the measurement as the product of a *number* and a *unit*. The unit indicates the standard against which the measured quantity is being compared. When we say that the length of the playing field in football is 100 yd, we mean that the field is 100 times as long as a standard of length called the yard (yd). In this section, we will introduce some basic units of measurement that are important to chemists.

The scientific system of measurement is called the *Système Internationale d'Unités* (International System of Units) and is abbreviated **SI**. It is a modern version of the metric system, a system based on the unit of length called a *meter* (m). The meter was originally defined as 1/10,000,000 of the distance from the equator to the North Pole and translated into the length of a metal bar kept in Paris. Unfortunately, the length of the bar is subject to change with temperature, and it cannot be exactly reproduced. The SI system substitutes for the

Table 1.1 SI Base Quantities

Physical Quantity	Unit	Symbol
Length	meter ^a	m
Mass	kilogram	kg
Time	second	s
Temperature	kelvin	K
Amount of substance	mole ^b	mol
Electric current ^c	ampere	A
Luminous intensity ^d	candela	cd

^aThe official spelling of this unit is “metre,” but we will use the American spelling.
^bThe mole is introduced in Section 2-7.
^cElectric current is described in Appendix B and in Chapter 20.
^dLuminous intensity is not discussed in this text.

standard meter bar an unchanging, reproducible quantity: 1 meter is the distance traveled by light in a vacuum in 1/299,792,458 of a second. Length is one of the seven fundamental quantities in the SI system (see Table 1.1). All other physical quantities have units that can be derived from these seven. SI is a *decimal* system. Quantities differing from the base unit by powers of ten are noted by the use of prefixes. For example, the prefix *kilo* means “one thousand” (10^3) times the base unit; it is abbreviated as k. Thus 1 *kilometer* = 1000 meters, or 1 km = 1000 m. The SI prefixes are listed in Table 1.2.

Most measurements in chemistry are made in SI units. Sometimes we must convert between SI units, as when converting kilometers to meters. At other times we must convert measurements expressed in non-SI units into SI units, or from SI units into non-SI units. In all these cases, we can use a *conversion factor* or a series of conversion factors in a scheme called a conversion pathway. Later in this chapter, we will apply conversion pathways in a method of problem solving known as *dimensional analysis*. The method itself is described in some detail in Appendix A.

Mass

Mass describes the quantity of matter in an object. In SI the standard of mass is 1 *kilogram* (kg), which is a fairly large unit for most applications in chemistry. More commonly we use the unit *gram* (g).

Weight is the force of gravity on an object. It is directly proportional to mass, as shown in the following mathematical expressions.

$$W \propto m \text{ and } W = g \times m \quad (1.1)$$

An object has a fixed mass (m), which is independent of where or how the mass is measured. Its weight (W), however, may vary because the acceleration caused by gravity (g) varies slightly from one point on Earth to another. Thus, an object that weighs 100.0 kg in St. Petersburg, Russia, weighs only 99.6 kg in Panama (about 0.4% less). The same object would weigh only about 17 kg on the moon. Although the weight of an object varies from place to place, its mass is the same in all locations. The terms *weight* and *mass* are often used interchangeably, but only mass is a measure of the quantity of matter. A common laboratory device for measuring mass is called a balance. A balance is often called, incorrectly, a scale.

The principle used in a balance is that of counteracting the force of gravity on an unknown mass with a force of equal magnitude that can be precisely measured. In older two-pan beam balances, the object whose mass is being determined is placed on one pan and counterbalancing is achieved through the force of gravity acting on *weights*, objects of precisely known mass, placed on the other pan. In the type of balance most commonly seen in laboratories today—the electronic balance—the counterbalancing force is a magnetic force produced by passing an electric current through an electromagnet. First, an initial balance condition is achieved when no object is present on the balance pan.

Table 1.2 SI Prefixes

Multiple	Prefix
10^{18}	exa (E)
10^{15}	peta (P)
10^{12}	tera (T)
10^9	giga (G)
10^6	mega (M)
10^3	kilo (k)
10^2	hecto (h)
10^1	deka (da)
10^{-1}	deci (d)
10^{-2}	centi (c)
10^{-3}	milli (m)
10^{-6}	micro (μ) ^a
10^{-9}	nano (n)
10^{-12}	pico (p)
10^{-15}	femto (f)
10^{-18}	atto (a)
10^{-21}	zepto (z)
10^{-24}	yocto (y)

^aThe Greek letter μ (pronounced “mew”).

◀ It is a good idea to *memorize the most common SI prefixes* (such as G, M, k, d, c, m, μ , n, and p) because you can’t survive in a world of science without knowing these SI prefixes.

◀ The symbol $\propto$ means “proportional to.” It can be replaced by an equality sign and a proportionality constant. In expression (1.1), the constant is the acceleration caused by gravity, g . (See Appendix B.)



▲ An electronic balance.

► Electromagnetic radiation is discussed in Section 8-1.

When the object to be weighed is placed on the pan, the initial balance condition is upset. To restore the balance condition, additional electric current must be passed through the electromagnet. The magnitude of this additional current is proportional to the mass of the object being weighed and is translated into a mass reading that is displayed on the balance. An electronic balance is shown in the margin.



1-2 Concept Assessment

Would either the two-pan beam balance or the electronic balance yield the same result for the mass of an object measured on the moon as that measured for the same object on Earth? Explain.

Time

In daily use we measure time in seconds, minutes, hours, and years, depending on whether we are dealing with short intervals (such as the time for a 100 m race) or long ones (such as the time before the next appearance of Halley's comet in 2062). We can use all these units in scientific work also, although in SI the standard of time is the *second* (s). A time interval of 1 second is not easily established. At one time it was based on the length of a day, but this is not constant because the rate of Earth's rotation undergoes slight variations. In 1956, the second was defined as $1/31,556,925.9747$ of the length of the year 1900. With the advent of atomic clocks, a more precise definition became possible. The second is now defined as the duration of 9,192,631,770 cycles of a particular radiation emitted by certain atoms of the element cesium (cesium-133).

Temperature

To establish a temperature scale, we arbitrarily set certain fixed points and temperature increments called degrees. Two commonly used fixed points are the temperature at which ice melts and the temperature at which water boils, both at standard atmospheric pressure.*

On the **Celsius** scale, the melting point of ice is 0°C , the boiling point of water is 100°C , and the interval between is divided into 100 equal parts called Celsius degrees. On the **Fahrenheit** temperature scale, the melting point of ice is 32°F , the boiling point of water is 212°F , and the interval between is divided into 180 equal parts called Fahrenheit degrees. Figure 1-8 compares the Fahrenheit and Celsius temperature scales.

The SI temperature scale, called the **Kelvin** scale, assigns a value of zero to the lowest possible temperature. The zero on the Kelvin scale is denoted 0 K and it comes at -273.15°C . We will discuss the Kelvin temperature scale in detail in Chapter 6. For now, it is enough to know the following:

- The interval on the Kelvin scale, called a *kelvin*, is the same size as the Celsius degree.
- When writing a Kelvin temperature, we do not use a degree symbol. That is, we write 0 K or 300 K, not 0°K or 300°K .
- The Kelvin scale is an absolute temperature scale; there are no negative Kelvin temperatures.

In the laboratory, temperature is most commonly measured in Celsius degrees; however, these temperatures must often be converted to the Kelvin scale (in describing the behavior of gases, for example). Occasionally, particularly in some engineering applications, temperatures must be converted

*Standard atmospheric pressure is defined in Section 6-1. The effect of pressure on melting and boiling points is described in Chapter 12.

► The SI symbol for Kelvin temperature is T and that for Celsius temperature is t but shown here as $t(^{\circ}\text{C})$. The Fahrenheit temperature, shown here as $t(^{\circ}\text{F})$, is not recognized in SI.

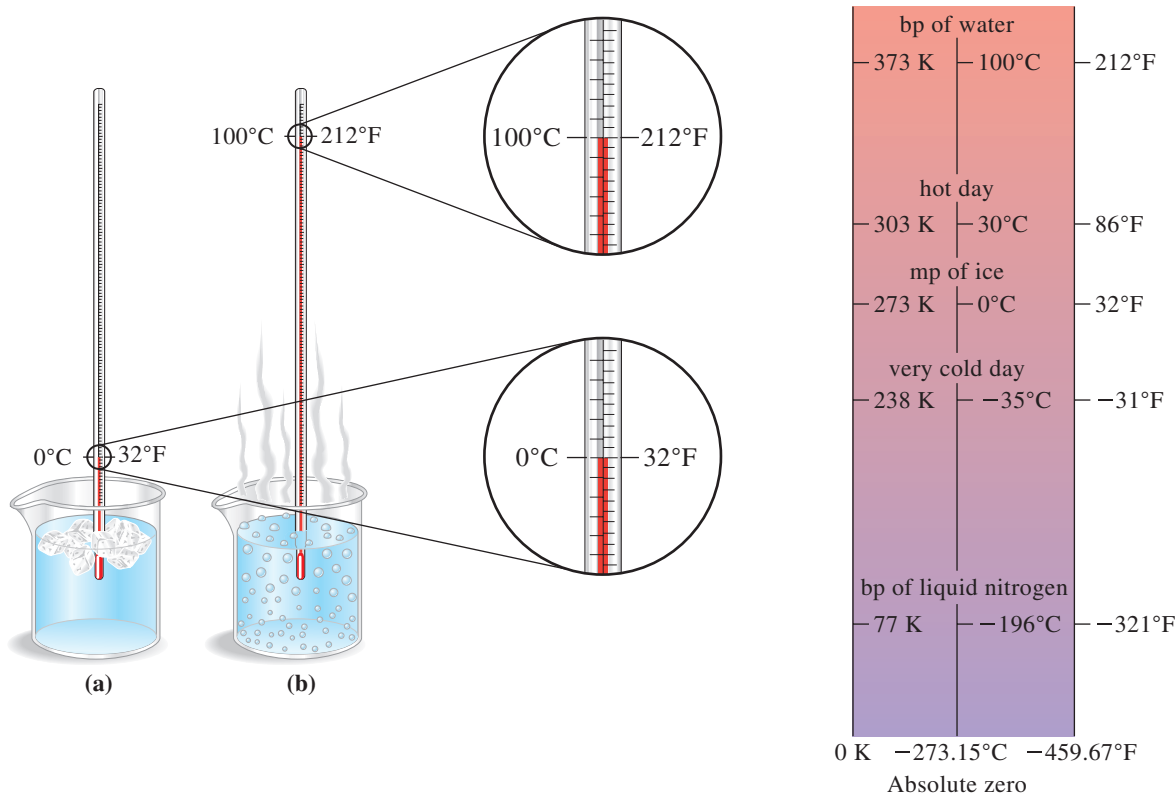


Figure 1-8 A comparison of temperature scales

(a) The melting point (mp) of ice. (b) The boiling point (bp) of water.

between the Celsius and Fahrenheit scales. Temperature conversions can be made in a straightforward way by using the algebraic equations shown below.

$$\text{Kelvin from Celsius} \quad T(\text{K}) = t(^{\circ}\text{C}) + 273.15$$

$$\text{Fahrenheit from Celsius} \quad t(^{\circ}\text{F}) = \frac{9}{5} t(^{\circ}\text{C}) + 32$$

$$\text{Celsius from Fahrenheit} \quad t(^{\circ}\text{C}) = \frac{5}{9} [t(^{\circ}\text{F}) - 32]$$

The factors $\frac{9}{5}$ and $\frac{5}{9}$ arise because the Celsius scale uses 100 degrees between the two chosen reference points and the Fahrenheit scale uses 180 degrees: $180/100 = \frac{9}{5}$ and $100/180 = \frac{5}{9}$. The diagram in Figure 1-8 illustrates the relationship among the three scales for several temperatures.

EXAMPLE 1-1

Converting Between Fahrenheit and Celsius Temperatures

The predicted high temperature for New Delhi, India, on a given day is 41°C. Is this temperature higher or lower than the predicted daytime high of 103°F for the same day in Phoenix, Arizona, reported by a newscaster?

Analyze

We are given a Celsius temperature and seek a comparison with a Fahrenheit temperature. To convert the given Celsius temperature to a Fahrenheit temperature, we use the equation given previously that expresses $t(^{\circ}\text{F})$ as a function of $t(^{\circ}\text{C})$.

Solve

$$t(^{\circ}\text{F}) = \frac{9}{5} t(^{\circ}\text{C}) + 32 = \frac{9}{5} (41) + 32 = 106^{\circ}\text{F}$$

The predicted temperature for New Delhi, 106°F, is 3°F higher than for Phoenix, 103°F.

(continued)

Assess

For temperatures at which $t(^{\circ}\text{C}) > -40^{\circ}\text{C}$, the Fahrenheit temperature is greater than the Celsius temperature. If the Celsius temperature is lower than -40°C , then $t(^{\circ}\text{F})$ is lower than (more negative than) $t(^{\circ}\text{C})$ (Fig. 1-8). Concept Assessment 1-3 asks you to think further about the relationship between $t(^{\circ}\text{C})$ and $t(^{\circ}\text{F})$.

PRACTICE EXAMPLE A: A recipe in an American cookbook calls for roasting a cut of meat at 350°F . What is this temperature on the Celsius scale?

PRACTICE EXAMPLE B: A particular automobile engine coolant has antifreeze protection to a temperature of -22°C . Will this coolant offer protection at temperatures as low as -15°F ?

Answers to Practice Examples are given in Appendix G.

**1-3 Concept Assessment**

Can there be a temperature at which $^{\circ}\text{C}$ and $^{\circ}\text{F}$ have the same value?
Can there be more than one such temperature? Explain.

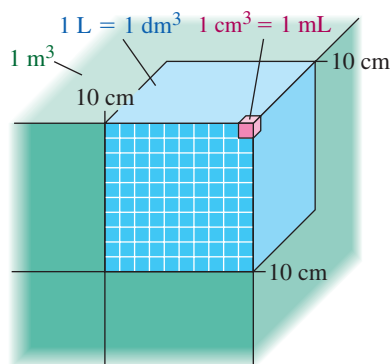


Figure 1-9 Some metric volume units compared

The largest volume, shown in part, is the SI standard of 1 cubic meter (m^3). A cube with a length of 10 cm (1 dm) on edge (in blue) has a volume of 1000 cm^3 (1 dm^3) and is called 1 liter (1 L). The smallest cube is 1 cm on edge (red) and has a volume of $1 \text{ cm}^3 = 1 \text{ mL}$.

► The official spelling is *litre*, but we will use the American spelling, *liter*.

Derived Units

The seven units listed in Table 1.1 are the SI units for the fundamental quantities of length, mass, time, and so on. Many measured properties are expressed as combinations of these fundamental, or base, quantities. We refer to the units of such properties as *derived units*. For example, velocity is a distance divided by the time required to travel that distance. The unit of velocity is length divided by time, such as m/s or ms^{-1} . Some derived units have special names. For example, the combination $\text{kg m}^{-1} \text{s}^{-2}$ is called the *pascal* (Chapter 6) and the combination $\text{kg m}^2 \text{s}^{-2}$ is called the *joule* (Chapter 7). Other examples are given in Appendix C. An important measurement that uses derived units is *volume*. Volume has the unit $(\text{length})^3$, and the SI standard unit of volume is the *cubic meter* (m^3). More commonly used volume units are the *cubic centimeter* (cm^3) and the *liter* (L). One liter is defined as a volume of 1000 cm^3 , which means that one *milliliter* (1 mL) is equal to 1 cm^3 . The liter is also equal to one *cubic decimeter* (1 dm^3). Several volume units are depicted in Figure 1-9.

Non-SI Units

Although its citizens are growing more accustomed to expressing distances in kilometers and volumes in liters, the United States is one of the few countries where most units used in everyday life are still non-SI. Masses are given in pounds, room dimensions in feet, and so on. In this book, we will not routinely use these non-SI units, but we will occasionally introduce them in examples and end-of-chapter exercises. In such cases, any necessary relationships between non-SI and SI units will be given or can be found on the inside back cover.

**1-1 ARE YOU WONDERING?****Why is it so important to attach units to a number?**

In 1993, NASA started the Mars Surveyor program to conduct an ongoing series of missions to explore Mars. In 1995, two missions were scheduled that would be launched in late 1998 and early 1999. The missions were the Mars Climate Orbiter (MCO) and the Mars Polar Lander (MPL). The MCO was launched December 11, 1998, and the MPL, January 3, 1999.

Nine and a half months after launch, the MCO was to fire its main engine to achieve an elliptical orbit around Mars. The MCO engine start occurred on September 23, 1999, but the MCO mission was lost when the orbiter entered the Martian atmosphere on a lower-than-expected trajectory. The MCO entered the low orbit because the computer on Earth used British Engineering units, whereas the MCO computer used SI units.

This error in units brought the MCO 56 km above the surface of Mars instead of the desired 250 km. At 250 km, the MCO would have successfully entered the desired elliptical orbit, and the \$168 million orbiter would probably not have been lost.

◀ The development of science requires careful *quantitative measurement*. Theories have stood or fallen based on their agreement or otherwise with experiments in the fourth significant figure or beyond. Problem solving, the handling of units, and the use of significant figures (Section 1-7) are important in all areas of science.

1-5

Density and Percent Composition: Their Use in Problem Solving

Throughout this text, we will encounter new concepts about the structure and behavior of matter. One means of firming up our understanding of these new concepts is to work problems that relate concepts that we already know to those we are trying to understand. In this section, we will introduce two quantities frequently required in problem solving: density and percent composition.

Density

Here is an old riddle: “What weighs more, a ton of bricks or a ton of cotton?” If you answer that they weigh the same, you demonstrate a clear understanding of the meaning of weight and, indirectly, of the quantity of matter to which weight is proportional, that is, mass. Anyone who answers that the bricks weigh more than the cotton has confused the concepts of weight and density. Matter in a brick is more concentrated than in cotton—that is, the matter in a brick is confined to a smaller volume. Bricks are more dense than cotton. **Density** is the ratio of mass to volume.

$$\text{density } (d) = \frac{\text{mass } (m)}{\text{volume } (V)} \quad (1.2)$$

Mass and volume are both extensive properties. An **extensive property** is *dependent* on the quantity of matter observed. However, if we divide the mass of a substance by its volume, we obtain density, an intensive property. An **intensive property** is *independent* of the amount of matter observed. Thus, the density of pure water at 25°C has a unique value, whether the sample fills a small beaker (small mass/small volume) or a swimming pool (large mass/large volume). Intensive properties are especially useful in chemical studies because they can often be used to identify substances.

The SI base units of mass and volume are kilograms and cubic meters, respectively, but chemists generally express mass in grams and volume in cubic centimeters or milliliters. Thus, the most commonly encountered density unit is grams per cubic centimeter (g/cm³) or the identical unit grams per milliliter (g/mL).

The mass of 1.000 L of water at 4°C is 1.000 kg. The density of water at 4°C is 1000 g/1000 mL, or 1.000 g/mL. At 20°C, the density of water is 0.9982 g/mL. Density is a function of temperature because volume varies with temperature, whereas mass remains constant. One reason that climate change is a concern is because as the average temperature of seawater increases, the seawater will become less dense, its volume will increase, and sea level will rise—even if no continental ice melts.

Like temperature, the state of matter affects the density of a substance. In general, solids are denser than liquids and both are denser than gases, but

Keep in Mind

that recognizing the *scale of things* is one important step in avoiding mistakes. If a solid density calculates out as 0.05 g/cm^3 , or a gas density as 5.0 g/cm^3 , review the work done up to that point!

there are notable overlaps in densities between solids and liquids. Following are the ranges of values generally observed for densities; this information should prove useful in solving problems.

- Solid densities: from about 0.2 g/cm^3 to 20 g/cm^3
- Liquid densities: from about 0.5 g/mL to $3\text{--}4 \text{ g/mL}$
- Gas densities: mostly in the range of a few grams per liter

In general, densities of liquids are known more precisely than those of solids (which may have imperfections in their microscopic structures). Also, densities of elements and compounds are known more precisely than densities of materials with variable compositions (such as wood or rubber).

The different densities of solids and liquids have several important consequences. A solid that is insoluble and floats on a liquid is *less* dense than the liquid, and it displaces a *mass* of liquid equal to its own mass. An insoluble solid that sinks to the bottom of a liquid is *more* dense than the liquid and displaces a *volume* of liquid equal to its own volume. Liquids that are immiscible in each other separate into distinct layers, with the most dense liquid at the bottom and the least dense liquid at the top.

**1-4 Concept Assessment**

Approximately what fraction of its volume is submerged when a 1.00 kg block of wood ($d = 0.68 \text{ g/cm}^3$) floats on water?

Keep in Mind

that in a conversion pathway, all units must cancel except for the desired unit in the final result (see Appendix A-5). Also, note that the quantities given and sought are typically *extensive* properties and the conversion factor(s) are often *intensive* properties (here, density).

Density in Conversion Pathways

If we measure the mass of an object and its volume, simple division gives us its density. Conversely, if we know the density of an object, we can use density as a conversion factor to determine the object's mass or volume. For example, a cube of osmium 1.000 cm on edge weighs 22.59 g . The density of osmium (the densest of the elements) is 22.59 g/cm^3 . What would be the mass of a cube of osmium that is 1.25 inches on edge ($1 \text{ in} = 2.54 \text{ cm}$)? To solve this problem, we begin by relating the volume of a cube to its length, that is, $V = l^3$. Then we can map out the *conversion pathway*:

inches of osmium $\longrightarrow$ cm osmium $\longrightarrow$ cm^3 osmium $\longrightarrow$ g osmium

$$\begin{array}{c} \text{(converts inches to cm)} \quad \text{(converts cm to cm}^3\text{)} \quad \text{(converts cm}^3\text{ to g osmium)} \\ \text{? g osmium} = \left[1.25 \cancel{\text{ in}} \times \frac{2.54 \cancel{\text{ cm}}}{1 \cancel{\text{ in}}} \right]^3 \times \frac{22.59 \text{ g osmium}}{1 \text{ cm}^3} = 723 \text{ g osmium} \end{array}$$

At 25°C the density of mercury, the only metal that is liquid at this temperature, is 13.5 g/mL . Suppose we want to know the volume, in mL, of 1.000 kg of mercury at 25°C . We proceed by (1) identifying the known information: 1.000 kg of mercury and $d = 13.5 \text{ g/mL}$ (at 25°C); (2) noting what we are trying to determine—a volume in milliliters (which we designate mL mercury); and (3) looking for the relevant conversion factors. Outlining the conversion pathway will help us find these conversion factors:

kg mercury $\longrightarrow$ g mercury $\longrightarrow$ mL mercury

We need the factor 1000 g/kg to convert from kilograms to grams. Density provides the factor to convert from mass to volume. But in this instance, we need to use density in the *inverted* form. That is,

$$\text{? mL mercury} = 1.000 \cancel{\text{ kg}} \times \frac{1000 \cancel{\text{ g}}}{1 \cancel{\text{ kg}}} \times \frac{1 \text{ mL mercury}}{13.5 \cancel{\text{ g}}} = 74.1 \text{ mL mercury}$$

Examples 1-2 and 1-3 further illustrate that numerical calculations involving density are generally of two types: determining density from mass and volume measurements and using density as a conversion factor to relate mass and volume.

EXAMPLE 1-2 Relating Mass, Volume, and Density

The stainless steel in the solid cylindrical rod pictured below has a density of 7.75 g/cm^3 . If we want a 1.00 kg mass of this rod, how long a section must we cut off? Refer to the inside back cover for the formula to calculate the volume of a cylinder.



Analyze

We are given the density, d , and the desired mass, m . Because $d = m/V$, we can solve for V and then use the formula for the volume of a cylinder, $V = \pi r^2 h$, to calculate h , the length of rod we seek. Two different mass units (g and kg) and two different length units (centimeters and inches) appear in the information given in this problem, so we anticipate having to make at least two unit conversions. To avoid errors, we include units in all steps.

Solve

Solve equation (1.2) for V . The reciprocal of density, $1/d$, is a conversion factor for converting from mass to volume.

$$V = \frac{m}{d} = m \times \frac{1}{d}$$

Calculate the volume of the rod that will have a mass of 1.00 kg . A conversion from kg to g is required in this step.

$$V = 1.00 \text{ kg} \times \frac{1000 \text{ g}}{1 \text{ kg}} \times \frac{1 \text{ cm}^3}{7.75 \text{ g}} = 129 \text{ cm}^3$$

Solve $V = \pi r^2 h$ for h and then calculate h . We must be certain to use the radius of the rod (one-half the diameter) and to express the radius in centimeters.

$$h = \frac{V}{\pi r^2} = \frac{129 \text{ cm}^3}{3.1416 \times (0.500 \text{ in} \times 2.54 \text{ cm/1 in})^2} = 25.5 \text{ cm}$$

Assess

One way to check whether our answer is correct is to work the problem in reverse. For example, we calculate $d = 1.00 \times 10^3 \text{ g} / [3.1416 \times (1.27 \text{ cm})^2 \times 25.5 \text{ cm}] = 7.74 \text{ g/cm}^3$, which is very close to the given density. We are confident that our answer, $h = 25.5 \text{ cm}$, is correct.

PRACTICE EXAMPLE A: To determine the density of trichloroethylene, a liquid used to degrease electronic components, a flask is first weighed empty (108.6 g). It is then filled with 125 mL of the trichloroethylene to give a total mass of 291.4 g . What is the density of trichloroethylene in grams per milliliter?

PRACTICE EXAMPLE B: Suppose that instead of using the cylindrical rod of Example 1-2 to prepare a 1.000 kg mass we were to use a solid spherical ball of copper ($d = 8.96 \text{ g/cm}^3$). What must be the radius of this ball?

EXAMPLE 1-3 Determining the Density of an Irregularly Shaped Solid

A chunk of coal is weighed twice while suspended from a spring scale (see Figure 1-10). When the coal is suspended in air, the scale registers 156 g ; when the coal is suspended underwater at 20°C , the scale registers 59 g . What is the density of the coal? The density of water at 20°C is 0.9982 g/cm^3 .

Analyze

We need the ratio of mass to volume of the chunk of coal. The mass of the coal is easily obtained; it is what registers on the scale when the coal is suspended in air: 156 g . But what is the volume of this chunk of coal? The key to this calculation is the weight measurement under water. The coal weighs less than 156 g when submerged in water because the water exerts a buoyant force on the coal. The buoyant force is the difference

(continued)

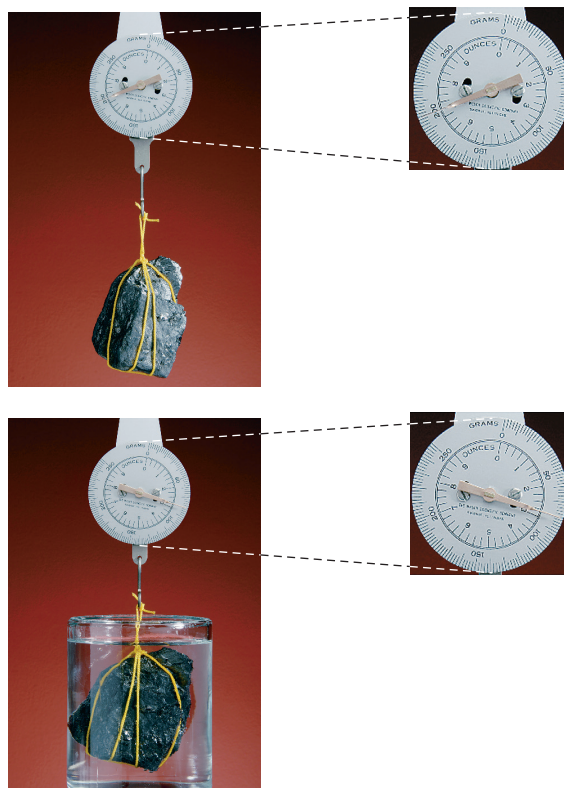


Figure 1-10 Measuring the volume of an irregularly shaped solid

When submerged in a liquid, an irregularly shaped solid displaces a volume of liquid equal to its own volume. The necessary data can be obtained by two mass measurements of the type illustrated here; the required calculations are like those in Example 1-3.

Kristen Brochmann/Fundamental Photographs

between the two weight measurements: $156 \text{ g} - 59 \text{ g} = 97 \text{ g}$. Recall the statement on page 42 that a submerged solid displaces a volume of water equal to its own volume. We don't know this volume of water directly, but we can use the mass of displaced water, 97 g , and its density, 0.9982 g/cm^3 , to calculate the volume of displaced water. The volume of the coal is equal to the volume of displaced water.

Solve

The mass of the chunk of coal is 156 g . If we use m_{water} to denote the mass of displaced water, then the volume of the displaced water is calculated as follows:

$$V = \frac{m_{\text{water}}}{d} = \frac{156 \text{ g} - 59 \text{ g}}{0.9982 \text{ g/cm}^3} = 97 \text{ cm}^3$$

The volume of the chunk of coal is the same as the volume of displaced water. Therefore, the density of the coal is

$$d = \frac{156 \text{ g}}{97 \text{ cm}^3} = 1.6 \text{ g/cm}^3$$

Assess

To determine the density of an object, we might think it is necessary to make measurements of both the mass and volume of the object. Example 1-3 shows that a volume measurement is not necessary. The steps in our calculation can be combined to give the following expression:

$$(\text{density of object})/(\text{density of water}) = (\text{weight in water})/(\text{weight in air} - \text{weight in water}).$$

The expression above clearly shows that the density of an object can be determined by making two weight measurements: one in air, and the other in a fluid (such as water) of known density.

PRACTICE EXAMPLE A: A graduated cylinder contains 33.8 mL of water. A stone with a mass of 28.4 g is placed in the cylinder and the water level rises to 44.1 mL . What is the density of the stone?

PRACTICE EXAMPLE B: In the situation shown in the photograph, when the ice cube melts completely, will the water overflow the container, will the water level in the container drop, or will the water level remain unchanged? Explain.



Richard Megna/Fundamental Photographs

Percent Composition as a Conversion Factor

In Section 1-2, we described composition as an identifying characteristic of a sample of matter. A common way of referring to composition is through percentages. Percent (*per centum*) is the Latin for *per* (meaning “for each”) and *centum* (meaning “100”). Thus, percent is the number of parts of a constituent in 100 parts of the whole. To say that a seawater sample contains 3.5% sodium chloride by mass means that there are 3.5 g of sodium chloride in every 100 g of the seawater. We make the statement in terms of grams because we are talking about percent *by mass*. We can express this percent by writing the following ratios:

$$\frac{3.5 \text{ g sodium chloride}}{100 \text{ g seawater}} \quad \text{and} \quad \frac{100 \text{ g seawater}}{3.5 \text{ g sodium chloride}} \quad (1.3)$$

In Example 1-4, we will use one of these ratios as a conversion factor.

EXAMPLE 1-4

Using Percent Composition as a Conversion Factor

A 75 g sample of sodium chloride (table salt) is to be produced by evaporating to dryness a quantity of seawater containing 3.5% sodium chloride by mass. What volume of seawater, in liters, must be taken for this purpose? Assume a density of 1.03 g/mL for seawater.

Analyze

The conversion pathway is g sodium chloride → g seawater → mL seawater → L seawater. To convert from g sodium chloride to g seawater, we need the conversion factor in expression (1.3), with g seawater in the numerator and g sodium chloride in the denominator. To convert from g seawater to mL seawater, we use the reciprocal of the density of seawater as the conversion factor. To make the final conversion, from mL seawater to L of seawater, we use the fact that 1 L = 1000 mL.

Solve

Following the conversion pathway described above, we obtain

$$\begin{aligned} ? \text{ L seawater} &= 75 \text{ g sodium chloride} \times \frac{100 \text{ g seawater}}{3.5 \text{ g sodium chloride}} \\ &\quad \times \frac{1 \text{ mL seawater}}{1.03 \text{ g seawater}} \times \frac{1 \text{ L seawater}}{1000 \text{ mL seawater}} \\ &= 2.1 \text{ L seawater} \end{aligned}$$

Assess

In solving this problem, we set up a conversion pathway, and then we thought about the conversion factors that were required. We will make use of this approach throughout the text.

PRACTICE EXAMPLE A: How many kilograms of ethanol are present in 25 L of a gasohol solution that is 90% gasoline to 10% ethanol by mass? The density of gasohol is 0.71 g/mL.

PRACTICE EXAMPLE B: Common rubbing alcohol is a solution of 70.0% isopropyl alcohol by mass in water. If a 25.0 mL sample of rubbing alcohol contains 15.0 g of isopropyl alcohol, what is the density of the rubbing alcohol?



1-2 ARE YOU WONDERING?

When do we multiply and when do we divide in solving problems with percentages?

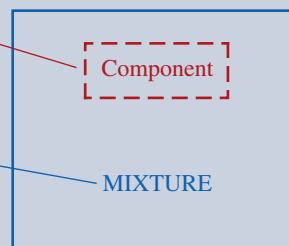
A common way of dealing with a percentage is to convert it to decimal form (3.5% becomes 0.035) and then to multiply or divide by this decimal, but students sometimes can't decide which to do. Expressing percentage as a conversion factor and

(continued)

using it to produce a necessary cancellation of units gets around this difficulty. Also, remember that

The quantity of a *component* must always be less than the quantity of the entire mixture. (*Multiply* by percentage.)

The quantity of the entire *mixture* must always be greater than the quantity of any of the components. (*Divide* by percentage.)



Keep in Mind

that a numerical answer that defies common sense is probably wrong.

If, in Example 1-4, we had not been careful about the cancellation of units and had multiplied by percentage (3.5/100) instead of dividing by it (100/3.5), we would have obtained the numerical answer 2.5×10^{-3} . This would be a 2.5 mL sample of seawater, weighing about 2.5 g. Clearly, a sample of seawater that *contains* 75 g of sodium chloride must have a mass *greater than* 75 g.

1-6 Uncertainties in Scientific Measurements

► *Random errors* are observed by scatter in the data and can be dealt with effectively by taking the average of many measurements. *Systematic errors*, conversely, are the bane of the experimental scientist. They are not readily apparent and must be avoided by carefully calibrating a method against a known sample or result. Systematic errors influence the *accuracy* of a measurement, whereas random errors are linked to the *precision* of measurements.

All measurements are subject to error. To some extent, measuring instruments have built-in, or inherent, errors, called **systematic errors**. (For example, a kitchen scale might consistently yield results that are 25 g too high or a thermometer a reading that is 2°C too low.) Limitations in an experimenter's skill or ability to read a scientific instrument also lead to errors and give results that may be either too high or too low. Such errors are called **random errors**.

Precision refers to the degree of reproducibility of a measured quantity—that is, the closeness of agreement when the same quantity is measured several times. The precision of a series of measurements is *high* (or good) if each of a series of measurements deviates by only a small amount from the average. Conversely, if there is wide deviation among the measurements, the precision is *poor* (or low). **Accuracy** refers to how close a measured value is to the accepted, or actual, value. High-precision measurements are not always accurate—a large systematic error could be present. (A tight cluster of three darts near the edge of a dart board can be considered precise but not very accurate if the intention was to strike the center of the board.) Still, scientists generally strive for high precision in measurements.

To illustrate these ideas, consider measuring the mass of an object by using the two balances shown on page 47. One of the balances is a single-pan balance that gives the mass in grams with only one decimal place. The other balance is a sophisticated analytical balance that gives the mass in grams with four decimal places. The accompanying table gives results obtained when the object is weighed three times on each balance. For the single-pan balance, the average of the measurements is 10.5 g, with measurements ranging from 10.4 g to 10.6 g. For the analytical balance, the average of the measurements is 10.4978 g, with measurements ranging from 10.4977 g to 10.4979 g. The scatter in the data obtained with the single-pan balance (± 0.1 g) is greater than that obtained with the analytical balance (± 0.0001 g). Thus, the results obtained by using the single-pan balance have lower (or poorer) precision than those obtained by using the analytical balance.



Richard Megna/Fundamental Photographs



Richard Megna/Fundamental Photographs

	Pan Balance	Analytical Balance
Three measurements	10.5, 10.4, 10.6 g	10.4978, 10.4979, 10.4977 g
Their average	10.5 g	10.4978 g
Reproducibility	± 0.1 g	± 0.0001 g
Precision	low or poor	high or good



1-5 Concept Assessment

Can a set of measurements be precise without being accurate? Can the average of a set of measurements be accurate and the individual measurements be imprecise? Explain.

1-7 Significant Figures

Consider these measurements made on a low-precision balance: 10.4, 10.2, and 10.3 g. The reported result is best expressed as their average, that is, 10.3 g.

A scientist would interpret these results to mean that the first two digits—10—are known with certainty and the last digit—3—is uncertain because it was estimated. That is, the mass is known only to the nearest 0.1 g, a fact that we could also express by writing 10.3 ± 0.1 g. To a scientist, the measurement 10.3 g is said to have *three significant figures*. If this mass is reported in kilograms rather than in grams, $10.3 \text{ g} = 0.0103 \text{ kg}$, the measurement is still expressed to *three significant figures* even though more than three digits are shown. When measured on an analytical balance, the corresponding reported value might be 10.3107 g—a value with *six significant figures*. The number of significant figures in a measured quantity gives an indication of the capabilities of the measuring device and the precision of the measurements.

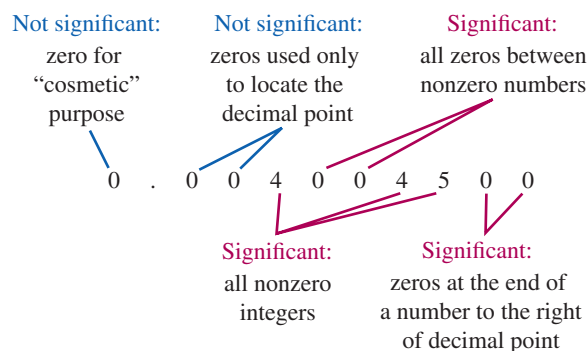
We will frequently need to determine the number of significant figures in a numerical quantity. The rules for doing this, outlined in Figure 1-11, are as follows:

- All nonzero digits are significant.
- Zeros are also significant, but with two important *exceptions* for quantities less than one. Any zeros (1) preceding the decimal point, or (2) following the decimal point and preceding the first nonzero digit, are *not* significant.
- The case of terminal zeros that precede the decimal point in quantities greater than one is *ambiguous*.

The quantity 7500 m is an example of an ambiguous case.

Figure 1-11 Determining the number of significant figures in a quantity

The quantity shown here, 0.004004500, has seven significant figures. All nonzero digits are significant, as are the indicated zeros.



Do we mean 7500 m, measured to the nearest meter? Nearest 10 meters? If all the zeros are significant—if the value has *four* significant figures—we can write 7500. m. That is, by writing a decimal point that is not otherwise needed, we show that all zeros preceding the decimal point are significant. This technique does not help if only one of the zeros, or if neither zero, is significant. The best approach here is to use exponential notation. (Review Appendix A if necessary.) The coefficient establishes the number of significant figures, and the power of ten locates the decimal point.

2 significant figures

$$7.5 \times 10^3 \text{ m}$$

3 significant figures

$$7.50 \times 10^3 \text{ m}$$

4 significant figures

$$7.500 \times 10^3 \text{ m}$$

Significant Figures in Numerical Calculations

Precision must neither be gained nor be lost in calculations involving measured quantities. There are several methods for determining how precisely to express the result of a calculation, but it is usually sufficient just to observe some simple rules involving significant figures.

The result of multiplication or division may contain only as many significant figures as the least precisely known quantity in the calculation.

In the following chain multiplication to determine the volume of a rectangular block of wood, we should round off the result to *three* significant figures. Figure 1-12 may help you to understand this.

$$14.79 \text{ cm} \times 12.11 \text{ cm} \times 5.05 \text{ cm} = 904 \text{ cm}^3$$

(4 sig. fig.) (4 sig. fig.) (3 sig. fig.) (3 sig. fig.)

In adding and subtracting numbers, the applicable rule is as follows:

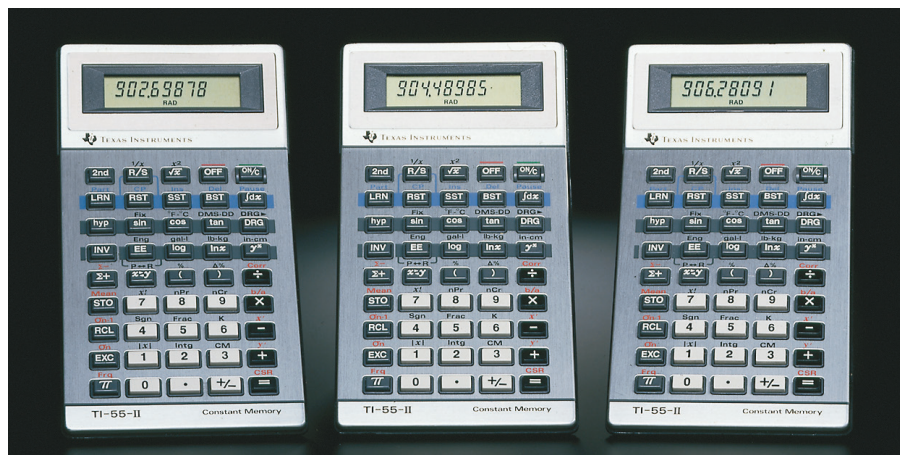
The result of addition or subtraction must be expressed with the same number of digits beyond the decimal point as the quantity carrying the *smallest* number of such digits.

Consider the following sum of masses.

$$\begin{array}{r} 15.02 \text{ g} \\ 9,986.0 \text{ g} \\ 3.518 \text{ g} \\ \hline 10,004.538 \text{ g} \end{array}$$

► A more exact rule on multiplication/division is that the result should have about the same relative error—for example, expressed as parts per hundred (percent) or parts per thousand—as the least precisely known quantity. Usually the significant figure rule conforms to this requirement; occasionally, it does not (see Exercise 67).

► In addition and subtraction the *absolute* error in the result can be no less than the absolute error in the least precisely known quantity. In the summation at the right, the absolute error in one quantity is ± 0.1 g; in another, ± 0.01 g; and in the third, ± 0.001 g. The sum *must* be expressed with an absolute error of ± 0.1 g.



Carey B. Van Loon

Figure 1-12 Significant figure rule in multiplication

In forming the product $14.79 \text{ cm} \times 12.11 \text{ cm} \times 5.05 \text{ cm}$, the least precisely known quantity is 5.05 cm. Shown on the calculators are the products of 14.79 and 12.11 with 5.04, 5.05, and 5.06, respectively. In the three results, only the first two digits, 90 ..., are identical; variations begin in the third digit. We are certainly not justified in carrying digits beyond the third. We express the volume as 904 cm^3 . Usually, instead of a detailed analysis of the type done here, we can use a simpler idea: *The result of a multiplication may contain only as many significant figures as does the least precisely known quantity.*

The sum has the same uncertainty, $\pm 0.1 \text{ g}$, as does the term with the *smallest* number of digits beyond the decimal point, 9986.0 g. Note that this calculation is *not* limited by significant figures. In fact, the sum has more significant figures (six) than do any of the terms in the addition.

There are two situations when a quantity appearing in a calculation may be *exact*, that is, not subject to errors in measurement. This may occur

- by definition (such as $1 \text{ min} = 60 \text{ s}$, or $1 \text{ in} = 2.54 \text{ cm}$)
- as a result of counting (such as *six* faces on a cube, or *two* hydrogen atoms in a water molecule)

Exact numbers can be considered to have an unlimited number of significant figures.



1-6 Concept Assessment

Which of the following is a more precise statement of the length 1 inch:
1 in = 2.54 cm or 1 m = 39.37 in? Explain.

Rounding Off Numerical Results

To three significant figures, we should express 15.453 as 15.5 and 14,775 as 1.48×10^4 . If we need to drop just one digit, that is, to round off a number, the rule that we will follow is to increase the final digit by one unit if the digit dropped is 5, 6, 7, 8, or 9 and to leave the final digit unchanged if the digit dropped is 0, 1, 2, 3, or 4.* To three significant figures, 15.44 rounds off to 15.4, and 15.45 rounds off to 15.5.

*C. J. Guare, *J. Chem. Educ.*, **68**, 818 (1991).

◀ Later in the text, we will need to apply ideas about significant figures to logarithms. This concept is discussed in Appendix A.

◀ As added practice in working with significant figures, review the calculations in Section 1-6. You will note that they conform to the significant figure rules presented here.

◀ Some people prefer the “round 5 to even” rule. Thus 15.55 rounds to 15.6, and 17.65 rounds to 17.6. In banking and with large data sets, rounding needs to be unbiased. With a small number of data, this is less important.

EXAMPLE 1-5**Applying Significant Figure Rules: Multiplication/Division**

Express the result of the following calculation with the correct number of significant figures.

$$\frac{0.225 \times 0.0035}{2.16 \times 10^{-2}} = ?$$

Analyze

By inspecting the three quantities, we see that the least precisely known quantity, 0.0035, has *two* significant figures. Our result must also contain only *two* significant figures.

Solve

When we carry out the calculation above by using an electronic calculator, the result is displayed as 0.0364583. In our analysis of this problem, we determined that the result must be rounded off to two significant figures, and so the result is properly expressed as **0.036** or as **3.6×10^{-2}** .

Assess

To check for any possible calculation error, we can estimate the correct answer through a quick mental calculation by using exponential numbers. The answer should be $(2 \times 10^{-1})(4 \times 10^{-3})/(2 \times 10^{-2}) \approx 4 \times 10^{-2}$, and it is. Expressing numbers in exponential notation can often help us quickly estimate what the result of a calculation should be.

PRACTICE EXAMPLE A: Perform the following calculation, and express the result with the appropriate number of significant figures.

$$\frac{62.356}{0.000456 \times 6.422 \times 10^3} = ?$$

PRACTICE EXAMPLE B: Perform the following calculation, and express the result with the appropriate number of significant figures.

$$\frac{8.21 \times 10^4 \times 1.3 \times 10^{-3}}{0.00236 \times 4.071 \times 10^{-2}} = ?$$

EXAMPLE 1-6**Applying Significant Figure Rules: Addition/Subtraction**

Express the result of the following calculation with the correct number of significant figures.

$$(2.06 \times 10^2) + (1.32 \times 10^4) - (1.26 \times 10^3) = ?$$

Analyze

If the calculation is performed with an electronic calculator, the quantities can be entered just as they are written, and the answer obtained can be adjusted to the correct number of significant figures. To determine the correct number of significant figures, identify the largest quantity, and then write the other quantities with the same power of ten as appears in the largest quantity. The answer can have no more digits beyond the decimal point than the quantity having the smallest number of such digits.

Solve

The largest quantity is 1.32×10^4 and thus, we write the other two quantities as 0.0206×10^4 and 0.126×10^4 . The result of the required calculation must be rounded off to two decimal places.

$$\begin{aligned} & (2.06 \times 10^2) + (1.32 \times 10^4) - (1.26 \times 10^3) \\ &= (0.0206 \times 10^4) + (1.32 \times 10^4) - (0.126 \times 10^4) \\ &= (0.0206 + 1.32 - 0.126) \times 10^4 \\ &= 1.2146 \times 10^4 \\ &= \mathbf{1.21 \times 10^4} \end{aligned}$$

Assess

If you refer back to the margin note on page 48, you will see that there is another way to approach this problem. To determine the absolute error in the least precisely known quantity, we write the three quantities as $(2.06 \pm 0.01) \times 10^2$, $(1.32 \pm 0.01) \times 10^4$ and $(1.26 \pm 0.01) \times 10^3$. We conclude that 1.32×10^4 has the largest absolute error ($\pm 0.01 \times 10^4$) and so, the absolute error in the result of the calculation above is also $\pm 0.01 \times 10^4$. Thus, 1.2146×10^4 is rounded to 1.21×10^4 .

PRACTICE EXAMPLE A: Express the result of the following calculation with the appropriate number of significant figures.

$$0.236 + 128.55 - 102.1 = ?$$

PRACTICE EXAMPLE B: Perform the following calculation, and express the result with the appropriate number of significant figures.

$$\frac{(1.302 \times 10^3) + 952.7}{(1.57 \times 10^2) - 12.22} = ?$$

In working through the preceding examples, you likely used an electronic calculator. What's nice about using electronic calculators is that we don't have to write down intermediate results. In general, disregard occasional situations where intermediate rounding may be justified and store all intermediate results in your electronic calculator without regard to significant figures. Then, round off to the correct number of significant figures only in the final answer.

Keep in Mind

that addition and subtraction are governed by one significant figure rule, and multiplication and division are governed by a different rule.



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In the late 1960s, scientists heatedly debated the reported discovery of a new form of water called polywater. For a discussion of the polywater debate and the importance of the scientific method in helping the scientific community reach a consensus, go to the Focus On feature for Chapter 1 (The Scientific Method at Work: Polywater) in Mastering.

Summary

1-1 The Scientific Method—The **scientific method** is a set of procedures used to develop explanations of natural phenomena and possibly to predict additional phenomena. The four basic stages of the scientific method are (1) gathering data through observations and experiments; (2) reducing the data to simple verbal or mathematical expressions known as **natural laws**; (3) offering a plausible explanation of the data through a **hypothesis**; (4) testing the hypothesis through predictions and further experimentation, leading ultimately to a conceptual model called a **theory** that explains the hypothesis, often together with other related hypotheses.

1-2 Properties of Matter—**Matter** is defined as anything that occupies space, possesses mass, and displays inertia. **Composition** refers to the component parts of a sample of matter and their relative proportions. **Properties** are the qualities or attributes that distinguish one sample of matter from another. Properties of matter can be grouped into two broad categories: **physical** and **chemical**. Matter can undergo

two types of changes: **chemical changes** or **reactions** are changes in composition; **physical changes** are changes in state or physical form and do not affect composition.

1-3 Classification of Matter—The basic building blocks of matter are called **atoms**. Matter that is composed of a collection of a single type of atom is known as an **element**. A sample of matter composed of two or more elements is known as a **compound**. A **molecule** is the smallest entity of a compound having the same proportions of the constituent atoms as does the compound as a whole. Collectively, elements and compounds compose the types of matter called **substances**. **Mixtures** of substances can be classified as **homogeneous** or **heterogeneous** (Fig. 1-4). The three states of matter are **solid**, **liquid**, and **gas**.

1-4 Measurement of Matter: SI (Metric) Units—Chemistry is a quantitative science, meaning that chemical measurements are usually expressed in terms of a number

and an accompanying unit. The scientific system of measurement, called the *Système Internationale d'Unités* (abbreviated **SI**), involves seven base quantities (Table 1.1). **Mass** describes a quantity of matter. Weight measures the force of gravity on an object; weight is related to, but different from, mass. The temperature scales used by chemists are the **Celsius** and **Kelvin** scales. The **Fahrenheit** temperature scale, commonly used in daily life in the United States, is also used in some industrial settings. The three temperature scales can be related algebraically (Fig. 1-8).

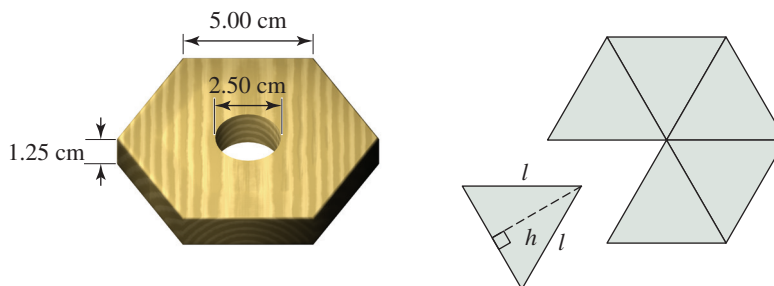
1-5 Density and Percent Composition: Their Use in Problem Solving—Mass and volume are **extensive properties**; they *depend* on the amount of matter in a sample. **Density**, the ratio of the mass of a sample to its volume, is an **intensive property**, a property *independent* of the amount of matter sampled. Density is used as a conversion factor in a variety of calculations.

1-6 Uncertainties in Scientific Measurements—Measurements are subject to **systematic** and **random errors**. In making a series of measurements, the degree to which the measurements agree with one another is known as the **precision** of the measurement, while the degree to which the measurement agrees with the actual value is referred to as the **accuracy** of the measurement.

1-7 Significant Figures—The proper use of **significant figures** is important in that it prevents the suggestion of a higher degree of precision in a calculated quantity than is warranted by the precision of the measured quantities used in the calculation. The precision of an answer cannot be greater than the precision of the numbers used in the calculation. In addition to reporting the correct number of significant figures in a calculated quantity, it is important to know the rules for rounding off numerical results.

Integrative Example

Consider a 58.35 g hexagonal block of wood that is 5.00 cm on edge and 1.25 cm thick, with a 2.50 cm diameter hole drilled through its center. Also given are the densities of the liquids hexane ($d = 0.667 \text{ g/mL}$) and decane ($d = 0.845 \text{ g/mL}$). Assume that the density of a mixture of the two liquids is a linear function of the volume percent composition of the solution. Determine the volume percent of hexane required in the solution so that the hexagonal block of wood will just barely float on the solution.



Analyze

The first task is to determine the density of the wood block, $d = m/V$. The mass is given, so the critical calculation is that of the volume. The key to the volume calculation is recognizing that the volume of the block is the *difference* between two volumes: The volume of the block if there were no hole *minus* the volume of the cylindrical hole. The second task is to write a simple equation relating density to the volume percent composition of the liquid solution, and then to solve that equation for the volume percent hexane that yields a solution density equal to the density calculated for the wood.

Solve

The solid hexagonal block can be divided into six smaller blocks, each an equilateral triangle of length, l , and height, h . The area of a triangle is given by the formula

$$A = \frac{1}{2}(\text{base} \times \text{height}) = \frac{1}{2} \times l \times h$$

Only the base, l , is given (5.00 cm). To express h in terms of l , we use the Pythagorean theorem for the right triangle pictured, that is, $a^2 + b^2 = c^2$, rearranged to the form $a^2 = c^2 - b^2$.

$$h^2 = l^2 - \left(\frac{l}{2}\right)^2 = l^2 - \frac{l^2}{4} = \frac{3l^2}{4} \text{ and}$$

$$h = \frac{\sqrt{3}}{2} \times l$$

Now, for the area of one of the six triangles we have

$$A = \frac{1}{2} \times l \times \frac{\sqrt{3}}{2} \times l = \frac{\sqrt{3}}{4} \times l^2$$

The volume of one of the triangular blocks is obtained by multiplying the cross-sectional area A by the thickness, 1.25 cm. The volume of the hexagonal block of wood, without the cylindrical hole, is that of six triangular blocks.

$$\begin{aligned} V &= 6 \times A \times 1.25 \text{ cm} \\ &= 6 \times \frac{\sqrt{3}}{4} \times (5.00 \text{ cm})^2 \times 1.25 \text{ cm} \\ &= 81.2 \text{ cm}^3 \end{aligned}$$

The volume of the cylindrical hole with a radius of 1.25 cm (one-half the 2.50 cm diameter) and a height of 1.25 cm is

$$\begin{aligned} V &= \pi r^2 h = 3.1416 \times (1.25 \text{ cm})^2 \times 1.25 \text{ cm} \\ &= 6.14 \text{ cm}^3 \end{aligned}$$

The volume of the block of wood is the difference

$$V = 81.2 \text{ cm}^3 - 6.14 \text{ cm}^3 = 75.1 \text{ cm}^3$$

The density of the wood is

$$d = \frac{m}{V} = \frac{58.35 \text{ g}}{75.1 \text{ cm}^3} = 0.777 \text{ g/cm}^3 \text{ or } 0.777 \text{ g/mL}$$

The general formula for a linear (straight-line) relationship is

$$y = mx + b$$

In the present case, let y represent the density, d , of a solution (in g/mL) and x , the volume fraction of hexane (volume percent/100). By substituting the density of pure decane into the equation, we note that $x = 0$ and find that $b = 0.845$.

$$d = 0.845 = (m \times 0) + b$$

Now, using the density of pure hexane, $d = 0.667$, and the values $x = 1.00$ and $b = 0.845$, we obtain the value of m .

$$\begin{aligned} d &= 0.667 = (m \times 1.00) + 0.845 \\ m &= 0.667 - 0.845 = -0.178 \end{aligned}$$

Our final step is to find the value of x for a solution having the same density as the wood: 0.777 g/mL.

$$\begin{aligned} d &= 0.777 = -0.178x + 0.845 \\ x &= \frac{0.845 - 0.777}{0.178} = 0.38 \end{aligned}$$

The volume fraction of hexane is 0.38, and the volume percent composition of the solution is 38% hexane and 62% decane.

Assess

There is an early point in this calculation where we can check the correctness of our work. We have two expectations for the density of the block of wood: (1) it should be less than 1 g/cm³ (practically all wood floats on water), and (2) it must fall between 0.667 g/cm³ and 0.845 g/cm³. If the calculated density of the wood were outside this range, the block of wood would either float on both liquids or sink in both of them, making the rest of the calculation impossible.

Another point to notice in this calculation is that we are justified in carrying three significant figures throughout the calculation up to the last step. There, because we must take the difference between two numbers of similar magnitudes, the number of significant figures drops from three to two.

PRACTICE EXAMPLE A: Magnalium is a solid mixture (an alloy) of aluminum metal and magnesium metal. An irregularly shaped chunk of a sample of magnalium is weighed twice, once in air and once in vegetable oil, by using a spring scale (see Figure 1-10). The weight in air is 211.5 g, and the weight in oil is 135.3 g. If the densities of pure aluminum, pure magnesium, and vegetable oil are 2.70 g/cm³, 1.74 g/cm³, and 0.926 g/cm³, respectively, then what is the mass percent of magnesium in this chunk of magnalium? Assume that the density of a mixture of the two metals is a linear function of the mass percent composition.

PRACTICE EXAMPLE B: A particular sample of seawater has a density of 1.027 g/cm³ at 10°C and is 2.67% sodium chloride by mass. Given that sodium chloride is 39.34% sodium by mass and that the mass of a single sodium atom is 3.817×10^{-26} kg, calculate the maximum mass of sodium and the maximum number of sodium atoms that can be extracted from a 1.5 L sample of this seawater.

Exercises

(see also Appendices A-1 and A-5)

The Scientific Method

1. What are the principal reasons that one theory might be adopted over a conflicting one?
2. Can one predict how many experiments are required to verify a natural law? Explain.
3. A common belief among scientists is that there exists an underlying order to nature. Einstein described this belief in the words “God is subtle, but He is not malicious.” What do you think Einstein meant by this remark?
4. Describe several ways in which a scientific law differs from a legislative law.
5. Describe the necessary characteristics of an experiment that is suitable to test a theory.
6. Describe the necessary characteristics of a scientific theory.

Properties and Classification of Matter

7. State whether the following properties of matter are physical or chemical.
 - (a) An iron nail is attracted to a magnet.
 - (b) A piece of paper spontaneously ignites when its temperature reaches 451°F.
 - (c) A bronze statue develops a green coating (patina) over time.
 - (d) A block of wood floats on water.
8. State whether the following properties are physical or chemical.
 - (a) A piece of sliced apple turns brown.
 - (b) A slab of marble feels cool to the touch.
 - (c) A sapphire is blue.
 - (d) A clay pot fired in a kiln becomes hard and covered by a glaze.
9. Indicate whether each sample of matter listed is a substance or a mixture; if it is a mixture, indicate whether it is homogeneous or heterogeneous.
 - (a) clean fresh air
 - (b) a silver-plated spoon
 - (c) garlic salt
 - (d) ice
10. Indicate whether each sample of matter listed is a substance or a mixture; if it is a mixture, indicate whether it is homogeneous or heterogeneous.
 - (a) a wooden beam
 - (b) red ink
 - (c) distilled water
 - (d) freshly squeezed orange juice
11. Suggest physical changes by which the following mixtures can be separated.
 - (a) iron filings and wood chips
 - (b) ground glass and sucrose (cane sugar)
 - (c) water and olive oil
 - (d) gold flakes and water
12. What type of change—physical or chemical—is necessary to separate the following?
[Hint: Refer to a listing of the elements.]
 - (a) sugar from a sand/sugar mixture
 - (b) iron from iron oxide (rust)
 - (c) pure water from seawater
 - (d) water from a slurry of sand in water

Exponential Arithmetic

13. Express each number in exponential notation. (a) 8950.; (b) 10,700.; (c) 0.0240; (d) 0.0047; (e) 938.3; (f) 275,482.
14. Express each number in common decimal form. (a) 5.12×10^{-3} ; (b) 8.05×10^{-5} ; (c) 291.1×10^{-4} ; (d) 72.1×10^{-2} .
15. Express each value in exponential form. Where appropriate, include units in your answer.
 - (a) speed of sound (sea level): 34,000 centimeters per second
 - (b) equatorial radius of Earth: 6378 kilometers
 - (c) the distance between the two hydrogen atoms in the hydrogen molecule: 74 trillionths of a meter
 - (d) $\frac{(2.2 \times 10^3) + (4.7 \times 10^2)}{5.8 \times 10^{-3}} =$
16. Express each value in exponential form. Where appropriate, include units in your answer.
 - (a) solar radiation received by Earth: 173 thousand trillion watts
 - (b) average human cell diameter: 1 ten-millionth of a meter
 - (c) the distance between the centers of the atoms in silver metal: 142 trillionths of a meter
 - (d) $\frac{(5.07 \times 10^4) \times (1.8 \times 10^{-3})^2}{0.065 + (3.3 \times 10^{-2})} =$

Significant Figures

17. Indicate whether each of the following is an exact number or a measured quantity subject to uncertainty.
- the number of sheets of paper in a ream of paper
 - the volume of milk in a liter bottle
 - the distance between Earth and the sun
 - the distance between the centers of the two oxygen atoms in the oxygen molecule
18. Indicate whether each of the following is an exact number or a measured quantity subject to uncertainty.
- the number of pages in this text
 - the number of days in the month of January
 - the area of a city lot
 - the distance between the centers of the atoms in a gold medal
19. Express each of the following to *four* significant figures. (a) 3984.6; (b) 422.04; (c) 186,000; (d) 33,900; (e) 6.321×10^4 ; (f) 5.0472×10^{-4}
20. How many significant figures are shown in each of the following? If this is indeterminate, explain why. (a) 450; (b) 98.6; (c) 0.0033; (d) 902.10; (e) 0.02173; (f) 7000; (g) 7.02; (h) 67,000,000
21. Perform the following calculations; express each answer in exponential form and with the appropriate number of significant figures.
- $0.406 \times 0.0023 =$
 - $0.1357 \times 16.80 \times 0.096 =$
 - $0.458 + 0.12 - 0.037 =$
 - $32.18 + 0.055 - 1.652 =$
22. Perform the following calculations; express each number and the answer in exponential form and with the appropriate number of significant figures.
- $\frac{320 \times 24.9}{0.080} =$
 - $\frac{432.7 \times 6.5 \times 0.002300}{62 \times 0.103} =$
 - $\frac{32.44 + 4.9 - 0.304}{82.94} =$
 - $\frac{8.002 + 0.3040}{13.4 - 0.066 + 1.02} =$
23. Perform the following calculations and retain the appropriate number of significant figures in each result.
- $(38.4 \times 10^{-3}) \times (6.36 \times 10^5) =$
 - $\frac{(1.45 \times 10^2) \times (8.76 \times 10^{-4})}{(9.2 \times 10^{-3})^2} =$
- $24.6 + 18.35 - 2.98 =$
 - $(1.646 \times 10^3) - (2.18 \times 10^2) + [(1.36 \times 10^4) \times (5.17 \times 10^{-2})] =$
 - $\frac{-7.29 \times 10^{-4} + \sqrt{(7.29 \times 10^{-4})^2 + 4(1.00)(2.7 \times 10^{-5})}}{2 \times (1.00)} =$
[Hint: The significant figure rule for the extraction of a root is the same as for multiplication.]
24. Express the result of each of the following calculations in exponential form and with the appropriate number of significant figures.
- $(4.65 \times 10^4) \times (2.95 \times 10^{-2}) \times (6.663 \times 10^{-3}) \times 8.2 =$
 - $\frac{1912 \times (0.0077 \times 10^4) \times (3.12 \times 10^{-3})}{(4.18 \times 10^{-4})^3} =$
 - $(3.46 \times 10^3) \times 0.087 \times 15.26 \times 1.0023 =$
 - $\frac{(4.505 \times 10^{-2})^2 \times 1.080 \times 1545.9}{0.03203 \times 10^3} =$
 - $\frac{(-3.61 \times 10^{-4}) + \sqrt{(3.61 \times 10^{-4})^2 + 4(1.00)(1.9 \times 10^{-5})}}{2 \times (1.00)} =$
[Hint: The significant figure rule for the extraction of a root is the same as for multiplication.]
25. An American press release describing the 1986 non-stop, round-the-world trip by the ultra-lightweight aircraft *Voyager* included the following data:
- flight distance: 25,012 mi
 - flight time: 9 days, 3 minutes, 44 seconds
 - fuel capacity: nearly 9000 lb
 - fuel remaining at end of flight: 14 gal
- To the maximum number of significant figures permitted, calculate
- the average speed of the aircraft in kilometers per hour
 - the fuel consumption in kilometers per kilogram of fuel (assume a density of 0.70 g/mL for the fuel)
26. Use the concept of significant figures to criticize the way in which the following information was presented. "The estimated proved reserve of natural gas as of January 1, 1982, was 2,911,346 trillion cubic feet."

Units of Measurement

27. Perform the following conversions.
- 0.127 L = ____ mL
 - 15.8 mL = ____ L
 - 981 cm³ = ____ L
 - 2.65 m³ = ____ cm³
28. Perform the following conversions.
- 2.35 kg = ____ g
 - 792 g = ____ kg
 - 3869 mm = ____ cm
 - 0.043 cm = ____ mm
29. Perform the following conversions from non-SI to SI units. (Use information from the inside back cover, as needed.)
- 68.4 in = ____ cm
 - 94 ft = ____ m
 - 1.42 lb = ____ g
 - 248 lb = ____ kg
 - 1.85 gal = ____ dm³
 - 3.72 qt = ____ mL
30. Determine the number of the following:
- square meters (m²) in 1 square kilometer (km²)
 - cubic centimeters (cm³) in 1 cubic meter (m³)
 - square meters (m²) in 1 square mile (mi²) (1 mi = 5280 ft)

31. Which is the greater mass, 3245 μg or 0.00515 mg? Explain.
32. Which is the greater mass, 3257 mg or 0.000475 kg? Explain.
33. The non-SI unit, the *hand* (used by equestrians), is 4 inches. What is the height, in meters, of a horse that stands 15 hands high?
34. The unit *furlong* is used in horse racing. The units *chain* and *link* are used in surveying. There are exactly 8 furlongs in 1 mi, 10 chains in 1 furlong, and 100 links in 1 chain. To three significant figures, what is the length of 1 link in centimeters?
35. A sprinter runs the 100 yd dash in 9.3 s. At this same rate,
 - (a) how long would it take the sprinter to run 100.0 m?
 - (b) what is the sprinter's speed in meters per second?
 - (c) how long would it take the sprinter to run a distance of 1.45 km?
36. A non-SI unit of mass used in pharmaceutical work is the *grain* (gr) (15 gr = 1.0 g). An aspirin tablet contains 5.0 gr of aspirin. A 161 lb arthritic individual takes two aspirin tablets per day.
 - (a) What is the quantity of aspirin in two tablets, expressed in milligrams?
 - (b) What is the dosage rate of aspirin, expressed in milligrams of aspirin per kilogram of body mass?
 - (c) At the given rate of consumption of aspirin tablets, how many days would it take to consume 1.0 kg of aspirin?
37. In SI units, land area is measured in *hectares* (1 hectare = 1 hm^2). The commonly used unit for land area in the United States is the *acre*. How many acres correspond to 1 hectare? (1 $\text{mi}^2 = 640$ acres, 1 mi = 5280 ft, 1 ft = 12 in).
38. In an engineering reference book, you find that the density of iron is 0.284 lb/in³. What is the density in g/cm³?
39. In a user's manual accompanying an American-made automobile, a typical gauge pressure for optimal performance of automobile tires is 32 lb/in². What is this pressure in grams per square centimeter and kilograms per square meter?
40. The volume of a red blood cell is about $90.0 \times 10^{-12} \text{ cm}^3$. Assuming that red blood cells are spherical, what is the diameter of a red blood cell in millimeters?

Temperature Scales

41. We want to mark off a thermometer in both Celsius and Fahrenheit temperatures. On the Celsius scale, the lowest temperature mark is at -10°C , and the highest temperature mark is at 50°C . What are the equivalent Fahrenheit temperatures?
42. The highest and lowest temperatures on record for San Bernardino, California, are 118°F and 17°F , respectively. What are these temperatures on the Celsius scale?
43. The absolute zero of temperature is -273.15°C . Should it be possible to achieve a temperature of -465°F ? Explain.
44. A family/consumer science class is given an assignment in candy-making that requires a sugar mixture to be brought to a "soft-ball" stage ($234\text{--}240^\circ\text{F}$).

A student borrows a thermometer having a range from -10°C to 110°C from the chemistry laboratory to do this assignment. Will this thermometer serve the purpose? Explain.
45. You decide to establish a new temperature scale on which the melting point of mercury (-38.9°C) is 0°M , and the boiling point of mercury (356.9°C) is 100°M . What would be (a) the boiling point of water in $^\circ\text{M}$; and (b) the temperature of absolute zero in $^\circ\text{M}$?
46. You decide to establish a new temperature scale on which the melting point of ammonia (-77.75°C) is 0°A and the boiling point of ammonia (-33.35°C) is 100°A . What would be (a) the boiling point of water in $^\circ\text{A}$; and (b) the temperature of absolute zero in $^\circ\text{A}$?

Density

47. A 2.18 L sample of butyric acid, a substance present in rancid butter, has a mass of 2088 g. What is the density of butyric acid in grams per milliliter?
48. A 15.2 L sample of chloroform at 20°C has a mass of 22.54 kg. What is the density of chloroform at 20°C , in grams per milliliter?
49. To determine the density of acetone, a 55.0 gal drum is weighed twice. The drum weighs 75.0 lb when empty and 437.5 lb when filled with acetone. What is the density of acetone expressed in grams per milliliter?
50. To determine the volume of an irregularly shaped glass vessel, the vessel is weighed empty (121.3 g) and when filled with carbon tetrachloride (283.2 g). What is the volume capacity of the vessel, in milliliters, given that the density of carbon tetrachloride is 1.59 g/mL?
51. A solution consisting of 8.50% acetone and 91.5% water by mass has a density of 0.9867 g/mL.

What mass of acetone, in kilograms, is present in 7.50 L of the solution?
52. A solution contains 10.05% sucrose (cane sugar) by mass. What mass of the solution, in grams, is needed for an application that requires 1.00 kg of sucrose?
53. A fertilizer contains 21% nitrogen by mass. What mass of this fertilizer, in kilograms, is required for an application requiring 225 g of nitrogen?
54. A vinegar sample is found to have a density of 1.006 g/mL and to contain 5.4% acetic acid by mass. How many grams of acetic acid are present in 1.00 L of this vinegar?
55. Calculate the mass of a block of iron ($d = 7.86 \text{ g/cm}^3$) with dimensions of $52.8 \text{ cm} \times 6.74 \text{ cm} \times 3.73 \text{ cm}$.
56. Calculate the mass of a cylinder of stainless steel ($d = 7.75 \text{ g/cm}^3$) with a height of 18.35 cm and a radius of 1.88 cm.

57. The following densities are given at 20°C: water, 0.998 g/cm³; iron, 7.86 g/cm³; aluminum, 2.70 g/cm³. Arrange the following items in terms of *increasing* mass.

(a) a rectangular bar of iron,

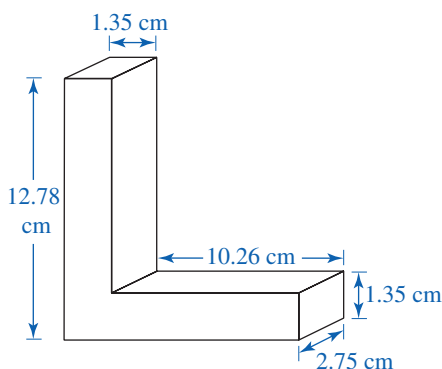
$$81.5 \text{ cm} \times 2.1 \text{ cm} \times 1.6 \text{ cm}$$

(b) a sheet of aluminum foil,

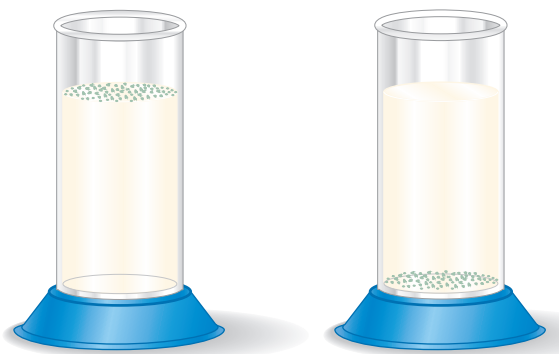
$$12.12 \text{ m} \times 3.62 \text{ m} \times 0.003 \text{ cm}$$

(c) 4.051 L of water

58. To determine the approximate mass of a small spherical shot of copper, the following experiment is performed. When 125 pieces of the shot are counted out and added to 8.4 mL of water in a graduated cylinder, the total volume becomes 8.9 mL. The density of copper is 8.92 g/cm³. Determine the approximate mass of a single piece of shot, assuming that all of the pieces are of the same dimensions.
59. The density of aluminum is 2.70 g/cm³. A square piece of aluminum foil, 22.86 cm on a side is found to weigh 2.568 g. What is the thickness of the foil, in millimeters?
60. The angle iron pictured here is made of steel with a density of 7.78 g/cm³. What is the mass, in grams, of this object?



61. In normal blood, there are about 5.4×10^9 red blood cells per milliliter. The volume of a red blood cell is about $90.0 \times 10^{-12} \text{ cm}^3$, and its density is 1.096 g/mL. How many liters of whole blood would be needed to collect 0.5 kg of red blood cells?
62. A technique once used by geologists to measure the density of a mineral is to mix two dense liquids in such proportions that the mineral grains just float. When a sample of the mixture in which the mineral calcite just floats is put in a special density bottle, the weight is 15.4448 g. When empty, the bottle weighs 12.4631 g, and when filled with water, it weighs 13.5441 g. What is the density of the calcite sample? (All measurements were carried out at 25°C, and the density of water at 25°C is 0.9970 g/mL.)



▲ At the left, grains of the mineral *calcite* float on the surface of the liquid bromoform ($d = 2.890 \text{ g/mL}$). At the right, the grains sink to the bottom of liquid chloroform ($d = 1.444 \text{ g/mL}$). By mixing bromoform and chloroform in just the proportions required so that the grains barely float, the density of the calcite can be determined (Exercise 62).

Percent Composition

63. In a class of 76 students, the results of a particular examination were 7 A's, 22 B's, 37 C's, 8 D's, 2 F's. What was the percent distribution of grades, that is, % A's, % B's, and so on?
64. A class of 84 students had a final grade distribution of 18% A's, 25% B's, 32% C's, 13% D's, 12% F's. How many students received each grade?

65. A solution of sucrose in water is 28.0% sucrose by mass and has a density of 1.118 g/mL. What mass of sucrose, in grams, is contained in 3.50 L of this solution?
66. A solution containing 12.0% sodium hydroxide by mass in water has a density of 1.131 g/mL. What volume of this solution, in liters, must be used in an application requiring 2.75 kg of sodium hydroxide?

Integrative and Advanced Exercises

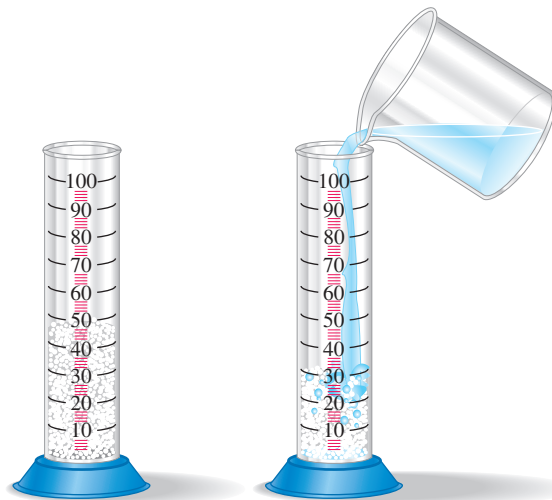
67. According to the rules on significant figures, the product of the measured quantities 99.9 m and 1.008 m should be expressed to three significant figures—101 m². Yet, in this case, it would be more appropriate to express the result to *four* significant figures—100.7 m². Explain why.

68. For a solution containing 6.38% para-dichlorobenzene by mass in benzene, the density of the solution as a function of temperature (t) in the temperature range 15 to 65°C is given by the equation

$$d(\text{g/mL}) = 1.5794 - 1.836 \times 10^{-3}(t - 15)$$

At what temperature will the solution have a density of 1.543 g/mL?

69. A solution used to chlorinate a home swimming pool contains 7% chlorine by mass. An ideal chlorine level for the pool is one part per million (1 ppm). (Think of 1 ppm as being 1 g chlorine per million grams of water.) If you assume densities of 1.10 g/mL for the chlorine solution and 1.00 g/mL for the swimming pool water, what volume of the chlorine solution, in liters, is required to produce a chlorine level of 1 ppm in an 18,000-gallon swimming pool?
70. A standard 1.000 kg mass is to be cut from a bar of steel having an equilateral triangular cross section with sides equal to 2.50 in. The density of the steel is 7.70 g/cm³. How many inches long must the section of bar be?
71. The volume of seawater on Earth is about 330,000,000 mi³. If seawater is 3.5% sodium chloride by mass and has a density of 1.03 g/mL, what is the approximate mass of sodium chloride, in tons, dissolved in the seawater on Earth (1 ton = 2000 lb)?
72. The diameter of metal wire is often referred to by its American wire-gauge number. A 16-gauge wire has a diameter of 0.05082 in. What length of wire, in meters, is found in a 1.00 lb spool of 16-gauge copper wire? The density of copper is 8.92 g/cm³.
73. Magnesium occurs in seawater to the extent of 1.4 g magnesium per kilogram of seawater. What volume of seawater, in cubic meters, would have to be processed to produce 1.50×10^5 tons of magnesium (1 ton = 2000 lb)? Assume a density of 1.025 g/mL for seawater.
74. A typical rate of deposit of dust ("dustfall") from unpolluted air was reported as 10 tons per square mile per month. (a) Express this dustfall in milligrams per square meter per hour. (b) If the dust has an average density of 2 g/cm³, how long would it take to accumulate a layer of dust 1 mm thick?
75. In the United States, the volume of irrigation water is usually expressed in acre-feet. One acre-foot is a volume of water sufficient to cover 1 acre of land to a depth of 1 ft (640 acres = 1 mi²; 1 mi = 5280 ft). The principal lake in the California Water Project is Lake Oroville, whose water storage capacity is listed as 3.54×10^6 acre-feet. Express the volume of Lake Oroville in (a) cubic feet; (b) cubic meters; (c) U.S. gallons.
76. A Fahrenheit and a Celsius thermometer are immersed in the same medium. At what Celsius temperature will the numerical reading on the Fahrenheit thermometer be
(a) 49° less than that on the Celsius thermometer;
(b) twice that on the Celsius thermometer;
(c) one-eighth that on the Celsius thermometer;
(d) 300° more than that on the Celsius thermometer?
77. The accompanying illustration shows a 100.0 mL graduated cylinder half-filled with 8.0 g of diatomaceous earth, a material consisting mostly of silica and used as a filtering medium in swimming pools. How many milliliters of water are required to fill the cylinder to the 100.0 mL mark? The diatomaceous earth is insoluble in water and has a density of 2.2 g/cm³.



78. The simple device pictured here, a pycnometer, is used for precise density determinations. From the data presented, together with the fact that the density of water at 20°C is 0.99821 g/mL, determine the density of methanol, in grams per milliliter.



Empty
25.601 g



Filled with water
at 20°C: 35.552 g



Filled with methanol
at 20°C: 33.490 g

79. If the pycnometer of Exercise 78 is filled with ethanol at 20°C instead of methanol, the observed mass is 33.470 g. What is the density of ethanol? How precisely could you determine the composition of an ethanol-methanol solution by measuring its density with a pycnometer? Assume that the density of the solution is a linear function of the volume percent composition.
80. A pycnometer (see Exercise 78) weighs 25.60 g empty and 35.55 g when filled with water at 20°C. The density of water at 20°C is 0.9982 g/mL. When 10.20 g of lead is placed in the pycnometer and the pycnometer is again filled with water at 20°C, the total mass is 44.83 g. What is the density of the lead in grams per cubic centimeter?
81. The Greater Vancouver Regional District (GVRD) chlorinates the water supply of the region at the rate of 1 ppm, that is, 1 kilogram of chlorine per million kilograms of water. The chlorine is introduced in the form of sodium hypochlorite, which is 47.62% chlorine. The population of the GVRD is 1.8 million persons. If each person uses 750 L of water per day, how many kilograms of sodium hypochlorite must be added to the water supply each week to produce the required chlorine level of 1 ppm?

82. A Boeing 767 due to fly from Montreal to Edmonton required refueling. Because the fuel gauge on the aircraft was not working, a mechanic used a dipstick to determine that 7682 L of fuel were left on the plane. The plane required 22,300 kg of fuel to make the trip. In order to determine the volume of fuel required, the pilot asked for the conversion factor needed to convert a volume of fuel to a mass of fuel. The mechanic gave the factor as 1.77. Assuming that this factor was in metric units (kg/L), the pilot calculated the volume to be added as 4916 L. This volume of fuel was added and the 767 subsequently ran out the fuel, but landed safely by gliding into Gimli Airport near Winnipeg. The error arose because the factor 1.77 was in units of pounds per liter. What volume of fuel should have been added?

83. The following equation can be used to relate the density of liquid water to Celsius temperature in the range from 0°C to about 20°C:

$$d(\text{g/cm}^3) = \frac{0.99984 + (1.6945 \times 10^{-2}t) - (7.987 \times 10^{-6}t^2)}{1 + (1.6880 \times 10^{-2}t)}$$

(a) To four significant figures, determine the density of water at 10°C.

(b) At what temperature does water have a density of 0.99860 g/cm³?

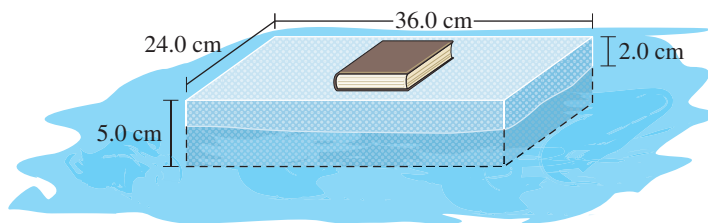
(c) In the following ways, show that the density passes through a maximum somewhere in the temperature range to which the equation applies.

(i) by estimation

(ii) by a graphical method

(iii) by a method based on differential calculus

84. A piece of high-density Styrofoam measuring 24.0 cm by 36.0 cm by 5.0 cm floats when placed in a tub of water. When a 1.5 kg book is placed on top of the Styrofoam, the Styrofoam partially sinks, as illustrated in the diagram below. Assuming that the density of water is 1.00 g/mL, what is the density of Styrofoam?



85. A tabulation of data lists the following equation for calculating the densities (d) of solutions of

naphthalene in benzene at 30°C as a function of the mass percent of naphthalene.

$$d(\text{g/cm}^3) = \frac{1}{1.153 - 1.82 \times 10^{-3}(\%N) + 1.08 \times 10^{-6}(\%N)^2}$$

Use the equation above to calculate (a) the density of pure benzene at 30°C; (b) the density of pure naphthalene at 30°C; (c) the density of a solution at 30°C that is 1.15% naphthalene; (d) the mass percent of naphthalene in a solution that has a density of 0.952 g/cm³ at 30°C. [Hint: For (d), you need to use the quadratic formula. See Section A-3 of Appendix A.]

86. The total volume of ice in the Antarctic is about 3.01×10^7 km³. If all the ice in the Antarctic were to melt completely, estimate the rise, h , in sea level that would result from the additional liquid water entering the oceans. The densities of ice and fresh water are 0.92 g/cm³ and 1.0 g/cm³, respectively. Assume that the oceans of the world cover an area, A , of about 3.62×10^8 km² and that the increase in volume of the oceans can be calculated as $A \times h$.

87. An empty 3.00 L bottle weighs 1.70 kg. Filled with a certain wine, it weighs 4.72 kg. The wine contains 11.5% ethyl alcohol by mass. How many grams of ethyl alcohol are there in 250.0 mL of this wine?

88. The filament in an incandescent light bulb is made from tungsten metal ($d = 19.3$ g/cm³) that has been drawn into a very thin wire. The diameter of the wire is difficult to measure directly, so it is sometimes estimated by measuring the mass of a fixed length of wire. If a 0.200 m length of tungsten wire weighs 42.9 mg, then what is the diameter of the wire? Express your answer in millimeters.

89. Blood alcohol content (BAC) is sometimes reported in weight-volume percent and, when it is, a BAC of 0.10% corresponds to 0.10 g of ethyl alcohol per 100 mL of blood. In many jurisdictions, a person is considered legally intoxicated if their BAC is 0.10%. Suppose that a 68 kg person has a total blood volume of 5.4 L and breaks down ethyl alcohol at a rate of 10.0 grams per hour.* How many 145 mL glasses of wine, consumed over three hours, will produce a BAC of 0.10% in this 68 kg person? Assume the wine has a density of 1.01 g/mL and is 11.5% ethyl alcohol by mass. (*The rate at which ethyl alcohol is broken down varies dramatically from person to person. The value given here for the rate is a realistic, but not necessarily accurate, value.)

Feature Problems

90. In an attempt to determine any possible relationship between the year in which a U.S. penny was minted and its current mass (in grams), students weighed an assortment of pennies and obtained the following data.

1968	1973	1977	1980	1982	1983	1985
3.11	3.14	3.13	3.12	3.12	2.51	2.54
3.08	3.06	3.10	3.11	2.53	2.49	2.53
3.09	3.07	3.06	3.08	2.54	2.47	2.53

What valid conclusion(s) might they have drawn about the relationship between the masses of the pennies within a given year and from year to year?

91. In the third century B.C., the Greek mathematician Archimedes is said to have discovered an important principle that is useful in density determinations. The story told is that King Hiero of Syracuse (in Sicily) asked Archimedes to verify that an ornate crown made for him by a goldsmith consisted of pure gold and not a gold-silver alloy. Archimedes

had to do this, of course, without damaging the crown in any way. Describe how Archimedes did this, or if you don't know the rest of the story, rediscover Archimedes's principle and explain how it can be used to settle the question.

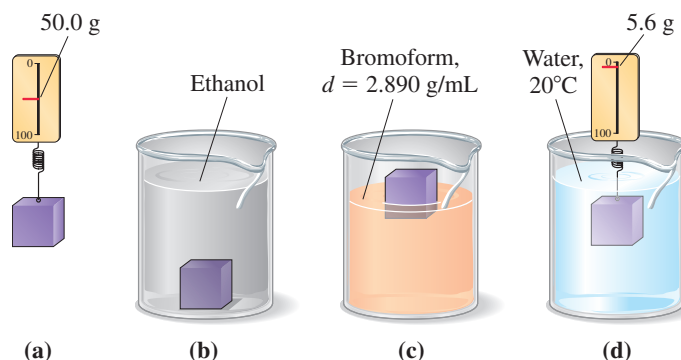
92. The Galileo thermometer shown in the photograph is based on the dependence of density on temperature. The liquid in the outer cylinder and the liquid in the partially filled floating glass balls are the same except that a colored dye has been added to the liquid in the balls. Explain how the Galileo thermometer works.



93. The canoe gliding gracefully along the water in the photograph is made of concrete, which has a density of about 2.4 g/cm^3 . Explain why the canoe does not sink.



94. The accompanying sketches suggest four observations made on a small block of plastic material. Tell what conclusions can be drawn from each sketch, and conclude by giving your best estimate of the density of the plastic.



95. As mentioned on page 41, the MCO was lost because of a mix-up in the units used to calculate the force needed to correct its trajectory. Ground-based computers generated the force correction file. On September 29, 1999, it was discovered that the forces reported by the ground-based computer for use in MCO navigation software were low by a factor of 4.45. The erroneous trajectory brought the MCO 56 km above the surface of Mars; the correct trajectory would have brought the MCO approximately 250 km above the surface. At 250 km, the MCO would have successfully entered the desired elliptic orbit. The data contained in the force correction file were delivered in lb-sec instead of the required SI units of newton-sec for the MCO navigation software. The newton is the SI unit of force and is described in Appendix B. The British Engineering (gravitational) system uses a pound (lb) as a unit of force and ft/s^2 as a unit of acceleration. In turn, the pound is defined as the pull of Earth on a unit of mass at a location where the acceleration caused by gravity is 32.174 ft/s^2 . The unit of mass in this case is the slug, which is 14.59 kg. Thus,

$$\text{BE unit of force} = 1 \text{ pound} = (\text{slug})(\text{ft/s}^2)$$

Use this information to confirm that

$$\text{BE unit of force} = 4.45 \times \text{SI unit of force}$$

$$1 \text{ pound} = 4.45 \text{ newton}$$

Self-Assessment Exercises

96. In your own words, define or explain the following terms or symbols: (a) mL; (b) % by mass; (c) $^{\circ}\text{C}$; (d) density; (e) element.
97. Briefly describe each of the following ideas: (a) SI base units; (b) significant figures; (c) natural law; (d) exponential notation.
98. Explain the important distinctions between each pair of terms: (a) mass and weight; (b) intensive and extensive properties; (c) substance and mixture; (d) systematic and random errors; (e) hypothesis and theory.
99. A procedure designed to test the truth or the validity of an explanation for many observations is called (a) a law; (b) a theory; (c) an experiment; (d) a hypothesis; (e) none of these.
100. The fact that the volume of a fixed amount of gas at a fixed temperature is inversely proportional to the gas pressure is an example of (a) a hypothesis; (b) a theory; (c) a paradigm; (d) the absolute truth; (e) a natural law.
101. If a sample of matter cannot be separated by physical means or decomposed by chemical means, the

sample is (a) a homogeneous mixture; (b) a heterogeneous mixture; (c) an element; (d) a compound; (e) a pure substance.

102. A good example of a homogeneous mixture is
(a) a cola drink in a tightly capped bottle
(b) distilled water leaving a distillation apparatus
(c) oxygen gas in a cylinder used in welding
(d) the material produced in a kitchen blender
103. Compared with its mass on Earth, the mass of the same object on the moon should be (a) less; (b) more; (c) the same; (d) nearly the same, but somewhat less.
104. Which answer has the correct number of significant figures?
(a) $(14.7 + 24.312) \times 87.27 = 3405$
(b) $(58 + 18 + 51)/3.000 = 42$
(c) $[97.2/(114 - 37)] = 1.26$
(d) $(1.172 - 0.4963)(4.193) = 2.83$
(e) none of these
105. Which two of the following masses are expressed to the nearest milligram? (a) 32.7 g; (b) 0.03271 kg; (c) 32.7068 g; (d) 32.707 g; (e) 30.7 mg; (f) $3 \times 10^3 \mu\text{g}$.
106. The highest temperature of the following group is (a) 217 K; (b) 273 K; (c) 217°F; (d) 105°C; (e) 373 K.
107. Which of the following quantities has the greatest mass?
(a) 752 mL of water at 20°C
(b) 1.05 L of ethanol at 20°C ($d = 0.789 \text{ g/mL}$)
(c) 750 g of chloroform at 20°C ($d = 1.483 \text{ g/mL}$)
(d) a cube of balsa wood ($d = 0.11 \text{ g/cm}^3$) that is 19.20 cm on edge
108. The density of silver is 10.5 g/mL. What is the volume of 475 g of silver?
(a) $475/10.5$; (b) $10.5/475$; (c) $1/[10.5 \times 475]$; (d) 475×10.5 ; (e) none of these
109. The density of water is 0.9982 g/cm^3 at 20°C. Express the density of water at 20°C in the following units: (a) g/L; (b) kg/m^3 ; (c) kg/km^3 .
110. Two students each made four measurements of the mass of an object. Their results are shown in the table below.

	Student A	Student B
Four measurements:	51.6, 50.8, 52.2, 50.2 g	50.1, 49.6, 51.0, 49.4 g
Their average:	51.3 g	50.0 g

The exact mass of the object is 51.0 g. Whose results are more precise, Student A's or Student B's? Whose results are more accurate?

111. The reported value for the volume of a rectangular piece of cardboard with the dimensions $36 \text{ cm} \times 20.2 \text{ cm} \times 9 \text{ mm}$ should be (a) $6.5 \times 10^3 \text{ cm}^3$; (b) $7 \times 10^2 \text{ cm}^3$; (c) 655 cm^3 ; (d) $6.5 \times 10^2 \text{ cm}^3$.

112. List the following in the order of increasing precision, indicating any quantities about which the precision is uncertain: (a) 1400 km; (b) 1516 kg; (c) 0.00304 g; (d) 125.34 cm; (e) 2000 mg.
113. Without doing detailed calculations, explain which of the following objects contains the greatest mass of the element iron.
(a) A 1.00 kg pile of pure iron filings.
(b) A cube of wrought iron, 5.0 cm on edge. Wrought iron contains 98.5% iron by mass and has a density of 7.7 g/cm^3 .
(c) A square sheet of stainless steel 0.30 m on edge and 1.0 mm thick. The stainless steel is an alloy (mixture) containing iron, together with 18% chromium, 8% nickel, and 0.18% carbon by mass. Its density is 7.7 g/cm^3 .
(d) 10.0 L of a solution characterized as follows: $d = 1.295 \text{ g/mL}$. This solution is 70.0% water and 30.0% of a compound of iron, by mass. The iron compound consists of 34.4% iron by mass.
114. A lump of pure copper weighs 25.305 g in air and 22.486 g when submerged in water ($d = 0.9982 \text{ g/mL}$) at 20.0°C. Suppose the copper is then rolled into a 248 cm^2 foil of uniform thickness. What will this thickness be, in millimeters?
115. Water, a compound, is a substance. Is there any circumstance under which a sample of pure water can exist as a heterogeneous mixture? Explain.
116. In the production of ammonia, the following processes occur.
(a) Air is liquefied at low temperature and high pressure.
(b) The temperature of the liquid air is raised gradually until the oxygen boils off. Essentially pure liquid nitrogen remains.
(c) Natural gas is treated with steam to produce carbon dioxide and hydrogen. The proportions of the products vary with the composition of the natural gas.
(d) Hydrogen and nitrogen (both gases) are combined at high temperature and pressures. They react to form ammonia gas. Some unreacted hydrogen and nitrogen remain.
(e) The gases are cooled until the ammonia liquefies, and then the gaseous hydrogen and nitrogen are recirculated to react again.
For each of (a) to (e), indicate whether a physical or chemical change occurs, and briefly explain your choice. For each italicized word or phrase, state whether it refers to an element, a compound, a mixture, or none of these, and briefly explain your choice.
117. Appendix E describes a useful study aid known as concept mapping. Using the method presented in Appendix E, construct a concept map illustrating the different concepts presented in Sections 1-2, 1-3, and 1-4.

2

Atoms and the Atomic Theory

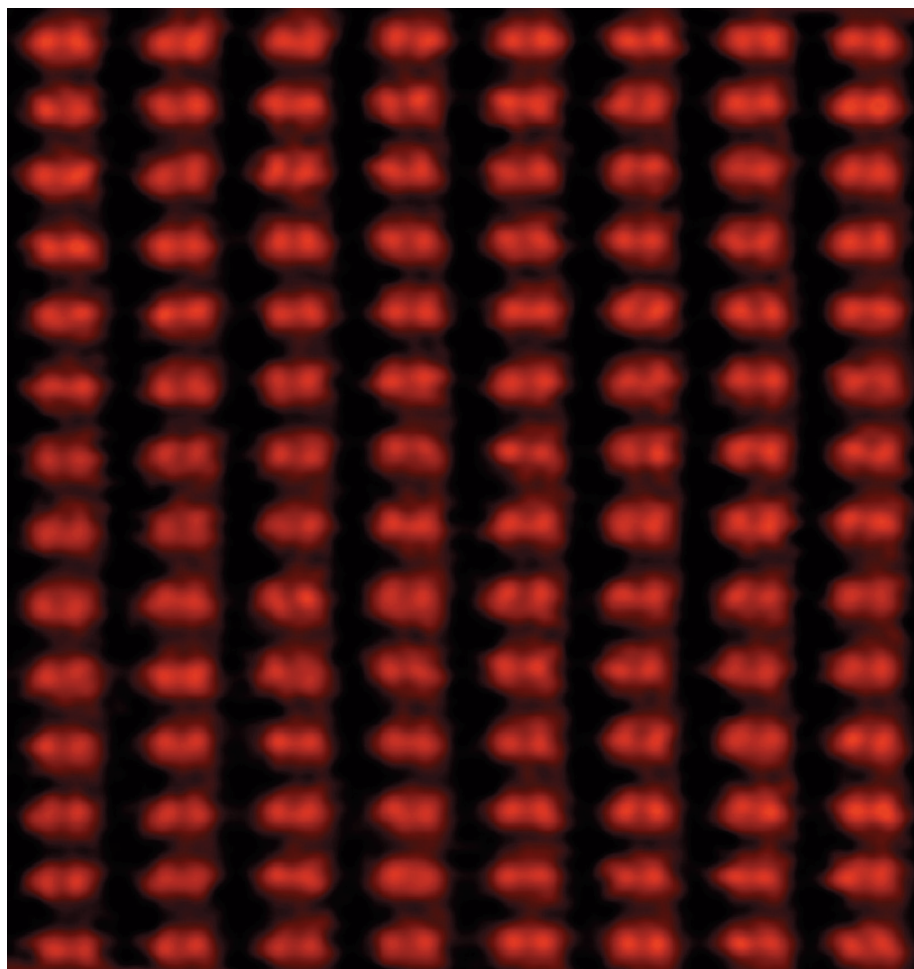
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- 2-2** Electrons and Other Discoveries in Atomic Physics
- 2-3** The Nuclear Atom
- 2-4** Chemical Elements
- 2-5** Atomic Mass
- 2-6** Introduction to the Periodic Table
- 2-7** The Concept of the Mole and the Avogadro Constant
- 2-8** Using the Mole Concept in Calculations

Learning Objectives

- 2.1** Describe and distinguish between the law of conservation of mass, the law of constant composition, and the law of multiple proportions.
- 2.2** Discuss the discovery of electrons, and describe their basic properties, such as charge and mass.
- 2.3** Identify the features of the nucleus of an atom, and discuss the properties of protons, neutrons, and electrons.
- 2.4** Describe the meaning of *isotope*, and discuss how the masses of all the elements were deduced.
- 2.5** Determine the atomic mass of an element from the experimentally determined masses and percent abundances of the isotopes.
- 2.6** Identify different regions of the periodic table, including metals and nonmetals, noble gases, metalloids, transition metals, and main-group elements.

We begin this chapter with a brief survey of early chemical discoveries, culminating in Dalton's atomic theory. This is followed by a description of the physical evidence leading to the modern picture of the *nuclear atom*, in which protons and neutrons are combined into a nucleus with electrons in space surrounding the nucleus. We will also introduce the periodic table as the primary means of organizing elements into groups with similar properties. Finally, we will introduce the



Courtesy of the Oak Ridge National Laboratory, managed by the U.S. Department of Energy by UT-Battelle, LLC.

▲ Image of silicon atoms that are only 78 pm apart; image produced by using a scanning transmission electron microscope (STEM). The hypothesis that all matter is made up of atoms has existed for more than 2000 years. It is only within the last few decades, however, that techniques have been developed that can render individual atoms visible.

concept of the mole and the Avogadro constant, which are the principal tools for counting atoms and molecules and measuring amounts of substances. We will use these tools throughout the text.

2-1

Early Chemical Discoveries and the Atomic Theory

Chemistry has been practiced for a very long time, even if its practitioners were much more interested in its applications than in its underlying principles. The blast furnace for extracting iron from iron ore appeared as early as A.D. 1300, and such important chemicals as sulfuric acid (oil of vitriol), nitric acid (aqua fortis), and sodium sulfate (Glauber's salt) were all well known and used several hundred years ago. Before the end of the eighteenth century, the principal gases of the atmosphere—nitrogen and oxygen—had been isolated, and natural laws had been proposed describing the physical behavior of gases. Yet chemistry cannot be said to have entered the modern age until the process of combustion was explained. In this section, we explore the direct link between the explanation of combustion and Dalton's atomic theory.

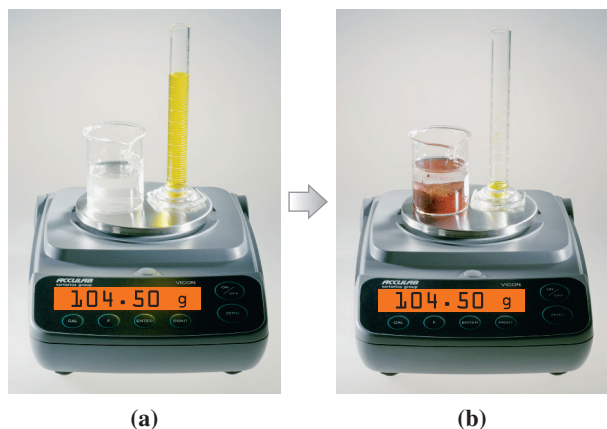
Law of Conservation of Mass

The process of combustion—burning—is so familiar that it is hard to realize what a difficult riddle it posed for early scientists. Some of the difficult-to-explain observations are described in Figure 2-1.

In 1774, Antoine Lavoisier (1743–1794) performed an experiment in which he heated a sealed glass vessel containing a sample of tin and some air. He found that the mass before heating (glass vessel + tin + air) and after heating (glass vessel + “tin calx” + remaining air) were the same. Through further experiments, he showed that the product of the reaction, tin calx (tin oxide), consisted of the original tin together with a portion of the air. Experiments like this proved to Lavoisier that oxygen from air is essential to combustion and also led him to formulate the **law of conservation of mass**:

The total mass of substances present after a chemical reaction is the same as the total mass of substances before the reaction.

This law is illustrated in Figure 2-2, where the reaction between silver nitrate and potassium chromate to give a red solid (silver chromate) is monitored by placing the reactants on a single-pan balance—the total mass does not change. Stated another way, the law of conservation of mass says that matter is neither created nor destroyed in a chemical reaction.



Richard Magnia/Fundamental Photographs

Figure 2-2 Mass is conserved during a chemical reaction

(a) Before the reaction, a beaker with a silver nitrate solution and a graduated cylinder with a potassium chromate solution are placed on a single-pan balance, which displays their combined mass—104.50 g. (b) When the solutions are mixed, a chemical reaction occurs that forms silver chromate (red precipitate) in a potassium nitrate solution. Note that the total mass—104.50 g—remains unchanged.

- 2.7** Identify the value of Avogadro's constant, and describe the meaning of molar mass.
- 2.8** Use Avogadro's constant to convert between values of mass, amount in moles, and number of atoms in a sample of element.



Carey B. Van Loon

Figure 2-1 Two combustion reactions

The apparent product of the combustion of the match—the ash—weighs *less* than the match. The product of the combustion of the magnesium ribbon (the “smoke”) weighs *more* than the ribbon. Actually, in each case, the total mass remains unchanged. To understand this, you have to know that oxygen gas enters into both combustions and that water and carbon dioxide are also products of the combustion of the match.

EXAMPLE 2-1 Applying the Law of Conservation of Mass

A 0.455 g sample of magnesium is allowed to burn in 2.315 g of oxygen gas. The sole product is magnesium oxide. After the reaction, no magnesium remains and the mass of unreacted oxygen is 2.015 g. What mass of magnesium oxide is produced?

Analyze

The total mass is unchanged. The total mass is the sum of the masses of the substances present initially. The mass of magnesium oxide is the total mass minus the mass of unreacted oxygen.

Solve

First, determine the total mass before the reaction.

$$\begin{aligned}\text{mass before reaction} &= 0.455 \text{ g magnesium} + 2.315 \text{ g oxygen} \\ &= 2.770 \text{ g mass before reaction}\end{aligned}$$

The total mass after the reaction is the same as before the reaction.

$$\begin{aligned}2.770 \text{ g mass after reaction} &= ? \text{ g magnesium oxide after reaction} \\ &\quad + 2.015 \text{ g oxygen after reaction}\end{aligned}$$

Solve for the mass of magnesium oxide.

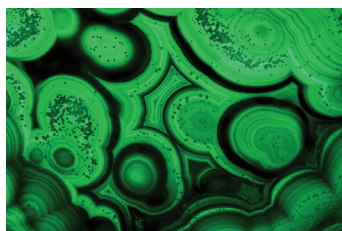
$$\begin{aligned}? \text{ g magnesium oxide after reaction} &= 2.770 \text{ g mass after reaction} \\ &\quad - 2.015 \text{ g oxygen after reaction} \\ &= 0.755 \text{ g magnesium oxide after reaction}\end{aligned}$$

Assess

Here is another approach. The mass of oxygen that reacted is $2.315 \text{ g} - 2.015 \text{ g} = 0.300 \text{ g}$. Thus, 0.300 g oxygen combined with 0.455 g magnesium to give $0.300 \text{ g} + 0.455 \text{ g} = 0.755 \text{ g}$ magnesium oxide.

PRACTICE EXAMPLE A: A 0.382 g sample of magnesium is allowed to react with 2.652 g of nitrogen gas. The sole product is magnesium nitride. After the reaction, the mass of unreacted nitrogen is 2.505 g. What mass of magnesium nitride is produced?

PRACTICE EXAMPLE B: A 7.12 g sample of magnesium is heated with 1.80 g of bromine. All the bromine is used up, and 2.07 g of magnesium bromide is the only product. What mass of magnesium remains *unreacted*?



(a)



(b)

▲ The mineral malachite (a) and the green patina on a copper roof (b) are both basic copper carbonate, just like the basic copper carbonate prepared by Proust in 1799.

**2-1 Concept Assessment**

Jan Baptista van Helmont (1579–1644) weighed a young willow tree and the soil in which the tree was planted. Five years later he found that the mass of soil had decreased by only 0.057 kg, while that of the tree had increased by 75 kg. During that period he had added only water to the bucket in which the tree was planted. Helmont concluded that essentially all the mass gained by the tree had come from the water. Was this a valid conclusion? Explain.

Law of Constant Composition

In 1799, Joseph Proust (1754–1826) reported, “One hundred pounds of copper, dissolved in sulfuric or nitric acids and precipitated by the carbonates of soda or potash, invariably gives 180 pounds of green carbonate.”* This and similar observations became the basis of the **law of constant composition**, or the **law of definite proportions**:

All samples of a compound have the same composition—the same proportions by mass of the constituent elements.

To see how the law of constant composition works, consider the compound water. Water is made up of two atoms of hydrogen (H) for every atom of oxygen (O), a fact that can be represented symbolically by a *chemical formula*, the familiar H_2O .

*The substance Proust produced is actually a more complex substance called *basic* copper carbonate. Proust’s results were valid because, like all compounds, basic copper carbonate has a constant composition.

The two samples described below have the same proportions of the two elements, expressed as percentages by mass. To determine the percent by mass of hydrogen, for example, simply divide the mass of hydrogen by the sample mass and multiply by 100%. For each sample, you will obtain the same result: 11.19% H.

Sample A and Its Composition		Sample B and Its Composition	
10.000 g		27.000 g	
1.119 g H	% H = 11.19	3.021 g H	% H = 11.19
8.881 g O	% O = 88.81	23.979 g O	% O = 88.81

EXAMPLE 2-2**Using the Law of Constant Composition**

In Example 2-1 we found that when 0.455 g of magnesium reacted with 2.315 g of oxygen, 0.755 g of magnesium oxide was obtained. Determine the mass of magnesium contained in a 0.500 g sample of magnesium oxide.

Analyze

We know that 0.755 g of magnesium oxide contains 0.455 g of magnesium. According to the law of constant composition, the mass ratio 0.455 g magnesium/0.755 g magnesium oxide should exist in all samples of magnesium oxide.

Solve

Application of the law of constant composition gives

$$\frac{0.455 \text{ g magnesium}}{0.755 \text{ g magnesium oxide}} = \frac{? \text{ g magnesium}}{0.500 \text{ g magnesium oxide}}$$

Solving the expression above, we obtain

$$\begin{aligned} ? \text{ g magnesium} &= 0.500 \text{ g magnesium oxide} \times \frac{0.455 \text{ g magnesium}}{0.755 \text{ g magnesium oxide}} \\ &= 0.301 \text{ g magnesium} \end{aligned}$$

Assess

You can also work this problem by using mass percentages. If 0.755 g of magnesium oxide contains 0.455 g of magnesium, then magnesium oxide is $(0.455 \text{ g}/0.755 \text{ g}) \times 100\% = 60.3\%$ magnesium by mass and $(100\% - 60.3\%) = 39.7\%$ oxygen by mass. Thus, a 0.500 g sample of magnesium oxide must contain $0.500 \text{ g} \times 60.3\% = 0.301 \text{ g}$ of magnesium and $0.500 \text{ g} \times 39.7\% = 0.199 \text{ g}$ of oxygen.

PRACTICE EXAMPLE A: What masses of magnesium and oxygen must be combined to make exactly 2.000 g of magnesium oxide?

PRACTICE EXAMPLE B: What substances are present, and what are their masses, after the reaction of 10.00 g of magnesium and 10.00 g of oxygen?

**2-2 Concept Assessment**

When 4.15 g magnesium and 82.6 g bromine react, (1) all the magnesium is used up, (2) some bromine remains unreacted, and (3) magnesium bromide is the only product. With this information alone, is it possible to deduce the mass of magnesium bromide produced? Explain.

Dalton's Atomic Theory

From 1803 to 1808, John Dalton, an English chemist, physicist, and meteorologist, used the two fundamental laws of chemical combination just described as the basis of an atomic theory. His theory involved three assumptions:

1. Each chemical element is composed of minute, indivisible particles called atoms. Atoms can be neither created nor destroyed during a chemical change.



▲ John Dalton (1766–1844), developer of the atomic theory. Dalton is also known for his research on color blindness (a condition he had); color blindness is referred to as Daltonism in several languages.

Keep in Mind

that all we know is that the second oxide is twice as rich in oxygen as the first. If the first is CO, the possibilities for the second are CO₂, C₂O₄, C₃O₆, and so on. (See also Exercise 18.)

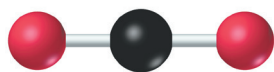


Figure 2-3 Molecules CO and CO₂ illustrating the law of multiple proportions

The mass of carbon is the same in the two molecules, but the mass of oxygen in CO₂ is twice the mass of oxygen in CO. Thus, in accordance with the law of multiple proportions, the masses of oxygen in the two compounds, relative to a fixed mass of carbon, are in a ratio of small whole numbers, 2:1.

- All atoms of an element are alike in mass (weight) and other properties, but the atoms of one element are different from those of all other elements.
- In each of their compounds, different elements combine in a simple numerical ratio, for example, one atom of A to one of B (AB), or one atom of A to two of B (AB₂).

If atoms of an element are indestructible (assumption 1), then the *same* atoms must be present after a chemical reaction as before. The total mass remains unchanged. Dalton's theory explains the law of conservation of mass. If all atoms of an element are alike in mass (assumption 2) and if atoms unite in *fixed* numerical ratios (assumption 3), the percent composition of a compound must have a unique value, regardless of the origin of the sample analyzed. Dalton's theory also explains the law of constant composition.

Like all good theories, Dalton's atomic theory led to a prediction—the **law of multiple proportions**.

If two elements form more than a single compound, the masses of one element combined with a fixed mass of the second are in the ratio of small whole numbers.

To illustrate, consider two oxides of carbon (an oxide is a combination of an element with oxygen). In one oxide, 1.000 g of carbon is combined with 1.333 g of oxygen, and in the other, with 2.667 g of oxygen. We see that the second oxide is richer in oxygen; in fact, it contains twice as much oxygen as the first, $2.667 \text{ g} / 1.333 \text{ g} = 2.00$. We now know that the first oxide corresponds to the formula CO and the second, CO₂ (Fig. 2-3).

The characteristic relative masses of the atoms of the various elements became known as atomic weights, and throughout the nineteenth century, chemists worked at establishing reliable values of relative atomic weights. Mostly, however, chemists directed their attention to discovering new elements, synthesizing new compounds, developing techniques for analyzing materials, and in general, building up a vast body of chemical knowledge. Efforts to unravel the structure of the atom became the focus of physicists, as we see in the next several sections.

2-2 Electrons and Other Discoveries in Atomic Physics

Fortunately, we can acquire a qualitative understanding of atomic structure without having to retrace all the discoveries that preceded atomic physics. We do, however, need a few key ideas about the interrelated phenomena of electricity and magnetism, which we briefly discuss here. Electricity and magnetism were used in the experiments that led to the current theory of atomic structure.

Certain objects display a property called electric charge, which can be either positive (+) or negative (–). Positive and negative charges attract each other, while two positive or two negative charges repel each other. As we learn in this section, all objects of matter are made up of charged particles. An object having equal numbers of positively and negatively charged particles carries no net charge and is electrically neutral. If the number of positive charges exceeds the number of negative charges, the object has a net positive charge. If negative charges exceed positive charges, the object has a net negative charge. Sometimes when one substance is rubbed against another, as in combing hair, net electric charges build up on the objects, implying that rubbing separates

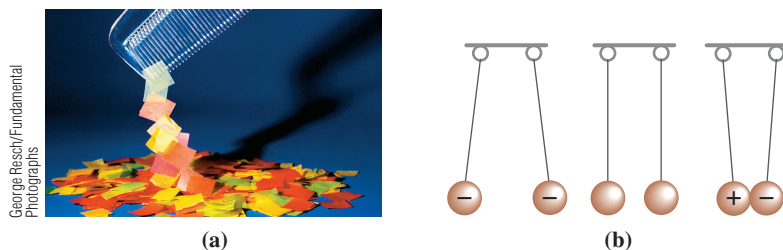


Figure 2-4 Forces between electrically charged objects

(a) Electrostatically charged comb. If one combs their hair on a dry day, a static charge develops on the comb and causes bits of paper to be attracted to the comb. (b) Both objects on the left carry a negative electric charge. Objects with like charge repel each other. The objects in the center lack any electric charge and exert no forces on each other. The objects on the right carry opposite charges—one positive and one negative—and attract each other.

some positive and negative charges (Fig. 2-4). Moreover, when a stationary (static) positive charge builds up in one place, a negative charge of equal size appears somewhere else; the charge is balanced.

Figure 2-5 shows how charged particles behave when they move through the field of a magnet. They are deflected from their straight-line path into a curved path in a plane perpendicular to the field. Think of the field or region of influence of the magnet as represented by a series of invisible “lines of force” running from the north pole to the south pole of the magnet.

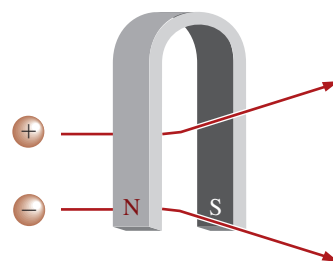


Figure 2-5 Effect of a magnetic field on charged particles

When charged particles travel through a magnetic field so that their path is perpendicular to the field, they are deflected by the field. Negatively charged particles are deflected in one direction, and positively charged particles in the opposite direction. Several phenomena described in this section depend on this behavior.

The Discovery of Electrons

CRT, the abbreviation for cathode-ray tube, was once a familiar acronym. Before liquid crystal display (LCD) was available, the CRT was the heart of computer monitors and TV sets. The first cathode-ray tube was made by Michael Faraday (1791–1867) about 150 years ago. When he passed electricity through glass tubes from which most of the air had been evacuated, Faraday discovered **cathode rays**, a type of radiation emitted by the negative terminal or *cathode*. The radiation crossed the evacuated tube to the positive terminal or *anode*. Later scientists found that cathode rays travel in straight lines and have properties that are independent of the cathode material (that is, whether it is iron, platinum, and so on). The construction of a CRT is shown in Figure 2-6. The cathode rays produced in the CRT are invisible, and they can be detected only by the light emitted by materials that they strike. These materials, called *phosphors*, are painted on the end of the CRT so that the path of the cathode rays can be revealed. (*Fluorescence* is the term used to describe the emission of light by a phosphor when it is struck by

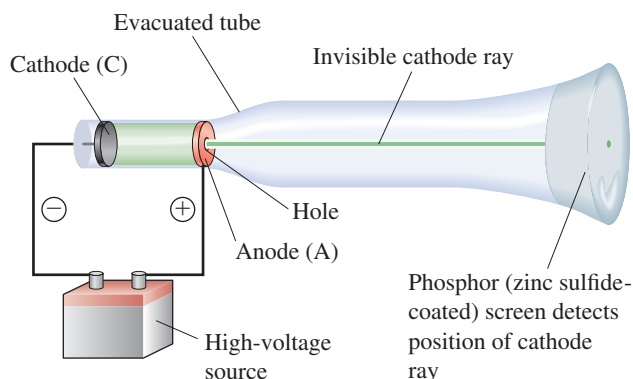


Figure 2-6 A cathode-ray tube

The high-voltage source of electricity creates a negative charge on the electrode at the left (cathode) and a positive charge on the electrode at the right (anode). Cathode rays pass from the cathode (C) to the anode (A), which is perforated to allow the passage of a narrow beam of cathode rays. The rays are visible only through the green fluorescence that they produce on the zinc sulfide-coated screen at the end of the tube. They are invisible in other parts of the tube.

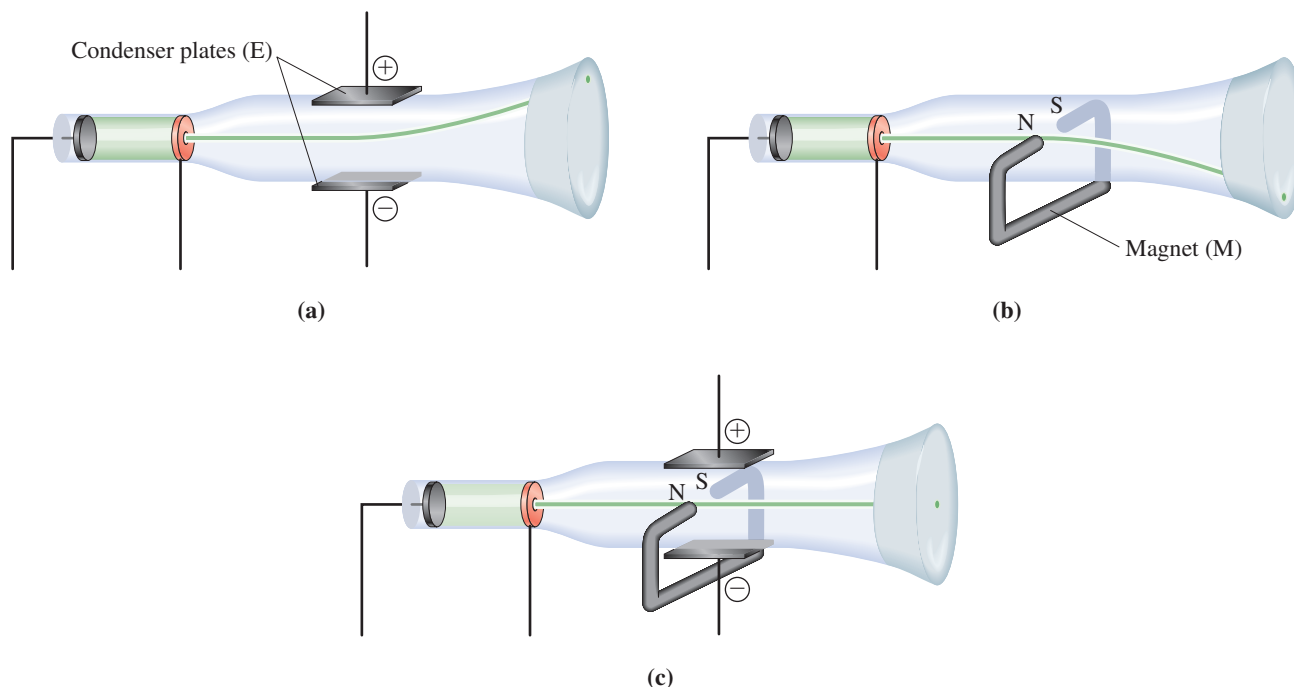


Figure 2-7 Cathode rays and their properties

(a) Deflection of cathode rays in an electric field. The beam of cathode rays is deflected as it travels from left to right in the field of the electrically charged condenser plates (E). The deflection corresponds to that expected of negatively charged particles.

(b) Deflection of cathode rays in a magnetic field. The beam of cathode rays is deflected as it travels from left to right in the field of the magnet (M). The deflection corresponds to that expected of negatively charged particles.

(c) Determining the mass-to-charge ratio, m/e , for cathode rays. The cathode-ray beam strikes the end screen undeflected if the forces exerted on it by the electric and magnetic fields are counterbalanced. By knowing the strengths of the electric and magnetic fields, together with other data, a value of m/e can be obtained. Precise measurements yield a value of -5.6857×10^{-9} g per coulomb. (Because cathode rays carry a negative charge, the sign of the mass-to-charge ratio is also negative.)

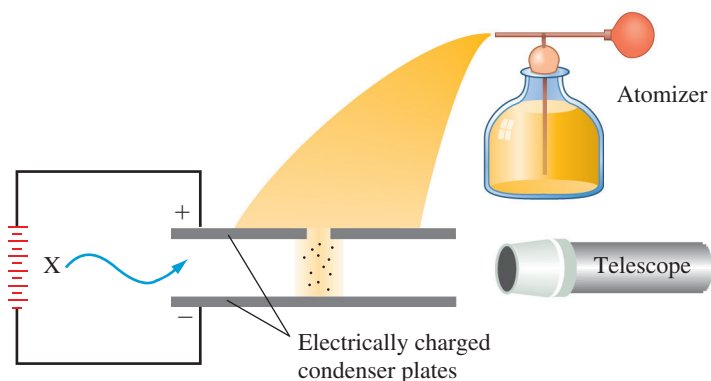
energetic radiation.) Another significant observation about cathode rays is that they are deflected by electric and magnetic fields in the manner expected for negatively charged particles (Fig. 2-7a, b).

In 1897, by the method outlined in Figure 2-7(c), J. J. Thomson (1856–1940) established the ratio of mass (m) to electric charge (e) for cathode rays, that is, m/e . Also, Thomson concluded that cathode rays are negatively charged *fundamental* particles of matter found in all atoms. (The properties of cathode rays are *independent* of the composition of the cathode.) Cathode rays subsequently became known as **electrons**, a term first proposed by George Stoney in 1874.

Robert Millikan (1868–1953) determined the electronic charge e through a series of oil-drop experiments (1906–1914), described in Figure 2-8. The currently accepted value of the electronic charge e , expressed in coulombs to five significant figures, is -1.6022×10^{-19} C. By combining this value with an accurate value of the mass-to-charge ratio for an electron, we find that the mass of an electron is 9.1094×10^{-28} g.

Once the electron was seen to be a fundamental particle of matter found in all atoms, atomic physicists began to speculate on how these particles were incorporated into atoms. The commonly accepted model was that proposed by J. J. Thomson. Thomson thought that the positive charge necessary to counterbalance the negative charges of electrons in a neutral atom was in the form of

► The coulomb (C) is the SI unit of electric charge (see also Appendix C).

**Figure 2-8** Millikan's oil-drop experiment

Ions (charged atoms or molecules) are produced by energetic radiation, such as X-rays (X). Some of these ions become attached to oil droplets, giving them a net charge. The fall of a droplet in the electric field between the condenser plates is speeded up or slowed down, depending on the magnitude and sign of the charge on the droplet. By analyzing data from a large number of droplets, Millikan concluded that the magnitude of the charge, q , on a droplet is an *integral* multiple of the electric charge, e . That is, $q = ne$ (where $n = 1, 2, 3, \dots$).

a nebulous cloud. Electrons, he suggested, floated in a diffuse cloud of positive charge (rather like a lump of gelatin with electron “fruit” embedded in it). This model became known as the plum-pudding model because of its similarity to a popular English dessert. The plum-pudding model is illustrated in Figure 2-9 for a neutral atom and for atomic species, called *ions*, which carry a net charge.

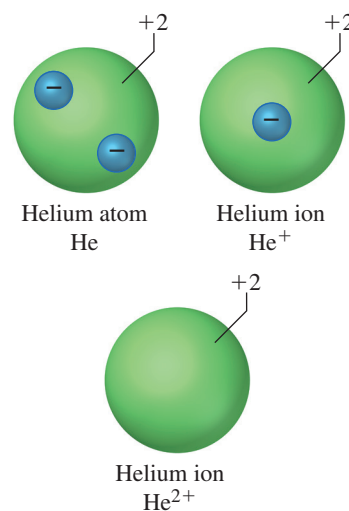
X-Rays and Radioactivity

Cathode-ray research had many important spin-offs. In particular, two natural phenomena of immense theoretical and practical significance were discovered in the course of other investigations.

In 1895, Wilhelm Roentgen (1845–1923) noticed that when cathode-ray tubes were operating, certain materials *outside* the tubes glowed or fluoresced. He showed that this fluorescence was caused by radiation emitted by the cathode-ray tubes. Because of the unknown nature of this radiation, Roentgen coined the term *X-ray*. We now recognize the X-ray as a form of high-energy electromagnetic radiation, which is discussed in Chapter 8.

Antoine Henri Becquerel (1852–1908) associated X-rays with fluorescence and wondered if naturally fluorescent materials produce X-rays. To test this idea, he wrapped a photographic plate with black paper, placed a coin on the paper, covered the coin with a uranium-containing fluorescent material, and exposed the entire assembly to sunlight. When he developed the film, a clear image of the coin could be seen. The fluorescent material had emitted radiation (presumably X-rays) that penetrated the paper and exposed the film. On one occasion, because the sky was overcast, Becquerel placed the experimental assembly inside a desk drawer for a few days while waiting for the weather to clear. On resuming the experiment, Becquerel decided to replace the original photographic film, expecting that it may have become slightly exposed. He developed the original film and found that instead of the expected feeble image, there was a very sharp one. The film had become strongly exposed because the uranium-containing material had emitted radiation continuously, even when it was not fluorescing. Becquerel had discovered **radioactivity**.

Ernest Rutherford (1871–1937) identified two types of radiation from radioactive materials, alpha (α) and beta (β). **Alpha particles** carry two fundamental units of positive charge and have essentially the same mass as helium atoms. In fact, alpha particles are identical to He^{2+} ions. **Beta particles** are negatively charged particles produced by changes occurring within the nuclei of radioactive atoms and have the same properties as electrons. A third form of radiation, which is not affected by electric or magnetic fields, was discovered in 1900 by Paul Villard (1860–1934). This radiation, called **gamma rays** (γ),

**Figure 2-9** The plum-pudding atomic model

According to this model, a helium atom would have a +2 cloud of positive charge and two electrons (−2). If a helium atom loses one electron, it becomes charged and is called an *ion*. This ion, referred to as He^+ , has a net charge of 1+. If the helium atom loses both electrons, the He^{2+} ion forms.

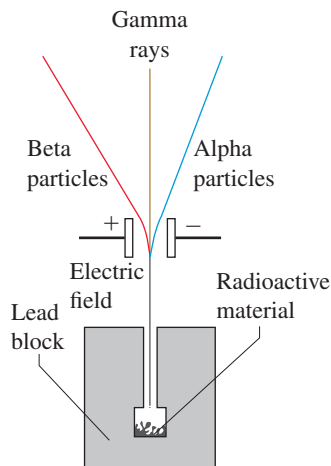


Figure 2-10 Three types of radiation from radioactive materials

The radioactive material is enclosed in a lead block. All the radiation except that passing through the narrow opening is absorbed by the lead. When the escaping radiation is passed through an electric field, it splits into three beams. One beam is undeflected—these are gamma (γ) rays. A second beam is attracted to the negatively charged plate. These are the positively charged alpha (α) particles. The third beam, of negatively charged beta (β) particles, is deflected toward the positive plate.

► Perhaps because he found it tedious to sit in the dark and count spots of light on a zinc sulfide screen, Geiger was motivated to develop an automatic radiation detector. The result was the well-known Geiger counter.

is not made up of particles; it is electromagnetic radiation of extremely high penetrating power. These three forms of radioactivity are illustrated in Figure 2-10.

By the early 1900s, additional radioactive elements were discovered, principally by Marie Curie (1867–1934) and Pierre Curie (1859–1906). Rutherford and Frederick Soddy (1877–1956) made another profound finding: The chemical properties of a radioactive element *change* as it undergoes radioactive decay. This observation suggests that radioactivity involves fundamental changes at the *subatomic* level—in radioactive decay, one element is changed into another, a process known as *transmutation*.

2-3 The Nuclear Atom

In 1909, Rutherford, with his assistant Hans Geiger, began a line of research using α particles as probes to study the inner structure of atoms. Based on Thomson's plum-pudding model, Rutherford expected that most particles in a beam of α particles would pass through thin sections of matter largely undeflected, but that some α particles would be slightly scattered or deflected as they encountered electrons. By studying these scattering patterns, he hoped to deduce something about the distribution of electrons in atoms.

The apparatus used for these studies is pictured in Figure 2-11. Alpha particles were detected by the flashes of light they produced when they struck a zinc sulfide screen mounted on the end of a telescope. When Geiger and Ernst Marsden, a student, bombarded very thin foils of gold with α particles, they observed the following:

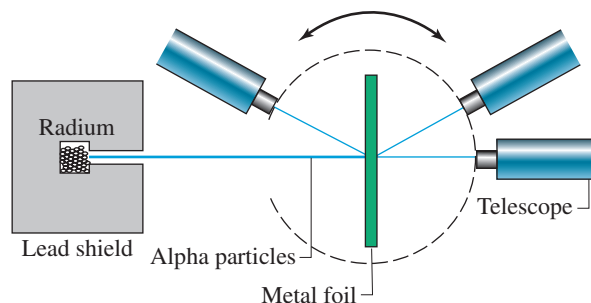
- The majority of α particles penetrated the foil undeflected.
- Some α particles experienced slight deflections.
- A few (about 1 in every 20,000) suffered rather serious deflections as they penetrated the foil.
- A similar number did not pass through the foil at all but bounced back in the direction from which they had come.

The large-angle scattering greatly puzzled Rutherford. As he commented some years later, this observation was “about as credible as if you had fired a 15-inch shell at a piece of tissue paper and it came back and hit you.” By 1911, though, Rutherford had an explanation. He based his explanation on a model of the atom known as the *nuclear atom* and having these features:

1. Most of the mass and all of the positive charge of an atom are centered in a very small region called the *nucleus*. The remainder of the atom is mostly *empty space*.
2. The magnitude of the positive charge is different for different atoms and is approximately one-half the atomic weight of the element.
3. There are as many electrons outside the nucleus as there are units of positive charge on the nucleus. The atom as a whole is electrically neutral.

Figure 2-11 The scattering of α particles by metal foil

The telescope travels in a circular track around an evacuated chamber containing the metal foil. Most α particles pass through the metal foil undeflected, but some are deflected through large angles.



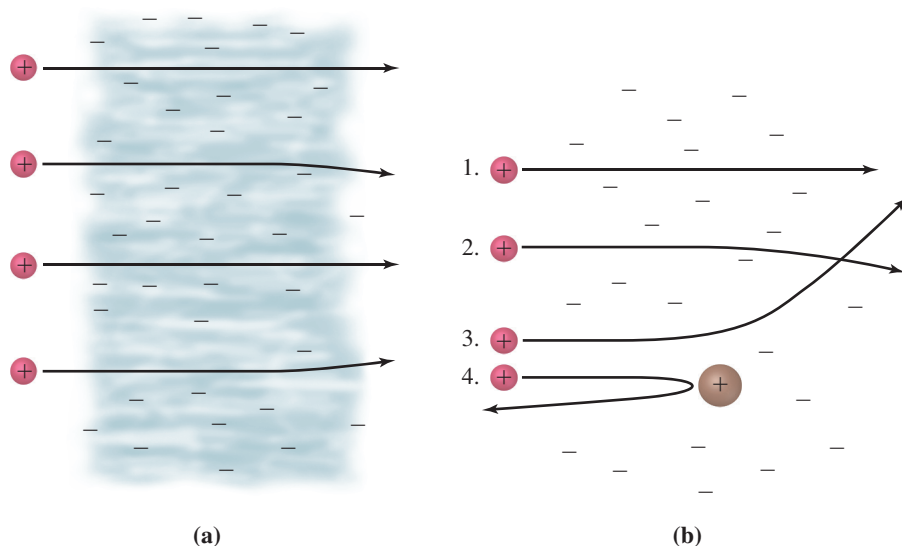


Figure 2-12 Explaining the results of α -particle scattering experiments

(a) Rutherford's expectation was that small, positively charged α particles should pass through the nebulous, positively charged cloud of the Thomson plum-pudding model largely undeflected. Some would be slightly deflected by passing near electrons (present to neutralize the positive charge of the cloud). (b) Rutherford's explanation was based on a nuclear atom. With an atomic model having a small, dense, positively charged nucleus and extranuclear electrons, we would expect the four different types of paths actually observed:

1. undeflected straight-line paths exhibited by most of the α particles
2. slight deflections of α particles passing close to electrons
3. severe deflections of α particles passing close to a nucleus
4. reflections from the foil of α particles approaching a nucleus head-on

Rutherford's initial expectation and his explanation of the α -particle experiments are described in Figure 2-12.

Discovery of Protons and Neutrons

Rutherford's nuclear atom suggested the existence of positively charged fundamental particles of matter in the nuclei of atoms. Rutherford himself discovered these particles, called **protons**, in 1919 in studies involving the scattering of α particles by nitrogen atoms in air. The protons were freed as a result of collisions between α particles and the nuclei of nitrogen atoms. At about this same time, Rutherford predicted the existence in the nucleus of electrically neutral fundamental particles. In 1932, James Chadwick (1891–1974) showed that a newly discovered penetrating radiation consisted of beams of *neutral* particles. These particles, called **neutrons**, originated from the nuclei of atoms. Thus, it has been only for about the past 100 years that we have had the atomic model suggested by Figure 2-13.

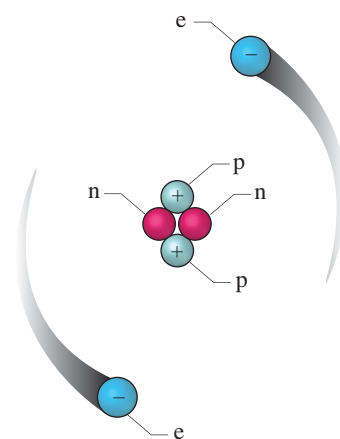


Figure 2-13 The nuclear atom—illustrated by the helium atom

In this drawing, electrons are shown much closer to the nucleus than is the case. The actual situation is more like this: If the entire atom were represented by a room, $5\text{ m} \times 5\text{ m} \times 5\text{ m}$, the nucleus would occupy only about as much space as the period at the end of this sentence.

2-3 Concept Assessment

In light of information presented to this point in the text, explain which of the three assumptions of Dalton's atomic theory (page 65) can still be considered correct and which cannot.

► The masses of the proton and neutron are different in the fourth significant figure. The charges of the proton and electron, however, are believed to be exactly equal in magnitude (but opposite in sign). The charges and masses are known much more precisely than suggested here. More precise values are given on the inside back cover.

Table 2.1 Properties of Three Fundamental Particles

	Electric Charge		Mass	
	SI (C)	Atomic	SI (g)	Atomic (u) ^a
Proton	$+1.6022 \times 10^{-19}$	+1	1.6726×10^{-24}	1.0073
Neutron	0	0	1.6749×10^{-24}	1.0087
Electron	-1.6022×10^{-19}	-1	9.1094×10^{-28}	0.00054858

^au is the SI symbol for atomic mass unit (abbreviated as amu).

Properties of Protons, Neutrons, and Electrons

The number of protons in a given atom is called the **atomic number**, or the **proton number**, **Z**. The number of electrons in the atom is also equal to **Z** because the atom is electrically neutral. The total number of protons and neutrons in an atom is called the **mass number**, **A**. The number of neutrons, the **neutron number**, is $A - Z$. An electron carries an atomic unit of negative charge, a proton carries an atomic unit of positive charge, and a neutron is electrically neutral. Table 2.1 presents the charges and masses of protons, neutrons, and electrons in two ways.

The **atomic mass unit** (described more fully on page 74) is defined as exactly 1/12 of the mass of the atom known as carbon-12 (read as carbon twelve). An atomic mass unit is abbreviated as amu and denoted by the symbol u. As we see from Table 2.1, the proton and neutron masses are just slightly greater than 1 u. By comparison, the mass of an electron is only about 1/2000th the mass of the proton or neutron.

The three subatomic particles considered in this section are the only ones involved in the phenomena of interest to us in this text. You should be aware, however, that a study of matter at its most fundamental level must consider many additional subatomic particles. The electron is believed to be a truly fundamental particle. However, modern particle physics now considers the neutron and proton to be composed of other, more fundamental particles.

2-4 Chemical Elements

Now that we have acquired some fundamental ideas about atomic structure, we can more thoroughly discuss the concept of chemical elements.

All atoms of a particular element have the same atomic number, **Z**, and, conversely, all atoms with the same number of protons are atoms of the same element. The elements shown on the inside front cover have atomic numbers from $Z = 1$ to $Z = 118$. Each element has a name and a distinctive symbol. **Chemical symbols** are one- or two-letter abbreviations of the name (usually the English name). The first (but never the second) letter of the symbol is capitalized; for example, carbon, C; oxygen, O; neon, Ne; and silicon, Si. Some elements known since ancient times have symbols based on their Latin names, such as Fe for iron (*ferrum*) and Pb for lead (*plumbum*). The element sodium has the symbol Na, based on the Latin *natrium* for sodium carbonate. Potassium has the symbol K, based on the Latin *kalium* for potassium carbonate. The symbol for tungsten, W, is based on the German *wolfram*.

Elements beyond uranium ($Z = 92$) do not occur naturally and must be synthesized in particle accelerators (described in Chapter 25). Elements of the very highest atomic numbers have been produced only on a limited number of occasions, a few atoms at a time. Inevitably, controversies have arisen about

► Other atomic symbols not based on English names include Cu, Ag, Sn, Sb, Au, and Hg.

which research team discovered a new element and, in fact, whether a discovery was made at all. However, international agreement has been reached on all of the 118 elements; each one has an official name and symbol.

Isotopes

To represent the composition of any particular atom, we need to specify its number of protons (p), neutrons (n), and electrons (e). We can do this with the symbolism

$$\begin{array}{c} \text{number p} + \text{number n} \xrightarrow{\hspace{1cm}} \\ \text{number p} \xrightarrow{\hspace{1cm}} \end{array} \begin{array}{c} A \\ Z \end{array} E \longleftarrow \text{symbol of element} \quad (2.1)$$

This symbolism indicates that the atom is element E and that it has atomic number Z and mass number A. For example, an atom of aluminum represented as ${}_{13}^{27}\text{Al}$ has 13 protons and 14 neutrons in its nucleus and 13 electrons outside the nucleus. (Recall that an atom has the same number of electrons as protons.)

Contrary to what Dalton thought, we now know that atoms of an element do not necessarily all have the same mass. In 1912, J. J. Thomson measured the mass-to-charge ratios of positive ions formed from neon atoms. From these ratios he deduced that about 91% of the atoms had one mass and that the remaining atoms were about 10% heavier. All neon atoms have 10 protons in their nuclei, and most have 10 neutrons as well. A very few neon atoms, however, have 11 neutrons and some have 12. We can represent these three different types of neon atoms as



Atoms that have the *same* atomic number (Z) but *different* mass numbers (A) are called **isotopes**. Of all Ne atoms on Earth, 90.51% are ${}_{10}^{20}\text{Ne}$. The percentages of ${}_{10}^{21}\text{Ne}$ and ${}_{10}^{22}\text{Ne}$ are 0.27% and 9.22%, respectively. These percentages—90.51%, 0.27%, 9.22%—are the **percent isotopic abundances** of the three neon isotopes. Sometimes the mass numbers of isotopes are incorporated into the names of elements, such as neon-20 (neon twenty). Percent isotopic abundances are always based on *numbers*, not masses. Thus, 9051 of every 10,000 neon atoms are neon-20 atoms. Some elements, as they exist in nature, consist of just a single type of atom and therefore do not have naturally occurring isotopes.* Aluminum, for example, consists only of aluminum-27 atoms.

Ions

When atoms lose or gain electrons, for example, in the course of a chemical reaction, the species formed are called **ions** and carry net charges. Because an electron is negatively charged, adding electrons to an electrically neutral atom produces a negatively charged ion. Removing electrons results in a positively charged ion. The number of protons does not change when an atom becomes an ion. For example, ${}^{20}\text{Ne}^+$ and ${}^{22}\text{Ne}^{2+}$ are ions. The first one has 10 protons, 10 neutrons, and 9 electrons. The second one also has 10 protons, but 12 neutrons and 8 electrons. The charge on an ion is equal to the number of protons *minus* the number of electrons. That is

$$\begin{array}{c} \text{number p} + \text{number n} \xrightarrow{\hspace{1cm}} \\ \text{number p} \xrightarrow{\hspace{1cm}} \end{array} \begin{array}{c} A \\ Z \end{array} E^{\# \pm} \longleftarrow \text{number p} - \text{number e} \quad (2.2)$$

Another example is the ${}^{16}\text{O}^{2-}$ ion. In this ion, there are 8 protons (atomic number 8), 8 neutrons (mass number – atomic number), and 10 electrons ($8 - 10 = -2$).

***Nuclide** is the general term used to describe an atom with a particular atomic number and mass number. Although there are several elements with only one naturally occurring nuclide, it is possible to produce additional nuclides of these elements—*isotopes*—by artificial means (Section 25-3). The artificial isotopes are radioactive, however. In all, the number of synthetic isotopes exceeds the number of naturally occurring ones by several fold.

◀ Because neon is the only element with $Z = 10$, the symbols ${}^{20}\text{Ne}$, ${}^{21}\text{Ne}$, and ${}^{22}\text{Ne}$ convey the same meaning as ${}_{10}^{20}\text{Ne}$, ${}_{10}^{21}\text{Ne}$, and ${}_{10}^{22}\text{Ne}$.

◀ Odd-numbered elements tend to have fewer isotopes than do even-numbered elements. Section 25-7 will explain why.

◀ Usually all the isotopes of an element share the same name and atomic symbol. The exception is hydrogen. Isotope ${}^2_1\text{H}$ is called deuterium (symbol D), and ${}^3_1\text{H}$ is tritium (T).

◀ In this expression, $\# \pm$ indicates that the charge is written with the number (#) *before* the + or – sign. However, when the charge is 1+ or 1–, the number 1 is not included.

EXAMPLE 2-3 Relating the Numbers of Protons, Neutrons, and Electrons in Atoms and Ions

Through an appropriate symbol, indicate the number of protons, neutrons, and electrons in (a) an atom of barium-135 and (b) the double negatively charged ion of selenium-80.

Analyze

Given the name of an element, we can find the symbol and the atomic number, Z , for that element from a list of elements or a periodic table. To determine the number of protons, neutrons, and electrons, we make use of the following relationships:

$$Z = \text{number } p \qquad A = \text{number } p + \text{number } n \qquad \text{charge} = \text{number } p - \text{number } e$$

The relationships above are summarized in expression (2.2).

Solve

- (a) We are given the name (barium) and the mass number of the atom (135). From a list of the elements or a periodic table we obtain the symbol (Ba) and the atomic number ($Z = 56$), leading to the symbolic representation



From this symbol one can deduce that the neutral atom has 56 protons; a neutron number of $A - Z = 135 - 56 = 79$ neutrons; and a number of electrons equal to Z , that is, 56 electrons.

- (b) We are given the name (selenium) and the mass number of the ion (80). From a list of the elements or a periodic table we obtain the symbol (Se) and the atomic number (34). Together with the fact that the ion carries a charge of $2-$, we have the data required to write the symbol



From this symbol, we can deduce that the ion has 34 protons; a neutron number of $A - Z = 80 - 34 = 46$ neutrons; and 36 electrons, leading to a net charge of $+34 - 36 = -2$.

Assess

When writing the symbol for a particular atom or ion, we often omit the atomic number. For example, for $^{135}_{56}\text{Ba}$ and $^{80}_{34}\text{Se}^{2-}$, we often use the simpler representations ^{135}Ba and $^{80}\text{Se}^{2-}$.

PRACTICE EXAMPLE A: Use the notation $^A_Z E$ to represent the isotope of silver having a neutron number of 62.

PRACTICE EXAMPLE B: Use the notation $^A_Z E$ to represent a tin ion having the same number of electrons as an atom of the isotope cadmium-112. Explain why there can be more than one answer.

**2-4 Concept Assessment**

What is the single exception to the statement that all atoms comprise protons, neutrons, and electrons?

► Ordinarily we expect like-charged objects (such as protons) to repel each other. The forces holding protons and neutrons together in the nucleus are very much stronger than ordinary electrical forces (Section 25-6).

► This definition also establishes that one atomic mass unit (1 u) is *exactly* $1/12$ the mass of a carbon-12 atom.

Isotopic Masses

We cannot determine the mass of an individual atom just by adding up the masses of its fundamental particles. When protons and neutrons combine to form a nucleus, a very small portion of their original mass is converted to energy and released. However, we cannot predict exactly how much this so-called nuclear binding energy will be. Determining the masses of individual atoms, then, is something that must be done by experiment, in the following way. By international agreement, one type of atom has been chosen and assigned a specific mass. This standard is an atom of the isotope carbon-12, which is assigned a mass of exactly 12 atomic mass units, that is, 12 u. Next, the masses of other atoms relative to carbon-12 are determined with a **mass spectrometer**. In this device, a beam of gaseous ions passing through electric and magnetic fields separates into components of differing masses. The

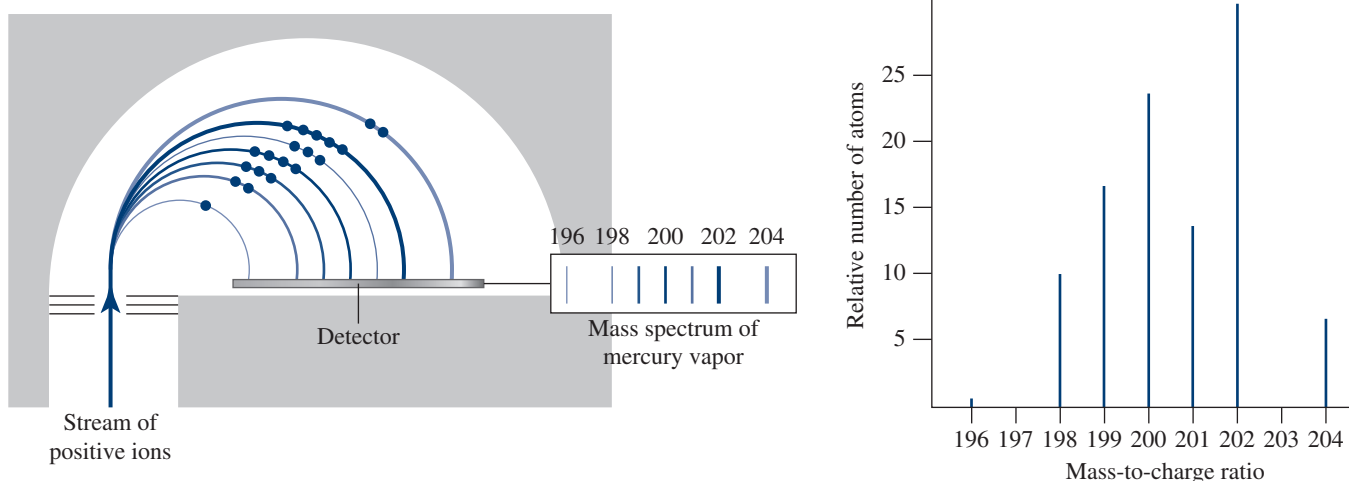


Figure 2-14 A mass spectrometer and mass spectrum

In this mass spectrometer, a gaseous sample is ionized by bombardment with electrons in the lower part of the apparatus (not shown). The positive ions thus formed are subjected to an electrical force by the electrically charged velocity selector plates and a magnetic force by a magnetic field perpendicular to the page. Only ions with a particular velocity pass through and are deflected into circular paths by the magnetic field. Ions with different masses strike the detector (here a photographic plate) in different regions. The more ions of a given type, the greater the response of the detector (intensity of line on the photographic plate). In the mass spectrum shown for mercury, the response of the ion detector (intensity of lines on photographic plate) has been converted to a scale of relative numbers of atoms. The percent isotopic abundances of the mercury isotopes are ^{196}Hg , 0.146%; ^{198}Hg , 10.02%; ^{199}Hg , 16.84%; ^{200}Hg , 23.13%; ^{201}Hg , 13.22%; ^{202}Hg , 29.80%; and ^{204}Hg , 6.85%.

◀ The primary standard for atomic masses has evolved over time. For example, Dalton originally assigned H a mass of 1 u. Later, chemists took naturally occurring oxygen at 16 u to be the definition of the atomic-weight scale. Concurrently, physicists defined the oxygen-16 isotope as 16 u. This resulted in conflicting values. In 1971 the adoption of carbon-12 as the universal standard resolved this disparity.

separated ions are focused on a measuring instrument, which records their presence and amounts. Figure 2-14 illustrates mass spectrometry and a typical mass spectrum.

Although mass numbers are whole numbers, the actual masses of individual atoms (in atomic mass units, u) are never whole numbers, except for carbon-12. However, they are very close in value to the corresponding mass numbers, as we can see for the isotope oxygen-16. From mass spectral data the ratio of the mass of ^{16}O to ^{12}C is found to be 1.33291. Thus, the mass of the oxygen-16 atom is

$$1.33291 \times 12 \text{ u} = 15.9949 \text{ u}$$

which is very nearly equal to the mass number of 16.

EXAMPLE 2-4

Establishing Isotopic Masses by Mass Spectrometry

With mass spectral data, the mass of an oxygen-16 atom is found to be 1.06632 times that of a nitrogen-15 atom. Given that ^{16}O has a mass of 15.9949 u (see above), what is the mass of a nitrogen-15 atom, in u?

Analyze

Given the ratio $(\text{mass of } ^{16}\text{O})/(\text{mass of } ^{15}\text{N}) = 1.06632$ and the mass of ^{16}O , 15.9949 u, we solve for the mass of ^{15}N .

Solve

We know that

$$\frac{\text{mass of } ^{16}\text{O}}{\text{mass of } ^{15}\text{N}} = 1.06632$$

(continued)

We solve the preceding expression for the mass of ^{15}N and then substitute 15.9949 u for the mass of ^{16}O . We obtain the result

$$\text{mass of } ^{15}\text{N} = \frac{\text{mass of } ^{16}\text{O}}{1.06632} = \frac{15.9949 \text{ u}}{1.06632} = 15.0001 \text{ u}$$

Assess

The mass of ^{15}N is very nearly 15, as we should expect. If we had mistakenly multiplied instead of dividing by the ratio 1.06632, the result would have been slightly larger than 16 and clearly incorrect.

PRACTICE EXAMPLE A: What is the ratio of masses for $^{202}\text{Hg}/^{12}\text{C}$, if the isotopic mass for ^{202}Hg is 201.97062 u?

PRACTICE EXAMPLE B: An isotope with atomic number 64 and mass number 158 is found to have a mass ratio relative to that of carbon-12 of 13.16034. What is the isotope, what is its atomic mass in u, and what is its mass relative to oxygen-16?

2-5 Atomic Mass

► Carbon-14, used for radiocarbon dating, is formed in the upper atmosphere. The amount of carbon-14 on Earth is too small to affect the atomic mass of carbon.

Keep in Mind

that the fractional abundance is the percent abundance divided by 100%. Thus, a 98.93% abundance is a 0.9893 abundance.

In a table of atomic masses, the value listed for carbon is about 12.01, yet the atomic mass standard is *exactly* 12. Why the difference? The atomic mass standard is based on a sample of carbon containing only atoms of carbon-12, whereas naturally occurring carbon contains some carbon-13 atoms as well. The existence of these two isotopes causes the observed atomic mass to be greater than 12. The **atomic mass (weight)*** of an element is the average of the isotopic masses, *weighted* according to the naturally occurring abundances of the isotopes of the element. In a weighted average, we must assign greater importance—give greater weight—to the quantity that occurs more frequently. Since carbon-12 atoms are much more abundant than carbon-13, the weighted average must lie much closer to 12 than to 13. This is the result that we get by applying the following general equation, where the right-hand side of the equation includes one term for each naturally occurring isotope.

$$\begin{array}{l} \text{at. mass} \\ \text{of an} \\ \text{element} \end{array} = \left(\begin{array}{l} \text{fractional} \\ \text{abundance of} \\ \text{isotope 1} \end{array} \times \begin{array}{l} \text{mass of} \\ \text{isotope 1} \end{array} \right) + \left(\begin{array}{l} \text{fractional} \\ \text{abundance of} \\ \text{isotope 2} \end{array} \times \begin{array}{l} \text{mass of} \\ \text{isotope 2} \end{array} \right) + \dots \quad (2.3)$$

The first term on the right side of equation (2.3) represents the contribution from isotope 1; the second term represents the contribution from isotope 2; and so on.

We will use equation (2.3), with appropriate data, in Example 2-6, but first let us illustrate the ideas of fractional abundance and a weighted average in a different way in establishing the atomic mass of carbon. The mass spectrum of a particular sample of carbon shows that 98.93% of the carbon atoms are carbon-12 with a mass of exactly 12 u; the rest are carbon-13 atoms with a mass of 13.0033548378 u. Therefore:

$$\begin{aligned} \text{atomic mass of carbon} &= 0.9893 \times 12 \text{ u} + (1 - 0.9893) \times 13.0033548378 \text{ u} \\ &= 13.0033548378 \text{ u} - 0.9893 \times (13.0033548378 \text{ u} - 12 \text{ u}) \\ &= 13.0033548378 \text{ u} - 0.9893 \times (1.0033548378 \text{ u}) \\ &= 13.0033548378 \text{ u} - 0.9926 \text{ u} \\ &= 12.0108 \text{ u} \end{aligned}$$

*Since Dalton's time, atomic masses have often been erroneously called atomic weights. They still are, yet what we are describing here is mass, not weight. Old habits die hard.

It is important to note that, in the setup just shown, 12 u and the “1” appearing in the factor $(1 - 0.9893)$ are exact numbers. Thus, by applying the rules for significant figures (see Chapter 1), the atomic mass of carbon can be reported with four decimal places.

To determine the atomic mass of an element having three naturally occurring isotopes, such as potassium, we would have to include three contributions in the weighted average, and so on.

Atomic Mass Intervals and Conventional Atomic Masses

With technological improvements in mass spectrometry, scientists can now determine atomic masses and isotopic abundances with a very high degree of precision. This ability has led to the discovery that, for certain elements, the isotopic abundances can vary significantly from one sample to another. For example, the highest reported value for the isotopic abundance of ^{13}C is 1.1466% (in samples of deep-sea pore water), and the lowest reported value is 0.9629% (from crocetene samples obtained from the ocean bottom in the North Pacific). Because of the variation of isotopic abundances, the experimentally determined atomic mass of carbon lies within an interval that has a lower bound of 12.0096 u and an upper bound of 12.0116 u. For this reason, the IUPAC has recommended that the atomic mass of carbon, and several other elements, be reported as an **atomic mass interval** rather than as a single specific value (see Table 2.2). The atomic mass interval for carbon is given as [12.0096, 12.0116]. Writing the standard atomic mass of carbon as [12.0096, 12.0116] indicates that its atomic mass in any normal material will be at least 12.0096 u and not more than 12.0116 u.

In general, an atomic mass interval is expressed in the form $[a, b]$, where a is the lower bound of the interval and b is the upper bound. The interval designation does not imply any statistical distribution of atomic mass values between the lower and upper bounds, nor does it represent a measure of the statistical uncertainty. For example, the average of a and b is neither the most likely value nor the most representative value. The difference $b - a$ does not represent the uncertainty.

For those elements with standard atomic masses given as intervals, the IUPAC also provides **conventional atomic mass** values (Table 2.2). These conventional values can be used when we need a specific, representative value of the atomic mass. The values have been selected so that, for materials normally encountered, the atomic mass would be within in an interval of plus or minus one in the last digit.

Table 2.2 Conventional Atomic Masses and Atomic Mass Intervals for Selected Elements

Atomic Number	Atomic Symbol	Atomic Mass, u	
		Conventional	Interval
1	H	1.008	[1.00784, 1.00811]
3	Li	6.94	[6.938, 6.997]
5	B	10.81	[10.806, 10.821]
6	C	12.011	[12.0096, 12.0116]
7	N	14.007	[14.00643, 14.00728]
8	O	15.999	[15.99903, 15.9997]
12	Mg	24.305	[24.304, 24.307]
14	Si	28.085	[28.084, 28.086]
16	S	32.06	[32.059, 32.076]
17	Cl	35.45	[35.446, 35.457]
35	Br	79.904	[79.901, 79.907]
81	Tl	204.38	[204.382, 204.385]

Some Representative Examples

Sometimes a qualitative understanding of the relationship between isotopic masses, percent isotopic abundances, and weighted-average atomic mass is all that we need, and no calculation is necessary, as illustrated in Example 2-5. Example 2-6 and the accompanying Practice Examples provide additional applications of equation (2.3).

The table of atomic masses (inside the front cover) shows that some atomic masses are stated more precisely than others. For example, the atomic mass of F is given as 18.998 u and that of Kr is given as 83.798 u. In fact, the atomic mass of fluorine is known even more precisely (18.9984032 u); the value of 18.998 u has been rounded off to five significant figures. Why is the atomic mass of F known so much more precisely than that of Kr? Only one type of fluorine atom occurs naturally: fluorine-19. Determining the atomic mass of fluorine means establishing the mass of this type of atom as precisely as possible. The atomic mass of krypton is known less precisely because krypton has six naturally occurring isotopes. Because the percent distribution of the isotopes of krypton differs very slightly from one sample to another, the weighted-average atomic mass of krypton cannot be stated with high precision.

EXAMPLE 2-5

Understanding the Meaning of a Weighted-Average Atomic Mass

The two naturally occurring isotopes of lithium, lithium-6 and lithium-7, have masses of 6.01512 u and 7.01600 u, respectively. Which of these two occurs in greater abundance?

Analyze

Look up the atomic mass of Li and compare it with the masses of ${}^6\text{Li}$ and ${}^7\text{Li}$. If the atomic mass of Li is closer to that of ${}^6\text{Li}$, then ${}^6\text{Li}$ is the more abundant isotope. If the atomic mass of Li is closer to that of ${}^7\text{Li}$, then ${}^7\text{Li}$ is the more abundant isotope.

Solve

From a table of atomic masses (inside the front cover), we see that the atomic mass of lithium is reported as an atomic mass interval [6.938, 6.997]. The conventional atomic mass value (from Table 2.2) is 6.94. Because the values in this range, and the conventional atomic mass value, are all much closer to 7.01600 u than to 6.01512 u, **lithium-7** must be the more abundant isotope.

Assess

Atomic masses of specific isotopes can be determined very precisely. The values given above for ${}^6\text{Li}$ and ${}^7\text{Li}$ have been rounded to five decimal places. The precise values are 6.015122795 u and 7.01600455 u.

PRACTICE EXAMPLE A: The two naturally occurring isotopes of boron, boron-10 and boron-11, have masses of 10.0129370 u and 11.0093054 u, respectively. Which of these two occurs in greater abundance?

PRACTICE EXAMPLE B: Indium has two naturally occurring isotopes and a weighted atomic mass of 114.818 u. One of the isotopes has a mass of 112.904058 u. Which of the following must be the second isotope: ${}^{111}\text{In}$, ${}^{112}\text{In}$, ${}^{114}\text{In}$, or ${}^{115}\text{In}$? Which of the two naturally occurring isotopes must be the more abundant?

EXAMPLE 2-6

Relating the Masses and Abundances of Isotopes to the Atomic Mass of an Element

Bromine has two naturally occurring isotopes, bromine-79 and bromine-81, with masses of 78.918338 u and 80.916291 u, respectively. Bromine has an atomic mass interval of [79.901, 79.907]. Estimate the percent isotopic abundances of ${}^{79}\text{Br}$ by using (a) the lower bound and (b) the upper bound of the atomic mass interval.

Analyze

We need to apply two key concepts: (1) the atomic mass of Br is a weighted average of the masses of ${}^{79}\text{Br}$ and ${}^{81}\text{Br}$, and (2) the percent isotopic abundances of ${}^{79}\text{Br}$ and ${}^{81}\text{Br}$ must add up to 100%. In effect, we use equation (2.3) twice, setting the atomic mass of Br equal to the lower and upper bounds of the interval.

Solve

The atomic mass of Br is calculated by using equation (2.3):

$$\text{atomic mass of Br} = \left(\begin{array}{c} \text{fractional} \\ \text{abundance of } ^{79}\text{Br} \times \text{mass of } ^{79}\text{Br} \end{array} \right) + \left(\begin{array}{c} \text{fractional} \\ \text{abundance of } ^{81}\text{Br} \times \text{mass of } ^{81}\text{Br} \end{array} \right)$$

The percent isotopic abundances must total 100%, and the fractional isotopic abundances must total 1. If we let f represent the fractional abundance of ^{79}Br , then the fractional abundance of ^{81}Br is $1 - f$. Therefore, we may write

$$\begin{aligned} \text{atomic mass of Br} &= f \times 78.91338 \text{ u} + (1 - f) \times 80.916291 \text{ u} \\ &= f \times (78.91338 \text{ u} - 80.916291 \text{ u}) + 80.916291 \text{ u} \\ &= \frac{\text{atomic mass of Br} - 80.916291 \text{ u}}{78.91338 \text{ u} - 80.916291 \text{ u}} \end{aligned}$$

- (a) By substituting 79.901 u for the atomic mass of Br (the lower bound of the interval), we obtain the following result for the fractional abundance of ^{79}Br :

$$f = \frac{79.901 \text{ u} - 80.916291 \text{ u}}{78.91338 \text{ u} - 80.916291 \text{ u}} = 0.50817$$

An atomic mass of 79.901 u implies that the percent isotopic abundance of ^{79}Br is 50.817%. Therefore, the percent isotopic abundance of ^{81}Br is $(100 - 50.817)\% = 49.183\%$.

- (b) When we set the atomic mass of Br equal to 79.907 u, the upper bound of the interval, we obtain

$$f = \frac{79.907 \text{ u} - 80.916291 \text{ u}}{78.91338 \text{ u} - 80.916291 \text{ u}} = 0.50516$$

The percent isotopic abundance of ^{79}Br is 50.516%, and the percent isotopic abundance of ^{81}Br is $(100 - 50.516)\% = 49.484\%$.

Assess

These results indicate that the percent isotopic abundance of ^{79}Br varies between 50.516% and 50.817%. Because of this variation, the atomic mass of bromine is best expressed as an atomic mass interval. When a representative value of the atomic mass of Br is required, we would use the conventional atomic mass (Table 2.2) of Br, which is 79.904 u.

PRACTICE EXAMPLE A: The masses and percent isotopic abundances of the three naturally occurring isotopes of silicon are ^{28}Si , 27.9769265325 u, 92.223%; ^{29}Si , 28.976494700 u, 4.685%; ^{30}Si , 29.973377017 u, 3.092%. Calculate the weighted-average atomic mass of silicon.

PRACTICE EXAMPLE B: Use data from Example 2-5 and the conventional atomic mass of Li (Table 2.2) to estimate the percent isotopic abundances of lithium-6 and lithium-7.

**2-5 Concept Assessment**

The value listed for chromium in the table of atomic masses inside the front cover is 51.996 u. Should we conclude that naturally occurring chromium atoms are all of the type ^{52}Cr ? The same table lists a value of 65.38 u for zinc. Should we conclude that zinc occurs as a mixture of isotopes? Explain.

2-6 Introduction to the Periodic Table

Scientists spend a lot of time organizing information into useful patterns. Before they can organize information, however, they must possess it, and it must be correct. Botanists had enough information about plants to organize their field in the eighteenth century. Because of uncertainties in atomic masses and because many elements remained undiscovered, chemists were not able to organize the elements until a century later.

Keep in Mind

that the periodic table shown in Figure 2-15 is the one currently recommended by IUPAC. In Figure 2-15, lutetium (Lu) and lawrencium (Lr) are the last members of the lanthanide and actinide series, respectively. A strong argument* has been made for placing Lu and Lr in group 3, meaning the lanthanide series would end with ytterbium (Yb) and the actinide series would end with nobelium (No). The question of precisely which elements should be placed in group 3 has been debated from time to time. An IUPAC project has been recently initiated to resolve the question surrounding if group 3 consists of Sc, Y, Lu, and Lr or of Sc, Y, La, and Ac.

*See W. B. Jensen, *J. Chem. Educ.*, **59**, 634 (1982).

1 1A																	13	14	15	16	17	18 8A
1 H (1.007, 1.009)	2 2A											3 3A	4 4A	5 5A	6 6A	7 7A	2 He 4.0026					
3 Li 6.94	4 Be 9.0122											5 B [10.80, 10.83]	6 C [12.00, 12.02]	7 N [14.00, 14.01]	8 O [15.99, 16.00]	9 F 18.998	10 Ne 20.180					
11 Na 22.990	12 Mg [24.30, 24.31]	3 3B	4 4B	5 5B	6 6B	7 7B	8 8B		10	11 1B	12 2B	13 Al 26.982	14 Si [28.08, 28.09]	15 P 30.974	16 S 32.06	17 Cl [35.44, 35.46]	18 Ar 39.948					
19 K 39.098	20 Ca 40.078	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr					
37 Rb 85.468	38 Sr 87.62	39 Y 88.906	40 Zr 91.224	41 Nb 92.906	42 Mo 95.95	43 Tc (98)	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.76	52 Te 127.60	53 I 126.90	54 Xe 131.29					
55 Cs 132.91	56 Ba 137.33	57–71 La–Lu		72 Hf 178.49	73 Ta 181.0	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl [204.3, 204.4]	82 Pb 207.2	83 Bi 208.98	84 Po (209)	85 At (210)	86 Rn (222)				
87 Fr (223)	88 Ra (226)	89–103 Ac–Lr		104 Rf (267)	105 Db (268)	106 Sg (269)	107 Bh (270)	108 Hs (271)	109 Mt (281)	110 Ds (282)	111 Rg (285)	112 Cn (286)	113 Nh (290)	114 Fl (290)	115 Mc (290)	116 Lv (293)	117 Ts (294)	118 Og (294)				
*Lanthanide series		57 La 138.91	58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm (145)	62 Sm 150.36	63 Eu 151.96	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.05	71 Lu 174.97						
†Actinide series		89 Ac (227)	90 Th 232.04	91 Pa 231.04	92 U 238.03	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (251)	98 Cf (252)	99 Es (257)	100 Fm (258)	101 Md (259)	102 No (262)	103 Lr (262)						

Figure 2-15 Periodic table of the elements

Atomic masses are relative to carbon-12. For 12 elements, the atomic mass is given as an interval (see Section 2-5). For certain radioactive elements, the numbers listed in parentheses are the mass numbers of the isotopes with the longest confirmed half life. Metals are shown in tan, nonmetals in blue, and metalloids in green. The noble gases (also nonmetals) are shown in pink.



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▲ Dmitri Mendeleev (1834–1907), in 1869, arranged the elements in an organized manner in what we call the periodic table today. Using his table, he was able to successfully predict the existence of three undiscovered elements.

► That elements in one group have similar properties is perhaps the most useful simplifying feature of atomic properties. Significant differences within a group do occur. The manner and reason for such differences is much of what we try to discover in studying chemistry.

We can distinguish one element from all others by its particular set of observable physical properties. For example, sodium has a low density of 0.971 g/cm^3 and a low melting point of 97.81°C . No other element has this same combination of density and melting point. Potassium, though, also has a low density (0.862 g/cm^3) and low melting point (63.65°C), much like sodium. Sodium and potassium further resemble each other in that both are good conductors of heat and electricity, and both react vigorously with water to liberate hydrogen gas. Gold, conversely, has a density (19.32 g/cm^3) and melting point (1064°C) that are very much higher than those of sodium or potassium, and gold does not react with water or even with ordinary acids. It does resemble sodium and potassium in its ability to conduct heat and electricity, however. Chlorine is very different still from sodium, potassium, and gold. It is a gas under ordinary conditions, which means that the melting point of solid chlorine (-101°C) is far below room temperature. Also, chlorine is a nonconductor of heat and electricity.

Even from these very limited data, we get an inkling of a useful classification scheme of the elements. If the scheme is to group together elements with similar properties, then sodium and potassium should appear in the same group. And if the classification scheme is in some way to distinguish between elements that are good conductors of heat and electricity and those that are not, chlorine should be set apart from sodium, potassium, and gold. The classification system we need is the one shown in Figure 2-15 (and inside the front cover), known as the **periodic table** of the elements. In Chapter 9, we will describe how the periodic table was formulated, and we will also learn its theoretical basis. For the present, we will consider only a few features of the table.

Features of the Periodic Table In the periodic table, elements are listed according to increasing atomic number starting at the upper left and arranged in a series of horizontal rows. This arrangement places similar elements in *vertical groups* or *families*. For example, sodium and potassium are found together in a group labeled 1 (called the *alkali metals*). We should expect other members of the

Group 1: Alkali metals
 Group 17: The halogens
 Period 3

group, such as cesium and rubidium, to have properties similar to sodium and potassium. Chlorine is found at the other end of the table in a group labeled 17. Some of the groups are given distinctive names, mostly related to an important property of the elements in the group. For example, the group 17 elements are called the *halogens*, a term derived from Greek, meaning “salt former.”

Each element is listed in the periodic table by placing its symbol in the middle of a box in the table. The atomic number (Z) of the element is shown above the symbol, and the weighted-average atomic mass of the element is shown below its symbol. Some periodic tables provide other information, such as density and melting point, but the atomic number and atomic mass are generally sufficient for our needs. Elements with atomic masses in parentheses, such as plutonium, Pu (244), are produced synthetically, and the number shown is the mass number of the most stable isotope.

It is customary also to divide the elements into two broad categories—**metals** and **nonmetals**. In Figure 2-15, colored backgrounds are used to distinguish the metals (tan) from the nonmetals (blue and pink). Except for mercury, a liquid, metals are solids at room temperature. They are generally malleable (capable of being flattened into thin sheets), ductile (capable of being drawn into fine wires), good conductors of heat and electricity, and have a lustrous or shiny appearance. The properties of nonmetals are generally opposite those of metals; for example, nonmetals are poor conductors of heat and electricity. Several of the nonmetals, such as nitrogen, oxygen, and chlorine, are gases at room temperature. Some, such as silicon and sulfur, are brittle solids. One—bromine—is a liquid.

Two other highlighted categories in Figure 2-15 are a special group of nonmetals known as the **noble gases** (pink), and a small group of elements, often called **metalloids** (green), that have some metallic and some nonmetallic properties.

The *horizontal* rows of the table are called **periods**. (The periods are numbered at the extreme left in the periodic table inside the front cover.) The first period of the table consists of just two elements, hydrogen and helium. This is followed by two periods of eight elements each, lithium through neon and sodium through argon. The fourth and fifth periods contain 18 elements each, ranging from potassium through krypton and from rubidium through xenon. The sixth period is a long one of 32 members. To fit this period in a table that is held to a maximum width of 18 members, 15 members of the period are placed at the bottom of the periodic table. This series of 15 elements start with lanthanum ($Z = 57$), and these elements are called the **lanthanides**. The seventh and final period is also a long one. A 15-member series is also extracted from the seventh period and placed at the bottom of the table. Because the elements in this series start with actinium ($Z = 89$), they are called the **actinides**.

The labeling of the groups of the periodic table has been a matter of some debate among chemists. The 1 to 18 numbering system used in Figure 2-15 is the one most recently adopted. Group labels previously used in the United States consisted of a letter and a number, closely following the method adopted by Mendeleev, the developer of the periodic table. As seen in Figure 2-15, the A groups 1 and 2 are separated from the remaining A groups (3 to 8) by B groups 1 through 8. The International Union of Pure and Applied Chemistry (IUPAC) recommended the simple 1 to 18 numbering scheme to

◀ There is lack of agreement on just which elements to label as metalloids. However, they are generally considered either to lie adjacent to the stair-step line or to be close by.

◀ Mendeleev’s arrangement of the elements in the original periodic table was based on observed chemical and physical properties of the elements and their compounds. The arrangement of the elements in the modern periodic table is based on atomic properties—atomic number and electron configuration.

avoid confusion between the American number and letter system and that used in Europe, where some of the A and B designations were switched! Currently, the IUPAC system is officially recommended by the American Chemical Society (ACS) and chemical societies in other nations. Because both numbering systems are in use, we show both in Figure 2-15 and in the periodic table inside the front cover. However, except for an occasional reminder of the earlier system, we will use the IUPAC numbering system in this text.

Useful Relationships from the Periodic Table

The periodic table helps chemists describe and predict the properties of chemical compounds and the outcomes of chemical reactions. Throughout this text, we will use it as an aid to understanding chemical concepts. One application of the table worth mentioning here is how it can be used to predict likely charges on simple monatomic ions.

Main-group elements are those in groups 1, 2, and 13 to 18. When main-group metal atoms in groups 1 and 2 form ions, they lose the same number of electrons as the IUPAC group number. Thus, Na atoms (group 1) lose one electron to become Na^+ , and Ca atoms (group 2) lose two electrons to become Ca^{2+} . Aluminum, in group 13, loses three electrons to form Al^{3+} (here the charge is “group number minus 10”). The few other metals in groups 13 and higher form more than one possible ion, a matter that we deal with in Chapter 9.

When nonmetal atoms form ions, they gain electrons. The number of electrons gained is normally 18 minus the IUPAC group number. Thus, an O atom gains $18 - 16 = 2$ electrons to become O^{2-} , and a Cl atom gains $18 - 17 = 1$ electron to become Cl^- . The “18 minus group number” rule suggests that an atom of Ne in group 18 gains no electrons: $18 - 18 = 0$. The very limited tendency of the noble gas atoms to form ions is one of several characteristics of this family of elements.

EXAMPLE 2-7

Describing Relationships Based on the Periodic Table

Refer to the periodic table on the inside front cover, and indicate

- (a) the element that is in group 14 and the fourth period;
- (b) two elements with properties similar to those of molybdenum (Mo);
- (c) the ion most likely formed from a strontium atom.

Analyze

For (a), the key concept is that the rows (periods) are numbered 1 through 7, starting from the top of the periodic table, and the groups are numbered 1 through 18, starting from the left side. For (b), the key concept is that elements in the same group have similar properties. For (c), the key concept is that main-group metal atoms in groups 1 and 2 form positive ions with charges of +1 and +2, respectively.

Solve

- (a) The elements in the fourth period range from K ($Z = 19$) to Kr ($Z = 36$). Those in group 14 are C, Si, Ge, Sn, and Pb. The only element that is common to both of these groupings is **Ge** ($Z = 32$).
- (b) Molybdenum is in group 6. Two other members of this group that should resemble it are **chromium (Cr)** and **tungsten (W)**.
- (c) Strontium (Sr) is in group 2. It should form the ion **Sr^{2+}** .

Assess

In Chapter 8, we will examine in greater detail reasons for the arrangement of the periodic table.

PRACTICE EXAMPLE A: Write a symbol for the ion most likely formed by an atom of each of the following: Li, S, Ra, F, I, and Al.

PRACTICE EXAMPLE B: Classify each of the following elements as a main-group or transition element. Also, specify whether they are metals, metalloids, or nonmetals: Na, Re, S, I, Kr, Mg, U, Si, B, Al, As, H.

The elements in groups 3 to 12 are the **transition elements**, and because all of them are metals, they are also called the **transition metals**. Like the main-group metals, the transition metals form positive ions, but the number of electrons lost is not related in any simple way to the group number, mostly because transition metals can form two or more ions of differing charge.

2-7 The Concept of the Mole and the Avogadro Constant

Starting with Dalton, chemists have recognized the importance of relative numbers of atoms, as in the statement that *two* hydrogen atoms and *one* oxygen atom combine to form *one* molecule of water. Yet it is physically impossible to count every atom in a macroscopic sample of matter. Instead, some other measurement must be employed, which requires a relationship between the measured quantity, usually mass, and some known, but uncountable, number of atoms. Consider a practical example of mass substituting for a desired number of items. Suppose a carpenter wants to nail down new floorboards on the deck of a mountain cabin, and they have calculated how many nails they will need. If they have an idea of how many nails there are in a pound, then they can buy the nails by the pound.

The SI quantity that describes an amount of substance by relating it to a number of particles of that substance is called the *mole* (abbreviated *mol*). A **mole** is the amount of a substance that contains the same number of elementary entities as there are atoms in exactly 12 g of pure carbon-12. The “number of elementary entities (atoms, molecules, and so on)” in a mole is the **Avogadro constant**, N_A .

$$N_A = 6.022140857 \times 10^{23} \text{ mol}^{-1} \quad (2.4)$$

The Avogadro constant consists of a number, $6.022140857 \times 10^{23}$, known as Avogadro’s *number*, and a unit, mol^{-1} . The unit mol^{-1} signifies that the entities being counted are those present in 1 mole.

The value of Avogadro’s number is based on both a definition and a measurement. A mole of carbon-12 is *defined* to be 12 g. If the mass of one carbon-12 atom is *measured* by using a mass spectrometer (see Figure 2-14), the mass would be about 1.9926×10^{-23} g. The ratio of these two masses provides an estimate of Avogadro’s number. In actual fact, accurate determinations of Avogadro’s number make use of other measurements, not the measurement of the mass of a single atom of carbon-12.

Often the value of N_A is rounded off to $6.022 \times 10^{23} \text{ mol}^{-1}$, or even to $6.02 \times 10^{23} \text{ mol}^{-1}$.

If a substance contains atoms of only a single isotope, then

$$1 \text{ mol } ^{12}\text{C} = 6.02214 \times 10^{23} \text{ } ^{12}\text{C atoms} = 12.0000 \text{ g}$$

$$1 \text{ mol } ^{16}\text{O} = 6.02214 \times 10^{23} \text{ } ^{16}\text{O atoms} = 15.9949 \text{ g (and so on)}$$

Most elements are composed of mixtures of two or more isotopes so that the atoms in a sample of the element are not all of the same mass but are present in their naturally occurring proportions. Thus, in one mole of carbon, most of the atoms are carbon-12, but some are carbon-13. In one mole of oxygen, most of the atoms are oxygen-16, but some are oxygen-17 and some are oxygen-18. As a result,

$$1 \text{ mol of C} = 6.02214 \times 10^{23} \text{ C atoms} = 12.011 \text{ g}$$

$$1 \text{ mol of O} = 6.02214 \times 10^{23} \text{ O atoms} = 15.999 \text{ g, and so on.}$$

The Avogadro constant was purposely chosen so that the mass of one mole of carbon-12 atoms—exactly 12 g—would have the same *numeric* value as the

◀ Because the value of Avogadro’s number depends, in part, on a measurement, the value has changed slightly over the years. The values recommended since 1986 by the Committee on Data for Science and Technology (CODATA) are listed below.

Year	Avogadro’s Number
1986	6.0221367×10^{23}
1998	$6.02214199 \times 10^{23}$
2002	6.0221415×10^{23}
2006	$6.02214179 \times 10^{23}$
2010	$6.02214129 \times 10^{23}$
2014	$6.022140857 \times 10^{23}$

◀ When rounding Avogadro’s number or any other accurately known value, keep one more significant figure than that of the least accurate number in the calculation to avoid rounding errors.

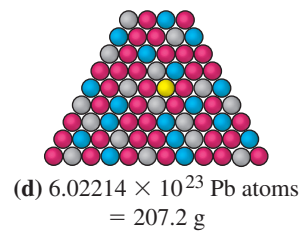
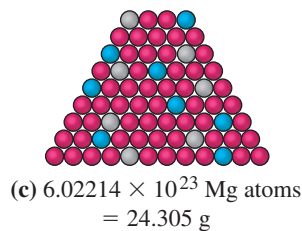
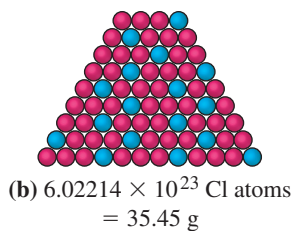
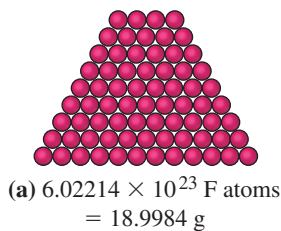


Figure 2-16 Distribution of isotopes in four elements

(a) There is only one type of fluorine atom, ^{19}F (shown in red). (b) In chlorine, 75.77% of the atoms are ^{35}Cl (red) and the remainder are ^{37}Cl (blue). (c) Magnesium has one principal isotope, ^{24}Mg (red), and two minor ones, ^{25}Mg (gray) and ^{26}Mg (blue). (d) Lead has four naturally occurring isotopes: 1.4% ^{204}Pb (yellow), 24.1% ^{206}Pb (blue), 22.1% ^{207}Pb (gray), and 52.4% ^{208}Pb (red).

► The weighted-average atomic mass of carbon was calculated on page 76.

Keep in Mind

that molar mass has the unit g/mol.

mass of a single carbon-12 atom—exactly 12 u. As a result, for all other elements the numeric value of the mass in grams of one mole of atoms and the weighted-average atomic mass in atomic mass units are equal. For example, the weighted-average atomic mass of iron is 55.845 u and the mass of one mole of iron atoms is 55.845 g. Thus, we can easily establish the mass of one mole of atoms, called the **molar mass**, M , from a table of atomic masses.* For example, the molar mass of iron is 55.845 g Fe/mol Fe. Figure 2-16 attempts to portray the distribution of isotopes of an element, and Figure 2-17 pictures one mole each of four common elements.



2-6 Concept Assessment

Dividing the molar mass of gold by the Avogadro constant yields the mass of any individual atom of naturally occurring gold. In contrast, no naturally occurring atom of silver has the mass obtained by dividing the molar mass of silver by the Avogadro constant. How can this be?

Thinking About Avogadro's Number

Avogadro's number (6.02214×10^{23}) is an enormously large number and practically inconceivable in terms of ordinary experience. Suppose we were counting garden peas instead of atoms. If the typical pea had a volume of about 0.1 cm^3 , the required pile of peas would cover the United States to a depth of about 6 km (4 mi). Or imagine that grains of wheat could be counted



Figure 2-17 One mole of an element

The watch glasses contain one mole of copper atoms (left) and one mole of sulfur atoms (right). The beaker contains one mole of mercury atoms as liquid mercury, and the balloon, of which only a small portion is visible here, contains one mole of helium atoms in the gaseous state.

*Atomic mass (atomic weight) values in tables are often written without units, especially if they are referred to as *relative* atomic masses. This simply means that the values listed are in relation to *exactly* 12 (rather than 12 u) for carbon-12. We will use the atomic mass unit (u) when referring to atomic masses (atomic weights). Most chemists do.

at the rate of 100 per minute. A given individual might be able to count out about 4 billion grains in a lifetime. Even so, if all the people currently on Earth were to spend their lives counting grains of wheat, they could not reach Avogadro's number. In fact, if all the people who ever lived on Earth had spent their lifetimes counting grains of wheat, the total would still be far less than Avogadro's number. (And Avogadro's number of wheat grains is far more wheat than has been produced in human history.) Now consider a much more efficient counting device, a modern personal computer; it is capable of counting at a rate of about 1 billion units per second. The task of counting out Avogadro's number would still take about 20 million years!

Avogadro's number is clearly not a useful number for counting ordinary objects. However, when this inconceivably large number is used to count inconceivably small objects, such as atoms and molecules, the result is a quantity of material that is easily within our grasp, essentially a "handful."

2-8 Using the Mole Concept in Calculations

Throughout the text, the mole concept will provide conversion factors for problem-solving situations. With each new situation, we will explore how the mole concept applies. For now, we will deal with the relationship between numbers of atoms and the mole. Consider the statement: $1 \text{ mol S} = 6.022 \times 10^{23} \text{ S atoms}$ $= 32.06 \text{ g S}$. This allows us to write the conversion factors

$$\frac{1 \text{ mol S}}{6.022 \times 10^{23} \text{ S atoms}} \quad \text{and} \quad \frac{32.06 \text{ g S}}{1 \text{ mol S}}$$

In calculations requiring the Avogadro constant, students often ask when to multiply and when to divide by N_A . One answer is always to use the constant in a way that gives the proper cancellation of units. Another answer is to think in terms of the expected result. In calculating a number of atoms, we expect the answer to be a very large number and certainly *never* smaller than one. The number of moles of atoms, conversely, is generally a number of more modest size and will often be less than one.

In the following examples, we use atomic masses and the Avogadro constant in calculations to determine the number of atoms present in a given sample. Atomic masses and the Avogadro constant are known rather precisely, and students often wonder how many significant figures to carry in atomic masses or the Avogadro constant when performing calculations. Here is a useful rule of thumb.

To ensure the maximum precision allowable, carry at least one more significant figure in well-known physical constants than in other measured quantities.

For example, in calculating the mass of 0.600 mol of sulfur, we should use the atomic mass of S with *at least* four significant figures. The answer $0.600 \text{ mol S} \times 32.06 \text{ g S/mol S} = 19.2 \text{ g S}$ is a more precise response than $0.600 \text{ mol S} \times 32.1 \text{ g S/mol S} = 19.3 \text{ g S}$.



Richard Megna/Fundamental Photographs

Figure 2-18

**Measurement of $7.64 \times 10^{22} \text{ S atoms}$ (0.127 mol S)—
Example 2-8 illustrated**

The balance was set to zero (tared) when just the weighing boat was present. The sample of sulfur weighs 4.07 g.

EXAMPLE 2-8

Relating Number of Atoms, Amount in Moles, and Mass in Grams

In the sample of sulfur weighing 4.07 g pictured in Figure 2-18, (a) how many moles of sulfur are present, and (b) what is the total number of sulfur atoms in the sample?

Analyze

For (a), the conversion pathway is $\text{g S} \rightarrow \text{mol S}$. To carry out this conversion, we multiply 4.07 g S by the conversion factor $(1 \text{ mol S}/32.06 \text{ g S})$. The conversion factor is the molar mass inverted. For (b), the conversion

(continued)

pathway is $\text{mol S} \rightarrow \text{atoms S}$. To carry out this conversion, we multiply the quantity in moles from part (a) by the conversion factor ($6.022 \times 10^{23} \text{ atoms S} / 1 \text{ mol S}$).

Solve

- (a) For the conversion $\text{g S} \rightarrow \text{mol S}$, using $(1/M)$ as a conversion factor achieves the proper cancellation of units. The result of this calculation should be stored without rounding it off because it is required in part (b).

$$? \text{ mol S} = 4.07 \text{ g S} \times \frac{1 \text{ mol S}}{32.06 \text{ g S}} = 0.127 \text{ mol S}$$

- (b) The conversion $\text{mol S} \rightarrow \text{atoms S}$ is carried out using the Avogadro constant as a conversion factor.

$$? \text{ atoms S} = 0.127 \text{ mol S} \times \frac{6.022 \times 10^{23} \text{ atoms S}}{1 \text{ mol S}} = 7.64 \times 10^{22} \text{ atoms S}$$

Assess

By including units in our calculations, we can check that proper cancellation of units occurs. Also, if our only concern is to calculate the number of sulfur atoms in the sample, the calculations carried out in parts (a) and (b) could be combined into a single calculation, as shown below.

$$? \text{ atoms S} = 4.07 \text{ g S} \times \frac{1 \text{ mol S}}{32.06 \text{ g S}} \times \frac{6.022 \times 10^{23} \text{ atoms S}}{1 \text{ mol S}} = 7.64 \times 10^{22} \text{ atoms S}$$

Had we rounded $4.07 \text{ g S} \times (1 \text{ mol S} / 32.06 \text{ g S})$ to 0.127 mol S and used the rounded result in part (b), we would have obtained a final answer of $7.65 \times 10^{22} \text{ atoms S}$. With a single line calculation, we do not have to write down an intermediate result and we avoid round-off errors.

PRACTICE EXAMPLE A: What is the mass of 2.35×10^{24} atoms of Cu?

PRACTICE EXAMPLE B: How many lead-206 atoms are present in a 22.6 g sample of lead metal? [Hint: See Figure 2-16.]

Example 2-9 is perhaps the most representative use of the mole concept. Here it is part of a larger problem that requires other unrelated conversion factors as well. One approach is to outline a conversion pathway to get from the given to the desired information.

EXAMPLE 2-9

Combining Several Factors in a Calculation—Molar Mass, the Avogadro Constant, Percent Abundances

Potassium-40 is one of the few naturally occurring radioactive isotopes of elements of low atomic number. Its percent isotopic abundance among K isotopes is 0.012%. How many ^{40}K atoms are present in 225 mL of whole milk containing 1.65 mg K/mL?

Analyze

Ultimately we need to complete the conversion $\text{mL milk} \rightarrow \text{atoms } ^{40}\text{K}$. There is no single conversion factor that allows us to complete this conversion in one step, so we anticipate having to complete several steps or conversions. We are told the milk contains $1.65 \text{ mg K/mL} = 1.65 \times 10^{-3} \text{ g K/mL}$, and this information can be used to carry out the conversion $\text{mL milk} \rightarrow \text{g K}$. We can carry out the conversions $\text{g K} \rightarrow \text{mol K} \rightarrow \text{atoms K}$ by using conversion factors based on the molar mass of K and the Avogadro constant. The final conversion, $\text{atoms K} \rightarrow \text{atoms } ^{40}\text{K}$, can be carried out by using a conversion factor based on the percent isotopic abundance of ^{40}K . A complete conversion pathway is shown below:



Solve

The required conversions can be carried out in a stepwise fashion, or they can be combined into a single line calculation. Let's use a stepwise approach. First, we convert from mL milk to g K.

$$? \text{ g K} = 225 \text{ mL milk} \times \frac{1.65 \text{ mg K}}{1 \text{ mL milk}} \times \frac{1 \text{ g K}}{1000 \text{ mg K}} = 0.371 \text{ g K}$$

Next, we convert from g K to mol K,

$$? \text{ mol K} = 0.371 \text{ g K} \times \frac{1 \text{ mol K}}{39.10 \text{ g K}} = 9.49 \times 10^{-3} \text{ mol K}$$

and then we convert from mol K to atoms K.

$$? \text{ atoms K} = 9.49 \times 10^{-3} \text{ mol K} \times \frac{6.022 \times 10^{23} \text{ atoms K}}{1 \text{ mol K}} = 5.71 \times 10^{21} \text{ atoms K}$$

Finally, we convert from atoms K to atoms ^{40}K .

$$? \text{ atoms } ^{40}\text{K} = 5.71 \times 10^{21} \text{ atoms K} \times \frac{0.012 \text{ atoms } ^{40}\text{K}}{100 \text{ atoms K}} = 6.9 \times 10^{17} \text{ atoms } ^{40}\text{K}$$

Assess

The final answer is rounded to two significant figures because the least precisely known quantity in the calculation, the percent isotopic abundance of ^{40}K , has two significant figures. It is possible to combine the steps above into a single line calculation.

$$\begin{aligned} ? \text{ atoms } ^{40}\text{K} &= 225 \text{ mL milk} \times \frac{1.65 \text{ mg K}}{1 \text{ mL milk}} \times \frac{1 \text{ g K}}{1000 \text{ mg K}} \times \frac{1 \text{ mol K}}{39.10 \text{ g K}} \\ &\quad \times \frac{6.022 \times 10^{23} \text{ atoms K}}{1 \text{ mol K}} \times \frac{0.012 \text{ atoms } ^{40}\text{K}}{100 \text{ atoms K}} \\ &= 6.9 \times 10^{17} \text{ atoms } ^{40}\text{K} \end{aligned}$$

PRACTICE EXAMPLE A: How many Pb atoms are present in a small piece of lead with a volume of 0.105 cm^3 ? The density of Pb = 11.34 g/cm^3 .

PRACTICE EXAMPLE B: Rhenium-187 is a radioactive isotope that can be used to determine the age of meteorites. A 0.100 mg sample of Re contains 2.02×10^{17} atoms of ^{187}Re . What is the percent isotopic abundance of rhenium-187 in the sample?



<https://mlm.pearson.com/global/>

What is the most abundant element? This seemingly simple question does not have a simple answer. To learn more about the abundances of elements in the universe and in the Earth's crust, go to the Focus On feature for Chapter 2, entitled Occurrence and Abundances of the Elements, in Mastering.

Summary

2-1 Early Chemical Discoveries and the Atomic Theory—Modern chemistry began with eighteenth-century discoveries leading to the formulation of two basic laws of chemical combination, the **law of conservation of mass** and the **law of constant composition (definite proportions)**. These discoveries led to Dalton's atomic theory—that matter is composed of indestructible particles called atoms, that the atoms of an element are identical to one another but different from atoms of all other elements, and that chemical compounds are combinations of atoms of different elements. Based on this theory, Dalton proposed still another law of chemical combination, the **law of multiple proportions**.

2-2 Electrons and Other Discoveries in Atomic Physics—The first clues to the structures of atoms came through the discovery and characterization of **cathode rays (electrons)**. Key experiments were those that established

the mass-to-charge ratio (Fig. 2-7) and then the charge on an electron (Fig. 2-8). Two important accidental discoveries made in the course of cathode-ray research were of X-rays and **radioactivity**. The principal types of radiation emitted by radioactive substances are **alpha (α) particles**, **beta (β) particles**, and **gamma (γ) rays** (Fig. 2-10).

2-3 The Nuclear Atom—Studies on the scattering of α particles by thin metal foils (Fig. 2-11) led to the concept of the nuclear atom—a tiny, but massive, positively charged nucleus surrounded by lightweight, negatively charged electrons (Fig. 2-12). A more complete description of the nucleus was made possible by the discovery of **protons** and **neutrons**. An individual atom is characterized in terms of its **atomic number (proton number) Z** and **mass number, A** . The difference, $A - Z$, is the **neutron number**. The masses of individual atoms and their component parts are expressed in **atomic mass units (u)**.

2-4 Chemical Elements—All elements from $Z = 1$ to $Z = 118$, have been characterized and all have been given a name and **chemical symbol**. Knowledge of the several elements following $Z = 112$ is more tenuous. **Nuclide** is the term used to describe an atom with a particular atomic number and a particular mass number. Atoms of the same element that differ in mass number are called **isotopes**. The **percent isotopic abundance** of an isotope and the precise mass of its atoms can be established with a **mass spectrometer** (Fig. 2-14). A special symbolism (expression 2.2) is used to represent the composition of an atom or an **ion** derived from the atom.

2-5 Atomic Mass—The **atomic mass (weight)** of an element is a weighted average based on an assigned value of exactly 12 u for the isotope carbon-12. This weighted average is calculated from the experimentally determined atomic masses and percent abundances of the naturally occurring isotopes of the element through expression (2.3). For certain elements, an **atomic mass interval** (Table 2.2) is used to indicate the range of values expected for the atomic mass because of observed variations in the isotopic abundances of these elements. For these elements, the **conventional atomic mass** can be used when a representative value of the atomic mass is required.

2-6 Introduction to the Periodic Table—The **periodic table** (Fig. 2-15) is an arrangement of the elements

in horizontal rows called **periods** and vertical columns called **groups** or **families**. Each group consists of elements with similar physical and chemical properties. The elements can also be subdivided into broad categories. One categorization is that of **metals**, **nonmetals**, **metalloids**, and **noble gases**. Another is that of **main-group elements** and **transition elements (transition metals)**. Included among the transition elements are the two subcategories **lanthanides** and **actinides**. The table has many uses, as will be seen throughout the text. Emphasis in this chapter is on the periodic table as an aid in writing symbols for simple ions.

2-7 The Concept of the Mole and the Avogadro Constant—The **Avogadro constant**, $N_A = 6.02214 \times 10^{23} \text{ mol}^{-1}$, represents the number of carbon-12 atoms in *exactly* 12 g of carbon-12. More generally, it is the number of elementary entities (for example, atoms or molecules) present in an amount known as one **mole** of substance. The mass of one mole of atoms of an element is called its **molar mass**, M .

2-8 Using the Mole Concept in Calculations—Molar mass and the Avogadro constant are used in a variety of calculations involving the mass, amount (in moles), and number of atoms in a sample of an element. Other conversion factors may also be involved in these calculations. The mole concept is encountered in ever broader contexts throughout the text.

Integrative Example

A stainless steel ball bearing has a radius of 6.35 mm and a density of 7.75 g/cm^3 . Iron is the principal element in steel. Carbon is a key minor element. The ball bearing contains 0.25% carbon, by mass. Assuming that the percent isotopic abundance of ^{13}C is 1.108%, how many ^{13}C atoms are present in the ball bearing?

Analyze

The goal is to determine the number of carbon-13 atoms found in a ball bearing with a particular composition. The critical point in this problem is recognizing that we can relate number of atoms to mass by using molar mass and Avogadro's constant. The first step is to use the radius of the ball bearing to determine its volume. The second step is to determine the mass of carbon present by using the density of steel along with the percent composition. The third step uses the molar mass of carbon to convert grams of carbon to moles of carbon; Avogadro's constant is then used to convert moles of carbon to the number of carbon atoms. In the final step, the isotopic abundance of carbon-13 atoms is used to find the number of carbon-13 atoms in the total number of carbon atoms in the ball bearing.

Solve

The ball-bearing volume in cubic centimeters is found by applying the formula for the volume of a sphere, $V = 4/3 \pi r^3$. Remember to convert the given radius from millimeters to centimeters, so that the volume will be in cubic centimeters.

$$V = \frac{4\pi}{3} \left[6.35 \text{ mm} \times \frac{1 \text{ cm}}{10 \text{ mm}} \right]^3 = 1.07 \text{ cm}^3$$

The product of the volume of the ball bearing and the density of steel equals the mass. The mass of the ball bearing multiplied by the percent carbon in the steel gives the mass of carbon present.

$$? \text{ g C} = 1.07 \text{ cm}^3 \times \frac{7.75 \text{ g steel}}{1 \text{ cm}^3 \text{ steel}} \times \frac{0.25 \text{ g C}}{100 \text{ g steel}} = 0.021 \text{ g C}$$

The mass of carbon is first converted to moles of carbon by using the inverse of the molar mass of carbon. Avogadro's constant is then used to convert moles of carbon to atoms of carbon.

$$\begin{aligned} ? \text{ C atoms} &= 0.021 \text{ g C} \times \frac{1 \text{ mol C}}{12.011 \text{ g C}} \times \frac{6.022 \times 10^{23} \text{ C atoms}}{1 \text{ mol C}} \\ &= 1.1 \times 10^{21} \text{ C atoms} \end{aligned}$$

The number of ^{13}C atoms is determined by using the percent isotopic abundance of carbon-13.

$$\begin{aligned} ? \text{ } ^{13}\text{C atoms} &= 1.1 \times 10^{21} \text{ C atoms} \times \frac{1.108 \text{ } ^{13}\text{C atoms}}{100 \text{ C atoms}} \\ &= 1.2 \times 10^{19} \text{ } ^{13}\text{C atoms} \end{aligned}$$

Assess

The number of carbon-13 atoms is smaller than the number of carbon atoms, which it should be, given that the isotopic abundance of carbon-13 is just 1.108%. To avoid mistakes, every quantity should be clearly labeled with its appropriate unit so that units cancel properly. Two points made by this problem are, first, that the relatively small ball bearing contains a large number of carbon-13 atoms even though carbon-13's abundance is only 1.108% of all carbon atoms. Second, the size of any atom must be very small.

PRACTICE EXAMPLE A: Calculate the number of ^{63}Cu atoms in a cubic crystal of copper that measures exactly 25 nm on edge. The density of copper is 8.92 g/cm^3 and the percent isotopic abundance of ^{63}Cu is 69.17%.

PRACTICE EXAMPLE B: The United States Food and Drug Administration (USFDA) suggests a daily value of 18 mg Fe for adults and for children over four years of age. The label on a particular brand of cereal states that one serving (55 g) of dry cereal contains 45% of the daily value of Fe. Given that the percent isotopic abundance of ^{58}Fe is 0.282%, how many full servings of dry cereal must be eaten to consume exactly one mole of ^{58}Fe ? The atomic weight of ^{58}Fe is 57.9333 u. Is it possible for a person to consume this much cereal in a lifetime, assuming that one full serving of cereal is eaten every day?

Exercises

Law of Conservation of Mass

- When an iron object rusts, its mass increases. When a match burns, its mass decreases. Do these observations violate the law of conservation of mass? Explain.
- When a strip of magnesium metal is burned in air (recall Figure 2-1), it produces a white powder that weighs more than the original metal. When a strip of magnesium is burned in a flashbulb, the bulb weighs the same before and after it is flashed. Explain the difference in these observations.
- A 0.406 g sample of magnesium reacts with oxygen, producing 0.674 g of magnesium oxide as the only product. What mass of oxygen was consumed in the reaction?
- A 1.446 g sample of potassium reacts with 8.178 g of chlorine to produce potassium chloride as the only product. After the reaction, 6.867 g of chlorine remains unreacted. What mass of potassium chloride was formed?
- When a solid mixture consisting of 10.500 g calcium hydroxide and 11.125 g ammonium chloride is strongly heated, gaseous products are evolved and 14.336 g of a solid residue remains. The gases are passed into 62.316 g of water, and the mass of the resulting solution is 69.605 g. Within the limits of experimental error, show that these data conform to the law of conservation of mass.
- Within the limits of experimental error, show that the law of conservation of mass was obeyed in the following experiment: 10.00 g calcium carbonate (found in limestone) was dissolved in 100.0 mL hydrochloric acid ($d = 1.148\text{ g/mL}$). The products were 120.40 g solution (a mixture of hydrochloric acid and calcium chloride) and 2.22 L carbon dioxide gas ($d = 1.9769\text{ g/L}$).

Law of Constant Composition

- In Example 2-1, we established that the mass ratio of magnesium to magnesium oxide is 0.455 g magnesium/0.755 g magnesium oxide.
 - What is the ratio of oxygen to magnesium oxide, by mass?
 - What is the mass ratio of oxygen to magnesium in magnesium oxide?
 - What is the percent by mass of magnesium in magnesium oxide?
- Samples of pure carbon weighing 3.62, 5.91, and 7.07 g were burned in an excess of air. The masses of carbon dioxide obtained (the sole product in each case) were 13.26, 21.66, and 25.91 g, respectively.
 - Do these data establish that carbon dioxide has a fixed composition?
 - What is the composition of carbon dioxide, expressed in % C and % O, by mass?
- In one experiment, 2.18 g sodium was allowed to react with 16.12 g chlorine. All the sodium was used up, and 5.54 g sodium chloride (salt) was produced. In a second experiment, 2.10 g chlorine was allowed to react with 10.00 g sodium. All the chlorine was used up, and 3.46 g sodium chloride was produced. Show that these results are consistent with the law of constant composition.
- When 3.06 g hydrogen was allowed to react with an excess of oxygen, 27.35 g water was obtained. In a second experiment, a sample of water was decomposed by electrolysis, resulting in 1.45 g hydrogen and 11.51 g oxygen. Are these results consistent with the law of constant composition? Demonstrate why or why not.
- In one experiment, the burning of 0.312 g sulfur produced 0.623 g sulfur dioxide as the sole product of the reaction. In a second experiment, 0.842 g sulfur dioxide was obtained. What mass of sulfur must have been burned in the second experiment?
- In one experiment, the reaction of 1.00 g mercury and an excess of sulfur yielded 1.16 g of a sulfide of mercury as the sole product. In a second experiment, the same sulfide was produced in the reaction of 1.50 g mercury and 1.00 g sulfur.
 - What mass of the sulfide of mercury was produced in the second experiment?
 - What mass of which element (mercury or sulfur) remained *unreacted* in the second experiment?

Law of Multiple Proportions

13. Sulfur forms two compounds with oxygen. In the first compound, 1.000 g sulfur is combined with 0.998 g oxygen, and in the second, 1.000 g sulfur is combined with 1.497 g oxygen. Show that these results are consistent with Dalton's law of multiple proportions.
14. Phosphorus forms two compounds with chlorine. In the first compound, 1.000 g of phosphorus is combined with 3.433 g chlorine, and in the second, 2.500 g phosphorus is combined with 14.308 g chlorine. Show that these results are consistent with Dalton's law of multiple proportions.
15. The following data were obtained for compounds of nitrogen and hydrogen:

Compound	Mass of Nitrogen, g	Mass of Hydrogen, g
A	0.500	0.108
B	1.000	0.0720
C	0.750	0.108

- (a) Show that these data are consistent with the law of multiple proportions.
- (b) If the formula of compound B is N_2H_2 , what are the formulas of compounds A and C?

16. The following data were obtained for compounds of iodine and fluorine:

Compound	Mass of Iodine, g	Mass of Fluorine, g
A	1.000	0.1497
B	0.500	0.2246
C	0.750	0.5614
D	1.000	1.0480

- (a) Show that these data are consistent with the law of multiple proportions.
- (b) If the formula for compound A is IF, what are the formulas for compounds B, C, and D?
17. There are two oxides of copper. One oxide has 20% oxygen, by mass. The second oxide has a *smaller* percent of oxygen than the first. What is the probable percent of oxygen in the second oxide?
18. The two oxides of carbon described on page 66 were CO and CO_2 . Another oxide of carbon has 1.106 g of oxygen in a 2.350 g sample. In what ratio are carbon and oxygen atoms combined in molecules of this third oxide? Explain.

Fundamental Charges and Mass-to-Charge Ratios

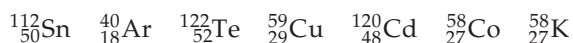
19. The following observations were made for a series of five oil drops in an experiment similar to Millikan's (see Figure 2-8). Drop 1 carried a charge of $1.28 \times 10^{-18} \text{ C}$; drops 2 and 3 each carried $\frac{1}{2}$ the charge of drop 1; drop 4 carried $\frac{1}{8}$ the charge of drop 1; drop 5 had a charge four times that of drop 1. Are these data consistent with the value of the electronic charge given in the text? Could Millikan have inferred the charge on the electron from this particular series of data? Explain.
20. In an experiment similar to that described in Exercise 19, drop 1 carried a charge of $6.41 \times 10^{-19} \text{ C}$; drop 2 had $\frac{1}{2}$ the charge of drop 1; drop 3 had twice the charge of drop 1; drop 4 had a charge of $1.44 \times 10^{-18} \text{ C}$; and drop 5 had $\frac{1}{3}$ the charge of drop 4. Are these data consistent with the value of the electronic charge given in the text? Could Millikan have inferred the charge on the electron from this particular series of data? Explain.
21. Use data from Table 2.1 to verify that
- (a) the mass of electrons is about 1/2000 that of H atoms;
- (b) the mass-to-charge ratio (m/e) for positive ions is considerably larger than that for electrons.
22. Determine the approximate value of m/e in grams per coulomb for the ions $^{127}_{53}\text{I}^-$ and $^{32}_{16}\text{S}^{2-}$. Why are these values only approximate?

Atomic Number, Mass Number, and Isotopes

23. The following radioactive isotopes have applications in medicine. Write their symbols in the form ^A_ZE . (a) cobalt-60; (b) phosphorus-32; (c) iron-59; (d) radium-226.
24. For the isotope ^{202}Hg , express the percentage of the fundamental particles in the nucleus that are neutrons.
25. Complete the following table. What minimum amount of information is required to completely characterize an atom or ion? [Hint: Not all rows can be completed.]

Name	Symbol	Number Protons	Number Electrons	Number Neutrons	Mass Number
Sodium	$^{23}_{11}\text{Na}$	11	11	12	23
Silicon	—	—	—	14	—
—	—	37	—	—	85
—	$^{40}_{19}\text{K}$	—	—	—	—
—	—	—	33	42	—
—	$^{20}\text{Ne}^{2+}$	—	—	—	—
—	—	—	—	—	80
—	—	—	—	126	—

26. Arrange the following species in order of increasing (a) number of electrons; (b) number of neutrons; (c) mass.



27. For the atom ${}^{108}\text{Pd}$ with mass 107.90389 u, determine
(a) the numbers of protons, neutrons, and electrons in the atom;
(b) the ratio of the mass of this atom to that of an atom of ${}^{12}_6\text{C}$.
28. For the ion ${}^{228}\text{Ra}^{2+}$ with a mass of 228.030 u, determine
(a) the numbers of protons, neutrons, and electrons in the ion;
(b) the ratio of the mass of this ion to that of an atom of ${}^{16}\text{O}$ (refer to page 75).
29. An isotope of silver has a mass that is 6.68374 times that of oxygen-16. What is the mass in u of this isotope? (Refer to page 75.)
30. The ratio of the masses of the two naturally occurring isotopes of indium is 1.0177:1. The heavier of the two isotopes has 7.1838 times the mass of ${}^{16}\text{O}$. What are the masses in u of the two isotopes? (Refer to page 75.)
31. The following data on isotopic masses are from a chemical handbook. What is the ratio of each of these masses to that of ${}^{12}_6\text{C}$? (a) ${}^{35}_{17}\text{Cl}$, 34.96885 u; (b) ${}^{26}_{12}\text{Mg}$, 25.98259 u; (c) ${}^{222}_{86}\text{Rn}$, 222.0175 u.
32. The following ratios of masses were obtained with a mass spectrometer: ${}^{19}\text{F}/{}^{12}_6\text{C} = 1.5832$; ${}^{35}\text{Cl}/{}^{19}\text{F} = 1.8406$; ${}^{81}\text{Br}/{}^{35}\text{Cl} = 2.3140$. Determine the mass of a ${}^{81}_{35}\text{Br}$ atom in amu.
33. Which of the following species has
(a) equal numbers of neutrons and electrons;
(b) protons, neutrons, and electrons in the ratio 9:11:8;
(c) a number of neutrons equal to the number of protons plus one-half the number of electrons?



34. Given the same species as listed in Exercise 33, which has
(a) equal numbers of neutrons and protons;
(b) protons contributing more than 50% of the mass;
(c) about 50% more neutrons than protons?
35. An isotope with mass number 44 has four more neutrons than protons. This is an isotope of what element?
36. Identify the isotope X that has one more neutron than protons and a mass number equal to nine times the charge on the ion X^{3+} .
37. Iodine has many radioactive isotopes. Iodine-123 is a radioactive isotope used for obtaining images of the thyroid gland. Iodine-123 is administered to patients in the form of sodium iodide capsules that contain ${}^{123}\text{I}^{-}$ ions. Determine the number of neutrons, protons, and electrons in a single ${}^{123}\text{I}^{-}$ ion.
38. Iodine-131 is a radioactive isotope that has important medical uses. Small doses of iodine-131 are used for treating hyperthyroidism (overactive thyroid) and larger doses are used for treating thyroid cancer. Iodine-131 is administered to patients in the form of sodium iodide capsules that contain ${}^{131}\text{I}^{-}$ ions. Determine the number of neutrons, protons, and electrons in a single ${}^{131}\text{I}^{-}$ ion.
39. Americium-241 is a radioactive isotope that is used in high-precision gas and smoke detectors. How many neutrons, protons, and electrons are there in an atom of americium-241?
40. Some foods are made safer to eat by being exposed to gamma rays from radioactive isotopes, such as cobalt-60. The energy from the gamma rays kills bacteria in the food. How many neutrons, protons, and electrons are there in an atom of cobalt-60?

Atomic Mass Units, Atomic Masses

41. Which statement is probably true concerning the masses of *individual* chlorine atoms: *All have, some have, or none has* a mass of 35.45 u? Explain.
42. The mass of a carbon-12 atom is taken to be exactly 12 u. Are there likely to be any other atoms with an *exact* integral (whole number) mass, expressed in u? Explain.
43. Magnesium has three naturally occurring isotopes. Their masses are 23.985042 u, 24.985837 u, and 25.982593 u. What is the weighted-average atomic mass of magnesium in a sample for which the percent isotopic abundances of these three isotopes are 78.99%, 10.00%, and 11.01%, respectively?
44. There are four naturally occurring isotopes of chromium. Their masses and percent isotopic abundances are 49.9461 u, 4.35%; 51.9405 u, 83.79%; 52.9407 u, 9.50%; and 53.9389 u, 2.36%. Calculate the weighted-average atomic mass of chromium.
45. The two naturally occurring isotopes of silver have the following abundances: ${}^{107}\text{Ag}$, 51.84%; ${}^{109}\text{Ag}$, 48.16%. The mass of ${}^{107}\text{Ag}$ is 106.905092 u. What is the mass of ${}^{109}\text{Ag}$?
46. Gallium has two naturally occurring isotopes. One of them, gallium-69, has a mass of 68.925581 u and a percent isotopic abundance of 60.11%. What must be the mass and percent isotopic abundance of the other isotope, gallium-71?
47. The three naturally occurring isotopes of potassium are ${}^{39}\text{K}$, 38.963707 u; ${}^{40}\text{K}$, 39.963999 u; and ${}^{41}\text{K}$. The percent isotopic abundances of ${}^{39}\text{K}$ and ${}^{41}\text{K}$ are 93.2581% and 6.7302%, respectively. Determine the isotopic mass of ${}^{41}\text{K}$.
48. Use the conventional atomic mass of boron to estimate the fractional isotopic abundances of the two naturally occurring isotopes, ${}^{10}\text{B}$ and ${}^{11}\text{B}$. These isotopes have masses of 10.012937 u and 11.009305 u, respectively.

Mass Spectrometry

49. A mass spectrum of germanium displayed peaks at mass numbers 70, 72, 73, 74, and 76, with relative heights of 20.5, 27.4, 7.8, 36.5, and 7.8, respectively.
- In the manner of Figure 2-14, sketch this mass spectrum.
 - Estimate the weighted-average atomic mass of germanium, and state why this result is only approximately correct.
50. Hydrogen and chlorine atoms react to form simple diatomic molecules in a 1:1 ratio, that is, HCl. The percent

isotopic abundances of the chlorine isotopes are ^{35}Cl and ^{37}Cl are estimated to be 75.77% and 24.23%, respectively. The percent isotopic abundances of ^2H and ^3H are estimated to be 0.015% and less than 0.001%, respectively.

- How many different HCl molecules are possible, and what are their mass numbers (that is, the sum of the mass numbers of the H and Cl atoms)?
- Which is the most abundant of the possible HCl molecules? Which is the second most abundant?

The Periodic Table

51. Refer to the periodic table inside the front cover and identify
- the element that is in group 14 and the fourth period
 - one element similar to and one unlike sulfur
 - the alkali metal in the fifth period
 - the halogen element in the sixth period
52. Refer to the periodic table inside the front cover and identify
- the element that is in group 11 and the sixth period
 - an element with atomic number greater than 50 that has properties similar to the element with atomic number 18

(c) the group number of an element E that forms an ion E^{2-}

(d) an element M that you would expect to form the ion M^{3+}

53. What should be the atomic number of the alkali metal following francium (Fr)?
54. Find the several pairs of elements that are “out of order” in terms of increasing atomic mass and explain why the reverse order is necessary.

The Avogadro Constant and the Mole

55. What is the total number of atoms in (a) 15.8 mol Fe; (b) 0.000467 mol Ag; (c) 8.5×10^{-11} mol Na?
56. *Without doing detailed calculations*, indicate which of the following quantities contains the greatest number of atoms: 6.022×10^{23} Ni atoms, 25.0 g nitrogen, 52.0 g Cr, 10.0 cm^3 Fe ($d = 7.86 \text{ g/cm}^3$). Explain your reasoning.
57. Determine
- the number of moles of Zn in a 415.0 g sample of zinc metal
 - the number of Cr atoms in 147.4 kg chromium
 - the mass of a one-trillion-atom (1.0×10^{12}) sample of metallic gold
 - the average mass of a fluorine atom
58. Determine
- the number of Kr atoms in a 5.25 mg sample of krypton
 - the molar mass, M , and identity of an element if the mass of a 2.80×10^{22} atom sample of the element is 2.09 g
 - the mass of a sample of phosphorus that contains the same number of atoms as 44.75 g of magnesium
59. How many Cu atoms are present in a piece of sterling-silver jewelry weighing 33.24 g? (Sterling silver is a silver-copper alloy containing 92.5% Ag by mass.)
60. How many atoms are present in a 50.0 cm^3 sample of plumber's solder, a lead-tin alloy containing 67% Pb by mass and having a density of 9.4 g/cm^3 ?
61. How many ^{204}Pb atoms are present in a piece of lead weighing 215 mg? The percent isotopic abundance of ^{204}Pb is 1.4%.

62. A particular lead-cadmium alloy is 8.0% cadmium by mass. What mass of this alloy, in grams, must you weigh out to obtain a sample containing 7.25×10^{23} Cd atoms?

63. Medical experts generally believe a level of $30 \mu\text{g}$ Pb per deciliter of blood poses a significant health risk (1 dL = 0.1 L). Express this level (a) in the unit mol Pb/L blood; (b) as the number of Pb atoms per milliliter of blood.

64. During a severe episode of air pollution, the concentration of lead in the air was observed to be $3.11 \mu\text{g Pb/m}^3$. How many Pb atoms would be present in a 0.500 L sample of this air (the volume of air displaced in the lungs between inhaling and exhaling)?

65. *Without doing detailed calculations*, determine which of the following samples has the greatest number of atoms:

- a cube of iron with a length of 10.0 cm ($d = 7.86 \text{ g/cm}^3$)
- 1.00 kg of hydrogen contained in a 10,000 L balloon
- a mound of sulfur weighing 20.0 kg
- a 76 lb sample of liquid mercury ($d = 13.5 \text{ g/mL}$)

66. *Without doing detailed calculations*, determine which of the following samples occupies the largest volume:

- 25.5 mol of sodium metal ($d = 0.971 \text{ g/cm}^3$)
- 0.725 L of liquid bromine ($d = 3.12 \text{ g/mL}$)
- 1.25×10^{25} atoms of chromium metal ($d = 9.4 \text{ g/cm}^3$)
- 2.15 kg of plumber's solder ($d = 9.4 \text{ g/cm}^3$), a lead-tin alloy with a 2:1 atom ratio of lead to tin

Integrative and Advanced Exercises

67. A solution was prepared by dissolving 2.50 g potassium chlorate (a substance used in fireworks and flares) in 100.0 mL of water at 40°C. When the solution was cooled to 20°C, its volume was still found to be 100.0 mL, but some of the potassium chlorate had crystallized (deposited from the solution as a solid). At 40°C, the density of water is 0.9922 g/mL, and at 20°C, the potassium chlorate solution had a density of 1.0085 g/mL.
- Estimate, to two significant figures, the mass of potassium chlorate that crystallized.
 - Why can't the answer in (a) be given more precisely?
68. William Prout (1815) proposed that all other atoms are built up of hydrogen atoms, suggesting that all elements should have integral atomic masses based on an atomic mass of one for hydrogen. This hypothesis appeared discredited by the discovery of atomic masses, such as 24.3 u for magnesium and 35.5 u for chlorine. In terms of modern knowledge, explain why Prout's hypothesis is actually quite reasonable.
69. Fluorine has a single atomic species, ^{19}F . Determine the atomic mass of ^{19}F by summing the masses of its protons, neutrons, and electrons, and compare your results with the value listed on the inside front cover. Explain why the agreement is poor.
70. Use 1×10^{-13} cm as the approximate diameter of the spherical nucleus of the hydrogen-1 atom, together with data from Table 2.1, to estimate the density of matter in a proton.
71. Use fundamental definitions and statements from Chapters 1 and 2 to establish the fact that $6.022 \times 10^{23} \text{ u} = 1.000 \text{ g}$.
72. In each case, identify the element in question.
- The mass number of an atom is 234, and the atom has 60.0% more neutrons than protons.
 - An ion with a +2 charge has 10.0% more protons than electrons.
 - An ion with a mass number of 110 and a 2+ charge has 25.0% more neutrons than electrons.
73. Determine the only possible +2 ion for which the following two conditions are both satisfied:
- The net ionic charge is *one-tenth* the nuclear charge.
 - The number of neutrons is *four* more than the number of electrons.
74. Determine the only possible isotope (E) for which the following conditions are met:
- The mass number of E is 2.50 times its atomic number.
 - The atomic number of E is equal to the mass number of another isotope (Y). In turn, isotope Y has a neutron number that is 1.33 times the atomic number of Y and equal to the neutron number of selenium-82.
75. Suppose we redefined the atomic mass scale by arbitrarily assigning to the naturally occurring *mixture* of chlorine isotopes an atomic mass of 35.00000 u.
- What would be the atomic masses of helium, sodium, and iodine on this new atomic mass scale?
 - Why do these three elements have nearly integral (whole-number) atomic masses based on carbon-12, but not based on naturally occurring chlorine?
76. The two naturally occurring isotopes of nitrogen have masses of 14.0031 and 15.0001 u, respectively. Use the conventional atomic mass of nitrogen to estimate the percentage of ^{15}N atoms in naturally occurring nitrogen.
77. The masses of the naturally occurring mercury isotopes are ^{196}Hg , 195.9658 u; ^{198}Hg , 197.9668 u; ^{199}Hg , 198.9683 u; ^{200}Hg , 199.9683 u; ^{201}Hg , 200.9703 u; ^{202}Hg , 201.9706 u; and ^{204}Hg , 203.9735 u. Use these data, together with data from Figure 2-14, to calculate the weighted-average atomic mass of mercury.
78. Germanium has three major naturally occurring isotopes: ^{70}Ge (69.92425 u, 20.85%), ^{72}Ge (71.92208 u, 27.54%), ^{74}Ge (73.92118 u, 36.29%). There are also two minor isotopes: ^{73}Ge (72.92346 u) and ^{76}Ge (75.92140 u). Calculate the percent isotopic abundances of the two minor isotopes. Comment on the precision of these calculations.
79. From the densities of the lines in the mass spectrum of krypton gas, the following observations were made:
- Somewhat more than 50% of the atoms were krypton-84.
 - The numbers of krypton-82 and krypton-83 atoms were essentially equal.
 - The number of krypton-86 atoms was 1.50 times as great as the number of krypton-82 atoms.
 - The number of krypton-80 atoms was 19.6% of the number of krypton-82 atoms.
 - The number of krypton-78 atoms was 3.0% of the number of krypton-82 atoms.
- The masses of the isotopes are
 ^{78}Kr , 77.9204 u ^{80}Kr , 79.9164 u ^{82}Kr , 81.9135 u
 ^{83}Kr , 82.9141 u ^{84}Kr , 83.9115 u ^{86}Kr , 85.9106 u
- The weighted-average atomic mass of Kr is 83.80. Use these data to calculate the percent isotopic abundances of the krypton isotopes.
80. The two naturally occurring isotopes of chlorine are ^{35}Cl (34.9689 u, 75.77%) and ^{37}Cl (36.9658 u, 24.23%). The two naturally occurring isotopes of bromine are ^{79}Br (78.9183 u, 50.69%) and ^{81}Br (80.9163 u, 49.31%). Chlorine and bromine combine to form bromine monochloride, BrCl . Sketch a mass spectrum for BrCl with the relative number of molecules plotted against molecular mass (similar to Figure 2-14).
81. How many atoms are present in a 1.50 m length of 20-gauge copper wire? A 20-gauge wire has a diameter of 0.03196 in., and the density of copper is 8.92 g/cm^3 .
82. Monel metal is a corrosion-resistant copper-nickel alloy used in the electronics industry. A particular alloy with a density of 8.80 g/cm^3 and containing 0.022% Si by mass is used to make a rectangular plate 15.0 cm long, 12.5 cm wide, 3.00 mm thick, and has a 2.50 cm diameter hole drilled through its center. How many silicon-30 atoms are found in this plate? The mass of a silicon-30 atom is 29.97376 u, and the percent isotopic abundance of silicon-30 is 3.10%.

83. Deuterium, ^2H (2.0140 u), is sometimes used to replace the principal hydrogen isotope ^1H in chemical studies. The percent isotopic abundance of deuterium is 0.015%. If it can be done with 100% efficiency, what mass of hydrogen gas would have to be processed to obtain a sample containing 2.50×10^{21} ^2H atoms?
84. An alloy that melts at about the boiling point of water has Bi, Pb, and Sn atoms in the ratio 10:6:5, respectively. What mass of alloy contains a total of one mole of atoms?
85. A particular silver solder (used in the electronics industry to join electrical components) is to have the atom ratio of 5.00 Ag/4.00 Cu/1.00 Zn. What masses of the three metals must be melted together to prepare 1.00 kg of the solder?
86. A low-melting Sn–Pb–Cd alloy called *eutectic alloy* is analyzed. The mole ratio of tin to lead is 2.73:1.00,

and the mass ratio of lead to cadmium is 1.78:1.00. What is the mass percent composition of this alloy?

87. In an experiment, 125 cm³ of zinc and 125 cm³ of iodine are mixed together and the iodine is completely converted to 164 cm³ of zinc iodide. What volume of zinc remains unreacted? The densities of zinc, iodine, and zinc iodide are 7.13 g/cm³, 4.93 g/cm³, and 4.74 g/cm³, respectively.
88. Atoms are spherical and so when silver atoms pack together to form silver metal, they cannot fill all the available space. In a sample of silver metal, approximately 26.0% of the sample is empty space. Given that the density of silver metal is 10.5 g/cm³, what is the radius of a silver atom? Express your answer in picometers.

Feature Problems

89. The data Lavoisier obtained in the experiment described on page 63 are as follows:

Before heating: glass vessel + tin + air

= 13 ounces, 2 gros, 2.50 grains

After heating: glass vessel + tin calx + remaining air

= 13 ounces, 2 gros, 5.62 grains

How closely did Lavoisier's results conform to the law of conservation of mass? (1 livre = 16 ounces; 1 ounce = 8 gros; 1 gros = 72 grains. In modern terms, 1 livre = 30.59 g)

90. Some of Millikan's oil-drop data are shown below. The measured quantities were not actual charges on oil drops but were proportional to these charges. Show that these data are consistent with the idea of a fundamental electronic charge.

Observation	Measured Quantity	Observation	Measured Quantity
1	19.66	8	53.91
2	24.60	9	59.12
3	29.62	10	63.68
4	34.47	11	68.65
5	39.38	12	78.34
6	44.42	13	83.22
7	49.41		

91. Before 1961, the standard for atomic masses was the isotope ^{16}O , to which physicists assigned a value of exactly 16. At the same time, chemists assigned a value of exactly 16 to the naturally occurring mixture of the isotopes ^{16}O , ^{17}O , and ^{18}O . Would you expect atomic masses listed in a 60-year-old text to be the same, generally higher, or generally lower than in this text? Explain.
92. German chemist Fritz Haber proposed paying off the reparations imposed against Germany after World War I by extracting gold from seawater. Given that

(1) the amount of the reparations was \$28.8 billion dollars, (2) the value of gold at the time was about \$21.25 per troy ounce (1 troy ounce = 31.103 g), and (3) gold occurs in seawater to the extent of 4.67×10^{17} atoms per ton of seawater (1 ton = 2000 lb), how many cubic kilometers of seawater would have had to be processed to obtain the required amount of gold? Assume that the density of seawater is 1.03 g/cm³. (Haber's scheme proved to be commercially infeasible, and the reparations were never fully paid.)

93. Mass spectrometry is one of the most versatile and powerful tools in chemical analysis because of its capacity to discriminate between atoms of different masses. When a sample containing a mixture of isotopes is introduced into a mass spectrometer, the ratio of the peaks observed reflects the ratio of the percent isotopic abundances. This ratio provides an internal standard from which the amount of a certain isotope present in a sample can be determined. This is accomplished by deliberately introducing a known quantity of a particular isotope into the sample to be analyzed. A comparison of the new isotope ratio to the first ratio allows the determination of the amount of the isotope present in the original sample.

An analysis was done on a rock sample to determine its rubidium content. The rubidium content of a portion of rock weighing 0.350 g was extracted, and to the extracted sample was added an additional 29.45 μg of ^{87}Rb . The mass spectrum of this spiked sample showed a ^{87}Rb peak that was 1.12 times as high as the peak for ^{85}Rb . Assuming that the two isotopes react identically, what is the Rb content of the rock (expressed in parts per million by mass)? The isotopic abundances and isotopic masses are shown in the table.

Isotope	% Isotopic Abundance	Atomic Mass, u
^{87}Rb	27.83	86.909
^{85}Rb	72.17	84.912

Self-Assessment Exercises

94. In your own words, define or explain these terms or symbols: (a) $\frac{A}{Z}E$; (b) β particle; (c) isotope; (d) ^{16}O ; (e) molar mass.
95. Briefly describe
 - (a) the law of conservation of mass
 - (b) Rutherford's nuclear atom
 - (c) weighted-average atomic mass
 - (d) a mass spectrum
96. Explain the important distinctions between each pair of terms:
 - (a) cathode rays and X-rays
 - (b) protons and neutrons
 - (c) nuclear charge and ionic charge
 - (d) periods and groups of the periodic table
 - (e) metal and nonmetal
 - (f) the Avogadro constant and the mole
97. A certain element contains one atom of mass 10.013 u for every four atoms of mass 11.009 u. Compute the atomic weight of the element.
98. When 10.0 g zinc and 8.0 g sulfur are allowed to react, all the zinc is consumed, 14.9 g zinc sulfide is produced, and the mass of unreacted sulfur remaining is
 - (a) 2.0 g
 - (b) 3.1 g
 - (c) 4.9 g
 - (d) impossible to predict from this information alone
99. One oxide of rubidium has 0.187 g O per gram of Rb. A possible O:Rb mass ratio for a second oxide of rubidium is (a) 16:85.5; (b) 8:42.7; (c) 1:2.674; (d) any of these.
100. An attempt was made to determine the atomic mass of element X. If X forms a compound with oxygen that contains 46.7% X by mass and has the formula XO, what is the atomic mass of X?
101. Cathode rays
 - (a) may be positively or negatively charged
 - (b) are a form of electromagnetic radiation similar to visible light
 - (c) have properties identical to β particles
 - (d) have masses that depend on the cathode that emits them
102. The scattering of α particles by thin metal foils established that
 - (a) the mass of an atom is concentrated in a positively charged nucleus
 - (b) electrons are fundamental particles of all matter
 - (c) all electrons carry the same charge
 - (d) atoms are electrically neutral
103. Which of the following have the same charge and approximately the same mass?
 - (a) an electron and a proton; (b) a proton and a neutron; (c) a hydrogen atom and a proton; (d) a neutron and a hydrogen atom; (e) an electron and an H^+ ion
104. Which of the following is *not* a fundamental particle?
 - (a) proton; (b) neutron; (c) beta particle; (d) alpha particle; (e) all are fundamental particles
105. Which of the following scientists did *not* contribute to determining the *structure* of the atom?
 - (a) Thomson; (b) Rutherford; (c) Millikan; (d) Dalton; (e) Becquerel
106. A subatomic particle that has about the same mass as the hydrogen atom and a negative charge is called
 - (a) a proton; (b) a neutron; (c) an electron; (d) an isotope; (e) none of these.
107. What is the correct symbol for the species that contains 18 neutrons, 17 protons, and 16 electrons?
108. The properties of magnesium will most resemble those of which of the following? (a) cesium; (b) sodium; (c) aluminum; (d) calcium; (e) manganese.
109. Which group in the main group of elements contains
 - (a) no metals or metalloids? (b) only one metal or metalloid? (c) only one nonmetal? (d) only nonmetals?
110. The two species that have the same number of electrons as ^{32}S are (a) ^{32}Cl ; (b) $^{34}\text{S}^+$; (c) $^{33}\text{P}^+$; (d) $^{28}\text{Si}^{2+}$; (e) $^{35}\text{S}^{2-}$; (f) $^{40}\text{Ar}^{2+}$; (g) $^{40}\text{Ca}^{2+}$.
111. To four significant figures, all of the following masses are possible for an individual titanium atom except one. The exception is (a) 45.95 u; (b) 46.95 u; (c) 47.87 u; (d) 47.95 u; (e) 48.95 u; (f) 49.94 u.
112. The mass of the isotope $^{84}_{36}\text{Xe}$ is 83.9115 u. If the atomic mass scale were redefined so that $^{84}_{36}\text{Xe} = 84$ u, *exactly*, the mass of the $^{12}_6\text{C}$ isotope would be (a) 11.9115 u; (b) 11.9874 u; (c) 12 u exactly; (d) 12.0127 u; (e) 12.0885 u.
113. A 5.585-kg sample of iron (Fe) contains
 - (a) 10.0 mol Fe
 - (b) twice as many atoms as does 600.6 g C
 - (c) 10 times as many atoms as does 52.00 g Cr
 - (d) 6.022×10^{24} atoms
114. A 91.84 g sample of Ti contains (a) 4.175 mol of Ti; (b) 6.022×10^{23} Ti atoms; (c) 1.155×10^{24} protons; (d) 2.542×10^{25} electrons; (e) none of these.
115. There are three common iron–oxygen compounds. The one with the greatest proportion of iron has one Fe atom for every O atom and the formula FeO. A second compound has 2.327 g Fe per 1.000 g O, and the third has 2.618 g Fe per 1.000 g O. What are the formulas of these other two iron–oxygen compounds?
116. The four naturally occurring isotopes of strontium have the atomic masses 83.9134 u; 85.9093 u; 86.9089 u; and 87.9056 u. The percent isotopic abundance of the lightest isotope is 0.56% and of the heaviest, 82.58%. Estimate the percent isotopic abundances of the other two. Why is this result only a rough approximation?
117. Gold is present in seawater to the extent of 0.15 mg/ton. Assume the density of the seawater is 1.03 g/mL and determine how many Au atoms could conceivably be extracted from 0.250 L of seawater (1 ton = 2.000×10^3 lb; 1 kg = 2.205 lb).
118. Appendix E describes a useful study aid known as concept mapping. Using the method presented in Appendix E, construct a concept map illustrating the different concepts in Sections 2-7 and 2-8.

3

Chemical Compounds

Contents

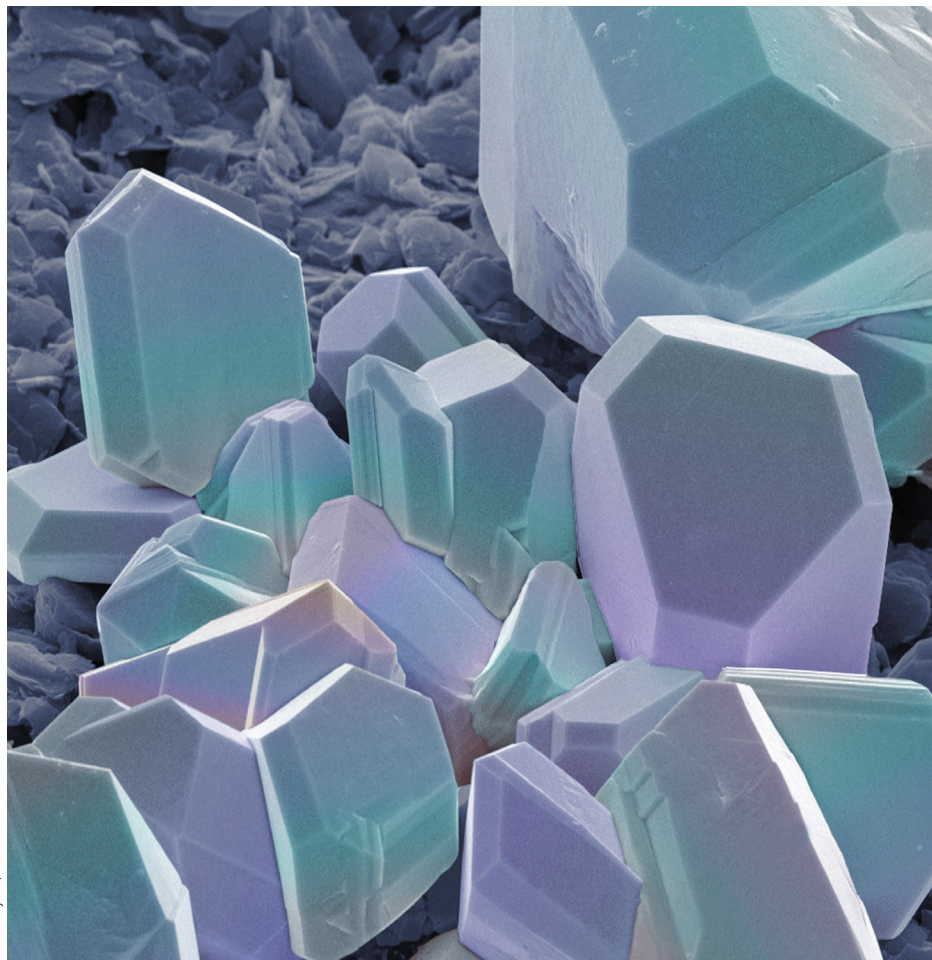
- 3-1** Types of Chemical Compounds and Their Formulas
- 3-2** The Mole Concept and Chemical Compounds
- 3-3** Composition of Chemical Compounds
- 3-4** Oxidation States: A Useful Tool in Describing Chemical Compounds
- 3-5** Naming Compounds: Organic and Inorganic Compounds
- 3-6** Names and Formulas of Inorganic Compounds
- 3-7** Names and Formulas of Organic Compounds

Learning Objectives

- 3.1** Distinguish between an empirical formula, a molecular formula, and the formula unit of an ionic compound.
- 3.2** Use Avogadro's constant to relate the mass and molar mass of molecules and ionic compounds to their elementary entities (atoms or ions).
- 3.3** Use percent composition data for a chemical compound to determine its empirical and molecular formula (and vice versa).
- 3.4** Use the rules for assigning oxidation states to determine the oxidation state of each element in a chemical compound.
- 3.5** Distinguish between organic and inorganic compounds.

Water, ammonia, carbon monoxide, and carbon dioxide—all familiar substances—are rather simple chemical compounds. Only slightly less familiar are sucrose (cane sugar), acetylsalicylic acid (aspirin), and ascorbic acid (vitamin C). They too are chemical compounds. In fact, the study of chemistry is mostly about chemical compounds, and, in this chapter, we will consider a number of ideas about compounds.

The common feature of all compounds is that they are composed of two or more elements. The full range of compounds can be divided into a few broad categories by applying ideas from the periodic table of the elements. Compounds are represented by chemical formulas, which in turn are derived from the symbols of their constituent elements. In this chapter, you will learn how to deduce and write chemical formulas and how to use the information



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▲ Scanning electron microscope image of sodium chloride crystals. Chemical compounds, their formulas, and their names are discussed in this chapter.

incorporated into chemical formulas. The chapter ends with an overview of the relationship between names and formulas—chemical nomenclature.

3-1 Types of Chemical Compounds and Their Formulas

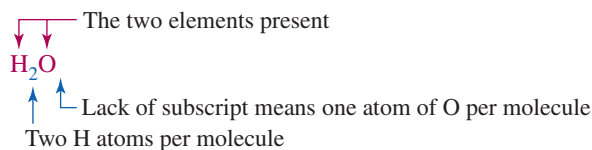
Generally speaking, two fundamental kinds of chemical bonds hold together the atoms in a compound. *Covalent* bonds, which involve a sharing of electrons between atoms, give rise to molecular compounds. *Ionic* bonds, which involve a transfer of electrons from one atom to another, give rise to ionic compounds. In this section, we consider only the basic features of molecular and ionic compounds that we need as background for the early chapters of the text. Our in-depth discussion of chemical bonding will come in Chapters 10 and 11.

Molecular Compounds

A **molecular compound** is made up of discrete units called molecules, which typically consist of a small number of *nonmetal* atoms held together by covalent bonds. Molecular compounds are represented by **chemical formulas**, symbolic representations that, at minimum, indicate

- the elements present
- the relative number of atoms of each element

In the formula for water, the constituent elements are denoted by their symbols. The relative numbers of atoms are indicated by *subscripts*. Where no subscript is written, the number 1 is understood.



Another example of a chemical formula is CCl_4 , which represents the compound carbon tetrachloride. The formulas H_2O and CCl_4 both represent distinct entities—*molecules*. Thus, we can refer to water and carbon tetrachloride as molecular compounds.

An **empirical formula** is the simplest formula for a compound; it shows the types of atoms present and their relative numbers. The subscripts in an empirical formula are reduced to their simplest whole-number ratio. For example, P_2O_5 is the empirical formula for a compound whose molecules have the formula P_4O_{10} . Generally, the empirical formula does not tell us a great deal about a compound. Acetic acid ($\text{C}_2\text{H}_4\text{O}_2$), formaldehyde (CH_2O , used to make certain plastics and resins), and glucose ($\text{C}_6\text{H}_{12}\text{O}_6$, blood sugar) all have the empirical formula CH_2O .

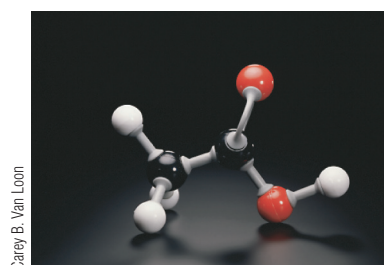
A **molecular formula** is based on an actual molecule of a compound. In some cases, the empirical and molecular formulas are identical, such as CH_2O for formaldehyde. In other cases, the molecular formula is a multiple of the empirical formula. A molecule of acetic acid, for example, consists of eight atoms—two C atoms, four H atoms, and two O atoms, so the molecular formula of acetic acid is $\text{C}_2\text{H}_4\text{O}_2$. This is twice the number of atoms in the formula unit (CH_2O). Empirical and molecular formulas tell us the combining ratio of the atoms in the compound, but they show nothing about how the atoms are attached to each other. Other types of formulas, however, do convey this information. Figure 3-1 shows several representations of acetic acid, the acid constituent that gives vinegar its sour taste.

A **structural formula** shows the order in which atoms are bonded together in a molecule and by what types of bonds. Thus, the structural formula of acetic acid tells us that three of the four H atoms are bonded to one of the C atoms, and

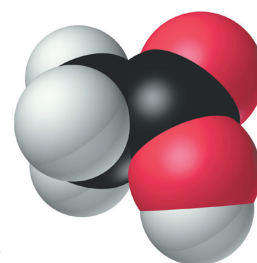
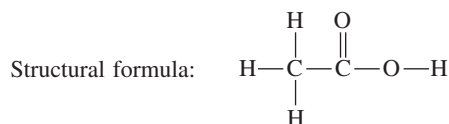
3.6 Use the general rules to name simple inorganic compounds, including binary compounds, binary acids, polyatomic ions, and oxoacids.

3.7 Use the general rules to name simple organic compounds, including basic branched alkanes and alkanes with functional groups, such as alcohols and carboxylic acids.

◀ In our later study of chemical bonding, we will find that the distinction between covalent and ionic bonding is not as clear-cut as these statements imply. We will consider this matter in Chapter 10.



Carey B. Van Loon

Molecular model
("ball and stick")Empirical formula: CH_2O Molecular formula: $\text{C}_2\text{H}_4\text{O}_2$ Molecular model
("space filling")**Figure 3-1** Several representations of the compound acetic acid

In the molecular model, the black spheres are carbon, the red are oxygen, and the white are hydrogen. To show that one H atom in the molecule is fundamentally different from the other three, the formula of acetic acid is often written as $\text{HC}_2\text{H}_3\text{O}_2$ (see Section 5-3). To show that this H atom is bonded to an O atom, the formulas CH_3COOH and $\text{CH}_3\text{CO}_2\text{H}$ are also used. For a few chemical compounds, you may find different versions of chemical formulas in different sources.

the remaining H atom is bonded to an O atom. Both of the O atoms are bonded to one of the C atoms, and the two C atoms are bonded to each other. The covalent bonds in the structural formula are represented by lines or dashes (—). One of the bonds is represented by a double dash (==) and is called a *double* covalent bond. Differences between single and double bonds are discussed later in the text. For now, just think of a double bond as being a stronger or tighter bond than a single bond.

A **condensed structural formula**, which is written on a single line, is an alternative, less cumbersome way of showing how the atoms of a molecule are connected. Thus, the acetic acid molecule is represented as either CH_3COOH or $\text{CH}_3\text{CO}_2\text{H}$. With this type of formula, the different ways in which the H atoms are attached are still apparent.

Condensed structural formulas can also be used to show how a group of atoms is attached to another atom. Consider methylpropane, C_4H_{10} , in Figure 3-2(b). The structural formula shows that there is a $-\text{CH}_3$ group of atoms attached to the central carbon atom. In the condensed structural formula, this is indicated by enclosing the CH_3 in parentheses to the right of the atom to which it is attached, thus $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_3$. Alternatively, because the central C atom is bonded to each of the other three C atoms, we can write the condensed structural formula $\text{CH}(\text{CH}_3)_3$.

Organic compounds are made up principally of carbon and hydrogen, with oxygen and/or nitrogen as important constituents in many of them. Each carbon atom forms *four covalent bonds*. Organic compounds can be very complex, and one way of simplifying their structural formulas is to write structures without showing the C and H atoms explicitly. We do this by using a **line-angle formula** (also referred to as a *line structure*), in which lines represent chemical bonds. A carbon atom exists wherever a line ends or meets another line, and the number of H atoms needed to complete each carbon atom's four bonds are assumed to be present. The symbols of other atoms or groups of atoms and the bond lines joining them to C atoms are written explicitly. The formula of the complex hormone molecule testosterone, seen in Figure 3-2(c), is a line-angle formula.

Molecules occupy space and have a three-dimensional shape, but empirical and molecular formulas do not convey any information about the spatial arrangements of atoms. Structural formulas can sometimes show this, but usually the only satisfactory way to represent the three-dimensional structure of molecules is with models. In a *ball-and-stick model*, atoms are represented by small balls, and the bonds between atoms by sticks (see Figure 3-1). Such models help us to visualize distances between the nuclei of atoms (bond lengths) and the geometrical shapes of molecules. Ball-and-stick models are easy to

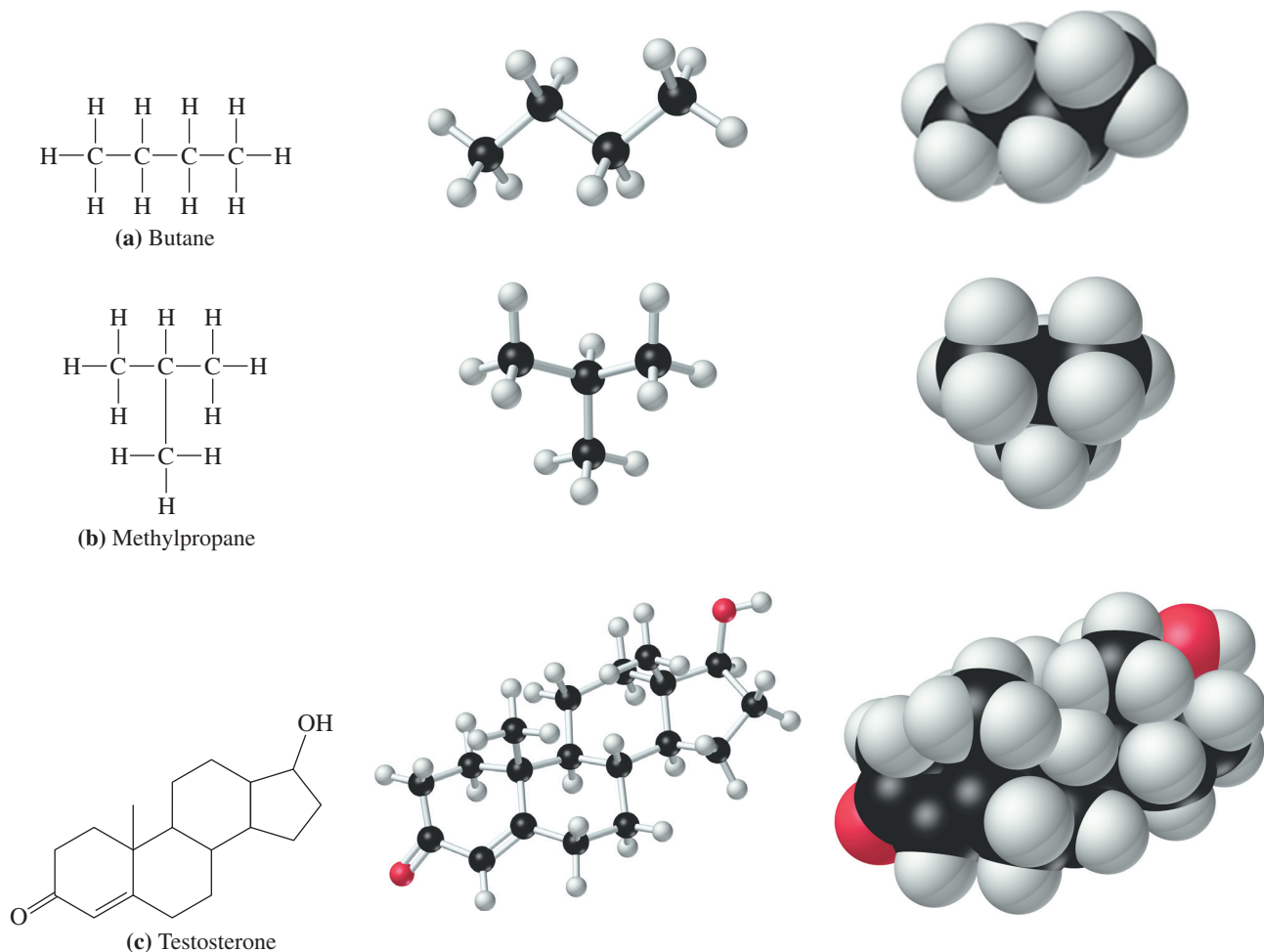


Figure 3-2 Visualizations of (a) butane, (b) methylpropane, and (c) testosterone

draw and interpret, but they can be somewhat misleading. Chemical bonds are forces that draw atoms in a molecule into direct contact. The atoms are not held apart as implied by a ball-and-stick model.

A *space-filling model* shows that the atoms in a molecule occupy space and that they are in actual contact with one another. Certain computer programs generate images of space-filling models such as those shown in Figures 3-1 and 3-2. A space-filling model is a more accurate representation of the size and shape of a molecule because it is constructed to scale (that is, a nanometer-size molecule is magnified to a millimeter or centimeter scale).

The acetic acid molecule is made up of three types of atoms (C, H, and O) and models of the molecule reflect this fact. Different colors are used to distinguish the various types of atoms in ball-and-stick and space-filling models (see Figure 3-3). You will notice that the colored spheres are of different sizes, which correspond to the size differences between the various atoms in the periodic table.

The depictions of molecules just discussed will be used throughout this book. In fact, visualization of the sizes and shapes of molecules and interpretation of the physical and chemical properties in terms of molecular sizes and shapes is one of the most important aspects of modern chemistry.

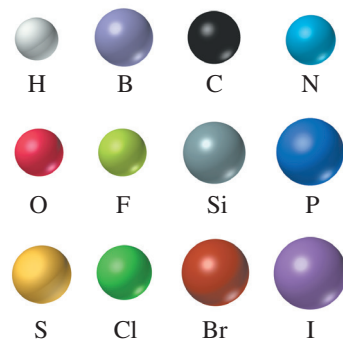


Figure 3-3 Color scheme for use in molecular models

The sizes of atoms, reflected in the various sizes of the colored spheres, are related to the locations of the elements in the periodic table, as discussed in Section 7-3.



3-1 Concept Assessment

Represent the succinic acid molecule, $\text{HOOCCH}_2\text{CH}_2\text{COOH}$, through empirical, molecular, structural, and line-angle formulas.

Ionic Compounds

Chemical combination of a metal and a nonmetal usually results in an ionic compound. An **ionic compound** is made up of positive and negative ions joined together by electrostatic forces of attraction (recall the attraction of oppositely charged objects pictured in Figure 2-4). The atoms of metallic elements tend to lose one or more electrons when they combine with nonmetal atoms, and the nonmetal atoms tend to gain one or more electrons. As a result of this electron transfer, the metal atom becomes a positive ion, or **cation**, and the nonmetal atom becomes a negative ion, or **anion**. We can usually deduce the charge on a main-group cation or anion from the group of the periodic table to which the element belongs (recall Section 2-6). Thus, the periodic table can help us to write the formulas of ionic compounds.

In the formation of sodium chloride—ordinary table salt—each sodium atom gives up one electron to become a sodium ion, Na^+ , and each chlorine atom gains one electron to become a chloride ion, Cl^- . This fact conforms to the relationship between locations of the elements in the periodic table and the charges on their simple ions (see page 82). For sodium chloride to be electrically neutral, there must be one Na^+ ion for each Cl^- ion ($+1 - 1 = 0$). Thus, the formula of sodium chloride is NaCl , and its structure, is shown in Figure 3-4.

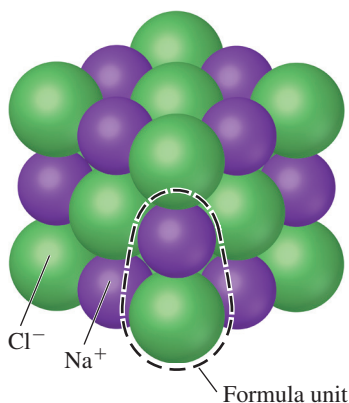


Figure 3-4 Portion of an ionic crystal and a formula unit of NaCl

Solid sodium chloride consists of enormous numbers of Na^+ and Cl^- ions in a network called a crystal. The combination of one Na^+ and one Cl^- ion is the smallest collection of ions from which we can deduce the formula NaCl . It is a formula unit.

We observe that each Na^+ ion in sodium chloride is surrounded by six Cl^- ions, and vice versa, and we cannot say that any one of these six Cl^- ions belongs exclusively to a given Na^+ ion. Yet, the ratio of Cl^- to Na^+ ions in sodium chloride is 1:1, and so we arbitrarily select a combination of one Na^+ ion and one Cl^- ion as a formula unit. The **formula unit** of an ionic compound is the smallest electrically neutral collection of ions. The ratio of atoms (ions) in the formula unit is the same as in the chemical formula. Because it is buried in a vast network of ions, called a crystal, a formula unit of an ionic compound does not exist as a distinct entity. Thus, it is inappropriate to call a formula unit of solid sodium chloride a molecule.

The situation with magnesium chloride is similar. In magnesium chloride, found in trace quantities in table salt, magnesium atoms lose two electrons to become magnesium ions, Mg^{2+} (Mg is in group 2). To obtain an electrically neutral formula unit, there must be two Cl^- ions, each with a charge of -1 , for every Mg^{2+} ion. The formula of magnesium chloride is MgCl_2 .

The ions Na^+ , Mg^{2+} , and Cl^- are *monatomic*, meaning that each consists of a single ionized atom. By contrast, a *polyatomic* ion is made up of two or more atoms. In the nitrate ion, NO_3^- , the subscripts signify that *three* O atoms and *one* N atom are joined by covalent bonds into the single ion NO_3^- . Magnesium nitrate is an ionic compound made up of magnesium and nitrate ions. An electrically neutral formula unit of this compound must consist of one Mg^{2+} ion and two NO_3^- ions. The formula based on this formula unit is denoted by enclosing NO_3 in parentheses, followed by the subscript 2; thus, $\text{Mg}(\text{NO}_3)_2$. Polyatomic ions are discussed further in Section 3-6.



3-1 ARE YOU WONDERING?

Can a compound be formed between different metal atoms?

In a metal, electrons of the atoms interact to form *metallic* bonds. The bonded atoms are usually of the same element, but they may also be of different elements, giving rise to *intermetallic* compounds. The metallic bond gives metals and intermetallic compounds their characteristic properties of electrical and heat conductivity.

3-2 The Mole Concept and Chemical Compounds

Once we know the chemical formula of a compound, we can determine its formula mass. **Formula mass** is the mass of a *formula unit* in atomic mass units. It is always appropriate to use the term formula mass, but, for a molecular compound, the formula unit is an actual molecule, so we can speak of molecular mass. **Molecular mass** is the mass of a molecule in atomic mass units.

Formula and molecular masses can be obtained just by adding up atomic masses (those on the inside front cover). Thus, for the molecular compound water, H_2O ,

$$\begin{aligned}\text{molecular mass H}_2\text{O} &= 2(\text{atomic mass H}) + (\text{atomic mass O}) \\ &= 2(1.008 \text{ u}) + 15.999 \text{ u} \\ &= 18.015 \text{ u}\end{aligned}$$

For the ionic compound magnesium chloride, MgCl_2 ,

$$\begin{aligned}\text{formula mass MgCl}_2 &= \text{atomic mass Mg} + 2(\text{atomic mass Cl}) \\ &= 24.305 \text{ u} + 2(35.45 \text{ u}) \\ &= 95.21 \text{ u}\end{aligned}$$

and for the ionic compound magnesium nitrate, $\text{Mg}(\text{NO}_3)_2$,

$$\begin{aligned}\text{formula mass Mg}(\text{NO}_3)_2 &= \text{atomic mass Mg} + 2[\text{atomic mass N} + 3(\text{atomic mass O})] \\ &= 24.305 \text{ u} + 2[14.007 \text{ u} + 3(15.999 \text{ u})] \\ &= 148.313 \text{ u}\end{aligned}$$

Mole of a Compound

Recall that in Chapter 2 a mole was defined as an amount of substance having the same number of elementary entities as there are atoms in exactly 12 g of pure carbon-12. This definition carefully avoids saying that the entities to be counted are always atoms. As a result, we can apply the concept of a mole to any quantity that we can represent by a symbol or formula—atoms, ions, formula units, or molecules. Specifically, a *mole of compound* is an amount of compound containing Avogadro's number (6.02214×10^{23}) of formula units or molecules. The **molar mass** is the mass of one mole of compound—one mole of molecules of a molecular compound and one mole of formula units of an ionic compound.

The weighted-average molecular mass of H_2O is 18.015 u, compared with a mass of exactly 12 u for a carbon-12 atom. If we compare samples of water molecules and carbon atoms by using Avogadro's number of each, we get a mass of 18.015 g H_2O , compared with exactly 12 g for carbon-12. The molar mass of H_2O is 18.015 g $\text{H}_2\text{O}/\text{mol H}_2\text{O}$. If we know the formula of a compound, we can equate the following terms, as illustrated for H_2O , MgCl_2 , and $\text{Mg}(\text{NO}_3)_2$.

$$\begin{aligned}1 \text{ mol H}_2\text{O} &= 18.015 \text{ g H}_2\text{O} = 6.02214 \times 10^{23} \text{ H}_2\text{O molecules} \\ 1 \text{ mol MgCl}_2 &= 95.21 \text{ g MgCl}_2 = 6.02214 \times 10^{23} \text{ MgCl}_2 \text{ formula units} \\ 1 \text{ mol Mg}(\text{NO}_3)_2 &= 148.313 \text{ g Mg}(\text{NO}_3)_2 = 6.02214 \times 10^{23} \text{ Mg}(\text{NO}_3)_2 \text{ formula units}\end{aligned}$$

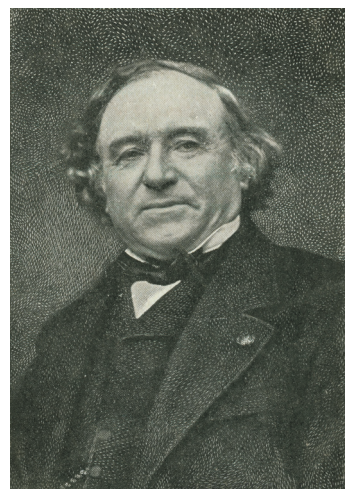
Such expressions as these provide several different types of conversion factors that can be applied in a variety of problem-solving situations. The strategy that works best for a particular problem will depend, in part, on how the necessary conversions are visualized. As we learned in Section 2-7, the most direct link to an amount in moles is through a mass in grams,

◀ The terms *formula weight* and *molecular weight* are often used in place of formula mass and molecular mass. This is similar to the situation described for atomic mass and atomic weight in the footnote on page 76.

◀ The terms *formula mass* and *molecular mass* have essentially the same meaning, although when referring to ionic compounds, such as NaCl and MgCl_2 , *formula mass* is the proper term.

Keep in Mind

that although *molecular mass* and *molar mass* sound similar and are related, they are not the same. Molecular mass is the weighted-average mass of one molecule expressed in atomic mass units, u. *Molar mass* is the mass of Avogadro's number of molecules expressed in grams per mole, g/mol. The two terms have the same numerical value but different units.



▲ Jean Baptiste André Dumas (1800–1884) was a French chemist known for his work on determining atomic and molecular weights. By careful measurements of vapor densities, he was able to recalculate the atomic mass of 30 elements.

so generally the central focus of a problem is the conversion of a mass in grams to an amount in moles, or vice versa. This conversion must often be preceded or followed by other conversions involving volumes, densities, percentages, and so on. As we saw in Chapter 2, one helpful tool in problem solving is to establish a conversion pathway. In Table 3.1, we summarize the roles that density, molar mass, and the Avogadro constant play in a conversion pathway.

Table 3.1 Density, Molar Mass, and the Avogadro Constant as Conversion Factors

Density, d	converts from volume to mass
Molar mass, M	converts from mass to amount (mol)
Avogadro constant, N_A	converts from amount (mol) to elementary entities

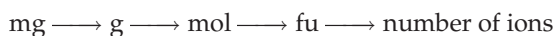
EXAMPLE 3-1

Relating Molar Mass, the Avogadro Constant, and Formula Units of an Ionic Compound

An analytical balance can detect a mass of 0.1 mg. How many ions are present in this minimally detectable quantity of MgCl_2 ?

Analyze

The central focus is the conversion of a measured quantity, 0.1 mg MgCl_2 , to an amount in moles. After making the mass conversion, $\text{mg} \longrightarrow \text{g}$, we can use the molar mass to convert from mass to amount in moles. Then, with the Avogadro constant as a conversion factor, we can convert from amount in moles to number of formula units. The final factor we need is based on the fact that there are *three* ions (*one* Mg^{2+} and *two* Cl^-) per formula unit (fu) of MgCl_2 . It is often helpful to map out a conversion pathway that starts with the information given and proceeds through a series of conversion factors to the information sought. For this problem, we can begin with milligrams of MgCl_2 and make the following conversions:



Solve

The required conversions can be carried out in a stepwise fashion (as was done in Example 2-9), or they can be combined into a single line calculation. To avoid having to write down intermediate results and to avoid rounding errors, we'll use a single line calculation this time.

$$\begin{aligned} ? \text{ ions} &= 0.1 \text{ mg MgCl}_2 \times \frac{1 \text{ g MgCl}_2}{1000 \text{ mg MgCl}_2} \times \frac{1 \text{ mol MgCl}_2}{95 \text{ g MgCl}_2} \\ &\quad \times \frac{6.0 \times 10^{23} \text{ fu MgCl}_2}{1 \text{ mol MgCl}_2} \times \frac{3 \text{ ions}}{1 \text{ fu MgCl}_2} \\ &= 2 \times 10^{18} \text{ ions} \end{aligned}$$

Assess

The mass of the sample (0.1 mg) is given with one significant figure, and so the final answer is rounded to one significant figure. In the calculation above, the molar mass of MgCl_2 and the Avogadro constant are rounded to two significant figures, that is, with one more significant figure than in the measured quantity.

PRACTICE EXAMPLE A: How many grams of MgCl_2 would you need to obtain $5.0 \times 10^{23} \text{ Cl}^-$ ions?

PRACTICE EXAMPLE B: How many nitrate ions, NO_3^- , and how many oxygen atoms are present in $1.00 \mu\text{g}$ of magnesium nitrate, $\text{Mg}(\text{NO}_3)_2$?

EXAMPLE 3-2

Combining Several Factors in a Calculation Involving Molar Mass

The volatile liquid ethyl mercaptan, $\text{C}_2\text{H}_6\text{S}$, is one of the most odoriferous substances known. It is sometimes added to natural gas to make gas leaks detectable. How many $\text{C}_2\text{H}_6\text{S}$ molecules are contained in a $1.0\ \mu\text{L}$ sample? The density of liquid ethyl mercaptan is $0.84\ \text{g/mL}$.

Analyze

The central focus is again the conversion of a measured quantity to an amount in moles. Because the density is given in g/mL , it will be helpful to convert the measured volume to milliliters. Then, density can be used as a conversion factor to obtain the mass in grams, and the molar mass can then be used to convert mass to amount in moles. Finally, the Avogadro constant can be used to convert the amount in moles to the number of molecules. In summary, the conversion pathway is $\mu\text{L} \rightarrow \text{L} \rightarrow \text{g} \rightarrow \text{mol} \rightarrow \text{molecules}$.

Solve

As always, the required conversions can be combined into a single line calculation. However, it is instructive to break the calculation into three steps: (1) a conversion from volume to mass, (2) a conversion from mass to amount in moles, and (3) a conversion from amount in moles to molecules. These three steps emphasize, respectively, the roles played by density, molar mass, and the Avogadro constant in the conversion pathway. (See Table 3.1.)

$$\begin{aligned}\text{Convert from volume to mass.} \quad ?\ \text{g C}_2\text{H}_6\text{S} &= 1.0\ \mu\text{L} \times \frac{1 \times 10^{-6}\ \text{L}}{1\ \mu\text{L}} \times \frac{1000\ \text{mL}}{1\ \text{L}} \times \frac{0.84\ \text{g C}_2\text{H}_6\text{S}}{1\ \text{mL}} \\ &= 8.4 \times 10^{-4}\ \text{g C}_2\text{H}_6\text{S}\end{aligned}$$

$$\begin{aligned}\text{Convert from mass to amount in moles.} \quad ?\ \text{mol C}_2\text{H}_6\text{S} &= 8.4 \times 10^{-4}\ \text{g C}_2\text{H}_6\text{S} \times \frac{1\ \text{mol C}_2\text{H}_6\text{S}}{62.1\ \text{g C}_2\text{H}_6\text{S}} \\ &= 1.4 \times 10^{-5}\ \text{mol C}_2\text{H}_6\text{S}\end{aligned}$$

$$\begin{aligned}\text{Convert from moles to molecules.} \quad ?\ \text{molecules C}_2\text{H}_6\text{S} &= 1.4 \times 10^{-5}\ \text{mol C}_2\text{H}_6\text{S} \times \frac{6.02 \times 10^{23}\ \text{molecules C}_2\text{H}_6\text{S}}{1\ \text{mol C}_2\text{H}_6\text{S}} \\ &= 8.1 \times 10^{18}\ \text{molecules C}_2\text{H}_6\text{S}\end{aligned}$$

Assess

Remember to store intermediate results in your calculator *without* rounding off. Round off at the end. The answer is rounded to two significant figures because the volume and density are given with two significant figures. Rounding errors are avoided if the required conversions are combined into a single line calculation.

$$\begin{aligned}?\ \text{molecules C}_2\text{H}_6\text{S} &= 1.0\ \mu\text{L} \times \frac{1 \times 10^{-6}\ \text{L}}{1\ \mu\text{L}} \times \frac{1000\ \text{mL}}{1\ \text{L}} \times \frac{0.84\ \text{g C}_2\text{H}_6\text{S}}{1\ \text{mL}} \\ &\quad \times \frac{1\ \text{mol C}_2\text{H}_6\text{S}}{62.1\ \text{g C}_2\text{H}_6\text{S}} \times \frac{6.02 \times 10^{23}\ \text{molecules C}_2\text{H}_6\text{S}}{1\ \text{mol C}_2\text{H}_6\text{S}} \\ &= 8.1 \times 10^{18}\ \text{molecules C}_2\text{H}_6\text{S}\end{aligned}$$

PRACTICE EXAMPLE A: Gold has a density of $19.32\ \text{g/cm}^3$. A piece of gold foil is $2.50\ \text{cm}$ on each side and $0.100\ \text{mm}$ thick. How many atoms of gold are in this piece of gold foil?

PRACTICE EXAMPLE B: If the $1.0\ \mu\text{L}$ sample of liquid ethyl mercaptan from Example 3-2 is allowed to evaporate and distribute itself throughout a chemistry lecture room with dimensions $62\ \text{ft} \times 35\ \text{ft} \times 14\ \text{ft}$, will the odor of the vapor be detectable in the room? The limit of detectability is $9 \times 10^{-4}\ \mu\text{mol/m}^3$.

Mole of an Element—A Second Look

In Chapter 2, we took one mole of an element to be 6.02214×10^{23} *atoms* of the element. This is the only definition possible for such elements as iron, magnesium, sodium, and copper, in which enormous numbers of individual spherical atoms are clustered together, much like marbles in a can. But the atoms of some elements are joined together to form molecules. Bulk samples of these elements are composed of collections of molecules. The molecules of P_4 and S_8 are represented in Figure 3-5. The molecular formulas of elements that you should become familiar with are



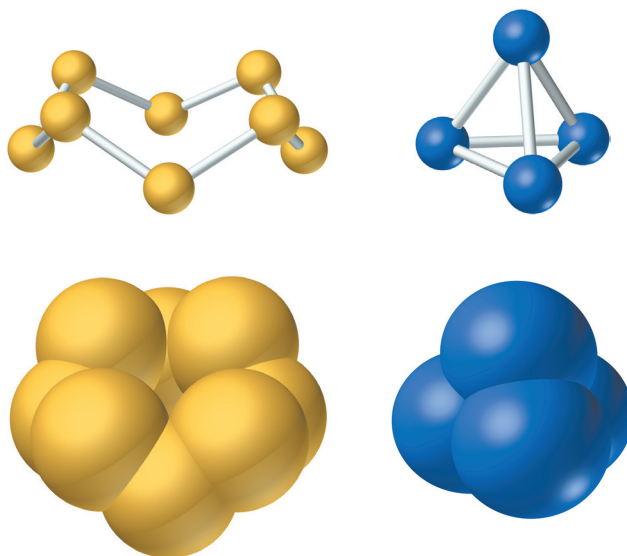


Figure 3-5 Molecular forms of elemental sulfur and phosphorus

In a sample of solid sulfur, there are eight sulfur atoms in a sulfur molecule. In solid white phosphorus, there are four phosphorus atoms per molecule.

For these elements, we speak of an *atomic* mass or a *molecular* mass, and molar mass can be expressed in two ways. Hydrogen, for example, has an atomic mass of 1.008 u and a molecular mass of 2.016 u; its molar mass can be expressed as 1.008 g H/mol H or 2.016 g H₂/mol H₂.

Another phenomenon occasionally encountered is the existence of an element in more than one molecular form, a situation referred to as *allotropy*. Thus, oxygen exists in two allotropic forms, the predominantly abundant diatomic oxygen, O₂, and the much less abundant allotrope *ozone*, O₃. The molar mass of ordinary dioxygen is 31.998 g O₂/mol O₂, and that of ozone is 47.997 g O₃/mol O₃.

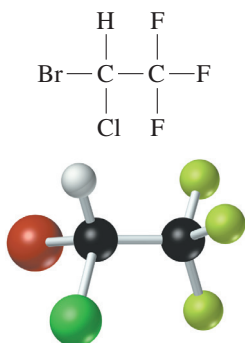


3-2 Concept Assessment

Without doing detailed calculations, determine which of the following quantities has the greatest mass and which has the smallest mass: (a) 0.50 mol O₂; (b) 2.0×10^{23} Cu atoms; (c) 1.0×10^{24} H₂O molecules; (d) a 20.000 g brass weight; (e) 1.0 mol Ne.

3-3 Composition of Chemical Compounds

▼ Two representations of halothane.



A chemical formula conveys considerable quantitative information about a compound and its constituent elements. We have already learned how to determine the molar mass of a compound, and, in this section, we consider some other types of calculations based on the chemical formula.

The colorless, volatile liquid halothane has been used as a fire extinguisher and also as an inhalation anesthetic. Both its empirical and molecular formulas are C₂HBrClF₃, its molecular mass is 197.38 u, and its molar mass is 197.38 g/mol, as calculated below:

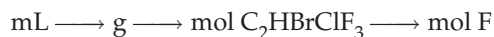
$$\begin{aligned} M_{\text{C}_2\text{HBrClF}_3} &= 2M_{\text{C}} + M_{\text{H}} + M_{\text{Br}} + M_{\text{Cl}} + 3M_{\text{F}} \\ &= [(2 \times 12.011) + 1.008 + 79.904 + 35.45 + (3 \times 18.9984)] \text{ g/mol} \\ &= 197.38 \text{ g/mol} \end{aligned}$$

The molecular formula of C₂HBrClF₃ tells us that *per mole* of halothane there are two moles of C atoms, one mole each of H, Br, and Cl atoms, and three moles of F atoms. This factual statement can be turned into conversion factors

to answer such questions as, “How many C atoms are present per mole of halothane?” In this case, the factor needed is 2 mol C/mol C₂HBrClF₃. That is,

$$\begin{aligned} ? \text{ C atoms} &= 1.000 \text{ mol C}_2\text{HBrClF}_3 \times \frac{2 \text{ mol C}}{1 \text{ mol C}_2\text{HBrClF}_3} \times \frac{6.022 \times 10^{23} \text{ C atoms}}{1 \text{ mol C}} \\ &= 1.204 \times 10^{24} \text{ C atoms} \end{aligned}$$

In Example 3-3, we use another conversion factor derived from the formula for halothane. This factor is shown in blue in the setup, which includes other familiar factors to make the conversion pathway:



EXAMPLE 3-3

Using Relationships Derived from a Chemical Formula

How many moles of F atoms are in a 75.0 mL sample of halothane ($d = 1.871 \text{ g/mL}$)?

Analyze

The conversion pathway for this problem is given above. First, convert the volume of the sample to mass; this requires density as a conversion factor. Next, convert the mass of halothane to its amount in moles; this requires the inverse of the molar mass as a conversion factor. The final conversion factor is based on the formula of halothane.

Solve

$$\begin{aligned} ? \text{ mol F} &= 75.0 \text{ mL C}_2\text{HBrClF}_3 \times \frac{1.871 \text{ g C}_2\text{HBrClF}_3}{1 \text{ mL C}_2\text{HBrClF}_3} \\ &\quad \times \frac{1 \text{ mol C}_2\text{HBrClF}_3}{197.4 \text{ g C}_2\text{HBrClF}_3} \times \frac{3 \text{ mol F}}{1 \text{ mol C}_2\text{HBrClF}_3} \\ &= 2.13 \text{ mol F} \end{aligned}$$

Assess

If we had been asked for the number of moles of C instead, the final conversion factor in the calculation above would have been (2 mol C/1 mol C₂HBrClF₃).

PRACTICE EXAMPLE A: How many grams of Br are contained in 25.00 mL of halothane ($d = 1.871 \text{ g/mL}$)?

PRACTICE EXAMPLE B: How many milliliters of halothane would contain 1.00×10^{24} Br atoms?



3-3 Concept Assessment

In hexachlorophene, C₁₃H₆Cl₆O₂, a compound used in germicidal soaps, which element contributes the greatest number of atoms and which contributes the greatest mass?

Calculating Percent Composition from a Chemical Formula

When chemists believe that they have synthesized a new compound, a sample is generally sent to an analytical laboratory where its percent composition is determined. This experimentally determined percent composition is then compared with the percent composition calculated from the formula of the expected compound. In this way, chemists can see if the compound obtained could be the one expected.

Equation (3.1) establishes how the mass percent of an element in a compound is calculated. In applying the equation, as in Example 3-4, think in terms of the following steps.

1. **Determine the molar mass of the compound.** This is the *denominator* in equation (3.1).
2. **Determine the contribution of the given element to the molar mass.** This product of the formula subscript and the molar mass of the element appears in the *numerator* of equation (3.1).
3. **Formulate the ratio of the mass of the given element to the mass of the compound as a whole.** This is the ratio of the numerator from step 2 to the denominator from step 1.
4. **Multiply this ratio by 100% to obtain the mass percent of the element.**

$$\text{mass \% element} = \frac{\left(\begin{array}{c} \text{number of} \\ \text{atoms of element} \\ \text{per formula unit} \end{array} \right) \times \left(\begin{array}{c} \text{molar mass} \\ \text{of element} \end{array} \right)}{\text{molar mass of compound}} \times 100\% \quad (3.1)$$

The mass composition of a compound is the collection of mass percentages of the individual elements in the compound.

EXAMPLE 3-4

Calculating the Mass Percent Composition of a Compound

What is the mass percent composition of halothane, $\text{C}_2\text{HBrClF}_3$?

Analyze

Apply the four-step method described above. First, determine the molar mass of $\text{C}_2\text{HBrClF}_3$. Then formulate the mass ratios and convert them to mass percents. If we use molar masses that are rounded to two decimal places, the calculated mass percents will be accurate to two decimal places.

Solve

The molar mass of $\text{C}_2\text{HBrClF}_3$ is 197.38 g/mol. The mass percents are

$$\% \text{C} = \frac{\left(2 \text{ mol C} \times \frac{12.01 \text{ g C}}{1 \text{ mol C}} \right)}{197.38 \text{ g C}_2\text{HBrClF}_3} \times 100\% = 12.17\% \text{C}$$

$$\% \text{H} = \frac{\left(1 \text{ mol H} \times \frac{1.01 \text{ g H}}{1 \text{ mol H}} \right)}{197.38 \text{ g C}_2\text{HBrClF}_3} \times 100\% = 0.51\% \text{H}$$

$$\% \text{Br} = \frac{\left(1 \text{ mol Br} \times \frac{79.90 \text{ g Br}}{1 \text{ mol Br}} \right)}{197.38 \text{ g C}_2\text{HBrClF}_3} \times 100\% = 40.48\% \text{Br}$$

$$\% \text{Cl} = \frac{\left(1 \text{ mol Cl} \times \frac{35.45 \text{ g Cl}}{1 \text{ mol Cl}} \right)}{197.38 \text{ g C}_2\text{HBrClF}_3} \times 100\% = 17.96\% \text{Cl}$$

$$\% \text{F} = \frac{\left(3 \text{ mol F} \times \frac{19.00 \text{ g F}}{1 \text{ mol F}} \right)}{197.38 \text{ g C}_2\text{HBrClF}_3} \times 100\% = 28.88\% \text{F}$$

Thus, the percent composition of halothane is 12.17% C, 0.51% H, 40.48% Br, 17.96% Cl, and 28.88% F.

Assess

The mass ratios appearing above are based on a sample that contains exactly *one mole* of halothane. Another approach is to calculate the mass of each element present in a sample that contains exactly 100 g of halothane. For example, in a 100 g sample of halothane,

$$? \text{ g C} = 100 \text{ g C}_2\text{HBrClF}_3 \times \frac{1 \text{ mol C}_2\text{HBrClF}_3}{197.38 \text{ g C}_2\text{HBrClF}_3} \times \frac{2 \text{ mol C}}{1 \text{ mol C}_2\text{HBrClF}_3} \times \frac{12.01 \text{ g C}}{1 \text{ mol C}} = 12.17 \text{ g C}$$

and so halothane is 12.17% C. The mass of carbon in a 100 g sample is numerically equal to the mass percent of carbon. If you compare the calculation of g C with that for % C given earlier, you will see that both calculations involve exactly the same factors but in a slightly different order.

PRACTICE EXAMPLE A: Adenosine triphosphate (ATP) is the main energy-storage molecule in cells. Its chemical formula is $\text{C}_{10}\text{H}_{16}\text{N}_5\text{P}_3\text{O}_{13}$. What is its mass percent composition?

PRACTICE EXAMPLE B: Without doing detailed calculations, determine which two compounds from the following list have the same percent oxygen by mass: (a) CO; (b) CH_3COOH ; (c) C_2O_3 ; (d) N_2O ; (e) $\text{C}_6\text{H}_{12}\text{O}_6$; (f) $\text{HOCH}_2\text{CH}_2\text{OH}$.

The percentages of the elements in a compound should add up to 100.00%, and we can use this fact in two ways.

1. Check the accuracy of the computations by ensuring that the percentages total 100.00%. As applied to the results of Example 3-4:

$$12.17\% + 0.51\% + 40.48\% + 17.96\% + 28.88\% = 100.00\%$$

2. Determine the percentages of all the elements but one. Obtain that one by difference (subtraction). From Example 3-4:

$$\begin{aligned}\% \text{H} &= 100.00\% - \% \text{C} - \% \text{Br} - \% \text{Cl} - \% \text{F} \\ &= 100.00\% - 12.17\% - 40.48\% - 17.96\% - 28.88\% \\ &= 0.51\%\end{aligned}$$

Establishing Formulas from the Experimentally Determined Percent Composition of Compounds

At times, a chemist isolates a chemical compound—say, from a tropical plant—and has no idea what it is. A report from an analytical laboratory on the percent composition of the compound yields data needed to determine its formula.

Percent composition establishes the relative proportions of the elements in a compound on a *mass* basis. A chemical formula requires these proportions to be on a *mole* basis, that is, in terms of *numbers* of atoms. Consider the following five-step approach to determining a formula from the experimentally determined percent composition of the compound 2-deoxyribose, a sugar that is a basic constituent of DNA (deoxyribonucleic acid). The mass percent composition of 2-deoxyribose is 44.77% C, 7.52% H, and 47.71% O.

1. Although we could choose any sample size, if we take one of *exactly* 100 g, the masses of the elements are numerically equal to their percentages, that is, 44.77 g C, 7.52 g H, and 47.71 g O.
2. Convert the masses of the elements in the 100.00 g sample to amounts in moles.

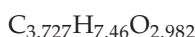
$$? \text{ mol C} = 44.77 \text{ g C} \times \frac{1 \text{ mol C}}{12.011 \text{ g C}} = 3.727 \text{ mol C}$$

$$? \text{ mol H} = 7.52 \text{ g H} \times \frac{1 \text{ mol H}}{1.008 \text{ g H}} = 7.46 \text{ mol H}$$

$$? \text{ mol O} = 47.71 \text{ g O} \times \frac{1 \text{ mol O}}{15.999 \text{ g O}} = 2.982 \text{ mol O}$$

◀ Use the molar mass with one more significant figure than in the mass of the element.

3. Write a tentative formula based on the numbers of moles just determined.

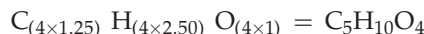


4. Attempt to convert the subscripts in the tentative formula to small whole numbers. This requires dividing each of the subscripts by the smallest one (2.982).

$$\text{C}_{\frac{3.727}{2.982}}\text{H}_{\frac{7.46}{2.982}}\text{O}_{\frac{2.982}{2.982}} = \text{C}_{1.25}\text{H}_{2.50}\text{O}$$

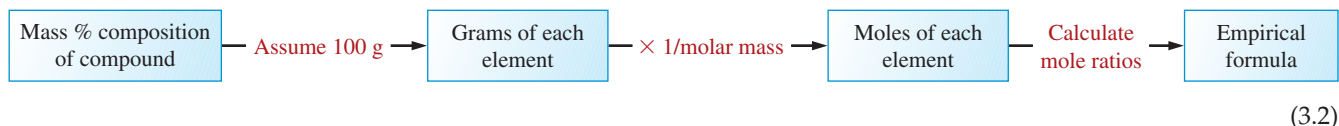
If all subscripts at this point differ only slightly from whole numbers—which is not the case here—round them off to whole numbers, concluding the calculation at this point.

5. If one or more subscripts is still not a whole number—which is the case here—multiply all subscripts by a small whole number that will make them all integral. Thus, multiply by 4 here.



The formula that we get by the method just outlined, $\text{C}_5\text{H}_{10}\text{O}_4$, is the simplest possible formula—the *empirical formula*. The actual *molecular formula* may be equal to, or some multiple of, the empirical formula, such as $\text{C}_{10}\text{H}_{20}\text{O}_8$, $\text{C}_{15}\text{H}_{30}\text{O}_{12}$, $\text{C}_{20}\text{H}_{40}\text{O}_{16}$, and so on. To find the multiplying factor, we must compare the formula mass based on the empirical formula with the true molecular mass of the compound. We can establish the molecular mass from a separate experiment (by methods introduced in Chapters 6 and 14). The experimentally determined molecular mass of 2-deoxyribose is 134 u. The formula mass based on the empirical formula, $\text{C}_5\text{H}_{10}\text{O}_4$, is 134.1 u. The measured molecular mass is the same as the empirical formula mass. The molecular formula is also $\text{C}_5\text{H}_{10}\text{O}_4$.

We outline the five-step approach described above in the flow diagram below and then apply the approach to Example 3-5, where we will find that the empirical formula and the molecular formula are not the same.



EXAMPLE 3-5

Determining the Empirical and Molecular Formulas of a Compound from Its Mass Percent Composition

Dibutyl succinate is an insect repellent used against household ants and roaches. Its composition is 62.58% C, 9.63% H, and 27.79% O. Its experimentally determined molecular mass is 230 u. What are the empirical and molecular formulas of dibutyl succinate?

Analyze

Use the five-step approach described above.

Solve

- | | |
|---|--|
| 1. Determine the mass of each element in a 100.00 g sample. | 62.58 g C, 9.63 g H, 27.79 g O |
| 2. Convert each of these masses to an amount in moles. | $? \text{ mol C} = 62.58 \text{ g C} \times \frac{1 \text{ mol C}}{12.011 \text{ g C}} = 5.210 \text{ mol C}$ $? \text{ mol H} = 9.63 \text{ g H} \times \frac{1 \text{ mol H}}{1.008 \text{ g H}} = 9.55 \text{ mol H}$ $? \text{ mol O} = 27.79 \text{ g O} \times \frac{1 \text{ mol O}}{15.999 \text{ g O}} = 1.737 \text{ mol O}$ |
| 3. Write a tentative formula based on these numbers of moles. | $\text{C}_{5.210}\text{H}_{9.55}\text{O}_{1.737}$ |

- Divide each of the subscripts of the tentative formula by the smallest subscript (1.737),
and round off any subscripts that differ only slightly from whole numbers; that is, round 2.99 to 3.
- Multiply all subscripts by a small whole number to make them integral (here by the factor 2), and write the empirical formula.

$$\frac{C_{5.210}}{1.737} \frac{H_{9.55}}{1.737} \frac{O_{1.737}}{1.737} = C_{2.99} H_{5.49} O$$



$$C_{2 \times 3} H_{2 \times 5.49} O_{2 \times 1} = C_6 H_{10.98} O_2$$

$$2 \times 5.49 = 10.98 \approx 11$$

Empirical formula: $C_6H_{11}O_2$

$$[(6 \times 12.0) + (11 \times 1.0) + (2 \times 16.0)]u = 115 u$$

To establish the molecular formula, first determine the empirical formula mass.

Since the experimentally determined formula mass (230 u) is twice the empirical formula mass, the molecular formula is twice the empirical formula.

Molecular formula: $C_{12}H_{22}O_4$

Assess

Check the result by working the problem in reverse and using numbers that are rounded off slightly. For $C_{12}H_{22}O_4$, % C $\approx (12 \times 12 u / 230 u) \times 100\% = 63\%$; % H $\approx (22 \times 1 u / 230 u) \times 100\% = 9.6\%$; and % O $\approx (4 \times 16 u / 230 u) \times 100\% = 28\%$. The calculated mass percents agree well with those given in the problem, so we can be confident that our answer is correct.

PRACTICE EXAMPLE A: Sorbitol, used as a sweetener in some sugar-free foods, has a molecular mass of 182 u and a mass percent composition of 39.56% C, 7.74% H, and 52.70% O. What are the empirical and molecular formulas of sorbitol?

PRACTICE EXAMPLE B: The chlorine-containing narcotic drug pentaerythritol chloral has 21.51% C, 2.22% H, and 17.64% O, by mass, as its other elements. Its molecular mass is 726 u. What are the empirical and molecular formulas of pentaerythritol chloral?



3-4 Concept Assessment

Explain why, if the percent composition *and* the molar mass of a compound are both known, the molecular formula can be determined much more readily by using the molar mass rather than a 100 g sample in the first step of the five-step procedure in Example 3-5. In this instance, how would you then obtain the empirical formula?



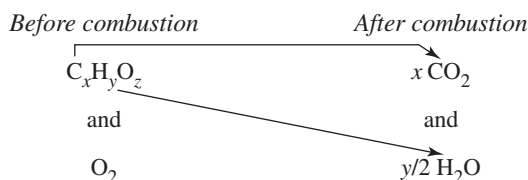
3-2 ARE YOU WONDERING?

How much rounding off should you do to get integral subscripts in an empirical formula and what factors should you use to convert fractional to whole numbers?

How much rounding off is justified depends on how precisely the elemental analysis is done. As a result, there is no ironclad rule on the matter. For the examples in this text, if you carry all the significant figures allowable in a calculation, you can generally round off a subscript that is within a few hundredths of a whole number (for example, 3.98 rounds off to 4). If the deviation is more than this, you will need to adjust subscripts to integral values by multiplying by the appropriate constant. If the appropriate constant is larger than a simple integer, such as 2, 3, 4, or 5, you may sometimes find it easier to make the adjustment by multiplying twice, for example, by 2 and then by 4 if the necessary constant is 8.

Combustion Analysis

Figure 3-6 illustrates an experimental method for establishing an empirical formula for compounds that are easily burned, such as compounds containing carbon and hydrogen with oxygen, nitrogen, and a few other elements. In *combustion analysis*, a weighed sample of a compound is burned in a stream of oxygen gas. The water vapor and carbon dioxide gas produced in the combustion are absorbed by appropriate substances. The increases in mass of these absorbers correspond to the masses of water and carbon dioxide. We can think of the matter as shown below. (The subscripts x , y , and z are integers whose values we do not know initially.)



After combustion, all the carbon atoms in the sample are found in the CO_2 . All the hydrogen atoms are in the H_2O . Moreover, the only source of the carbon and hydrogen atoms was the sample being analyzed. Oxygen atoms in the CO_2 and H_2O could have come partly from the sample and partly from the oxygen gas consumed in the combustion. Thus, the quantity of oxygen in the sample has to be determined indirectly. These ideas are applied in Example 3-6.

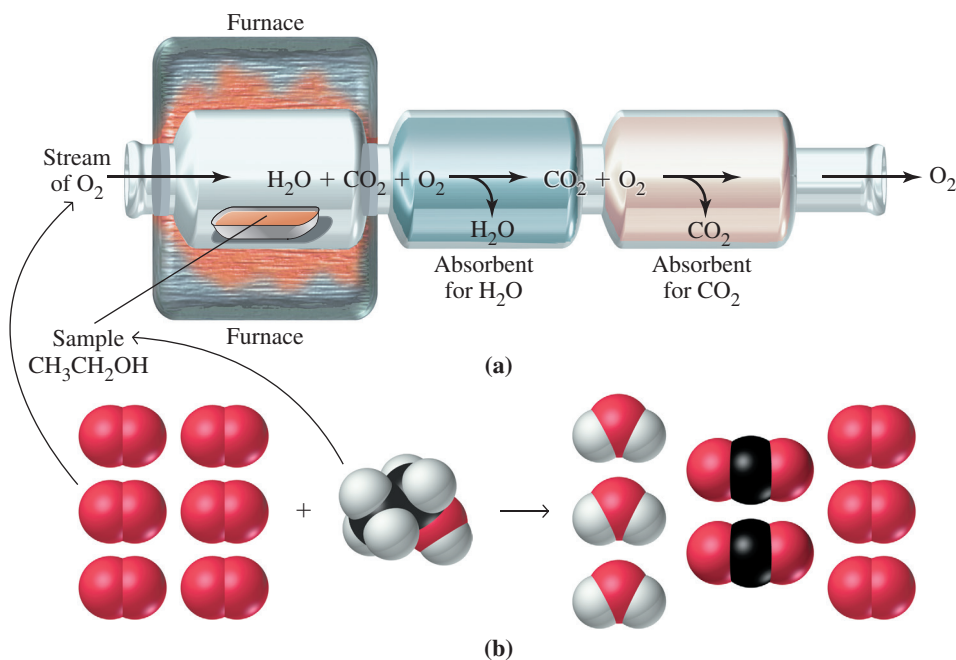


Figure 3-6 Apparatus for combustion analysis

(a) Oxygen gas passes through the combustion tube containing the sample being analyzed. This portion of the apparatus is enclosed in a high-temperature furnace. Products of the combustion are absorbed as they leave the furnace—water vapor by magnesium perchlorate, and carbon dioxide gas by sodium hydroxide (producing sodium carbonate). The differences in mass of the absorbers, after and before the combustion, yield the masses of H_2O and CO_2 produced in the combustion reaction. **(b)** A molecular picture of the combustion of ethanol. Each molecule of ethanol produces two CO_2 molecules and three H_2O molecules. Combustion takes place in an excess of oxygen, so that oxygen molecules are present at the end of the reaction. Note the conservation of mass.

EXAMPLE 3-6

Determining an Empirical Formula from Combustion Analysis Data

Vitamin C is essential for the prevention of scurvy. Combustion of a 0.2000 g sample of this carbon–hydrogen–oxygen compound yields 0.2998 g CO₂ and 0.0819 g H₂O. What are the percent composition and the empirical formula of vitamin C?

Analyze

After combustion, all the carbon atoms from the vitamin C sample are in CO₂ and all the hydrogen atoms are in H₂O. However, the oxygen atoms in CO₂ and H₂O come partly from the sample and partly from the oxygen gas consumed in the combustion. So, in the determination of the percent composition, we focus first on carbon and hydrogen and then on oxygen. To determine the empirical formula, we must calculate the amounts of C, H, and O in moles, and then calculate the mole ratios.

Solve

Percent Composition

First, determine the mass of carbon in 0.2998 g CO₂, by converting to mol C,

$$? \text{ mol C} = 0.2998 \text{ g CO}_2 \times \frac{1 \text{ mol CO}_2}{44.009 \text{ g CO}_2} \times \frac{1 \text{ mol C}}{1 \text{ mol CO}_2} = 0.006812 \text{ mol C}$$

and then to g C.

$$? \text{ g C} = 0.006812 \text{ mol C} \times \frac{12.011 \text{ g C}}{1 \text{ mol C}} = 0.08182 \text{ g C}$$

Proceed in a similar fashion for 0.0819 g H₂O to obtain

$$? \text{ mol H} = 0.0819 \text{ g H}_2\text{O} \times \frac{1 \text{ mol H}_2\text{O}}{18.02 \text{ g H}_2\text{O}} \times \frac{2 \text{ mol H}}{1 \text{ mol H}_2\text{O}} = 0.00909 \text{ mol H}$$

and

$$? \text{ g H} = 0.00909 \text{ mol H} \times \frac{1.008 \text{ g H}}{1 \text{ mol H}} = 0.00916 \text{ g H}$$

Obtain the mass of O in the 0.2000 g sample as the difference

$$? \text{ g O} = 0.2000 \text{ g sample} - 0.08182 \text{ g C} - 0.00916 \text{ g H} = 0.1090 \text{ g O}$$

Finally, multiply the mass fractions of the three elements by 100% to obtain mass percentages.

$$\% \text{ C} = \frac{0.08182 \text{ g C}}{0.2000 \text{ g sample}} \times 100\% = 40.91\% \text{ C}$$

$$\% \text{ H} = \frac{0.00916 \text{ g H}}{0.2000 \text{ g sample}} \times 100\% = 4.58\% \text{ H}$$

$$\% \text{ O} = \frac{0.1090 \text{ g O}}{0.2000 \text{ g sample}} \times 100\% = 54.50\% \text{ O}$$

Empirical Formula

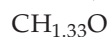
At this point we can choose either of two alternatives. The first is to obtain the empirical formula from the mass percent composition, in the same manner illustrated in Example 3-5. The second is to note that we have already determined the number of moles of C and H in the 0.2000 g sample. The number of moles of O is

$$? \text{ mol O} = 0.1090 \text{ g O} \times \frac{1 \text{ mol O}}{15.999 \text{ g O}} = 0.006813 \text{ mol O}$$

From the numbers of moles of C, H, and O in the 0.2000 g sample, we obtain the tentative empirical formula



Next, divide each subscript by the smallest (0.006812) to obtain



(continued)

Finally, multiply all the subscripts by 3 to obtain

Empirical formula of vitamin C: $\text{C}_3\text{H}_4\text{O}_3$

Assess

The determination of the empirical formula does not require determining the mass percent composition as a preliminary calculation. The empirical formula can be based on a sample of any size, as long as the numbers of moles of the different atoms in that sample can be determined.

PRACTICE EXAMPLE A: Isobutyl propionate is the substance that flavors rum extract. Combustion of a 1.152 g sample of this carbon–hydrogen–oxygen compound yields 2.726 g CO_2 and 1.116 g H_2O . What is the empirical formula of isobutyl propionate?

PRACTICE EXAMPLE B: Combustion of a 1.505 g sample of thiophene, a carbon–hydrogen–sulfur compound, yields 3.149 g CO_2 , 0.645 g H_2O , and 1.146 g SO_2 as the only products of the combustion. What is the empirical formula of thiophene?



3-5 Concept Assessment

When combustion is carried out in an excess of oxygen, the quantity producing the greatest mass of both CO_2 and H_2O is (a) 0.50 mol C_{10}H_8 ; (b) 1.25 mol CH_4 ; (c) 0.500 mol $\text{C}_2\text{H}_5\text{OH}$; (d) 1.00 mol $\text{C}_6\text{H}_5\text{OH}$.

We have just seen how combustion reactions can be used to analyze chemical substances, but not all samples can be easily burned. Fortunately, several other types of reactions can be used for chemical analyses. Also, modern methods in chemistry rely much more on physical measurements with instruments than on chemical reactions. We will cite some of these methods later in the text.

3-4

Oxidation States: A Useful Tool in Describing Chemical Compounds

► The oxidation state (O.S.) can be described as “a sometimes fictional charge.” With monatomic ions, O.S. and charge are the same thing. For polyatomic ions and molecules, O.S. is fictional but equal to what the charge would be if the compounds were entirely ionic.

Most basic concepts in chemistry deal with measurable properties or phenomena. In a few instances, though, a concept has been devised more for convenience than because of any fundamental significance. This is the case with the **oxidation state** (oxidation number),* which is related to the number of electrons that an atom loses, gains, or otherwise appears to use in joining with other atoms in compounds.

Consider NaCl . In this compound, an Na atom, a metal, loses one electron to a Cl atom, a nonmetal. The compound consists of the ions Na^+ and Cl^- (see Figure 3-4). Na^+ is in a +1 oxidation state and Cl^- is in a −1 state.

In MgCl_2 , an Mg atom loses two electrons to become Mg^{2+} , and each Cl atom gains one electron to become Cl^- . As in NaCl , the oxidation state of Cl is −1, but that of Mg is +2. If we take the *total* of the oxidation states of all the atoms (ions) in a formula unit of MgCl_2 , we get $+2 - 1 - 1 = 0$.

In the molecule Cl_2 , the two Cl atoms are identical and should have the same oxidation state. But if their total is to be zero, each oxidation state must itself be 0. Thus, the oxidation state of an atom can vary, depending on the compounds in which it occurs. In the molecule H_2O , we arbitrarily assign H the oxidation state of +1. Then, because the total of the oxidation states of the atoms must be zero, the oxidation state of oxygen must be −2.

*Because oxidation state refers to a number, the term oxidation number is often used synonymously. We will use the two terms interchangeably.

From these examples, you can see that we need some conventions or rules for assigning oxidation states. The seven rules in Table 3.2 are sufficient to deal with most cases in this text, with this understanding: *The rules must be applied in the numerical order listed, and whenever two rules appear to contradict each other (which they sometimes will), follow the lower numbered rule.* Some examples are given for each rule, and the rules are applied in Example 3-7.

Table 3.2 Rules for Assigning Oxidation States

1. The oxidation state (O.S.) of an individual atom in a free element (uncombined with other elements) is 0.
[Examples: The O.S. of an isolated Cl atom is 0; the two Cl atoms in the molecule Cl_2 both have an O.S. of 0.]
2. The sum of the O.S. of all the atoms in
 - (a) neutral species, such as isolated atoms, molecules, and formula units, is 0;
[Examples: The sum of the O.S. of all the atoms in CH_3OH and of all the ions in MgCl_2 is 0.]
 - (b) an ion is equal to the charge on the ion.
[Examples: The O.S. of Fe in Fe^{3+} is +3. The sum of the O.S. of all atoms in MnO_4^- is -1.]
3. In their compounds, the group 1 metals have an O.S. of +1 and the group 2 metals have an O.S. of +2.
[Examples: The O.S. of K is +1 in KCl and K_2CO_3 ; the O.S. of Mg is +2 in MgBr_2 and $\text{Mg}(\text{NO}_3)_2$.]
4. In its compounds, the O.S. of fluorine is -1.
[Examples: The O.S. of F is -1 in HF , ClF_3 , and SF_6 .]
5. In its compounds, hydrogen usually has an O.S. of +1.
[Examples: The O.S. of H is +1 in HI , H_2S , NH_3 , and CH_4 .]
6. In its compounds, oxygen usually has an O.S. of -2.
[Examples: The O.S. of O is -2 in H_2O , CO_2 and KMnO_4 .]
7. In binary (two-element) compounds with metals, group 17 elements have an O.S. of -1; group 16 elements, -2; and group 15 elements, -3.
[Examples: The O.S. of Br is -1 in MgBr_2 ; the O.S. of S is -2 in Li_2S ; and the O.S. of N is -3 in Li_3N .]

◀ The principal exceptions to rule 5 occur when H is bonded to metals, as in LiH , NaH , and CaH_2 ; exceptions to rule 6 occur in compounds with O—F bonds, such as OF_2 , and in compounds where O atoms are bonded to one another, as in H_2O_2 , and KO_2 .

EXAMPLE 3-7 Assigning Oxidation States

What is the oxidation state of the underlined element in (a) P_4 ; (b) Al_2O_3 ; (c) MnO_4^- ; (d) NaH ; (e) H_2O_2 ; (f) Fe_3O_4 ?

Analyze

Apply the rules in Table 3.2.

Solve

- (a) P_4 : This formula represents a molecule of elemental phosphorus. For an atom of a free element, the O.S. = 0 (rule 1). The O.S. of P in P_4 is 0.
- (b) Al_2O_3 : The total of the oxidation states of all the atoms in this formula unit is 0 (rule 2). The O.S. of oxygen is -2 (rule 6). The total for three O atoms is -6. The total for two Al atoms is +6. The O.S. of Al is +3.
- (c) MnO_4^- : This is the formula for permanganate ion. The sum of the oxidation states of all the atoms in the ion is -1 (rule 2). The total for the four O atoms is -8. The O.S. of Mn is +7.
- (d) NaH : This is a formula unit of the ionic compound sodium hydride. Rule 3 states that the O.S. of Na is +1. Rule 5 indicates that H should also have an O.S. of +1. If both atoms had an O.S. of +1, the total for the formula unit would be +2. This violates rule 2. Rules 2 and 3 take precedence over rule 5. Na has an O.S. of +1; the total for the formula unit is 0; and the O.S. of H must be -1.

(continued)

- (e) H_2O_2 : This is hydrogen peroxide. Rule 5, stating that H has an O.S. of +1, takes precedence over rule 6 (which says that oxygen has an O.S. of -2). The sum of the oxidation states of the two H atoms is +2 and that of the two O atoms must be -2 . The O.S. of O must be -1 .
- (f) Fe_3O_4 : The total of the oxidation states of four O atoms is -8 . For three Fe atoms, the sum of the oxidation states must be +8. The O.S. per Fe atom is $\frac{8}{3}$ or $+2\frac{2}{3}$.

Assess

With practice, you should be able to do the arithmetic associated with assigning oxidation states in your head, that is, without writing down any arithmetic expressions. Also, when you have determined an oxidation state by using arithmetic (as required by rule 2), check your result by making sure the sum of the oxidation states is equal to the charge on the atom, molecule, or ion. For example, in part (c), we determined that the oxidation state of Mn in MnO_4^- is +7. We know this result is correct because the sum of the oxidation states is $7 + 4(-2) = -1$, which is equal to the charge on the MnO_4^- ion.

PRACTICE EXAMPLE A: What is the oxidation state of the underlined element in each of the following: $\underline{\text{S}}_8$; $\underline{\text{Cr}}_2\text{O}_7^{2-}$; $\underline{\text{Cl}}_2\text{O}$; $\underline{\text{K}}\underline{\text{O}}_2$?

PRACTICE EXAMPLE B: What is the oxidation state of the underlined element in each of the following: $\underline{\text{S}}_2\text{O}_3^{2-}$; $\underline{\text{Hg}}_2\text{Cl}_2$; $\underline{\text{KMn}}\underline{\text{O}}_4$; $\underline{\text{H}}_2\underline{\text{C}}\underline{\text{O}}$?

In part (f) of Example 3-7, we got the somewhat surprising answer of $+2\frac{2}{3}$ for the oxidation state of the iron atoms in Fe_3O_4 . Before that, we saw only integral values for oxidation states. How does this fractional value come about? Generally, it comes from the assumption that all the atoms of an element have the same oxidation state in a given compound. Usually they do, but not always. Fe_3O_4 , for example, is probably better represented as $\text{FeO} \cdot \text{Fe}_2\text{O}_3$, that is, through a combination of two simpler formula units. In FeO , the O.S. of the Fe atom is +2. In Fe_2O_3 , the O.S. of each of *two* Fe atoms is +3. When we *average* the oxidation states over all three Fe atoms, we get a nonintegral value: $(2 + 3 + 3)/3 = \frac{8}{3} = 2\frac{2}{3}$.

Also, we may at times need to “fragment” a formula into its constituent parts before assigning oxidation states. The ionic compound NH_4NO_3 , for instance, consists of the ions NH_4^+ and NO_3^- . The oxidation state of N in NH_4^+ is -3 , and in NO_3^- , +5, and we do *not* want to average them. It is far more useful to know the oxidation states of the individual N atoms than it is to deal with an average oxidation state of +1 for the two N atoms.

Our first use of oxidation states comes in the naming of chemical compounds in the next section.



3-6 Concept Assessment

A nitrogen–hydrogen compound with molar mass 32 g/mol has N in a higher oxidation state than in NH_3 . What is a plausible formula for that compound?



Figure 3-7 Two oxides of lead

These two compounds contain the same elements—lead and oxygen—but in different proportions. Their names and formulas must convey this fact: lead(IV) oxide = PbO_2 (red-brown); lead(II) oxide = PbO (yellow).

3-5

Naming Compounds: Organic and Inorganic Compounds

Throughout this chapter, we have referred to compounds mostly by their formulas, but we do need to give them names. When we know the name of a compound, we can look up its properties in a handbook, locate a chemical on a storeroom shelf, or discuss an experiment with a colleague. Later in the text, we will see cases in which different compounds have the same formula. In these instances, we will find it essential to distinguish among compounds by name. We cannot give two substances the same name, yet we do want some similarities in the names of similar substances (Fig. 3-7). If all compounds were referred to by a common or trivial name, such as water (H_2O), ammonia (NH_3), or glucose

($\text{C}_6\text{H}_{12}\text{O}_6$), we would have to learn millions of unrelated names—an impossibility. What we need is a systematic method of assigning names—a system of *nomenclature*. Several systems are used, and we will introduce each at an appropriate point in the text. Compounds formed by carbon and hydrogen or carbon and hydrogen together with oxygen, nitrogen, and a few other elements are **organic compounds**. They are generally considered in a special branch of chemistry—*organic chemistry*—that has its own set of nomenclature rules. Compounds that do not fit this description are **inorganic compounds**. The branch of chemistry that concerns itself with the study of these compounds is called *inorganic chemistry*. In the next section, we will consider the naming of inorganic compounds, and in Section 3-7, we will introduce the naming of organic compounds.

◀ *Nomenclature of Organic Chemistry* (Blue Book) and *Nomenclature of Inorganic Chemistry* (Red Book) can be found at www.iupac.org.

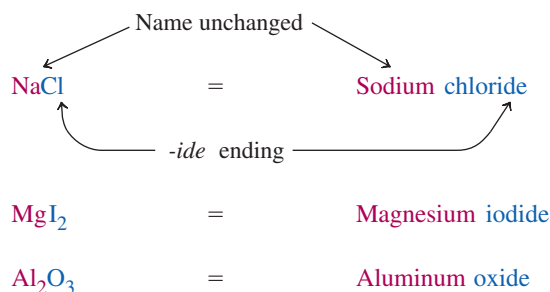
3-6 Names and Formulas of Inorganic Compounds

Binary Compounds of Metals and Nonmetals

Binary compounds are those formed between *two elements*. If one of the elements is a metal and the other a nonmetal, the binary compound is usually made up of ions; that is, it is a binary *ionic* compound. To name a binary compound of a metal and a nonmetal,

- write the *unmodified* name of the metal
- then write the name of the nonmetal, modified to end in *-ide*

The approach is illustrated below for NaCl , MgI_2 , and Al_2O_3 .



Ionic compounds, though made up of positive and negative ions, must be *electrically neutral*. The net, or total, charge of the ions in a formula unit must be zero. This means one Na^+ to one Cl^- in NaCl ; one Mg^{2+} to two I^- in MgI_2 ; two Al^{3+} to three O^{2-} in Al_2O_3 ; and so on. Table 3.3 lists the names and symbols of simple ions formed by metals and nonmetals. You will find this list useful when writing names and formulas of binary compounds of metals and nonmetals. Because some metals may form several ions, it is important to distinguish between them in naming their compounds. The metal iron, for example, forms *two* common ions, Fe^{2+} and Fe^{3+} . The first is called the *iron(II)* ion, and the second is the *iron(III)* ion. The Roman numeral immediately following the name of the metal indicates its oxidation state or simply the charge on the ion. Thus, FeCl_2 is *iron(II)* chloride, while FeCl_3 is *iron(III)* chloride.

An earlier system of nomenclature that is still used to some extent applies two different word endings to distinguish between two binary compounds containing the same two elements but in different proportions, such as Cu_2O and CuO . In Cu_2O , the oxidation state of copper is +1, and in CuO it is +2. Cu_2O is assigned the name *cuprous* oxide, and CuO is *cupric* oxide. Similarly, FeCl_2 is *ferrous* chloride, and FeCl_3 is *ferric* chloride. The idea is to use the *-ous* ending for the lower oxidation state of the metal and *-ic* for the higher oxidation state. The *ous/ic* system has several inadequacies though, and we will not

► Table 3.3 simplifies considerably once you understand the pattern. All group 1 ions are +1, group 2 ions are +2, and Al^{3+} , Zn^{2+} , and Ag^{+} have only the one common charge. Most other metal ions have variable charges. Monatomic halogen ions are -1 and oxygen is -2 . Almost everything else can be worked out from these basics and the rule that the sum of all O.S. equals the total charge.

Table 3.3 Some Simple Ions

Name	Symbol	Name	Symbol
Positive ions (cations)			
Lithium ion	Li^{+}	Chromium(II) ion	Cr^{2+}
Sodium ion	Na^{+}	Chromium(III) ion	Cr^{3+}
Potassium ion	K^{+}	Iron(II) ion	Fe^{2+}
Rubidium ion	Rb^{+}	Iron(III) ion	Fe^{3+}
Cesium ion	Cs^{+}	Cobalt(II) ion	Co^{2+}
Magnesium ion	Mg^{2+}	Cobalt(III) ion	Co^{3+}
Calcium ion	Ca^{2+}	Copper(I) ion	Cu^{+}
Strontium ion	Sr^{2+}	Copper(II) ion	Cu^{2+}
Barium ion	Ba^{2+}	Mercury(I) ion	Hg_2^{2+}
Aluminum ion	Al^{3+}	Mercury(II) ion	Hg^{2+}
Zinc ion	Zn^{2+}	Tin(II) ion	Sn^{2+}
Silver ion	Ag^{+}	Lead(II) ion	Pb^{2+}
Negative ions (anions)			
Hydride ion	H^{-}	Iodide ion	I^{-}
Fluoride ion	F^{-}	Oxide ion	O^{2-}
Chloride ion	Cl^{-}	Sulfide ion	S^{2-}
Bromide ion	Br^{-}	Nitride ion	N^{3-}

use it in this text. For example, the *-ous* and *-ic* endings do not help in naming the four oxides of vanadium: VO , V_2O_3 , VO_2 , and V_2O_5 .

Binary Compounds of Two Nonmetals

If the two elements in a binary compound are both nonmetals instead of a metal and a nonmetal, the compound is a molecular compound. The method of naming these compounds is similar to that just discussed. For example,

HCl = hydrogen chloride

In both the formula and the name, we write the element with the positive oxidation state first: HCl and *not* ClH .

EXAMPLE 3-8

Writing Formulas When Names of Compounds Are Given

Write formulas for the compounds barium oxide, calcium fluoride, and iron(III) sulfide.

Analyze

In each case, identify the cations and their charges, based on periodic table group numbers or on oxidation states appearing as Roman numerals in names: Ba^{2+} , Ca^{2+} , and Fe^{3+} . Then identify the anions and their charges: O^{2-} , F^{-} , and S^{2-} . Combine the cations and anions in the relative numbers required to produce electrically *neutral* formula units.

Solve

barium oxide: one Ba^{2+} and one O^{2-} = BaO
 calcium fluoride: one Ca^{2+} and two F^{-} = CaF_2
 iron(III) sulfide: two Fe^{3+} and three S^{2-} = Fe_2S_3

Assess

In the first case, an electrically neutral formula unit results from the combination of the charges $+2$ and -2 ; in the second case, $+2$ and $2 \times (-1)$; and in the third case, $2 \times (+3)$ and $3 \times (-2)$.

PRACTICE EXAMPLE A: Write formulas for lithium oxide, tin(II) fluoride, and lithium nitride.

PRACTICE EXAMPLE B: Write formulas for the compounds aluminum sulfide, magnesium nitride, and vanadium(III) oxide.

EXAMPLE 3-9**Naming Compounds When Their Formulas Are Given**

Write acceptable names for the compounds Na_2S , AlF_3 , Cu_2O .

Analyze

This task is generally easier than that of Example 3-8 because all you need to do is name the ions present. However, you must recognize that copper forms two different ions and that the cation in Cu_2O is Cu^+ , copper(I).

Solve

Na_2S : sodium sulfide

AlF_3 : aluminum fluoride

Cu_2O : copper(I) oxide

Assess

Knowing when to use Roman numerals and when not to use them is a tricky aspect of naming ionic compounds. Because the metals of groups 1 and 2 have only one ionic form (one oxidation state), Roman numerals are not used in naming their compounds.

PRACTICE EXAMPLE A: Write acceptable names for CsI , CaF_2 , FeO , CrCl_3 .

PRACTICE EXAMPLE B: Write acceptable names for CaH_2 , CuCl , Ag_2S , Hg_2Cl_2 .

Some pairs of nonmetals form more than one binary molecular compound, and we need to distinguish among them. Generally, we indicate relative numbers of atoms through *prefixes*: *mono* = 1, *di* = 2, *tri* = 3, *tetra* = 4, *penta* = 5, *hexa* = 6, *hepta* = 7, *octa* = 8, *nona* = 9, *deca* = 10, and so on. Thus, for the two principal oxides of sulfur we write

SO_2 = sulfur *dioxide*

SO_3 = sulfur *trioxide*

and for the following boron–bromine compound

B_2Br_4 = *diboron tetrabromide*

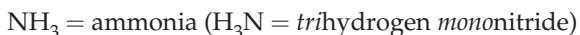
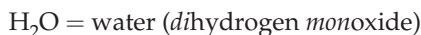
Additional examples are given in Table 3.4. Note that in these examples the prefix *mono-* is treated in a special way. We do not use it for the first named

Table 3.4 Naming Binary Molecular Compounds

Formula	Name ^a
BCl_3	Boron trichloride
CCl_4	Carbon tetrachloride
CO	Carbon monoxide
CO_2	Carbon dioxide
NO	Nitrogen monoxide
NO_2	Nitrogen dioxide
N_2O	Dinitrogen monoxide
N_2O_3	Dinitrogen trioxide
N_2O_4	Dinitrogen tetroxide
N_2O_5	Dinitrogen pentoxide
PCl_3	Phosphorus trichloride
PCl_5	Phosphorus pentachloride
SF_6	Sulfur hexafluoride

^aWhen the prefix ends in *a* or *o* and the element name begins with *a* or *o*, the final vowel of the prefix is dropped for ease of pronunciation. For example, carbon *monoxide*, not carbon *monoxide*, and dinitrogen *tetroxide*, not dinitrogen *tetraoxide*. However, PI_3 is phosphorus *triiodide*, not phosphorus *trioxide*.

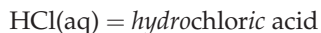
element. Thus, NO is called nitrogen monoxide, *not mononitrogen monoxide*. Finally, several substances have common or trivial names that are so well established that their systematic names are almost never used. For example,



Binary Acids

Even though we use names like hydrogen chloride for pure binary molecular compounds, we sometimes want to emphasize that their aqueous solutions are acids. Acids will be discussed in detail later in the text. For now, so that we can recognize these substances and name them when we see their formulas, let us say that an *acid* is a substance that *ionizes* or breaks down in water to produce hydrogen ions (H^+)* and anions. A *binary acid* is a two-element compound of hydrogen and a nonmetal. For example, HCl ionizes into hydrogen ions (H^+) and chloride ions (Cl^-) in water; it is a binary acid. NH_3 in water is *not* an acid. It shows practically no tendency to produce H^+ under any conditions. NH_3 belongs to a complementary category of substances called *bases*. As we will see in Chapter 5, bases yield hydroxide ions (OH^-) in aqueous solutions.

In naming binary acids we use the prefix *hydro-* followed by the name of the other nonmetal modified with an *-ic* ending. The most important binary acids are listed below.

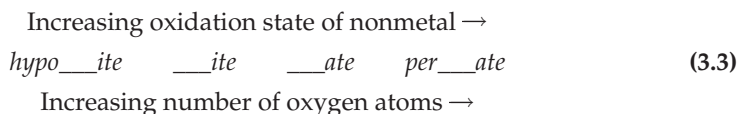


► The symbol (aq) signifies a substance in aqueous (water) solution.

Polyatomic Ions

With the exception of Hg_2^{2+} , the ions listed in Table 3.3 are *monatomic*—each consists of a single atom. In **polyatomic ions**, two or more atoms are joined together by covalent bonds. These ions are common, especially among the nonmetals. A number of polyatomic ions and compounds containing them are listed in Table 3.5. From this table, you can see the following:

1. Polyatomic anions are more common than polyatomic cations. The most familiar polyatomic cation is the ammonium ion, NH_4^+ .
2. Very few polyatomic anions carry the *-ide* ending in their names. Of those listed, only OH^- (hydroxide ion) and CN^- (cyanide ion) do. The common endings are *-ite* and *-ate*, and some names carry prefixes, *hypo-* or *per-*.
3. An element common to many polyatomic anions is *oxygen*, usually in combination with another nonmetal. Such anions are called **oxoanions**.
4. Certain nonmetals (such as Cl, N, P, and S) form a series of oxoanions containing different numbers of oxygen atoms. Their names are related to the oxidation state of the nonmetal atom to which the O atoms are bonded, ranging from *hypo-* (lowest) to *per-* (highest) according to the following scheme.



*The species produced in aqueous solution is actually more complex than the simple ion H^+ . The H^+ combines with an H_2O molecule to produce an ion known as the hydronium ion, H_3O^+ . Chemists often use H^+ in place of H_3O^+ , and that is what we will do until we discuss this matter more fully in Chapter 5.

Table 3.5 Some Common Polyatomic Ions

Name	Formula	Typical Compound
Cation		
Ammonium ion	NH_4^+	NH_4Cl
Anions		
Acetate ion	CH_3COO^-	NaCH_3COO
Carbonate ion	CO_3^{2-}	Na_2CO_3
Hydrogen carbonate ion ^a (or bicarbonate ion)	HCO_3^-	NaHCO_3
Hypochlorite ion	ClO^-	NaClO
Chlorite ion	ClO_2^-	NaClO_2
Chlorate ion	ClO_3^-	NaClO_3
Perchlorate ion	ClO_4^-	NaClO_4
Chromate ion	CrO_4^{2-}	Na_2CrO_4
Dichromate ion	$\text{Cr}_2\text{O}_7^{2-}$	$\text{Na}_2\text{Cr}_2\text{O}_7$
Cyanide ion	CN^-	NaCN
Hydroxide ion	OH^-	NaOH
Nitrite ion	NO_2^-	NaNO_2
Nitrate ion	NO_3^-	NaNO_3
Oxalate ion	$\text{C}_2\text{O}_4^{2-}$	$\text{Na}_2\text{C}_2\text{O}_4$
Permanganate ion	MnO_4^-	NaMnO_4
Phosphate ion	PO_4^{3-}	Na_3PO_4
Hydrogen phosphate ion ^a	HPO_4^{2-}	Na_2HPO_4
Dihydrogen phosphate ion ^a	H_2PO_4^-	NaH_2PO_4
Sulfite ion	SO_3^{2-}	Na_2SO_3
Hydrogen sulfite ion ^a (or bisulfite ion)	HSO_3^-	NaHSO_3
Sulfate ion	SO_4^{2-}	Na_2SO_4
Hydrogen sulfate ion ^a (or bisulfate ion)	HSO_4^-	NaHSO_4
Thiosulfate ion	$\text{S}_2\text{O}_3^{2-}$	$\text{Na}_2\text{S}_2\text{O}_3$

^aThese anion names are sometimes written as a single word—for example, hydrogencarbonate, hydrogenphosphate, and so forth.

◀ Learn the most common ions first, such as NH_4^+ , CO_3^{2-} , OH^- , NO_3^- , PO_4^{3-} , SO_4^{2-} , and ClO_3^- . When you understand the scheme in expression (3.3), the names of several others, such as NO_2^- , SO_3^{2-} , ClO^- , ClO_2^- , and ClO_4^- will become obvious. Over time, the rest will become more familiar to you.

5. All the common oxoanions of Cl, Br, and I carry a charge of -1 .
6. Some series of oxoanions also contain various numbers of H atoms and are named accordingly. For example, HPO_4^{2-} is the *hydrogen phosphate* ion and H_2PO_4^- , the *dihydrogen phosphate* ion.
7. The prefix *thio-* signifies that a sulfur atom has been substituted for an oxygen atom. (The sulfate ion has *one* S and *four* O atoms; thiosulfate ion has *two* S and *three* O atoms.)

Oxoacids

The majority of acids are **ternary compounds**. They contain *three* different elements—hydrogen and two other nonmetals. If one of the nonmetals is oxygen, the acid is called an **oxoacid**. Think of oxoacids as combinations of hydrogen ions (H^+) and oxoanions. The scheme for naming oxoacids is similar to that outlined for oxoanions, except that the ending *-ous* is used instead of *-ite* and *-ic* instead of *-ate*. Several oxoacids are listed in Table 3.6. Also listed are the names and formulas of compounds in which the hydrogen of the oxoacid has been replaced by a metal, such as sodium. These compounds are called **salts**; we will say much more about them in later chapters, beginning in Chapter 5. Acids are molecular compounds, and salts are ionic compounds.

Table 3.6 Nomenclature of Some Oxoacids and Their Salts

Oxidation State	Formula of Acid ^a	Name of Acid ^b	Formula of Salt	Name of Salt ^b
Cl: +1	HClO	<i>Hypochlorous acid</i>	NaClO	Sodium <i>hypochlorite</i>
Cl: +3	HClO ₂	<i>Chlorous acid</i>	NaClO ₂	Sodium <i>chlorite</i>
Cl: +5	HClO ₃	<i>Chloric acid</i>	NaClO ₃	Sodium <i>chlorate</i>
Cl: +7	HClO ₄	<i>Perchloric acid</i>	NaClO ₄	Sodium <i>perchlorate</i>
N: +3	HNO ₂	<i>Nitrous acid</i>	NaNO ₂	Sodium <i>nitrite</i>
N: +5	HNO ₃	<i>Nitric acid</i>	NaNO ₃	Sodium <i>nitrate</i>
S: +4	H ₂ SO ₃	<i>Sulfurous acid</i>	Na ₂ SO ₃	Sodium <i>sulfite</i>
S: +6	H ₂ SO ₄	<i>Sulfuric acid</i>	Na ₂ SO ₄	Sodium <i>sulfate</i>

^aIn all these acids, H atoms are bonded to O atoms, not the central nonmetal atom. Often formulas are written to reflect this fact, for instance, HOCl instead of HClO and HOCIO instead of HClO₂.

^bIn general, the *-ic* and *-ate* names are assigned to compounds in which the central nonmetal atom has an oxidation state equal to the periodic table group number minus 10. Halogen compounds are exceptional in that the *-ic* and *-ate* names are assigned to compounds in which the halogen has an oxidation state of +5 (even though the group number is 17).

► When we refer to perchloric acid, for example, we generally mean an aqueous solution of HClO₄.

EXAMPLE 3-10 Applying Various Rules for Naming Compounds

Name the compounds (a) CuCl₂; (b) ClO₂; (c) HIO₄; (d) Ca(H₂PO₄)₂.

Analyze

CuCl₂ and Ca(H₂PO₄)₂ are ionic compounds. To name these compounds, we must identify and name the ions. ClO₂ and HIO₄ are molecular compounds. ClO₂ is a binary compound of two nonmetals, and HIO₄ is an oxoacid.

Solve

- In this compound, the oxidation state of Cu is +2. Because Cu can also exist in the oxidation state of +1, we must clearly distinguish between the two possible chlorides. CuCl₂ is **copper(II) chloride**.
- Both Cl and O are nonmetals. ClO₂ is a binary molecular compound called **chlorine dioxide**.
- The oxidation state of I is +7. By analogy to the chlorine-containing oxoacids in Table 3.6, we should name this compound **periodic acid** (pronounced “purr-eye-oh-dic” acid).
- The polyatomic anion H₂PO₄[−] is dihydrogen phosphate ion. Two of these ions are present for every Ca²⁺ ion in the compound **calcium dihydrogen phosphate**.

Assess

A fair bit of memorization is associated with naming compounds correctly. Mastery of this subject usually requires a lot of practice.

PRACTICE EXAMPLE A: Name the compounds SF₆, HNO₂, Ca(HCO₃)₂, FeSO₄.

PRACTICE EXAMPLE B: Name the compounds NH₄NO₃, PCl₃, HBrO, AgClO₄, Fe₂(SO₄)₃.

► In general, centered dots (·) show that a formula is a composite of two or more simpler formulas.

Some Compounds of Greater Complexity

The copper compound that Joseph Proust (1754–1826) used to establish the law of constant composition (page 64) is referred to in different ways. If you look up Proust’s compound in a handbook of minerals, you will find it listed as *malachite*, with the formula Cu₂(OH)₂CO₃. In a handbook that specializes in pharmaceutical applications, this same compound is listed as *basic cupric carbonate*, with the formula CH₂Cu₂O₅. In a chemistry handbook, it is listed as *copper(II) carbonate dihydroxide*, with the formula CuCO₃ · Cu(OH)₂. All you need to understand at this point is that regardless of the formula you use, you

EXAMPLE 3-11

Applying Various Rules in Writing Formulas

Write the formula of the compound (a) tetranitrogen tetrasulfide; (b) ammonium chromate; (c) bromic acid; (d) calcium hypochlorite.

Analyze

We must apply our knowledge of prefixes (such as *tetra-*) and endings (such as *-ic*), as well as the names of common polyatomic ions (such as *ammonium*, *chromate*, and *hypochlorite*).

Solve

- (a) Molecules of this compound consist of *four* N atoms and *four* S atoms. The formula is N_4S_4 .
- (b) Two ammonium ions (NH_4^+) must be present for every chromate ion (CrO_4^{2-}). Place parentheses around NH_4^+ , followed by the subscript 2. The formula is $(\text{NH}_4)_2\text{CrO}_4$. (This formula is read as “N-H-4, taken twice, C-R-O-4.”)
- (c) The *-ic* acid for the oxoacids of the halogens (group 17) has the halogen in the oxidation state of +5. Bromic acid is HBrO_3 (analogous to HClO_3 in Table 3.6).
- (d) Here there are one Ca^{2+} and two ClO^- ions in a formula unit. This leads to the formula $\text{Ca}(\text{ClO})_2$.

Assess

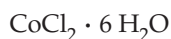
Notice that in writing formulas for compounds containing two or more polyatomic ions of the *same type*, as in (b) and (d), we put parentheses around the formula of the ion (without the charge), followed by a subscript indicating the number of ions of that type. Proper use and placement of parentheses in writing formulas is important.

PRACTICE EXAMPLE A: Write formulas for the compounds (a) boron trifluoride, (b) potassium dichromate, (c) sulfuric acid, (d) calcium chloride.

PRACTICE EXAMPLE B: Write formulas for the compounds (a) aluminum nitrate, (b) tetraphosphorus decoxide, (c) chromium(III) hydroxide, (d) iodic acid.

should obtain the same molar mass (221.116 g/mol), the same mass percent copper (57.48% Cu), the same H:O mole ratio (2 mol H/5 mol O), and so on. In short, you should be able to interpret a formula, no matter how complex its appearance.

Some complex substances you are certain to encounter are known as hydrates. In a **hydrate**, each formula unit of the compound has associated with it a certain number of water molecules. This does not mean that the compounds are “wet,” however. The water molecules are incorporated in the solid structure of the compound. The formula shown below signifies six H_2O molecules per formula unit of CoCl_2 .



The prefix for six is *hexa-* and this compound is cobalt(II) chloride *hexahydrate*. Its formula mass is that of CoCl_2 *plus* that associated with six H_2O : $129.83 \text{ u} + (6 \times 18.015 \text{ u}) = 237.92 \text{ u}$. We can speak of the mass percent water in a hydrate; for $\text{CoCl}_2 \cdot 6 \text{H}_2\text{O}$ this is

$$\% \text{H}_2\text{O} = \frac{(6 \times 18.015) \text{ g H}_2\text{O}}{237.92 \text{ g CoCl}_2 \cdot 6 \text{H}_2\text{O}} \times 100\% = 45.43\%$$

The water present in compounds as water of hydration can generally be removed, in part or totally, by heating. When the water is totally removed, the resulting compound is said to be *anhydrous* (without water). Anhydrous compounds can be used as water absorbers, as in the use of anhydrous magnesium perchlorate in combustion analysis (recall Figure 3-6). CoCl_2 gains and loses water quite readily and indicates this through a color change. Anhydrous CoCl_2 is blue, whereas the hexahydrate is pink. This fact can be used to make a simple moisture detector (Fig. 3-8).

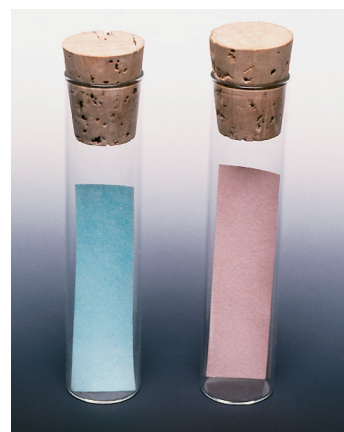


Figure 3-8 Effect of moisture on CoCl_2

The piece of filter paper was soaked in a water solution of cobalt(II) chloride and then allowed to dry. When kept in dry air, the paper is blue (anhydrous CoCl_2). In humid air, the paper changes to pink ($\text{CoCl}_2 \cdot 6 \text{H}_2\text{O}$).

3-7 Names and Formulas of Organic Compounds

Keep in Mind

that each carbon atom forms four covalent bonds.

Organic compounds abound in nature. The foods we eat are made up almost exclusively of organic compounds, including not only energy-producing fats and carbohydrates and muscle-building proteins but also trace compounds that impart color, odor, and flavor to these foods. Almost all fuels, whether used to power automobiles, trucks, trains, or airplanes, are mixtures of organic compounds of a type called hydrocarbons. Most of the drugs produced by pharmaceutical companies are complex organic compounds, as are common plastics. The multiplicity of organic compounds is so vast that *organic chemistry* exists as a distinctive field of chemistry.

The great diversity of organic compounds arises from the ability of carbon atoms to combine readily with other carbon atoms and with atoms of a number of other elements. Carbon atoms join together to form a framework of chains or rings to which other atoms are attached. All organic compounds contain carbon atoms; almost all contain hydrogen atoms; and many common ones also have oxygen, nitrogen, or sulfur atoms. These possibilities allow for an almost limitless number of different organic compounds. Organic compounds are mostly molecular; a few are ionic.

There are millions of organic compounds, many comprising highly complex molecules. Their names are equally complicated. A systematic approach to naming these compounds is crucial, and the rules for naming inorganic compounds are of little use here. The usual name, often called the common or *trivial* name, for a familiar sweetener is sucrose (sugar). The systematic name is α -D-glucopyranosyl- β -D-fructofuranoside. At this point, however, we only need to recognize organic compounds and use their common names, together with an occasional systematic name. We will look at the systematic nomenclature of organic compounds in more detail in Chapter 26.

Hydrocarbons

Compounds containing only carbon and hydrogen are called **hydrocarbons**. The simplest hydrocarbon contains one carbon atom and four hydrogen atoms—methane, CH_4 (Fig. 3-9a). As the number of carbon atoms increases, the number of hydrogen atoms also increases in a systematic way, depending on the type of hydrocarbon. The complexity of organic chemistry arises because carbon atoms can form chains and rings, and the nature of the chemical bonds between the carbon atoms can vary. Hydrocarbons containing only single bonds are called **alkanes**. The simplest alkane is methane, followed by ethane, C_2H_6 (Fig. 3-9b), and then propane, C_3H_8 (Fig. 3-9c). The fourth member of the series, butane, C_4H_{10} , was shown in Figure 3-2(a). Notice that each succeeding member of the alkane series is formed by the addition of one C atom and two H atoms to the preceding member.

The names of the alkanes are composed of two parts: a word stem (or prefix) and the ending (or suffix) *-ane* indicating that the molecule is an *alkane*. The first four word stems in Table 3.7 reflect common names, while the rest indicate the number of carbon atoms in the alkane. Thus C_5H_{12} is pentane and C_7H_{16} is heptane (hept = 7).

Hydrocarbon molecules with one or more double bonds between carbon atoms are called **alkenes**. The simplest of the alkenes is ethene (Fig. 3-9d); its name consists of the stem *eth-* and the ending *-ene*. Benzene, C_6H_6 (Fig. 3-9e), is a molecule with six carbon atoms arranged in a hexagonal ring. Molecules with structures related to benzene make up a large proportion of known organic compounds.

Refer to Figure 3-2 (page 99) and you will notice that butane and methylpropane have the same molecular formula, C_4H_{10} , but different structural formulas. Butane is based on a four-carbon chain, whereas in methylpropane,

Table 3.7

Word Stem
(or Prefix)
Indicating the
Number of
Carbon Atoms
in Simple
Organic
Molecules

Stem (or prefix)	Number of C Atoms
Meth-	1
Eth-	2
Prop-	3
But-	4
Pent-	5
Hex-	6
Hept-	7
Oct-	8
Non-	9
Dec-	10

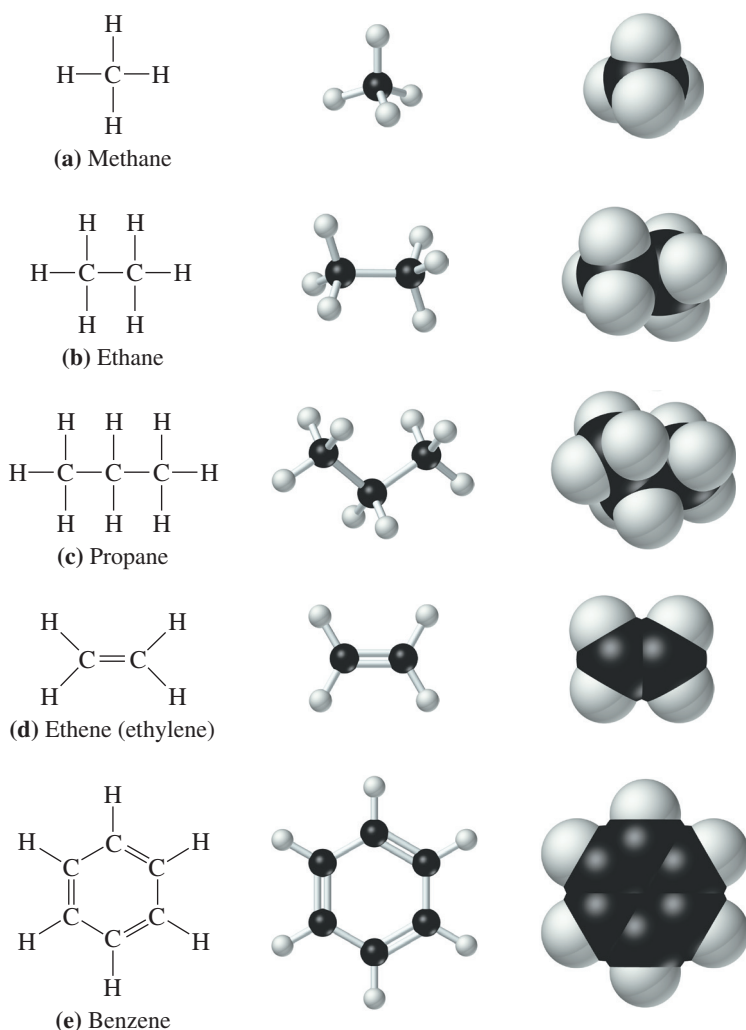


Figure 3-9 Visualizations of some hydrocarbons

a —CH_3 group, called a *methyl* group, is attached to the middle carbon atom of the three-carbon propane chain. Butane and methylpropane are *isomers*. **Isomers** are molecules that have the same molecular formula but different arrangements of atoms in space. As organic molecules become more complex, the possibilities for isomerism increase very rapidly.

EXAMPLE 3-12 Recognizing Isomers

Are the following pairs of molecules isomers or not?

- (a) $\text{CH}_3\text{CH}(\text{CH}_3)(\text{CH}_2)_3\text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)(\text{CH}_2)_3\text{CH}_3$
 (b) $\begin{array}{c} \text{CH}_3-\text{CH}-\text{CH}_2-\text{CH}_3 \\ | \\ \text{CH}_2-\text{CH}_3 \end{array}$ and $\begin{array}{c} \text{CH}_3-\text{CH}-\text{CH}_2-\text{CH}_2-\text{CH}_3 \\ | \\ \text{CH}_3 \end{array}$

Analyze

Isomers have the same molecular formula but different structures. We first check to see if the molecular formulas are the same. If the formulas are not the same, they represent different compounds. If the formulas are the same, the compounds *may* be isomers, but only if their structures are different.

Solve

- (a) The molecular formula of the first compound is C_7H_{16} , while that of the second compound is C_8H_{18} . **The molecules are not isomers.**

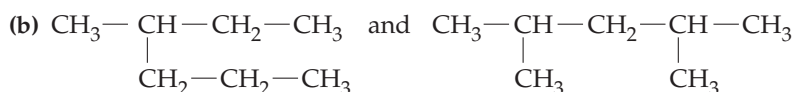
(continued)

- (b) These molecules have the same formula, C_6H_{14} , but they differ in structure. They are **isomers**. The difference in structure is that in the first structure, a methyl side chain is on the middle carbon atom of a five-carbon chain, and in the second structure it is on the second carbon atom from the end of the chain.

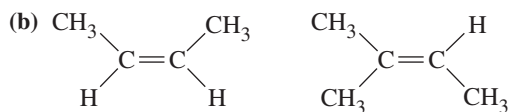
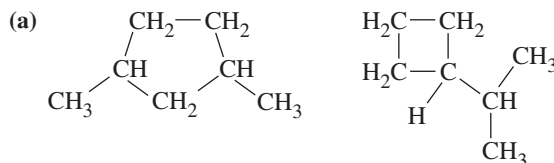
Assess

The compounds shown in part (a) have different molecular formulas and, thus, are clearly different compounds. We expect these compounds to have markedly different properties. The compounds shown in part (b) have the same molecular formula but different molecular structures. They are different compounds and have different properties. For example, the compound shown on the left in part (b) has a slightly higher boiling point than the compound shown on the right (63°C versus 60°C).

PRACTICE EXAMPLE A: Are the following pairs of molecules isomers?



PRACTICE EXAMPLE B: Are the pairs of molecules represented by the following structural formulas isomers?



3-7 Concept Assessment

In the combustion of a hydrocarbon, can the mass of H_2O produced ever exceed that of the CO_2 ? Explain.

Keep in Mind

that alcohols are molecular compounds in which oxygen atoms are covalently bonded to carbon atoms. They are not ionic compounds that contain hydroxide ions.

Functional Groups

Carbon chains provide the framework of organic compounds; other atoms or groups of atoms replace one or more of the hydrogen atoms to form different compounds. We can illustrate this with the common alcohol molecule that occurs in beer, wine, and spirits. The molecule is *ethanol*, $\text{CH}_3\text{CH}_2\text{OH}$, in which one of the H atoms of ethane is replaced by an $-\text{OH}$ group (Fig. 3-10a). The systematic name ethanol is derived from the name of the alkane, ethane, with the final $-e$ replaced by the suffix $-ol$. The suffix $-ol$ designates the presence of the OH group in a class of organic molecules called **alcohols**.

Ethyl alcohol, the common name of ethanol, also indicates attachment of the $-\text{OH}$ group to the ethane hydrocarbon chain. To name the alkane chain as a group, replace the final $-e$ with $-yl$, so that ethane becomes *ethyl*, thus *ethyl alcohol* for $\text{CH}_3\text{CH}_2\text{OH}$. It is often the case that the common name of one compound, alcohol in this case, will provide the generic name for a complete class of compounds; that is, all alcohols contain at least one $-\text{OH}$ group.

Another common alcohol is *methanol*, or wood alcohol, which has the formula CH_3OH (Fig. 3-10b). The common name for methanol is methyl alcohol. It is interesting to note that wood alcohol is a dangerous poison, whereas the grain alcohol in beer and wine is safe to consume in moderate quantities.

The $-\text{OH}$ group in alcohols is one of the many *functional groups* found in organic compounds. **Functional groups** are individual atoms or groupings of

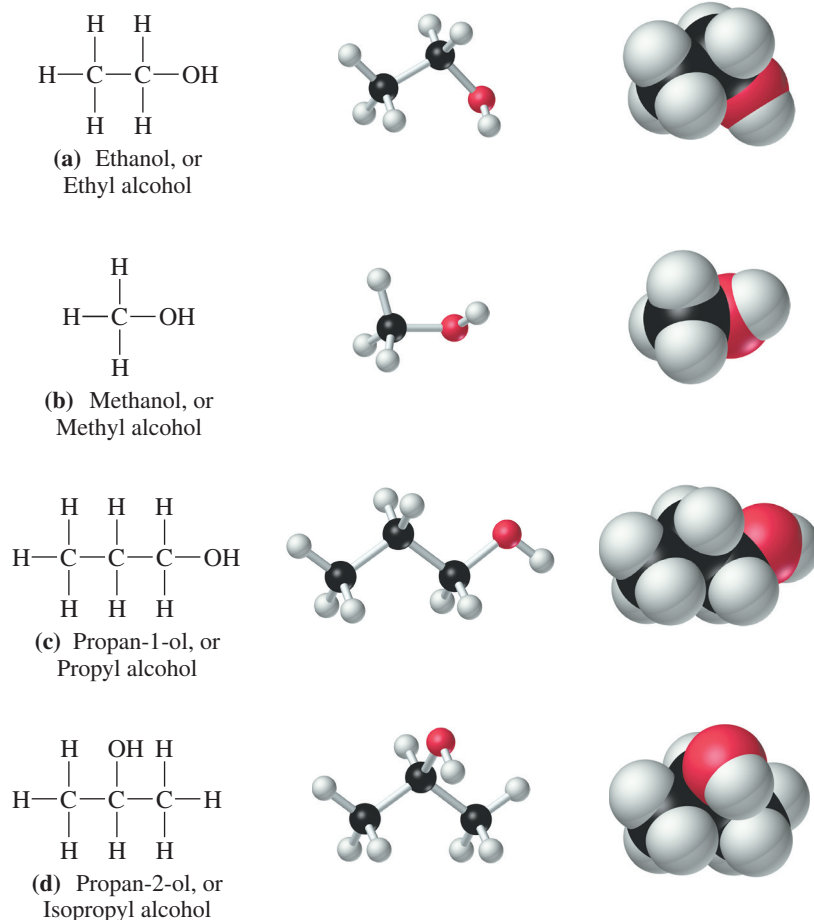


Figure 3-10 Visualizations of some alcohols

atoms that are attached to the carbon chains or rings of organic molecules and give the molecules their characteristic properties. Compounds with the same functional group generally have similar properties. The —OH group is called the *hydroxyl group*. At this point only a few functional groups will be introduced; functional groups are discussed in more detail in Chapter 26.

The presence of functional groups also increases the possibility of isomers. For example, there is only one propane molecule, C_3H_8 . However, if one of the H atoms is replaced by a hydroxyl group, two possibilities exist for the point of attachment: at one of the end C atoms or at the middle C atom (Fig. 3-10c, d). This leads to two isomers. The alcohol with the —OH group attached to the end carbon atom is commonly called propyl alcohol or, systematically, propan-1-ol; the 1 indicates that the —OH group is on the first or end C atom. The alcohol with the —OH group attached to the middle carbon atom is commonly called isopropyl alcohol or, systematically, propan-2-ol; the 2 indicates that the —OH group is on the second C atom from the end. Notice that, in the systematic name of an alcohol, the number specifying the position of the —OH group is placed immediately before the suffix *-ol*.

Another important functional group is the *carboxyl* group, —COOH , or $\text{—CO}_2\text{H}$ which confers acidic properties on a molecule. The C atom in the carboxyl group is bound to the two O atoms in two ways. One bond is a single bond to an oxygen atom that is also attached to a H atom, and the other is a double bond to a lone O atom (Fig. 3-11a). The hydrogen attached to one of the O atoms in a carboxyl group is *ionizable* or *acidic*. Compounds containing the carboxyl group are called **carboxylic acids**. The first carboxylic acid based on alkanes is *methanoic* acid, HCOOH (Fig. 3-11b). In the systematic name, the *methan-* indicates one carbon atom and the *-oic* acid indicates a carboxylic acid. The common name for methanoic acid is formic acid, deriving from the Latin word *formica*, meaning “ant.” Formic acid is injected by an ant when it bites; this leads to the burning sensation that accompanies the bite.

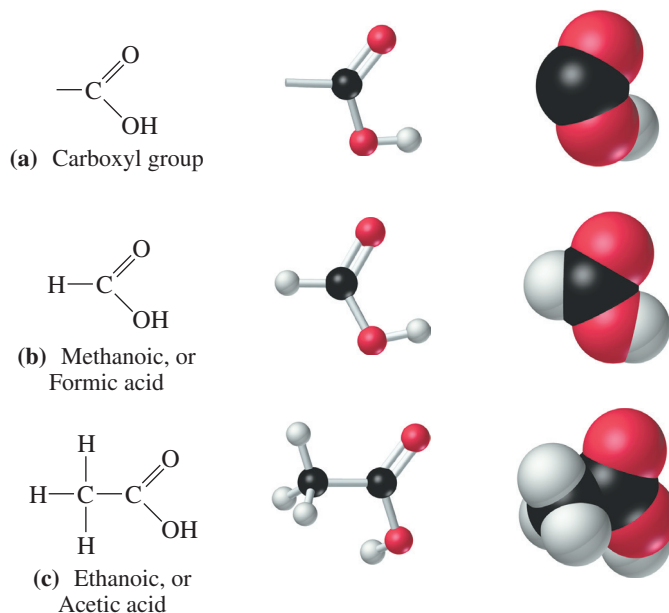


Figure 3-11 The carboxyl group and visualizations of two carboxylic acids

The simplest carboxylic acid containing two carbon atoms is *ethanoic* acid, more commonly known as *acetic* acid. The molecular formula is CH_3COOH , and the structure is shown in Figure 3-11(c). Vinegar is a solution of acetic acid in water. An additional functional group is introduced in the examples that follow—a halogen atom (F, Cl, Br, I) substituting for one or more H atoms. When present as functional groups, the halogens carry the names, *fluoro-*, *chloro-*, *bromo-*, and *iodo-*.

EXAMPLE 3-13 Recognizing Types of Organic Compounds

What type of compound is each of the following?

- (a) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$ (b) $\text{CH}_3\text{CHClCH}_2\text{CH}_3$
 (c) $\text{CH}_3\text{CH}_2\text{CO}_2\text{H}$ (d) $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$

Analyze

For each compound, examine the formula to determine which functional group, if any, is present. Also consider whether or not all the carbon–carbon bonds are single bonds.

Solve

- (a) The carbon–carbon bonds are all single bonds in this hydrocarbon. **This compound is an alkane.**
 (b) There are only single bonds in its molecules, and one H atom has been replaced by a Cl atom. **This compound is a chloroalkane.**
 (c) The presence of the carboxyl group, $-\text{CO}_2\text{H}$, in its molecules means that this compound is a **carboxylic acid**.
 (d) The presence of the hydroxyl group, $-\text{OH}$, in its molecules means that this compound is an **alcohol**.

Assess

Carbon–hydrogen and carbon–carbon bonds, especially carbon–carbon single bonds, are relatively unreactive. Because of this, functional groups are most important in determining the characteristic properties of organic compounds. We will encounter other functional groups and examine organic compounds in more detail in Chapters 26 and 27.

PRACTICE EXAMPLE A: What types of compounds correspond to each of the following formulas?

- (a) $\text{CH}_3\text{CH}_2\text{CH}_3$ (b) $\text{ClCH}_2\text{CH}_2\text{CH}_3$
 (c) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CO}_2\text{H}$ (d) $\text{CH}_3\text{CHCHCH}_3$

PRACTICE EXAMPLE B: What types of compounds correspond to these formulas?

- (a) $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$ (b) $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CO}_2\text{H}$
 (c) $\text{CH}_2\text{ClCH}_2\text{CO}_2\text{H}$ (d) BrCHCHCH_3

EXAMPLE 3-14 Naming Organic Compounds

Name these compounds.

- (a) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ (b) $\text{CH}_3\text{CHFCH}_2\text{CH}_3$
 (c) $\text{CH}_3\text{CH}_2\text{CO}_2\text{H}$ (d) $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$

Analyze

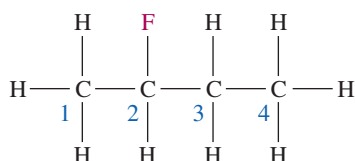
First, determine the type of compound. Then, count the number of carbon atoms and select the appropriate stem (or prefix) from Table 3.7 to form the name. If the compound is an alcohol, change the final *-e* in the name to *-ol*; if it is a carboxylic acid, change the final *-e* to *-oic acid*. If it is necessary to specify the position of the functional group, the carbon number is placed immediately before that part of the name to which it relates. Thus, the carbon number is placed immediately before the *-ol* suffix in the name of an alcohol and immediately before the prefix *fluoro-* in the name of a fluoroalkane.

Solve

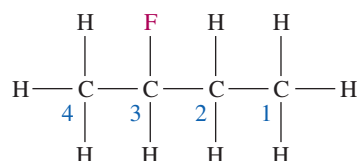
- (a) The structure is that of an alkane molecule with a five-carbon chain, so the compound is **pentane**.
 (b) The structure is that of a fluoroalkane molecule with the F atom on the *second* C atom of a four-carbon chain. The compound is called **2-fluorobutane**.
 (c) The carbon chain in this structure is three C atoms long, with the end C atom in a carboxyl group. The compound is **propanoic acid**.
 (d) This structure is that of an alcohol with the hydroxyl group on the *third* C atom of a five-carbon chain. The compound is called **pentan-3-ol**.

Assess

In naming the compound in part (b), we stated that the F atom was bonded to the second C atom in a four-carbon chain. There is some ambiguity in that statement because, as illustrated below, we can number the C atoms in two different ways.



2-fluorobutane
(correct)

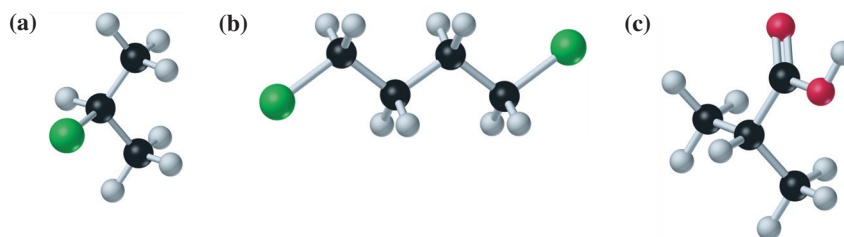


3-fluorobutane
(incorrect)

By convention, we always number the carbon atoms so that the position of the functional group is designated by the smallest possible number. Thus, 2-fluorobutane is the correct name.

PRACTICE EXAMPLE A: Name the following compounds: (a) $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$; (b) $\text{ICH}_2\text{CH}_2\text{CH}_3$; (c) $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_2\text{CO}_2\text{H}$; (d) CH_3CHCH_2 .

PRACTICE EXAMPLE B: Give plausible names for the molecules that correspond to the following ball-and-stick models.



EXAMPLE 3-15 Writing Structural Formulas from the Names of Organic Compounds

Write the condensed structural formula for the organic compounds (a) butane; (b) butanoic acid; (c) 1-chloropentane; (d) hexan-1-ol.

Analyze

First, identify the number of carbon atoms in the chain, and then determine the type and position of the functional group, if any.

Solve

- (a) The word stem *but-* indicates a structure with a four-carbon chain, and the suffix *-ane* indicates an alkane. No functional groups are indicated; hence, the condensed structural formula is $\text{CH}_3(\text{CH}_2)_2\text{CH}_3$.
- (b) The *-oic* ending indicates that the end carbon atom of the four-carbon chain is part of a carboxylic acid group. The condensed structural formula is $\text{CH}_3(\text{CH}_2)_2\text{CO}_2\text{H}$.
- (c) The prefix *chloro-* indicates the substitution of a chlorine atom for a H atom, and the 1- designates that it is on the first C atom of the carbon chain. The carbon chain is five C atoms long, as signified by the word stem *pent-*. The condensed structural formula is $\text{CH}_3(\text{CH}_2)_3\text{CH}_2\text{Cl}$.
- (d) The suffix *-ol* indicates the presence of a hydroxyl group in place of a H atom, and the 1 designates that it is on the first C atom of the carbon chain. The word stem *hex-* signifies that the carbon chain is six C atoms long. The condensed structural formula is $\text{CH}_3(\text{CH}_2)_4\text{CH}_2\text{OH}$.

Assess

In summary, to obtain a structural formula from the name, we split the name into its component pieces: stem, prefix, and suffix. All three components provide information about the structure of the molecule.

PRACTICE EXAMPLE A: Write the condensed structural formula for the organic compounds (a) pentane; (b) ethanoic acid; (c) 1-iodooctane (pronounced “eye-oh-dough-octane”); (d) pentan-1-ol.

PRACTICE EXAMPLE B: Write the line-angle formula for the organic compounds (a) propene; (b) heptan-1-ol; (c) chloroacetic acid; (d) hexanoic acid.



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For a discussion of how chemists use mass spectrometry to establish both molecular and structural formulas, go to the Focus On feature for Chapter 3, entitled Mass Spectrometry—Determining Molecular and Structural Formulas, in Mastering.

Summary

3-1 Types of Chemical Compounds and Their Formulas

The two main classes of chemical compounds are **molecular compounds** and **ionic compounds**. The fundamental unit of a molecular compound is a molecule and that of an ionic compound is a **formula unit**. A formula unit is the smallest collection of positively charged ions—called **cations**—and negatively charged ions—called **anions**—that is electrically neutral overall. A **chemical formula** is a symbolic representation of a compound that can be written in several ways (Fig. 3-1). If the formula has the smallest integral subscripts possible, it is an **empirical formula**; if the formula represents an actual molecule, it is a **molecular formula**; and if the formula is written to show how individual atoms are joined together into molecules, it is a **structural formula**. Abbreviated structural formulas, called **condensed structural formulas**, are often used for organic molecules. Also used for organic molecules is the **line-angle formula**, in which all bond lines are shown except those between C and H atoms and in which the symbols C and H are mostly omitted. The relative

sizes and positions of the atoms in molecules can be depicted by ball-and-stick and space-filling molecular models.

3-2 The Mole Concept and Chemical Compounds

In this section, the concept of atomic mass is extended to **molecular mass**, the mass in atomic mass units of a molecule of a molecular compound, and **formula mass**, the mass in atomic mass units of a formula unit of an ionic compound. Likewise, the concept of the Avogadro constant and the mole is now applied to compounds, with emphasis on quantitative applications involving the mass of a mole of compound—the **molar mass** M . For several elements, we can distinguish between a mole of molecules (for example, P_4) and a mole of atoms (that is, P).

3-3 Composition of Chemical Compounds—The mass percent composition of a compound can be established from its formula (equation 3.1). Conversely, a chemical formula can be deduced from the experimentally determined percent composition of the compound. For organic

compounds, this often involves combustion analysis (Fig. 3-6). Formulas determined from experimental percent composition data are empirical formulas—the simplest formulas that can be written. Molecular formulas can be related to empirical formulas when experimentally determined molecular masses are available.

3-4 Oxidation States: A Useful Tool in Describing Chemical Compounds—The **oxidation state** of an atom is roughly related to the number of electrons involved in the formation of a bond between that atom and another atom. Oxidation states are expressed as numbers assigned according to a set of conventions (Table 3.2). The oxidation state concept has several uses, one of which is to aid in naming chemical compounds and in writing chemical formulas.

3-5 Naming Compounds: Organic and Inorganic Compounds—Assigning names to the formulas of chemical compounds—nomenclature—is an important activity in chemistry. The topic is introduced in this section and applied to two broad categories: (1) **organic compounds**, which are compounds formed by the elements carbon and hydrogen, often together with a few other elements, such as oxygen and nitrogen; and (2) **inorganic compounds**, a category that includes the remaining chemical compounds. The subject of nomenclature is extended to additional contexts throughout the text.

3-6 Names and Formulas of Inorganic Compounds—In this section on the nomenclature of inorganic compounds, the names and formulas of two-element or **binary compounds** are considered first. The

names and symbols of some simple ions (Table 3.3) and the names of a few typical binary molecular compounds (Table 3.4) are listed. Also listed are some common ions comprising two or more atoms, **polyatomic ions**, and typical ionic compounds containing them (Table 3.5). The naming of binary acids is related to the names of the corresponding binary hydrogen compounds. The majority of acids, however, are **ternary compounds**; they consist of three elements—hydrogen and two other nonmetals. Typically, one of the elements in ternary acids is oxygen, giving rise to the name **oxoacids**. The polyatomic anions derived from oxoacids are **oxoanions**. A scheme for relating the names and formulas of oxoanions is given (expression 3.3), and the nomenclature of oxoacids and their salts is summarized (Table 3.6). Also described in this section are **hydrates**, ionic compounds having fixed numbers of water molecules associated with their formula units.

3-7 Names and Formulas of Organic Compounds—Organic compounds are based on the element carbon. **Hydrocarbons** contain only carbon and hydrogen (Fig. 3-9). **Alkane** hydrocarbon molecules contain only single bonds, and **alkenes** contain at least one double bond. A common phenomenon found in organic compounds is **isomerism**—the existence of different compounds, called **isomers**, having identical molecular formulas but different structural formulas. **Functional groups** confer distinctive properties on organic molecules when they are substituted for H atoms on carbon chains or rings. The hydroxyl group —OH is present in **alcohols** (Fig. 3-10) and the carboxyl group —COOH, in **carboxylic acids** (Fig. 3-11).

Integrative Example

Molecules of a dicarboxylic acid have *two* carboxyl groups (—COOH). A 2.250 g sample of a dicarboxylic acid was burned in an excess of oxygen and yielded 4.548 g CO₂ and 1.629 g H₂O. In a separate experiment, the molecular mass of the acid was found to be 174 u. From these data, what can we deduce about the structural formula of this acid?

Analyze

Our approach will require several steps: (1) Use the combustion data to determine the percent composition of the compound (similar to Example 3-6). (2) Determine the empirical formula from the percent composition (similar to Example 3-5). (3) Obtain the molecular formula from the empirical formula and the molecular mass. (4) Determine how the C, H, and O atoms represented in the molecular formula might be assembled into a dicarboxylic acid. Use molar masses with (at least) one more significant figure than in the measured masses where possible; store intermediate results in your calculator without rounding off.

Solve

1. *Determine the percent composition.* Calculate the mass of H in 1.629 g H₂O

and then the mass percent H in the 2.250 g sample of the dicarboxylic acid.

Also, calculate the mass of C in 4.548 g CO₂,

$$\begin{aligned} ? \text{ g H} &= 1.629 \text{ g H}_2\text{O} \times \frac{1 \text{ mol H}_2\text{O}}{18.015 \text{ g H}_2\text{O}} \\ &\times \frac{2 \text{ mol H}}{1 \text{ mol H}_2\text{O}} \times \frac{1.008 \text{ g H}}{1 \text{ mol H}} = 0.1823 \text{ g H} \end{aligned}$$

$$\% \text{ H} = \frac{0.1823 \text{ g H}}{2.250 \text{ g compd.}} \times 100\% = 8.102\% \text{ H}$$

$$\begin{aligned} ? \text{ g C} &= 4.548 \text{ g CO}_2 \times \frac{1 \text{ mol CO}_2}{44.009 \text{ g CO}_2} \\ &\times \frac{1 \text{ mol C}}{1 \text{ mol CO}_2} \times \frac{12.011 \text{ g C}}{1 \text{ mol C}} = 1.241 \text{ g C} \end{aligned}$$

followed by the mass percent C in the 2.250 g sample of dicarboxylic acid.

The mass percent O in the compound is obtained as a difference, that is,

2. *Obtain the empirical formula from the percent composition.* The masses of the elements in 100.0 g of the compound are

The numbers of moles of the elements in 100.0 g of the compound are

The tentative formula based on these numbers is

Divide all the subscripts by 2.296 to obtain

Multiply all subscripts by two to obtain the empirical formula,

and then determine the empirical formula mass.

3. *Obtain the molecular formula.* The experimentally determined molecular mass of 174 u is twice the empirical formula mass. The molecular formula is

4. *Assemble the atoms in $C_8H_{14}O_4$ into a plausible structural formula.* The dicarboxylic acid must contain two $-COOH$ groups. This accounts for the two C atoms, two H atoms, and all four O atoms. The remainder of the structure is based on C_6H_{12} . For example, arrange the six $-CH_2$ segments into a six-carbon chain and attach the $-COOH$ groups at the ends of the chain.

However, there are other possibilities based on shorter chains with branches, for example

$$\% C = \frac{1.241 \text{ g C}}{2.250 \text{ g compd.}} \times 100\% = 55.16\% C$$

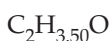
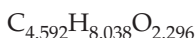
$$\% O = 100.00\% - 55.16\% C - 8.102\% H = 36.74\% O$$

$$55.16 \text{ g C} \quad 8.102 \text{ g H} \quad 36.74 \text{ g O}$$

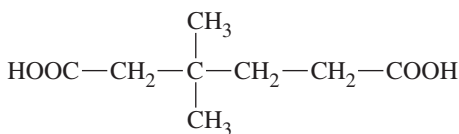
$$55.16 \text{ g C} \times \frac{1 \text{ mol C}}{12.011 \text{ g C}} = 4.592 \text{ mol C}$$

$$8.102 \text{ g H} \times \frac{1 \text{ mol H}}{1.008 \text{ g H}} = 8.038 \text{ mol H}$$

$$36.74 \text{ g O} \times \frac{1 \text{ mol O}}{15.999 \text{ g O}} = 2.296 \text{ mol O}$$

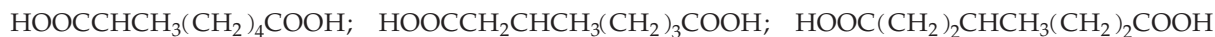


$$(4 \times 12.011 \text{ u}) + (7 \times 1.008 \text{ u}) + (2 \times 15.999 \text{ u}) = 87.098 \text{ u}$$



Assess

We have found a plausible structural formula, but there are many other possibilities. For example, the following three isomers have a seven-carbon chain with one methyl group ($-CH_3$) substituted for an H atom on the chain:



In conclusion, we cannot identify a specific isomer with only the data given.

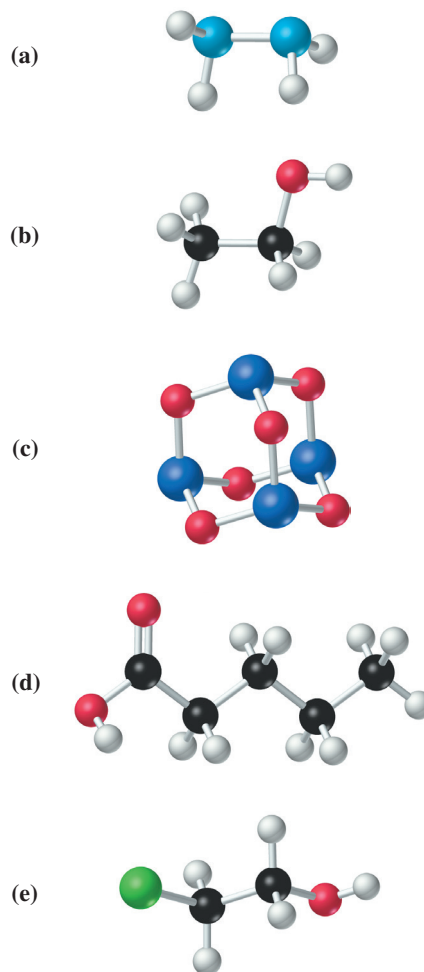
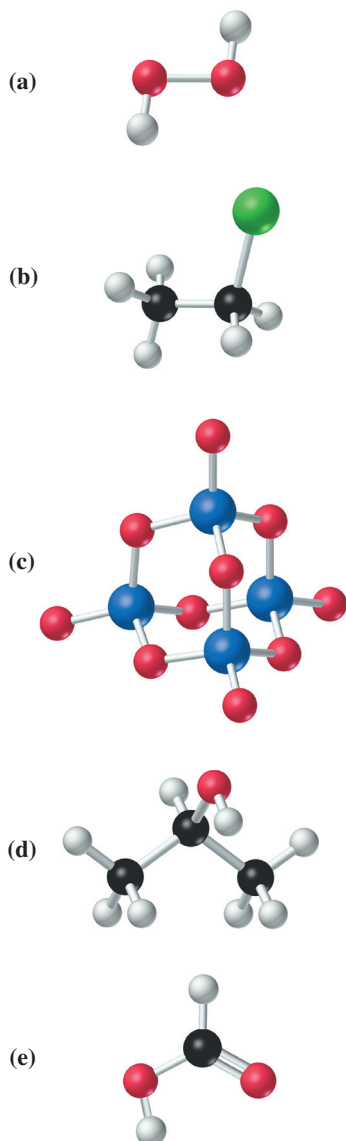
PRACTICE EXAMPLE A: A 2.4917 g sample of an unknown solid hydrate was heated to drive off all the water of hydration. The remaining solid, which weighed 1.8558 g, was analyzed and found to be 27.74% Mg, 23.57% P, and 48.69% O, by mass. What is the formula and name of the unknown solid hydrate?

PRACTICE EXAMPLE B: An unknown solid hydrate was analyzed and found to be 17.15% Cu, 19.14% Cl, and 60.45% O, by mass; the remainder was hydrogen. What are the oxidation states of copper and chlorine in this compound? What is the name of this compound?

Exercises

Representing Molecules

1. Refer to the color scheme given in Figure 3-3, and give the molecular formulas for the molecules whose ball-and-stick models are given here.
2. Give the molecular formulas for the molecules whose ball-and-stick models are given here. Refer to the color scheme in Figure 3-3.



3. Give the structural formulas of the molecules shown in Exercise 1 (b), (d), and (e).
4. Give the structural formulas of the molecules shown in Exercise 2 (b), (d), and (e).

The Avogadro Constant and the Mole

5. Calculate the total number of (a) atoms in one molecule of trinitrotoluene (TNT), $\text{CH}_3\text{C}_6\text{H}_2(\text{NO}_2)_3$; (b) atoms in $0.00102 \text{ mol CH}_3(\text{CH}_2)_4\text{CH}_2\text{OH}$; (c) F atoms in $12.15 \text{ mol C}_2\text{HBrClF}_3$.
6. Determine the mass, in grams, of (a) 7.34 mol NO_2 ; (b) $4.220 \times 10^{25} \text{ molecules}$; (c) $15.5 \text{ mol CuSO}_4 \cdot 5 \text{ H}_2\text{O}$; (d) $2.25 \times 10^{24} \text{ molecules of C}_2\text{H}_4(\text{OH})_2$.
7. The amino acid methionine, which is essential in human diets, has the molecular formula $\text{C}_5\text{H}_{11}\text{NO}_2\text{S}$. Determine (a) its molecular mass; (b) the number of moles of H atoms per mole of methionine;

- (c) the number of grams of C per mole of methionine;
 (d) the number of C atoms in 9.07 mol methionine.
8. Determine the number of moles of Br_2 in a sample consisting of (a) 4.04×10^{22} Br_2 molecules; (b) 5.78×10^{24} Br atoms (assuming that the Br atoms react completely to form Br_2); (c) 7.82 kg bromine; (d) 3.56 L liquid bromine ($d = 3.10$ g/mL).
9. *Without doing detailed calculations*, explain which of the following has the greatest number of N atoms (a) 50.0 g N_2O ; (b) 17.0 g NH_3 ; (c) 150 mL of liquid pyridine, $\text{C}_5\text{H}_5\text{N}$ ($d = 0.983$ g/mL); (d) 1.0 mol N_2 .
10. *Without doing detailed calculations*, determine which of the following has the greatest number of S atoms (a) 0.12 mol of solid sulfur, S_8 ; (b) 0.50 mol of gaseous S_2O ; (c) 65 g of gaseous SO_2 ; (d) 75 mL of liquid thiophene, $\text{C}_4\text{H}_4\text{S}$ ($d = 1.064$ g/mL).
11. Determine the number of moles of (a) N_2O_4 in a 115 g sample (b) N atoms in 43.5 g of $\text{Mg}(\text{NO}_3)_2$ (c) N atoms in a sample of $\text{C}_7\text{H}_5(\text{NO}_2)_3$ that has the same number of O atoms as 12.4 g $\text{C}_6\text{H}_{12}\text{O}_6$
12. Determine the mass, in grams, of (a) 2.10×10^2 mol S_8 (b) 5.02×10^{22} molecules of palmitic acid, $\text{C}_{16}\text{H}_{32}\text{O}_2$ (c) a quantity of the amino acid histidine, $\text{C}_6\text{H}_9\text{N}_3\text{O}_2$, containing 2.95 mol N atoms
13. The hemoglobin content of blood is about 15.5 g/100 mL blood. The molar mass of hemoglobin is about 64,500 g/mol, and there are four iron (Fe) atoms in a hemoglobin molecule. Approximately how many Fe atoms are present in the 6 L of blood in a typical adult?
14. In white phosphorus, P atoms are joined into P_4 molecules (see Figure 3-5). White phosphorus is commonly supplied in chalk-like cylindrical form. Its density is 1.823 g/cm³. For a cylinder of white phosphorus 6.50 cm long and 1.22 cm in diameter, determine (a) the number of moles of P_4 present; (b) the total number of P atoms.

Chemical Formulas

15. Explain which of the following statement(s) is (are) correct concerning glucose (blood sugar), $\text{C}_6\text{H}_{12}\text{O}_6$.
 (a) The percentages, by mass, of C and O are the same as in CO .
 (b) The ratio of C:H:O atoms is the same as in dihydroxyacetone, $(\text{CH}_2\text{OH})_2\text{CO}$.
 (c) The proportions, by mass, of C and O are equal.
 (d) The highest percentage, by mass, is that of H.
16. Explain which of the following statement(s) is (are) correct for sorbic acid, $\text{C}_6\text{H}_8\text{O}_2$, an inhibitor of mold and yeast.
 (a) It has a C:H:O mass ratio of 3:4:1.
 (b) It has the same mass percent composition as the aquatic herbicide, acrolein, $\text{C}_3\text{H}_4\text{O}$.
 (c) It has the same empirical formula as aspidinol, $\text{C}_{12}\text{H}_{16}\text{O}_4$, a drug used to kill parasitic worms.
 (d) It has four times as many H atoms as O atoms, but four times as much O as H by mass.
17. For the mineral torbernite, $\text{Cu}(\text{UO}_2)_2(\text{PO}_4)_2 \cdot 8 \text{H}_2\text{O}$, determine (a) the total number of atoms in one formula unit (b) the ratio, by number, of H atoms to O atoms (c) the ratio, by mass, of Cu to P (d) the element present in the greatest mass percent (e) the mass required to contain 1.00 g P
18. For the compound $\text{Ge}[\text{S}(\text{CH}_2)_4\text{CH}_3]_4$, determine (a) the total number of atoms in one formula unit (b) the ratio, by number, of C atoms to H atoms (c) the ratio, by mass, of Ge to S (d) the number of g S in 1 mol of the compound (e) the number of C atoms in 33.10 g of the compound

Percent Composition of Compounds

19. Determine the mass percent H in the hydrocarbon decane, $\text{C}_{10}\text{H}_{22}$.
20. Determine the mass percent O in the mineral malachite, $\text{Cu}_2(\text{OH})_2\text{CO}_3$.
21. Determine the mass percent H in the hydrocarbon isooctane, $\text{C}(\text{CH}_3)_3\text{CH}_2\text{CH}(\text{CH}_3)_2$.
22. Determine the mass percent H_2O in the hydrate $\text{Cr}(\text{NO}_3)_3 \cdot 9 \text{H}_2\text{O}$.
23. Determine the mass percent of each of the elements in the antimalarial drug quinine, $\text{C}_{20}\text{H}_{24}\text{N}_2\text{O}_2$.
24. Determine the mass percent of each of the elements in the fungicide copper(II) oleate, $\text{Cu}(\text{C}_{18}\text{H}_{33}\text{O}_2)_2$.
25. Determine the percent, by mass, of the indicated element:
 (a) Pb in tetraethyl lead, $\text{Pb}(\text{C}_2\text{H}_5)_4$, once extensively used as an additive to gasoline to prevent engine knocking
 (b) Fe in Prussian blue, $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$, a pigment used in paints and printing inks
 (c) Mg in chlorophyll, $\text{C}_{55}\text{H}_{72}\text{MgN}_4\text{O}_5$, the green pigment in plant cells
26. All of the following minerals are semiprecious or precious stones. Determine the mass percent of the indicated element.
 (a) Zr in zircon, ZrSiO_4
 (b) Be in beryl (emerald), $\text{Be}_3\text{Al}_2\text{Si}_6\text{O}_{18}$
 (c) Fe in almandine (garnet), $\text{Fe}_3\text{Al}_2\text{Si}_3\text{O}_{12}$
 (d) S in lazurite (lapis lazuli), $\text{Na}_4\text{S}_3\text{Si}_3\text{Al}_3\text{O}_{12}$
27. *Without doing detailed calculations*, arrange the following in order of increasing % Cr, by mass, and explain your reasoning: CrO , Cr_2O_3 , CrO_2 , CrO_3 .
28. *Without doing detailed calculations*, explain which of the following has the greatest mass percent of sulfur: SO_2 , S_2Cl_2 , Na_2S , $\text{Na}_2\text{S}_2\text{O}_3$, or $\text{CH}_3\text{CH}_2\text{SH}$.

Chemical Formulas from Percent Composition

29. Two oxides of sulfur have nearly identical molecular masses. One oxide consists of 40.05% S. What are the simplest possible formulas for the two oxides?
30. An oxide of chromium used in chrome plating has a formula mass of 100.0 u and contains *four* atoms per formula unit. Establish the formula of this compound, *with a minimum of calculation*.
31. Diethylene glycol, used to de-ice aircraft, is a carbon–hydrogen–oxygen compound with 45.27% C and 9.50% H by mass. What is its empirical formula?
32. The food flavor enhancer monosodium glutamate (MSG) has the composition 13.6% Na, 35.5% C, 4.8% H, 8.3% N, 37.8% O, by mass. What is the empirical formula of MSG?
33. Determine the empirical formula of (a) the rodenticide (rat killer) warfarin, which consists of 74.01% C, 5.23% H, and 20.76% O, by mass; (b) the antibacterial agent sulfamethizole, which consists of 39.98% C, 3.73% H, 20.73% N, 11.84% O, and 23.72% S, by mass.
34. Determine the empirical formula of (a) benzo[*a*]pyrene, a suspected carcinogen found in cigarette smoke, consisting of 95.21% C and 4.79% H, by mass; (b) hexachlorophene, used in germicidal soaps, which consists of 38.37% C, 1.49% H, 52.28% Cl, and 7.86% O by mass.
35. A compound of carbon and hydrogen consists of 94.34% C and 5.66% H, by mass. The molecular mass of the compound is found to be 178 u. What is its molecular formula?
36. Selenium, an element used in the manufacture of photoelectric cells and solar energy devices, forms two oxides. One has 28.8% O, by mass, and the other, 37.8% O. What are the formulas of these oxides? Propose acceptable names for them.
37. Indigo, the dye for blue jeans, has a percent composition, by mass, of 73.27% C, 3.84% H, 10.68% N, and the remainder is oxygen. The molecular mass of indigo is 262.3 u. What is the molecular formula of indigo?
38. Adenine, a component of nucleic acids, has the mass percent composition: 44.45% C, 3.73% H, 51.82% N. Its molecular mass is 135.14 u. What is its molecular formula?
39. The element X forms the chloride XCl_4 containing 75.0% Cl, by mass. What is element X?
40. The element X forms the compound XOCl_2 containing 59.6% Cl. What is element X?
41. Chlorophyll contains 2.72% Mg by mass. Assuming one Mg atom per chlorophyll molecule, what is the molecular mass of chlorophyll?
42. Two compounds of Cl and X are found to have molecular masses and % Cl, by mass, as follows: 137 u, 77.5% Cl; 208 u, 85.1% Cl. What is element X? What is the formula for each compound?

Combustion Analysis

43. A 0.1888 g sample of a hydrocarbon produces 0.6260 g CO_2 and 0.1602 g H_2O in combustion analysis. Its molecular mass is found to be 106 u. For this hydrocarbon, determine its (a) mass percent composition; (b) empirical formula; (c) molecular formula.
44. *Para*-cresol (*p*-cresol) is used as a disinfectant and in the manufacture of herbicides. A 0.4039 g sample of this carbon–hydrogen–oxygen compound yields 1.1518 g CO_2 and 0.2694 g H_2O in combustion analysis. Its molecular mass is 108.1 u. For *p*-cresol, determine its (a) mass percent composition; (b) empirical formula; (c) molecular formula.
45. Dimethylhydrazine is a carbon–hydrogen–nitrogen compound used in rocket fuels. When burned in an excess of oxygen, a 0.312 g sample yields 0.458 g CO_2 and 0.374 g H_2O . The nitrogen content of a 0.486 g sample is converted to 0.226 g N_2 . What is the empirical formula of dimethylhydrazine?
46. The organic solvent thiophene is a carbon–hydrogen–sulfur compound that yields CO_2 , H_2O , and SO_2 when burned in an excess of oxygen. When subjected to combustion analysis, a 1.3020 g sample of thiophene produces 2.7224 g CO_2 , 0.5575 g H_2O , and 0.9915 g SO_2 . What is the empirical formula of thiophene?
47. *Without doing detailed calculations*, explain which of these compounds produces the *greatest* mass of CO_2 when 1.00 mol of the compound is burned in an excess of oxygen: CH_4 , $\text{C}_2\text{H}_5\text{OH}$, C_{10}H_8 , $\text{C}_6\text{H}_5\text{OH}$.
48. *Without doing detailed calculations*, explain which of these compounds produces the *greatest* mass of H_2O when 1.00 g of the compound is burned in an excess of oxygen: CH_4 , $\text{C}_2\text{H}_5\text{OH}$, C_{10}H_8 , $\text{C}_6\text{H}_5\text{OH}$.
49. A 1.562 g sample of the alcohol $\text{CH}_3\text{CHOHCH}_2\text{CH}_3$ is burned in an excess of oxygen. What masses of CO_2 and H_2O should be obtained?
50. Liquid ethyl mercaptan, $\text{C}_2\text{H}_6\text{S}$, has a density of 0.84 g/mL. Assuming that the combustion of this compound produces only CO_2 , H_2O , and SO_2 , what masses of each of these three products would be produced in the combustion of 3.15 mL of ethyl mercaptan?

Oxidation States

51. Indicate the oxidation state of the underlined element in (a) $\underline{\text{C}}\text{H}_4$; (b) $\underline{\text{S}}\text{F}_4$; (c) $\text{Na}_2\underline{\text{O}}_2$; (d) $\underline{\text{C}}_2\text{H}_3\underline{\text{O}}_2^-$; (e) $\underline{\text{Fe}}\text{O}_4^{2-}$.
52. Indicate the oxidation state of S in (a) SO_3^{2-} ; (b) $\text{S}_2\text{O}_3^{2-}$; (c) $\text{S}_2\text{O}_8^{2-}$; (d) HSO_4^- ; (e) $\text{S}_4\text{O}_6^{2-}$.
53. Chromium forms three principal oxides. Write appropriate formulas for these compounds in which the oxidation states of Cr are +3, +4, and +6, respectively.
54. Nitrogen forms five oxides in which its oxidation states are +1, +2, +3, +4, and +5, respectively. Write appropriate formulas for these compounds.

55. In many of its compounds, oxygen has an oxidation state of -2 . However, there are exceptions. What is the oxidation state of oxygen in each of the following compounds? (a) OF_2 ; (b) O_2F_2 ; (c) CsO_2 ; (d) BaO_2 .
56. Hydrogen and oxygen usually have oxidation states of $+1$ and -2 , respectively, in their compounds.

Nomenclature

57. Name these compounds: (a) SrO ; (b) ZnS ; (c) K_2CrO_4 ; (d) Cs_2SO_4 ; (e) Cr_2O_3 ; (f) $\text{Fe}_2(\text{SO}_4)_3$; (g) $\text{Mg}(\text{HCO}_3)_2$; (h) $(\text{NH}_4)_2\text{HPO}_4$; (i) $\text{Ca}(\text{HSO}_3)_2$; (j) $\text{Cu}(\text{OH})_2$; (k) HNO_3 ; (l) KClO_4 ; (m) HBrO_3 ; (n) H_3PO_3 .
58. Name these compounds: (a) $\text{Ba}(\text{NO}_3)_2$; (b) HNO_2 ; (c) CrO_2 ; (d) KIO_3 ; (e) LiCN ; (f) KIO ; (g) $\text{Fe}(\text{OH})_2$; (h) $\text{Ca}(\text{H}_2\text{PO}_4)_2$; (i) H_3PO_4 ; (j) NaHSO_4 ; (k) $\text{Na}_2\text{Cr}_2\text{O}_7$; (l) $\text{NH}_4\text{C}_2\text{H}_3\text{O}_2$; (m) MgC_2O_4 ; (n) $\text{Na}_2\text{C}_2\text{O}_4$.
59. Assign suitable names to the compounds (a) CS_2 ; (b) SiF_4 ; (c) ClF_5 ; (d) N_2O_5 ; (e) SF_6 ; (f) I_2Cl_6 .
60. Assign suitable names to the compounds (a) ICl ; (b) ClF_3 ; (c) SiF_4 ; (d) PF_5 ; (e) NO_2 ; (f) S_4N_4 .
61. Write formulas for the compounds: (a) aluminum sulfate; (b) ammonium dichromate; (c) silicon tetrafluoride; (d) iron(III) oxide; (e) tricarbon disulfide; (f) cobalt(II) nitrate; (g) strontium nitrite; (h) hydrobromic acid; (i) iodic acid; (j) phosphorus dichloride trifluoride.
62. Write formulas for the compounds: (a) magnesium perchlorate; (b) lead(II) acetate; (c) tin(IV) oxide; (d) hydroiodic acid; (e) chlorous acid; (f) sodium hydrogen sulfite; (g) calcium dihydrogen phosphate; (h) aluminum phosphate; (i) dinitrogen tetroxide; (j) disulfur dichloride.

The following cases serve to remind us that there are exceptions. What are the oxidation states of the atoms in each of the following compounds? (a) MgH_2 ; (b) CsO_3 ; (c) HOF ; (d) NaAlH_4 .

63. Write a formula for (a) the chloride of titanium having Ti in the O.S. $+4$; (b) the sulfate of iron having Fe in the O.S. $+3$; (c) an oxide of chlorine with Cl in the O.S. $+7$; (d) an oxoanion of sulfur in which the apparent O.S. of S is $+7$ and the ionic charge is -2 .
64. Write a formula for (a) an oxide of nitrogen with N in the O.S. $+5$; (b) an oxoacid of nitrogen with N in the O.S. $+3$; (c) an oxide of carbon in which the apparent O.S. of C is $+4/3$; (d) a sulfur-containing oxoanion in which the apparent O.S. of S is $+2.5$; and the ionic charge is -2 .
65. Name the acids: (a) HClO_2 ; (b) H_2SO_3 ; (c) H_2Se ; (d) HNO_2 .
66. Supply the formula for the acids: (a) hydrofluoric acid; (b) nitric acid; (c) phosphorous acid; (d) sulfuric acid.
67. Name the following compounds and specify which ones are best described as ionic: (a) OF_2 ; (b) XeF_2 ; (c) CuSO_3 ; (d) $(\text{NH}_4)_2\text{HPO}_4$.
68. Name the following compounds and specify which ones are best described as ionic: (a) KNO_2 ; (b) BrF_3 ; (c) S_2Cl_2 ; (d) $\text{Mg}(\text{ClO})_2$; (e) Cl_2O .

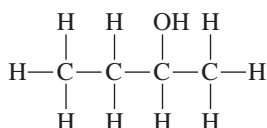
Hydrates

69. Without performing detailed calculations, indicate which of the following hydrates has the greatest % H_2O by mass: $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, $\text{Cr}_2(\text{SO}_4)_3 \cdot 18\text{H}_2\text{O}$, $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$, and $\text{LiC}_2\text{H}_3\text{O}_2 \cdot 2\text{H}_2\text{O}$.
70. Without performing detailed calculations, determine the hydrate of Na_2SO_3 that contains almost exactly 50% H_2O , by mass.
71. Anhydrous CuSO_4 can be used to dry liquids in which it is insoluble. The CuSO_4 is converted to $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, which can be filtered off from the liquid. What is the minimum mass of anhydrous CuSO_4 needed to remove 12.6 g H_2O from a tankful of gasoline?

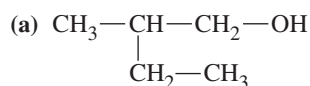
72. Anhydrous sodium sulfate, Na_2SO_4 , absorbs water vapor and is converted to the decahydrate, $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$. How much would the mass of 36.15 g of anhydrous Na_2SO_4 increase if converted completely to the decahydrate?
73. A certain hydrate is found to have the composition 20.3% Cu, 8.95% Si, 36.3% F, and 34.5% H_2O by mass. What is the empirical formula of this hydrate?
74. An 8.129 g sample of $\text{MgSO}_4 \cdot x\text{H}_2\text{O}$ is heated until all the water of hydration is driven off. The resulting anhydrous compound, MgSO_4 , weighs 3.967 g. What is the formula of the hydrate?

Organic Compounds and Organic Nomenclature

75. Which of the following names is most appropriate for the molecule with the structure shown below? (a) butyl alcohol; (b) butan-2-ol; (c) butan-1-ol; (d) isopentyl alcohol.



76. Which of the following names is most appropriate for the molecule $\text{CH}_3(\text{CH}_2)_2\text{COOH}$? (a) dimethyleneacetic acid; (b) propanoic acid; (c) butanoic acid; (d) oxobutylalcohol.
77. Which of the following structures are isomers?



89. Is it possible to have a sample of S_8 that weighs 1.00×10^{-23} g? What is the smallest possible mass that a sample of S_8 can have? Express your answer to the second question in appropriate SI units so that your answer has a numerical value greater than 1. (See Table 1.2 for a list of SI prefixes.)

90. What is the molecular formula of a hydrocarbon containing n carbon atoms and only one double bond? Can such a hydrocarbon yield a greater mass of H_2O than CO_2 when burned in an excess of oxygen?
91. A hydrocarbon mixture consists of 60.0% by mass of C_3H_8 and 40.0% of C_xH_y . When 10.0 g of this mixture is burned, 29.0 g CO_2 and 18.8 g H_2O are the only products. What is the formula of the unknown hydrocarbon?
92. A 0.732 g mixture of methane, CH_4 , and ethane, C_2H_6 , is burned, yielding 2.064 g CO_2 . What is the percent composition of this mixture (a) by mass; (b) on a mole basis?
93. The density of a mixture of H_2SO_4 and water is 1.78 g/mL. The percent composition of the mixture is to be determined by converting H_2SO_4 to $(\text{NH}_4)_2\text{SO}_4$. If 32.0 mL of the mixture gives 65.2 g $(\text{NH}_4)_2\text{SO}_4$, then what is the percent composition of the mixture?
94. In 2013, the IUPAC recommended that the atomic masses of 12 elements be expressed as an atomic mass interval rather than as a single invariant value. (See Section 2-5 and Table 2.2.) For example, the IUPAC recommends that the atomic mass of Cl be given as [35.446, 35.457]. Consequently, the results of calculations involving the atomic mass of chlorine should, in principle, be reported as a range of values. Demonstrate this approach by calculating the range of values possible for the mass percent of silver in an impure sample if all the silver in a 26.39 g sample is converted to 31.56 g of silver chloride. [Hint: Perform two calculations, using first the lower bound and then the upper bound of the atomic mass interval of Cl.]
95. In the year 2000, the *Guinness Book of World Records* called ethyl mercaptan, $\text{C}_2\text{H}_6\text{S}$, the smelliest substance known. The average person can detect its presence in air at levels as low as $9 \times 10^{-4} \mu\text{mol}/\text{m}^3$. Express the limit of detectability of ethyl mercaptan in parts per billion (ppb). (Note: 1 ppb $\text{C}_2\text{H}_6\text{S}$ means there is 1 g $\text{C}_2\text{H}_6\text{S}$ per billion grams of air.) The density of air is approximately 1.2 g/L at room temperature.
96. Dry air is essentially a mixture of the following entities: N_2 , O_2 , Ar, and CO_2 . The composition of dry air, in mole percent, is 78.08% N_2 , 20.95% O_2 , 0.93% Ar, and 0.04% CO_2 . (a) What is the mass, in grams, of a sample of air that contains exactly one mole of the entities? (b) Dry air also contains other entities in much smaller amounts. For example, the mole percent of krypton (Kr) is about $1.14 \times 10^{-4}\%$. Given that the density of dry air is about 1.2 g/L at room temperature, what mass of krypton could be obtained from exactly one cubic meter of dry air?
97. The mineral jade has the formula $\text{NaAlSi}_3\text{O}_6$. (a) What is the molar mass of jade to five significant figures? (b) What is the mass percent of silicon in jade? (c) How many silicon atoms are in 10.00 g of jade?
98. A sample of the compound MSO_4 weighing 0.1131 g reacts with barium chloride and yields 0.2193 g BaSO_4 . What must be the atomic mass of the metal M? [Hint: All the SO_4^{2-} from the MSO_4 appears in the BaSO_4 .]
99. The metal M forms the sulfate $\text{M}_2(\text{SO}_4)_3$. A 0.738 g sample of this sulfate is converted to 1.511 g BaSO_4 . What is the atomic mass of M? [Hint: Refer to Exercise 98.]
100. A 0.622 g sample of a metal oxide with the formula M_2O_3 is converted to 0.685 g of the sulfide, MS . What is the atomic mass of the metal M?
101. MgCl_2 often occurs in table salt (NaCl) and is responsible for caking of the salt. A 0.5200 g sample of table salt is found to contain 61.10% Cl, by mass. What is the % MgCl_2 in the sample? Why is the precision of this calculation so poor?
102. When 2.750 g of the oxide of lead Pb_3O_4 is strongly heated, it decomposes and produces 0.0640 g of oxygen gas and 2.686 g of a second oxide of lead. What is the empirical formula of this second oxide?
103. A 1.013 g sample of $\text{ZnSO}_4 \cdot x \text{H}_2\text{O}$ is dissolved in water and the sulfate ion precipitated as BaSO_4 . The mass of pure, dry BaSO_4 obtained is 0.8223 g. What is the formula of the zinc sulfate hydrate?
104. The iodide ion in a 1.552 g sample of the ionic compound MI is removed through precipitation. The precipitate is found to contain 1.186 g I. What is the element M?
105. An oxoacid with the formula $\text{H}_x\text{E}_y\text{O}_z$ has a formula mass of 178 u, has 13 atoms in its formula unit, contains 34.80% by mass, and 15.38% by number of atoms, of the element E. What is the element E, and what is the formula of this oxoacid?
106. The insecticide dieldrin contains carbon, hydrogen, oxygen, and chlorine. When burned in an excess of oxygen, a 1.510 g sample yields 2.094 g CO_2 and 0.286 g H_2O . The compound has a molecular mass of 381 u and has half as many chlorine atoms as carbon atoms. What is the molecular formula of dieldrin?
107. A thoroughly dried 1.271 g sample of Na_2SO_4 is exposed to the atmosphere and found to gain 0.387 g in mass. What is the percent, by mass, of $\text{Na}_2\text{SO}_4 \cdot 10 \text{H}_2\text{O}$ in the resulting mixture of anhydrous Na_2SO_4 and the decahydrate?
108. The atomic mass of Bi is to be determined by converting the compound $\text{Bi}(\text{C}_6\text{H}_5)_3$ to Bi_2O_3 . If 5.610 g of $\text{Bi}(\text{C}_6\text{H}_5)_3$ yields 2.969 g Bi_2O_3 , what is the atomic mass of Bi?
109. A piece of gold (Au) foil measuring 0.25 mm $\times$ 15 mm $\times$ 15 mm is treated with fluorine gas. The treatment converts all the gold in the foil to 1.400 g of a gold fluoride. What is the formula and name of the fluoride? The density of gold is 19.3 g/cm³.
110. In an experiment, 244 mL of chlorine gas (Cl_2 , $d = 2.898 \text{ g/L}$) combines with iodine to give 1.553 g of a binary compound. In a separate experiment, the molar mass of the compound is found to be about 467 g/mol. What is the molecular formula of this compound?
111. Placing a 0.725 g copper strip in the presence of iodine vapor produced a yellowish-white coating on the metal strip. The mass of the copper strip and coating was 0.733 g. The coating was removed by rinsing the coated metal strip in a potassium thiocyanate (KSCN) solution, yielding a clean copper strip of mass 0.721 g. What is the empirical formula of the yellowish-white compound?

Feature Problems

- 112.** All-purpose fertilizers contain the essential elements nitrogen, phosphorus, and potassium. A typical fertilizer carries numbers on its label, such as “5-10-5”. These numbers represent the % N, % P_2O_5 , and % K_2O , respectively. The N is contained in the form of a nitrogen compound, such as $(NH_4)_2SO_4$, NH_4NO_3 , or $CO(NH_2)_2$ (urea). The P is generally present as a phosphate, and the K as KCl. The expressions % P_2O_5 and % K_2O were devised in the nineteenth century, before the nature of chemical compounds was fully understood. To convert from % P_2O_5 to % P and from % K_2O to % K, the factors 2 mol P/mol P_2O_5 and 2 mol K/mol K_2O must be used, together with molar masses.
- (a) Assuming three-significant-figure precision, what is the percent composition of the “5-10-5” fertilizer in % N, % P, and % K?
- (b) What is the % P_2O_5 in the following compounds (both common fertilizers)? (i) $Ca(H_2PO_4)_2$; (ii) $(NH_4)_2HPO_4$.
- (c) In a similar manner to the “5-10-5” fertilizer described in this exercise, how would you describe a fertilizer in which the mass ratio of $(NH_4)_2HPO_4$ to KCl is 5.00:1.00?
- (d) Can a “5-10-5” fertilizer be prepared in which $(NH_4)_2HPO_4$ and KCl are the sole fertilizer components, with or without inert nonfertilizer additives? If so, what should be the proportions of the constituents of the fertilizer mixture? If this “5-10-5” fertilizer cannot be prepared, why not?
- 113.** A hydrate of copper(II) sulfate, when heated, goes through the succession of changes suggested by the photograph. In this photograph, (a) is the original fully hydrated copper(II) sulfate; (b) is the product obtained by heating the original hydrate to 140°C; (c) is the product obtained by further heating to 400°C; and (d) is the product obtained at 1000°C.



(a) (b) (c) (d)

A 2.574 g sample of $CuSO_4 \cdot x H_2O$ was heated to 140°C, cooled, and reweighed. The resulting solid was reheated to 400°C, cooled, and reweighed. Finally, this solid was heated to 1000°C, cooled, and reweighed for the last time.

Original sample	2.574 g
After heating to 140°C	1.833 g
After reheating to 400°C	1.647 g
After reheating to 1000°C	0.812 g

- (a) Assuming that all the water of hydration is driven off at 400°C, what is the formula of the original hydrate?
- (b) What is the formula of the hydrate obtained when the original hydrate is heated to only 140°C?
- (c) The black residue obtained at 1000°C is an oxide of copper. What is its percent composition and empirical formula?

- 114.** Some substances that are only very slightly soluble in water will spread over the surface of water to produce a film that is called a *monolayer* because it is only one molecule thick. A practical use of this phenomenon is to cover ponds to reduce the loss of water by evaporation. Stearic acid forms a monolayer on water. The molecules are arranged upright and in contact with one another, rather like pencils tightly packed and standing upright in a coffee mug. The model below represents an individual stearic acid molecule in the monolayer.

(a) How many square meters of water surface would be covered by a monolayer made from 10.0 g of stearic acid?

[Hint: What is the formula of stearic acid?]

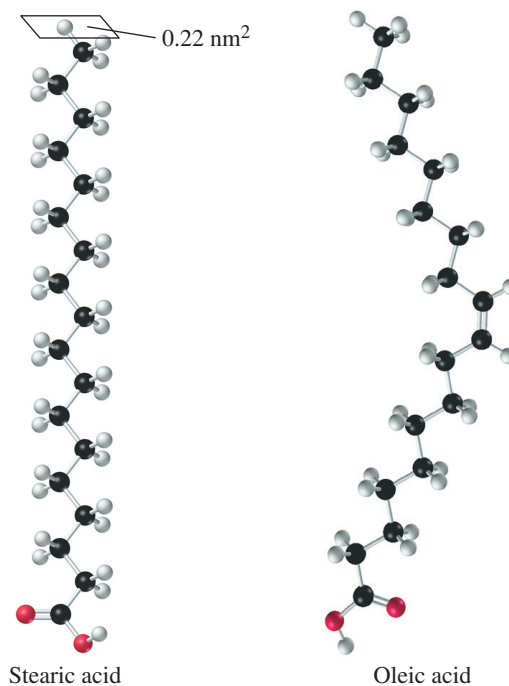
(b) If stearic acid has a density of 0.85 g/cm³, estimate the length (in nanometers) of a stearic acid molecule.

[Hint: What is the thickness of the monolayer described in part (a)?]

(c) A very dilute solution of oleic acid in liquid pentane is prepared in the following way:

- 1.00 mL oleic acid + 9.00 mL pentane → solution (1);
 1.00 mL solution (1) + 9.00 mL pentane → solution (2);
 1.00 mL solution (2) + 9.00 mL pentane → solution (3);
 1.00 mL solution (3) + 9.00 mL pentane → solution (4);

A 0.10 mL sample of solution (4) is spread in a monolayer on water. The area covered by the monolayer is 85 cm². Assume that oleic acid molecules are arranged in the same way as described for stearic acid, and that the cross-sectional area of the molecule is 4.6×10^{-15} cm². The density of oleic acid is 0.895 g/mL. Use these data to obtain an approximate value of Avogadro's number.



Self-Assessment Exercises

115. The mass, in grams, of one atomic mass unit (amu) is 1.66054×10^{-24} g. How many atomic mass units are there in 1.000 gram?
116. What is the mass (in grams) of one molecule of propylamine, $\text{C}_3\text{H}_7\text{NH}_2$?
117. Caffeine has a molecular mass of 194.19 amu and contains 28.85% N by mass. How many nitrogen atoms are in each caffeine molecule?
118. A piece of pure iron weighing 2.790 g is strongly heated in a stream of oxygen until all of it is converted to a red iron oxide. The resultant red powder weighs 3.990 g. What is the formula of the iron oxide?
119. Which answer is correct? One mole of liquid bromine, Br_2 , (a) has a mass of 79.9 g; (b) contains 6.022×10^{23} Br atoms; (c) contains the same number of atoms as in 12.01 g H_2O ; (d) has twice the mass of 0.500 mole of gaseous Cl_2 .
120. Three of the following formulas might be either an empirical or a molecular formula. The formula that must be a molecular formula is (a) N_2O ; (b) N_2H_4 ; (c) NaCl ; (d) NH_3 .
121. The compound $\text{C}_7\text{H}_7\text{NO}_2$ contains (a) 17 atoms per mole; (b) equal percents by mass of C and H; (c) about twice the percent by mass of O as of N; (d) about twice the percent by mass of N as of H.
122. The greatest number of N atoms is found in (a) 50.0 g N_2O ; (b) 17.0 g NH_3 ; (c) 150 mL of liquid pyridine, $\text{C}_5\text{H}_5\text{N}$ ($d = 0.983$ g/mL); (d) 1.0 mol N_2 .
123. Iron is present in red blood cells and acts to carry oxygen to the organs. Without oxygen, these organs will die. There are about 2.6×10^{13} red blood cells in the blood of an adult human, and the blood contains a total of 2.9 g of iron. How many atoms are there in each blood cell?
124. XF_3 consists of 65% F by mass. The atomic mass of the element X must be (a) 8 u; (b) 11 u; (c) 31 u; (d) 35 u.
125. The oxidation state of I in the ion H_4IO_6^- is (a) -1 ; (b) $+1$; (c) $+7$; (d) $+8$.
126. The oxidation state of Mn in MgMnO_4 is (a) $+2$; (b) $+7$; (c) $+6$; (d) $+4$; (e) $+3$.
127. The name of which compound ends with *-ate*? (a) HIO_4 ; (b) Na_2SO_3 ; (c) KClO_2 ; (d) HFO ; (e) NO_2 .
128. The name of $\text{Sr}(\text{HCO}_3)_2$ is (a) strontium oxalate; (b) strontium carbonate; (c) sodium bicarbonate; (d) strontium bicarbonate; (e) none of these.
129. The formula for calcium chlorite is (a) CaClO_2 ; (b) $\text{Ca}(\text{ClO}_2)_2$; (c) CaClO_3 ; (d) $\text{Ca}(\text{ClO}_4)_2$.
130. Which compound has a molar mass of 51.79 g mol^{-1} ? (a) NaCl ; (b) KF ; (c) MgS ; (d) Li_3P ; (e) none of these.
131. A formula unit of the compound $[\text{Cu}(\text{NH}_3)_4]\text{SO}_4$ has nearly equal masses of (a) S and O; (b) N and O; (c) H and N; (d) Cu and O.
132. An isomer of the compound $\text{CH}_3\text{CH}_2\text{CHOHCH}_3$ is (a) $\text{C}_4\text{H}_{10}\text{O}$; (b) $\text{CH}_3\text{CHOHCH}_2\text{CH}_3$; (c) $\text{CH}_3(\text{CH}_2)_2\text{OH}$; (d) $\text{CH}_3\text{CH}_2\text{OCH}_2\text{CH}_3$.
133. A hydrate of Na_2SO_3 contains almost exactly 50% H_2O by mass. What is the formula of this hydrate?
134. Malachite is a common copper-containing mineral with the formula $\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$. (a) What is the mass percent copper in malachite? (b) When malachite is strongly heated, carbon dioxide and water are driven off, yielding copper(II) oxide as the sole product. What mass of copper(II) oxide is produced per kg of malachite?
135. Acetaminophen, an analgesic and antipyretic drug, has a molecular mass of 151.2 u and a mass percent composition of 63.56% C, 6.00% H, 9.27% N, and 21.17% O. What is the molecular formula of acetaminophen?
136. Ibuprofen is a compound used in painkillers. When a 2.174 g sample is burned in an excess of oxygen, it yields 6.029 g CO_2 and 1.709 g H_2O as the sole products. (a) What is the percent composition, by mass, of ibuprofen? (b) What is the empirical formula of ibuprofen?
137. Appendix E describes a useful study aid known as concept mapping. Using the method presented in Appendix E, construct a concept map illustrating the different concepts in Sections 3-2 and 3-3.

Chemical Reactions

4

We are all aware that iron rusts and natural gas burns. These processes are chemical reactions. Chemical reactions are the central concern not just of this chapter but of the entire science of chemistry. In this chapter, we will establish quantitative (numerical) relationships among the substances involved in a reaction, a topic known as *reaction stoichiometry*. Because many chemical reactions occur in solution, we will also consider *solution stoichiometry* and introduce a method of describing the composition of a solution called *solution molarity*.

In describing chemical reactions, we often take a microscopic view and focus on the entities—atoms, ions, or molecules—that make up the substances involved. However, when *doing* chemistry, we often think of reactions in more macroscopic terms because, in the laboratory, we handle quantities of substances—grams or liters—that can be easily measured or manipulated. In large part, reaction stoichiometry provides the relationships we need to relate macroscopic amounts of substances to our microscopic view of chemical reactions.

To some, stoichiometry is no more exciting than the law of conservation of mass, but make no mistake—stoichiometry is important. Chemists use stoichiometric principles routinely to plan experiments, analyze their results, and make predictions, all of which contribute to making new discoveries and

Contents

- 4-1 Chemical Reactions and Chemical Equations
- 4-2 Chemical Equations and Stoichiometry
- 4-3 Chemical Reactions in Solution
- 4-4 Determining the Limiting Reactant
- 4-5 Other Practical Matters in Reaction Stoichiometry
- 4-6 The Extent of Reaction

Learning Objectives

- 4.1 Write a balanced chemical equation for a chemical reaction, specifying states of matter or reaction conditions, as appropriate.
- 4.2 Use the methodology of converting to, between, and from moles to solve stoichiometry problems.
- 4.3 Determine the molarity of a solution from its measured quantities, and determine the volume of solution used in a solution dilution or a chemical reaction.
- 4.4 Define the terms *limiting* and *excess reactant*, and describe how to determine which reactant is the limiting one in a chemical reaction.
- 4.5 Determine the theoretical and percent yield of a given reaction, and distinguish between consecutive and simultaneous reactions.
- 4.6 Express the changes in amount for reactions that occur to a limited extent in terms of the stoichiometric numbers and the extent of reaction.



ABACAPRESS, Abaca Press/Alamy Stock Photo

▲ The space shuttle *Discovery* lifts off on mission STS-26. Combustion reactions in the solid-fuel rocket engines provide the thrust to lift the shuttle off the launch pad. In this chapter, we learn to write and use balanced chemical equations for a wide variety of chemical reactions, including combustion reactions.

expanding our knowledge of the microscopic world of atoms, molecules, and ions.

In this chapter, we will first learn to represent chemical reactions by chemical equations, and then we will use chemical equations—and ideas from earlier chapters—to establish the quantitative relationships we seek. Throughout the chapter, we will discuss new aspects of problem solving and more uses for the mole concept.

4-1 Chemical Reactions and Chemical Equations



Richard Megna/Fundamental Photographs, NYC

Figure 4-1 Precipitation of silver chromate

When aqueous solutions of silver nitrate and potassium chromate are mixed, the disappearance of the distinctive yellow color of chromate ion and the appearance of the red-brown solid, silver chromate, provide physical evidence of a reaction.

A **chemical reaction** is a process in which one set of substances, called **reactants**, is converted to a new set of substances, called **products**. In other words, a chemical reaction is the process by which a chemical change occurs. In many cases, though, nothing happens when substances are mixed; each retains its original composition and properties. We need evidence before we can say that a reaction has occurred. Some of the types of physical evidence to look for are shown here:

- a color change (Fig. 4-1)
- formation of a solid (precipitate) within a clear solution (Fig. 4-1)
- evolution of a gas (Fig. 4-2a)
- evolution or absorption of heat (Fig. 4-2b)

Although such observations as these usually signify that a reaction has occurred, conclusive evidence still requires a detailed chemical analysis of the reaction mixture to identify all the substances present. Moreover, a chemical analysis may reveal that a chemical reaction has occurred even in the absence of obvious physical signs.

Just as there are symbols for elements and formulas for compounds, there is a symbolic, or shorthand, way of representing a chemical reaction—the **chemical equation**. In a chemical equation, formulas for the reactants are written on the left side of the equation and formulas for the products are written on the right. The two sides of the equation are joined by an arrow ($\longrightarrow$). We say that the reactants *yield* the products. Consider the reaction of colorless nitrogen monoxide and oxygen gases to form red-brown nitrogen dioxide gas, a reaction that occurs in the manufacture of nitric acid.



(a) Richard Megna/Fundamental Photographs, NYC

(a)



(b) Tom Bochsler/Pearson Education

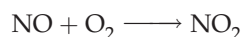
(b)

Figure 4-2 Evidence of a chemical reaction

(a) Evolution of a gas: When a copper penny reacts with nitric acid, the red-brown gas NO_2 is evolved. (b) Evolution of heat: When iron gauze (steel wool) is ignited in an oxygen atmosphere, evolved heat and light provide physical evidence of a reaction.

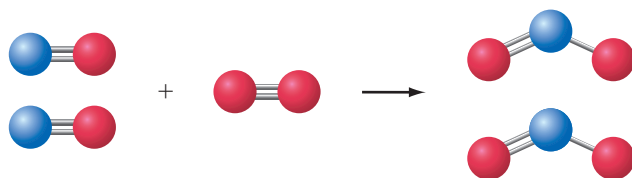
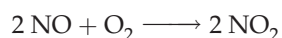
To complete the shorthand representation of this reaction, we must do two things:

1. Substitute chemical formulas for names, to obtain the following expression.



In this expression, there are three O atoms on the left side (one in the molecule NO and two in the molecule O₂), but only two O atoms (in the molecule NO₂) on the right. Because atoms are neither created nor destroyed in a chemical reaction, this expression needs to be *balanced*.

2. Balance the numbers of atoms of each kind on both sides of the expression to obtain a balanced chemical equation.* In this step, the coefficient 2 is placed in front of the formulas NO and NO₂. This means that two molecules of NO are consumed and two molecules of NO₂ are produced for every molecule of O₂ consumed. In the balanced equation there are two N atoms and four O atoms on each side. In a **balanced equation**, the total number of atoms of each element present is the same on both sides of the equation. We see this below, both in the symbolic equation and in the molecular representation of the reaction.



The coefficients required to balance a chemical equation are called **stoichiometric coefficients**. These coefficients are essential in relating the amounts of reactants used and products formed in a chemical reaction, through a variety of calculations. In balancing a chemical equation, keep the following point in mind.

An equation can be balanced only by adjusting the coefficients of formulas.

The method of equation balancing described above is called *balancing by inspection*. Balancing by inspection means to adjust stoichiometric coefficients by trial and error until a balanced condition is found. Although the elements can generally be balanced in any order, equation balancing need not be a hit-or-miss affair. Here are some useful strategies for balancing equations.

- If an element occurs in only one compound on each side of the equation, try balancing this element first.
- When one of the reactants or products exists as the free element, balance this element last.
- In some reactions, certain groups of atoms (for example, polyatomic ions) remain unchanged. In such cases, balance these groups as a unit.
- It is permissible to use fractional as well as integral numbers as coefficients. At times, an equation can be balanced most easily by using one or more fractional coefficients and then, if desired, clearing the fractions by multiplying *all* coefficients by a common multiplier.

*An equation—whether mathematical or chemical—must have the left and right sides equal. We should not call an expression an equation until it is balanced. The term *chemical equation* automatically signifies that this balance exists. Although unnecessary, the term *balanced* is commonly used when referring to a chemical equation.

◀ Sometimes products react, partially or completely, to re-form the original reactants. Such reactions are called *reversible reactions* and are designated by a double arrow ($\rightleftharpoons$). In this chapter, we assume that any reverse reactions are negligible and that reactions go only in the forward direction. We consider reversible reactions in detail in Chapter 16.

◀ The reaction of NO and O₂ is of considerable environmental interest. Automobile engines—and all other combustion devices—make NO. One purpose of a catalytic converter is to reduce the amount of NO emitted from a tailpipe. Once in the atmosphere, NO is converted into brown NO₂ by the reaction illustrated on this page. This is why many urban atmospheres have a brown tinge.

◀ We will encounter a few situations in Chapters 7 and 20 where some fractional coefficients are actually required.

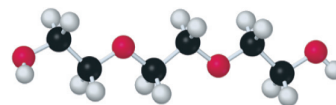
Keep in Mind

that the strategies described here work well for simple reactions. However, some reactions cannot be balanced by inspection and systematic methods must be employed. We will encounter such situations in Chapter 5.

In this chapter you should concentrate on learning to write formulas for the reactants and products of a reaction and to balance the equation representing the reaction. A third task, which you will face in later chapters, is to predict the products formed when certain reactants are brought together under the appropriate conditions. Even now, however, based on concepts presented in the previous chapter, you should be able to predict the products of a *combustion reaction*. In a plentiful supply of oxygen gas, the combustion of hydrocarbons and of carbon–hydrogen–oxygen compounds produces carbon dioxide and water as the only products. If the compound contains sulfur as well, sulfur dioxide will also be a product. These ideas and an equation-balancing strategy are illustrated in Example 4-1.

EXAMPLE 4-1**Writing and Balancing an Equation: The Combustion of a Carbon–Hydrogen–Oxygen Compound**

Liquid triethylene glycol is used as a solvent and plasticizer for vinyl and polyurethane plastics. Write a balanced chemical equation for the combustion of this compound in a plentiful supply of oxygen. A ball-and-stick model of triethylene glycol is shown here.



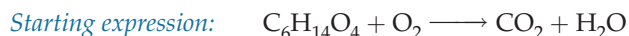
Triethylene glycol

Analyze

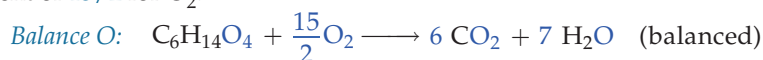
We deduce the formula of triethylene glycol from the molecular model. (Refer to the color scheme given on the inside back cover.) We see 6 C atoms (black), 4 O atoms (red), and 14 H atoms. The formula is $C_6H_{14}O_4$. When a carbon–hydrogen–oxygen compound is burned in excess oxygen, O_2 , the products are CO_2 and H_2O .

Solve

Having identified the reactants and products, we write down an unbalanced chemical expression for the reaction, showing all reactants and products, and then we balance the expression with respect to each kind of atom.



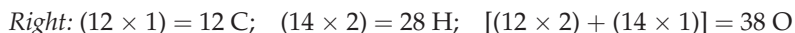
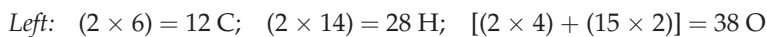
At this point, the right side of the expression has 19 O atoms (12 in six CO_2 molecules and 7 in seven H_2O molecules), and the left side, only 4 O atoms (in $C_6H_{14}O_4$). To obtain 15 more O atoms requires a fractional coefficient of $15/2$ for O_2 .



To remove the fractional coefficient, multiply all coefficients by two:

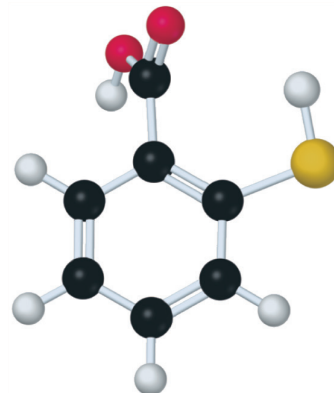
**Assess**

To check that the equation is balanced, determine the numbers of C, H, and O atoms that appear on the each side of the equation.



PRACTICE EXAMPLE A: Write a balanced equation to represent the reaction of mercury(II) sulfide and calcium oxide to produce calcium sulfide, calcium sulfate, and mercury metal.

PRACTICE EXAMPLE B: Write a balanced equation for the combustion of thiosalicylic acid, $C_7H_6O_2S$, used in the manufacture of indigo dyes. A ball-and-stick model of $C_7H_6O_2S$ is shown here.



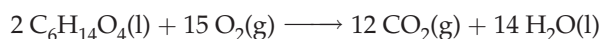
Thiosalicylic acid

States of Matter

In the reaction in Example 4-1, triethylene glycol is a liquid, oxygen and carbon dioxide are gases, and water is a liquid. Such facts as these are inconsequential if our interest is only in balancing an equation. Still, we convey a more complete representation of the reaction by including this information, and sometimes it is essential to include such information in a chemical equation. The state of matter or physical form of reactants and products is shown by symbols in parentheses.

(g) gas (l) liquid (s) solid

Thus, the equation for combustion of triethylene glycol can be written as



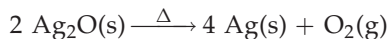
Another commonly used symbol for reactants or products dissolved in water is

(aq) *aqueous* solution

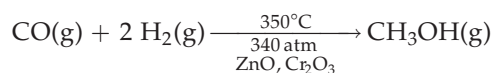
◀ At the high combustion temperature, water is present as $\text{H}_2\text{O}(\text{g})$. But when the reaction products are returned to the initial temperature, the water condenses to a liquid, $\text{H}_2\text{O}(\text{l})$.

Reaction Conditions

The equation for a chemical reaction does not provide enough information to enable you to carry out the reaction in a laboratory or chemical plant. An important aspect of modern chemical research involves working out the conditions for a reaction. The reaction conditions are often written above or below the arrow in an equation. For example, the Greek capital letter delta, Δ , means that a high temperature is required—that is, the reaction mixture must be heated, as in the decomposition of silver oxide.



An even more explicit statement of reaction conditions is shown below for the BASF (Badische Anilin & Soda-Fabrik) process for the synthesis of methanol from CO and H_2 . This reaction occurs at 350°C , under a total gas pressure that is 340 times as great as the normal pressure of the atmosphere, and on the surface of a mixture of ZnO and Cr_2O_3 acting as a *catalyst*. As we will learn later in the text, a catalyst is a substance that enters into a reaction in such a way that it speeds up the reaction without itself being consumed or changed by the reaction.



◀ In a *decomposition reaction*, a substance is broken down into simpler substances (for instance, into its elements).

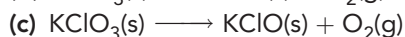
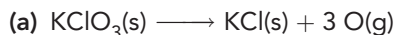
◀ In a *synthesis reaction*, a new compound is formed from the reaction of two or more simpler substances, usually called the reactants or starting materials.

It is important to be able to calculate how much of a particular product will be produced when certain quantities of the reactants are consumed. In the next section, we will see how to use chemical equations to set up conversion factors that we can use for these and related calculations.



4-1 Concept Assessment

For the elements K, Cl, and O in the following equations, the requirement that the number of atoms be equal on either side of the equation is met. Why is none of them an acceptable balanced equation for the decomposition of solid potassium chlorate yielding solid potassium chloride and oxygen gas?



4-2 Chemical Equations and Stoichiometry

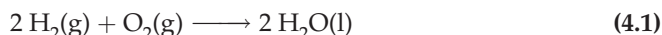
In Greek, the word *stoicheion* means element. The term **stoichiometry** (stoy-key-om'-eh-tree) means, literally, to measure the elements—but from a practical standpoint, it includes all the quantitative relationships involving atomic



▲ Jeremias Benjamin Richter (1762–1807) discovered that reactions always occurred in the same ratios. It is likely that Joseph Proust knew of his work when he proposed the law of definite proportions.

and formula masses, chemical formulas, and chemical equations. We considered the quantitative meaning of chemical formulas in Chapter 3, and now we will explore some additional quantitative aspects of chemical equations.

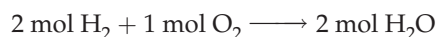
The coefficients in the chemical equation



mean that



Suppose we let $x = 6.02214 \times 10^{23}$ (Avogadro's number). Then x molecules represents *1 mole*. Thus the chemical equation also means that



The coefficients in the chemical equation allow us to make statements such as

- *Two moles of H₂O are produced for every two moles of H₂ consumed.*
- *Two moles of H₂O are produced for every one mole of O₂ consumed.*
- *Two moles of H₂ are consumed for every one mole of O₂ consumed.*

Moreover, we can turn such statements into conversion factors, called **stoichiometric factors**. A **stoichiometric factor** relates the amounts, on a mole basis, of any two substances involved in a chemical reaction; thus a stoichiometric factor is a mole ratio. In the following examples, stoichiometric factors are printed in blue.

EXAMPLE 4-2

Relating the Numbers of Moles of Reactant and Product

How many moles of CO₂ are produced in the combustion of 2.72 mol of triethylene glycol, C₆H₁₄O₄, in an excess of O₂?

Analyze

“An excess of O₂” means that there is more than enough O₂ available to permit the complete conversion of the triethylene glycol to CO₂ and H₂O. The factor for converting from moles of C₆H₁₄O₄ to moles of CO₂ is obtained from the balanced equation for the combustion reaction.

Solve

The first step in a stoichiometric calculation is to write a balanced equation for the reaction. The balanced chemical equation for the reaction is given below.



Thus, 12 mol CO₂ are produced for every 2 mol C₆H₁₄O₄ burned. The production of 12 mol CO₂ is equivalent to the consumption of 2 mol C₆H₁₄O₄; thus, the ratio 12 mol CO₂/2 mol C₆H₁₄O₄ converts from mol C₆H₁₄O₄ to mol CO₂.

$$? \text{ mol CO}_2 = 2.72 \text{ mol C}_6\text{H}_{14}\text{O}_4 \times \frac{12 \text{ mol CO}_2}{2 \text{ mol C}_6\text{H}_{14}\text{O}_4} = 16.3 \text{ mol CO}_2$$

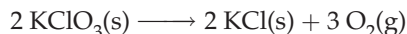
Assess

The expression above can be written in terms of two equal ratios:

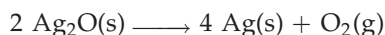
$$\frac{? \text{ mol CO}_2}{2.72 \text{ mol C}_6\text{H}_{14}\text{O}_4} = \frac{12 \text{ mol CO}_2}{2 \text{ mol C}_6\text{H}_{14}\text{O}_4}$$

You may find it easier to set up an expression in terms of ratios and then solve it for the unknown quantity.

PRACTICE EXAMPLE A: How many moles of O₂ are produced from the decomposition of 1.76 moles of potassium chlorate?



PRACTICE EXAMPLE B: How many moles of Ag are produced in the decomposition of 1.00 kg of silver oxide?



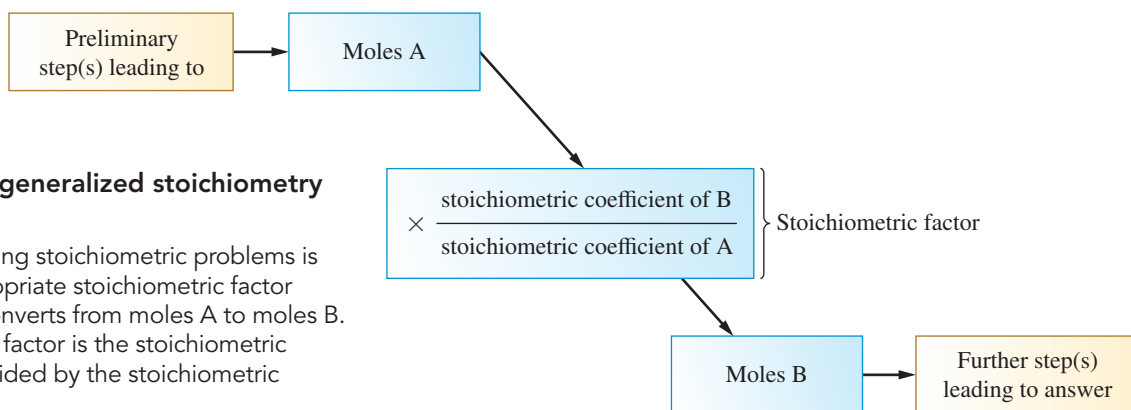


Figure 4-3 A generalized stoichiometry diagram

A key step in working stoichiometric problems is applying the appropriate stoichiometric factor (mole ratio) that converts from moles A to moles B. The stoichiometric factor is the stoichiometric coefficient of B divided by the stoichiometric coefficient of A.

Reaction stoichiometry problems range from relatively simple to complex, yet all can be solved by the same general strategy. This strategy, outlined in Figure 4-3, yields information about one substance, B, from information given about a second substance, A. Thus, the conversion pathway is from the *known* A to the *unknown* B. The heart of the calculation, shown in blue in Figure 4-3, is conversion from the number of moles of A to the number of moles of B using a factor based on stoichiometric coefficients from the balanced chemical equation—that is, (stoichiometric coefficient of B)/(stoichiometric coefficient of A). Example 4-2 was a single-step calculation requiring only the stoichiometric factor. Examples 4-3 through 4-6 require additional steps, both before and after the stoichiometric factor is employed. In short, this strategy involves the following conversions: “to moles,” “between moles,” and “from moles.” The conversions can be done separately or they can be combined into a single-step calculation, as shown in the following examples.

Keep in Mind

that it is important to include units and to work from a balanced chemical equation when solving stoichiometry problems.

EXAMPLE 4-3

Relating the Mass of a Reactant and a Product

What mass of CO_2 is formed in the reaction of 4.16 g triethylene glycol, $\text{C}_6\text{H}_{14}\text{O}_4$, with an excess of O_2 ?

Analyze

The general strategy involves the following conversions: (1) to moles, (2) between moles, and (3) from moles. In this example, the required conversions are $\text{g C}_6\text{H}_{14}\text{O}_4 \xrightarrow{1} \text{mol C}_6\text{H}_{14}\text{O}_4 \xrightarrow{2} \text{mol CO}_2 \xrightarrow{3} \text{g CO}_2$. Each numbered arrow refers to a conversion factor that changes the unit on the left to the one on the right.

Solve

The conversions can be carried out by using either a stepwise approach or the conversion pathway approach. Using a *stepwise approach*, we proceed as follows.

Convert from grams of $\text{C}_6\text{H}_{14}\text{O}_4$ to moles of $\text{C}_6\text{H}_{14}\text{O}_4$ by using the molar mass of $\text{C}_6\text{H}_{14}\text{O}_4$ as a conversion factor.

$$\begin{aligned} ? \text{ mol C}_6\text{H}_{14}\text{O}_4 &= 4.16 \text{ g C}_6\text{H}_{14}\text{O}_4 \times \frac{1 \text{ mol C}_6\text{H}_{14}\text{O}_4}{150.2 \text{ g C}_6\text{H}_{14}\text{O}_4} \\ &= 0.0277 \text{ mol C}_6\text{H}_{14}\text{O}_4 \end{aligned}$$

Convert from moles of $\text{C}_6\text{H}_{14}\text{O}_4$ to moles of CO_2 by using the stoichiometric factor.

$$\begin{aligned} ? \text{ mol CO}_2 &= 0.0277 \text{ mol C}_6\text{H}_{14}\text{O}_4 \times \frac{12 \text{ mol CO}_2}{2 \text{ mol C}_6\text{H}_{14}\text{O}_4} \\ &= 0.166 \text{ mol CO}_2 \end{aligned}$$

Convert from moles of CO_2 to grams of CO_2 by using the molar mass of CO_2 as a conversion factor.

$$\begin{aligned} ? \text{ g CO}_2 &= 0.166 \text{ mol CO}_2 \times \frac{44.01 \text{ g CO}_2}{1 \text{ mol CO}_2} \\ &= 7.31 \text{ g CO}_2 \end{aligned}$$

(continued)

In the *conversion pathway approach*, the individual steps are combined into a single line calculation, as shown below.

$$? \text{ g CO}_2 = 4.16 \text{ g C}_6\text{H}_{14}\text{O}_4 \times \underbrace{\frac{1 \text{ mol C}_6\text{H}_{14}\text{O}_4}{150.2 \text{ g C}_6\text{H}_{14}\text{O}_4}}_{\text{converts to moles of C}_6\text{H}_{14}\text{O}_4} \times \underbrace{\frac{12 \text{ mol CO}_2}{2 \text{ mol C}_6\text{H}_{14}\text{O}_4}}_{\text{converts to moles of CO}_2} \times \underbrace{\frac{44.01 \text{ g CO}_2}{1 \text{ mol CO}_2}}_{\text{converts to grams of CO}_2} = 7.31 \text{ g CO}_2$$

Note the connection between the stepwise approach and the conversion pathway approach for this problem. In the conversion pathway approach, the first conversion factor converts from grams of $\text{C}_6\text{H}_{14}\text{O}_4$ to moles of $\text{C}_6\text{H}_{14}\text{O}_4$. The second conversion factor is the stoichiometric factor, and it converts from moles of $\text{C}_6\text{H}_{14}\text{O}_4$ to moles of CO_2 . The third conversion factor converts from moles of CO_2 to grams of CO_2 . The conversion pathway approach combines all three calculations into a single line.

Assess

A quick scan of the numbers to the right of 4.16 g $\text{C}_6\text{H}_{14}\text{O}_4$ indicates that the stoichiometric factor has a value of 6; 6×44.01 is between 250 and 300 which, when divided by 150.2, yields a factor somewhat smaller than 2. The mass of CO_2 should be somewhat less than twice that of the $\text{C}_6\text{H}_{14}\text{O}_4$, and it is (compare 7.31 to 2×4.16). Also, note that all units cancel properly in the ultimate conversion from g $\text{C}_6\text{H}_{14}\text{O}_4$ to g CO_2 .

PRACTICE EXAMPLE A: How many grams of magnesium nitride, Mg_3N_2 , are produced by the reaction of 3.82 g Mg with an excess of N_2 ?

PRACTICE EXAMPLE B: How many grams of H_2 are required to produce 1.00 kg methanol, CH_3OH , by the reaction $\text{CO} + 2 \text{H}_2 \longrightarrow \text{CH}_3\text{OH}$?

EXAMPLE 4-4

Relating the Masses of Two Reactants to Each Other

What mass of O_2 is consumed in the complete combustion of 6.86 g of triethylene glycol, $\text{C}_6\text{H}_{14}\text{O}_4$?

Analyze

The required conversions are $\text{g C}_6\text{H}_{14}\text{O}_4 \xrightarrow{1} \text{mol C}_6\text{H}_{14}\text{O}_4 \xrightarrow{2} \text{mol O}_2 \xrightarrow{3} \text{g O}_2$.

Solve

We will first use a stepwise approach to solve this problem.

Convert from grams of $\text{C}_6\text{H}_{14}\text{O}_4$ to moles of $\text{C}_6\text{H}_{14}\text{O}_4$ by using the molar mass of $\text{C}_6\text{H}_{14}\text{O}_4$ as a conversion factor.

$$\begin{aligned} ? \text{ mol C}_6\text{H}_{14}\text{O}_4 &= 6.86 \text{ g C}_6\text{H}_{14}\text{O}_4 \times \frac{1 \text{ mol C}_6\text{H}_{14}\text{O}_4}{150.2 \text{ g C}_6\text{H}_{14}\text{O}_4} \\ &= 0.0457 \text{ mol C}_6\text{H}_{14}\text{O}_4 \end{aligned}$$

Convert from moles of $\text{C}_6\text{H}_{14}\text{O}_4$ to moles of O_2 by using the stoichiometric factor.

$$\begin{aligned} ? \text{ mol O}_2 &= 0.0457 \text{ mol C}_6\text{H}_{14}\text{O}_4 \times \frac{15 \text{ mol O}_2}{2 \text{ mol C}_6\text{H}_{14}\text{O}_4} \\ &= 0.343 \text{ mol O}_2 \end{aligned}$$

Convert from moles of O_2 to grams of O_2 by using the molar mass of O_2 as a conversion factor.

$$\begin{aligned} ? \text{ g O}_2 &= 0.343 \text{ mol O}_2 \times \frac{32.00 \text{ g O}_2}{1 \text{ mol O}_2} \\ &= 11.0 \text{ g O}_2 \end{aligned}$$

As in Example 4-3, the three steps can be combined into a single calculation, as shown below.

$$? \text{ g O}_2 = 6.86 \text{ g C}_6\text{H}_{14}\text{O}_4 \times \frac{1 \text{ mol C}_6\text{H}_{14}\text{O}_4}{150.2 \text{ g C}_6\text{H}_{14}\text{O}_4} \times \frac{15 \text{ mol O}_2}{2 \text{ mol C}_6\text{H}_{14}\text{O}_4} \times \frac{32.00 \text{ g O}_2}{1 \text{ mol O}_2} = 11.0 \text{ g O}_2$$

Assess

Focus on the single line calculation shown above. A quick scan of the numbers to the right of 6.86 g $\text{C}_6\text{H}_{14}\text{O}_4$ indicates that the stoichiometric factor has a value of 7.5; the product, 7.5×32.00 , is about 250,

which, when divided by 150.2, yields a factor of about $250/150 = 5/3$. The mass of O_2 should be about $5/3$ that of $C_6H_{14}O_4$, and it is—that is, compare 11.0 with $5/3$ of 6.86, which is about $35/3$ or somewhat less than 12. As in Example 4-3, note that the proper cancellation of units occurs.

PRACTICE EXAMPLE A: For the reaction $4 NH_3 + 3 O_3 \rightarrow 2 N_2 + 6 H_2O$, how many grams of NH_3 are consumed per gram of O_2 ?

PRACTICE EXAMPLE B: In the combustion of octane, C_8H_{18} , how many grams of O_2 are consumed per gram of octane?



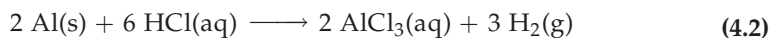
4-2 Concept Assessment

Which statements are correct for the reaction $2 H_2S + SO_2 \longrightarrow 3 S + 2 H_2O$? Explain your reasoning.

- (a) 3 mol S is produced per mole of H_2S consumed.
- (b) 3 g S is produced for every gram of SO_2 consumed.
- (c) 1 mol H_2O is produced per mole of H_2S consumed.
- (d) Two-thirds of the S produced comes from H_2S .
- (e) The number of moles of products formed equals the number of moles of reactants consumed.
- (f) The number of moles of atoms present after the reaction is the same as the number of moles of atoms before the reaction.

What lends great variety to stoichiometric calculations is that many other conversions may be required before and after the mol A $\longrightarrow$ mol B step at the center of the stoichiometric scheme shown in Figure 4-3. In Examples 4-3 and 4-4, the additional conversions involved molar mass. Other common conversions may require such factors as volume, density, and percent composition. In every case, however, we must always use the appropriate stoichiometric factor from the chemical equation as a key conversion factor.

The reaction between solid aluminum, $Al(s)$, and aqueous hydrochloric acid, $HCl(aq)$, can be used for preparing small volumes of hydrogen gas, $H_2(g)$, in the laboratory. A balanced chemical equation for the reaction is shown below.



A simple laboratory setup for collecting the hydrogen gas is pictured in Figure 4-4. The reaction between $Al(s)$ and $HCl(aq)$ provides a range of possibilities for calculations. Examples 4-5 and 4-6 are based on this reaction.

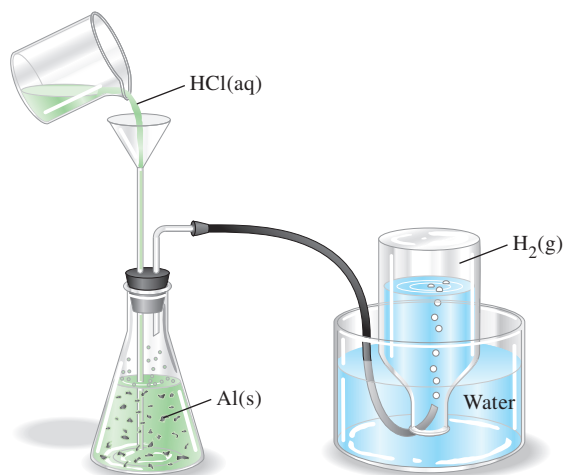


Figure 4-4 The reaction $2 Al(s) + 6 HCl(aq) \longrightarrow 2 AlCl_3(aq) + 3 H_2(g)$

$HCl(aq)$ is added to the flask on the left. The reaction occurs within the flask. The liberated $H_2(g)$ flows into a gas-collection apparatus, where it displaces water. Hydrogen is only very slightly soluble in water.

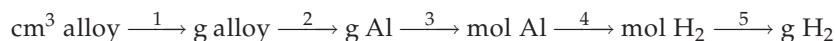
EXAMPLE 4-5

Additional Conversion Factors in a Stoichiometric Calculation:
Volume, Density, and Percent Composition

An alloy used in aircraft structures consists of 93.7% Al and 6.3% Cu by mass. The alloy has a density of 2.85 g/cm³. A 0.691 cm³ piece of the alloy reacts with an excess of HCl(aq). If we assume that *all* the Al but *none* of the Cu reacts with HCl(aq), what is the mass of H₂ obtained? Refer to reaction (4.2).

Analyze

A simple approach to this calculation is outlined below. Each numbered arrow refers to a conversion factor that changes the unit on the left to the one on the right.



The calculation can be done in five distinct steps, or with a single setup in which the five conversions are performed in sequence.

Solve

Using a stepwise approach, we proceed as follows.

Convert from volume of alloy to grams of alloy by using the density as a conversion factor.

$$\begin{aligned} ? \text{ g alloy} &= 0.691 \text{ cm}^3 \text{ alloy} \times \frac{2.85 \text{ g alloy}}{1 \text{ cm}^3 \text{ alloy}} \\ &= 1.97 \text{ g alloy} \end{aligned}$$

Convert from grams of alloy to grams of Al by using the percentage by mass of Al as a conversion factor.

$$\begin{aligned} ? \text{ g Al} &= 1.97 \text{ g alloy} \times \frac{93.7 \text{ g Al}}{100 \text{ g alloy}} \\ &= 1.85 \text{ g Al} \end{aligned}$$

Convert from grams of Al to moles of Al by using the molar mass of Al as a conversion factor.

$$\begin{aligned} ? \text{ mol Al} &= 1.85 \text{ g Al} \times \frac{1 \text{ mol Al}}{26.98 \text{ g Al}} \\ &= 0.0684 \text{ mol Al} \end{aligned}$$

Convert from moles of Al to moles of H₂ by using the stoichiometric factor.

$$\begin{aligned} ? \text{ mol H}_2 &= 0.0684 \text{ mol Al} \times \frac{3 \text{ mol H}_2}{2 \text{ mol Al}} \\ &= 0.103 \text{ mol H}_2 \end{aligned}$$

Convert from moles of H₂ to grams of H₂ using the molar mass of H₂ as a conversion factor.

$$\begin{aligned} ? \text{ g H}_2 &= 0.103 \text{ mol H}_2 \times \frac{2.016 \text{ g H}_2}{1 \text{ mol H}_2} \\ &= 0.207 \text{ g H}_2 \end{aligned}$$

Remember to store intermediate results without rounding off. When all of the steps are combined into a single calculation, we do not have to write down intermediate results and we reduce the likelihood of rounding errors.

$$\begin{aligned} ? \text{ g H}_2 &= 0.691 \text{ cm}^3 \text{ alloy} \times \frac{2.85 \text{ g alloy}}{1 \text{ cm}^3 \text{ alloy}} \times \frac{93.7 \text{ g Al}}{100 \text{ g alloy}} \times \frac{1 \text{ mol Al}}{26.98 \text{ g Al}} \times \frac{3 \text{ mol H}_2}{2 \text{ mol Al}} \times \frac{2.016 \text{ g H}_2}{1 \text{ mol H}_2} \\ &= 0.207 \text{ g H}_2 \end{aligned}$$

Assess

The units work out properly, but we must evaluate whether the answer is a reasonable one. The molar masses of Al and H₂ are approximately 27 g/mol and 2 g/mol, respectively. Equation (4.2) tells us that 1 mole of Al, which weighs approximately 27 g, produces 1.5 mol of H₂, which weighs 1.5 × 2 = 3 g. Thus, 27 g Al produces approximately 3 g H₂ and, thus, 2.7 g Al produces approximately 0.3 g H₂. In this example, we are dealing with less than 2.7 g of Al; therefore, we expect less than 0.3 g of H₂. The answer, 0.207 g H₂, is reasonable.

PRACTICE EXAMPLE A: What volume of the aluminum-copper alloy described in Example 4-5 must be dissolved in an excess of HCl(aq) to produce 1.00 g H₂? [*Hint:* Think of this as the “inverse” of Example 4-5.]

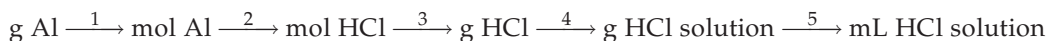
PRACTICE EXAMPLE B: A fresh sample of the aluminum-copper alloy described in Example 4-5 yielded 1.31 g H₂. How many grams of copper were present in the sample?

EXAMPLE 4-6**Additional Conversion Factors in a Stoichiometric Calculation:
Volume, Density, and Percent Composition of a Solution**

A hydrochloric acid solution consists of 28.0% HCl by mass and has a density of 1.14 g/mL. What volume of this solution is required to react completely with 1.87 g Al in reaction (4.2)?

Analyze

The first challenge here is to determine where to begin. Although the problem refers to 28.0% HCl and a density of 1.14 g/mL, the appropriate starting point is with the given information—1.87 g Al. The goal of our calculation is a solution volume—mL HCl solution.



The conversion factors in the calculation involve (1) the molar mass of Al, (2) stoichiometric coefficients from equation (4.2), (3) the molar mass of HCl, (4) the percent composition of the HCl solution, and (5) the density of the HCl solution.

Solve

Using a stepwise approach, we proceed as follows.

Convert from grams of Al to moles of Al by using the molar mass of Al. $? \text{ mol Al} = 1.87 \text{ g Al} \times \frac{1 \text{ mol Al}}{26.98 \text{ g Al}} = 0.0693 \text{ mol Al}$

Convert from moles of Al to moles of HCl by using the stoichiometric factor. $? \text{ mol HCl} = 0.0693 \text{ mol Al} \times \frac{6 \text{ mol HCl}}{2 \text{ mol Al}} = 0.208 \text{ mol HCl}$

Convert from moles of HCl to grams of HCl by using the molar mass of HCl. $? \text{ g HCl} = 0.208 \text{ mol HCl} \times \frac{36.46 \text{ g HCl}}{1 \text{ mol HCl}} = 7.58 \text{ g HCl}$

Convert from grams of HCl to grams of HCl solution by using the percentage by mass. $? \text{ g HCl soln} = 7.58 \text{ g HCl} \times \frac{100 \text{ g HCl soln}}{28.0 \text{ g HCl}} = 27.1 \text{ g HCl soln}$

Convert from grams of HCl solution to milliliters of HCl solution by using the density $? \text{ mL HCl soln} = 27.1 \text{ g HCl soln} \times \frac{1 \text{ mL HCl soln}}{1.14 \text{ g HCl soln}} = 23.8 \text{ mL HCl soln}$

In the conversion pathway approach, we combine the individual steps into a single line.

$$\begin{aligned} & (\text{g Al} \xrightarrow{1} \text{mol Al} \xrightarrow{2} \text{mol HCl} \xrightarrow{3} \text{g HCl} \\ ? \text{ mL HCl soln} &= 1.87 \text{ g Al} \times \frac{1 \text{ mol Al}}{26.98 \text{ g Al}} \times \frac{6 \text{ mol HCl}}{2 \text{ mol Al}} \times \frac{36.46 \text{ g HCl}}{1 \text{ mol HCl}} \\ & \xrightarrow{4} \text{g HCl soln} \xrightarrow{5} \text{mL HCl soln}) \\ & \times \frac{100.0 \text{ g HCl soln}}{28.0 \text{ g HCl}} \times \frac{1 \text{ mL HCl soln}}{1.14 \text{ g HCl soln}} \\ &= 23.8 \text{ mL HCl soln} \end{aligned}$$

Assess

Let's attempt to establish whether the answer is reasonable by working the problem in reverse and using numbers that are rounded off slightly. Because the density of the solution is approximately 1 g/mL and the solution is approximately 30% HCl by mass, a 24 mL sample of the solution will contain approximately $24 \times 0.30 = 7.2 \text{ g}$ of HCl. Equation (4.2) tells us that 1 mol Al, or 27 g Al, reacts with 3 mol HCl, or 108 g HCl. Stated another way, 4 g HCl reacts with 1 g Al. In this example, we are using approximately 7.2 g HCl; thus, we consume approximately $7.2 \text{ g HCl} \times 1 \text{ g Al}/4 \text{ g HCl} = 1.8 \text{ g Al}$. This value is close to the actual amount of Al consumed and we conclude that our answer, 23.8 mL of HCl solution, is reasonable.

PRACTICE EXAMPLE A: How many milligrams of H_2 are produced when one drop (0.05 mL) of the hydrochloric acid solution described in Example 4-6 reacts with an excess of aluminum in reaction (4.2)?

PRACTICE EXAMPLE B: A particular vinegar contains 4.0% CH_3COOH by mass. It reacts with sodium carbonate to produce sodium acetate, carbon dioxide, and water. How many grams of carbon dioxide are produced by the reaction of 5.00 mL of this vinegar with an excess of sodium carbonate? The density of the vinegar is 1.01 g/mL.



4-3 Concept Assessment

Without performing detailed calculations, determine which reaction produces the maximum quantity of $\text{O}_2(\text{g})$ per gram of reactant.

- (a) $2 \text{NH}_4\text{NO}_3(\text{s}) \xrightarrow{\Delta} 2 \text{N}_2(\text{g}) + 4 \text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g})$
 (b) $2 \text{Ag}_2\text{O}(\text{s}) \xrightarrow{\Delta} 4 \text{Ag}(\text{s}) + \text{O}_2(\text{g})$
 (c) $2 \text{HgO}(\text{s}) \xrightarrow{\Delta} 2 \text{Hg}(\text{l}) + \text{O}_2(\text{g})$
 (d) $2 \text{Pb}(\text{NO}_3)_2(\text{s}) \xrightarrow{\Delta} 2 \text{PbO}(\text{s}) + 4 \text{NO}_2(\text{g}) + \text{O}_2(\text{g})$

4-3 Chemical Reactions in Solution

Most reactions in the general chemistry laboratory are carried out in solution. This is partly because mixing the reactants in solution helps to achieve the close contact between atoms, ions, or molecules necessary for a reaction to occur. The stoichiometry of reactions in solutions can be described in the same way as the stoichiometry of other reactions, as we saw in Example 4-6. A few new ideas that apply specifically to solution stoichiometry are also helpful.

One component of a solution, called the **solvent**, determines whether the solution exists as a solid, liquid, or gas. In this discussion we will limit ourselves to *aqueous solutions*—solutions in which liquid water is the solvent. The other components of a solution, called **solutes**, are dissolved in the solvent. We use the notation $\text{NaCl}(\text{aq})$, for example, to describe a solution in which liquid water is the solvent and NaCl is the solute. The term *aqueous* does not convey any information, however, about the relative proportions of NaCl and H_2O in the solution. For this purpose, the property called **molarity** is commonly used.

Molarity

The composition of a solution may be specified by giving its molar concentration (or molarity), which is defined as the amount of solute, in moles, per liter of solution:

$$\text{molarity} = \frac{\text{amount of solute (in moles)}}{\text{volume of solution (in liters)}}$$

The expression above can be written more compactly as

$$c = \frac{n}{V} \quad (4.3)$$

where c is the molarity in moles per liter (mol/L), n is the amount of solute in moles (mol), and V is the volume of the solution in liters (L).

If 0.440 mol urea, $\text{CO}(\text{NH}_2)_2$, is dissolved in enough water to make 1.000 L of solution, the solution concentration, or molarity, is

$$\frac{0.440 \text{ mol CO}(\text{NH}_2)_2}{1.000 \text{ L soln}} = 0.440 \text{ M CO}(\text{NH}_2)_2$$

The symbol M stands for the term *molar*, or mol/L . Thus, a solution that has $0.440 \text{ mol CO}(\text{NH}_2)_2/\text{L}$ is $0.440 \text{ M CO}(\text{NH}_2)_2$ or $0.440 \text{ molar CO}(\text{NH}_2)_2$.

Alternatively, if 0.110 mol urea is present in 250.0 mL of solution, the solution is also 0.440 M .

$$\frac{0.110 \text{ mol CO}(\text{NH}_2)_2}{0.2500 \text{ L soln}} = 0.440 \text{ M CO}(\text{NH}_2)_2$$

When calculating concentration, we must determine the amount of solute in moles from other quantities that can be readily measured, such as the mass

► The IUPAC-preferred term for molarity is *amount concentration*. The use of the term *molarity* is still widespread.

of solute or the volume of a liquid solute. In Example 4-7, the mass of a liquid solute is related to its volume by using density as a conversion factor. Then, molar mass is used to convert from the mass of solute to an amount of solute in moles. Also, the solution volume is converted from milliliters (mL) to liters (L).

Figure 4-5 illustrates a method commonly used to prepare a solution. A solid sample is weighed out and dissolved in sufficient water to produce a solution of known volume—250.0 mL. To fill a beaker to a 250 mL mark is not nearly precise enough as a volume measurement. As discussed in Section 1-6, there would be a systematic error because the beaker is not calibrated with sufficient precision. (The error in volume could be 10 to 20 mL or more.) To dissolve the solute in water and bring the solution level to the 250 mL mark in a graduated cylinder is also not sufficiently precise. Although the graduated cylinder is calibrated more precisely than the beaker, the error might still be 1 to 2 mL or more. However, when filled to the calibration mark, the volumetric flask pictured in Figure 4-5 contains 250.0 mL with only about 0.1 mL of error.

When the molarity of a solution is known accurately, we can calculate the number of moles of solute in a carefully measured volume of solution by using the equation below.

$$n = c \times V \quad (4.4)$$

The expression above is equivalent to equation (4.3), but equation (4.4) emphasizes that the molarity, c , is a conversion factor for converting from liters of solution to moles of solute. In Example 4-8, we use molarity as a conversion factor in a calculation to determine the mass of solute needed to produce the solution in Figure 4-5.

Keep in Mind

that the volume used in molarity is that of the *solution*. It is neither that of the solvent used, nor the sum of the volumes of the solvent and solute. For example, if we add 25.0 mL ethanol to 250.0 mL water, the solution volume will be neither 250.0 mL nor 275.0 mL. We learn why in Chapter 14.

EXAMPLE 4-7

Calculating Molarity from Measured Quantities

A solution is prepared by dissolving 25.0 mL ethanol, $\text{CH}_3\text{CH}_2\text{OH}$ ($d = 0.789 \text{ g/mL}$), in enough water to produce 250.0 mL solution. What is the molarity of ethanol in the solution?

Analyze

We must first calculate how many moles of ethanol are in a 25.0 mL sample of pure ethanol. This calculation requires the following conversions: $\text{mL ethanol} \xrightarrow{1} \text{g ethanol} \xrightarrow{2} \text{mol ethanol}$. The first conversion uses the density as a conversion factor and the second conversion uses molar mass as a conversion factor. The molarity of the solution is then calculated by using equation (4.3).

Solve

The number of moles of ethanol in a 25.0 mL sample of pure ethanol is calculated below in a single line.

$$\begin{aligned} ? \text{ mol CH}_3\text{CH}_2\text{OH} &= 25.0 \text{ mL CH}_3\text{CH}_2\text{OH} \times \frac{0.789 \text{ g CH}_3\text{CH}_2\text{OH}}{1 \text{ mL CH}_3\text{CH}_2\text{OH}} \times \frac{1 \text{ mol CH}_3\text{CH}_2\text{OH}}{46.07 \text{ g CH}_3\text{CH}_2\text{OH}} \\ &= 0.428 \text{ mol CH}_3\text{CH}_2\text{OH} \end{aligned}$$

To apply the definition of molarity given in expression (4.3), note that $250.0 \text{ mL} = 0.2500 \text{ L}$.

$$\text{molarity} = \frac{0.428 \text{ mol CH}_3\text{CH}_2\text{OH}}{0.2500 \text{ L soln}} = 1.71 \text{ M CH}_3\text{CH}_2\text{OH}$$

Assess

It is important to include the units in this calculation to ensure that we obtain the correct units for the final answer. When dealing with liquid solutes, be careful to distinguish between mL solute and mL soln.

PRACTICE EXAMPLE A: A 22.3 g sample of acetone (see the model here) is dissolved in enough water to produce 1.25 L of solution. What is the molarity of acetone in this solution?

PRACTICE EXAMPLE B: If 15.0 mL of acetic acid, CH_3COOH ($d = 1.048 \text{ g/mL}$), is dissolved in enough water to produce 500.0 mL of solution, then what is the molarity of acetic acid in the solution?

